

Chapter 39

H264 Encoder

39.1 Overview

The H264 encoder is based on the H264 standard. It is used for real-time compression of video sequences, which can significantly reduce the total amount of data with minimal loss of video quality.

39.2 Terminology

bitstream	A sequence of bits that forms the representation of one or more coded video sequences.
profile	A subset of the entire bitstream syntax that is specified by the H264 standard.
level	Specified within each profile. A specified set of constraints imposed on values of the syntax elements in the bitstream.
MB	The full name is macroblock. It is a 16 x 16 block of luma samples and two corresponding blocks of chroma samples of a picture that has three sample arrays.
slice	An integer number of MBs.
I frame	Intra frame.
P frame	Inter frame.
I slice	Slice in I frame.
P slice	Slice in P frame.
P-skip MBs	MBs in P frame that skip the encoding.
CAVLC	Context-based adaptive variable length coding.
MV	Motion vector.
GOP	Group of pictures. In a GOP, the first frame is an I frame, and subsequent frames are P frames.
QP	Quantization parameter.
ROI	Region of interest.
AXI	Advanced extensible interface.
TX	Transmit.
RX	Receive.
DB	Deblocking filter.
NUM_H_MB	Number of horizontal MBs in a picture.

39.3 Features

The H264 encoder supports the following features:

- Baseline profile and level 4.1.
- 8-bit YUV 4:2:0 progressive video with a maximum resolution of 1920 × 1088 @ 30fps.
- I and P frames.
- Two working modes: GOP mode and dual stream mode. In dual stream mode, the total bandwidth of the two video sequences to be encoded should not exceed 1920 x 1088 @ 30fps.
- 4 x 4 and 16 x 16 intra frame luma MB segmentation.
- All 9 prediction modes for intra 4 x 4 luma MBs, and all 4 prediction modes for intra 16 x 16 luma MBs.
- All 4 prediction modes for intra chroma MBs.
- All partition size for inter luma MBs: 4 x 4, 4 x 8, 8 x 4, 8 x 8, 8 x 16, 16 x 8, 16 x 16.
- 1/2 and 1/4 pixel precision motion estimation.
- Horizontal motion search range [-29.75, +16.75] and vertical motion search range [-13.75, +13.75].
- Deblocking filter.
- CAVLC.
- P-skip MBs.
- I MB in P slices.
- Decimation operation of luma and chroma component quantization results.
- Fixed QP and MB level rate control.
- MV merge function. If the MV of MB is greater than 0, it can be output to memory.
- ROI function, with up to 8 rectangular ROI configurable at any position (allowing overlap, fixed priority). Each ROI can be configured with a constant QP or QP offset, and non-ROI can be configured with a QP offset.

39.4 Architecture

The architecture of H264 is shown in Figure [39.4-1](#).

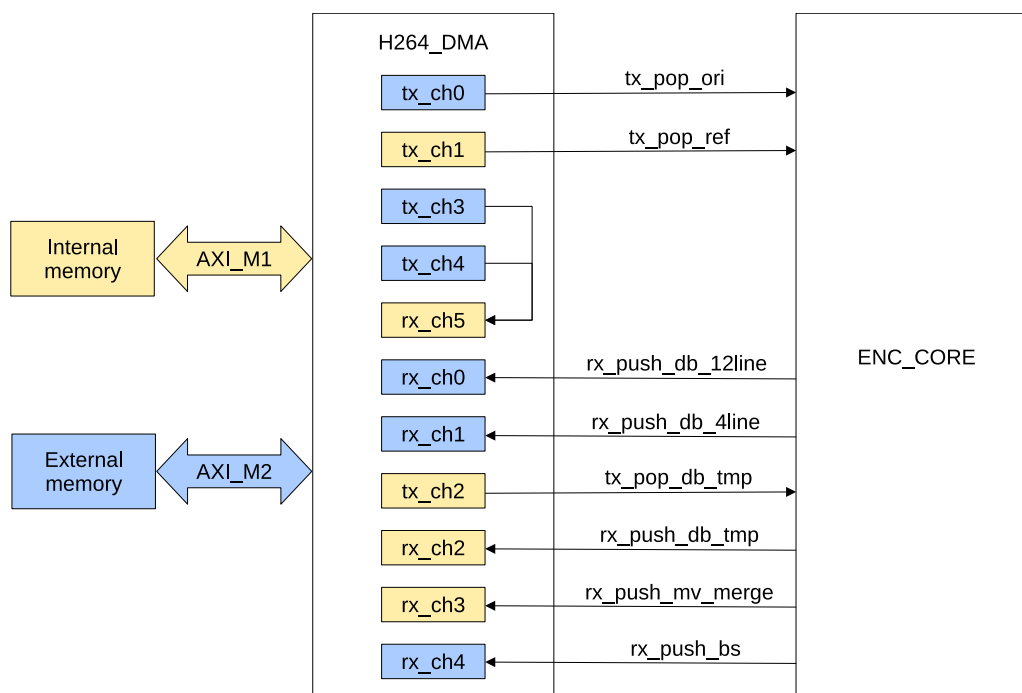


Figure 39.4-1. H264 Video Encoder Architecture

The H264 encoder can be divided into two parts, H264 dedicated DMA (H264_DMA) and encoder algorithm core (ENC_CORE). H264_DMA is used for data transfer, and is connected to the internal/external memory through two sets of AXI buses. Meanwhile, it is connected to the ENC_CORE through multiple DMA sending/receiving channels (H264_DMA has 5 sending channels (TX channel *n*) and 6 receiving channels (RX channel *n*)). TX channel 3, TX channel 4, and RX channel 5 are the internal channels of H264_DMA, which are not connected to the ENC_CORE. The ENC_CORE is used to compress video sequences.

The function of each H264_DMA sending and receiving channel is as follows:

- **TX channel 0** sends the original picture to ENC_CORE.
- **TX channel 1** sends the reference picture to ENC_CORE.
- **TX channel 2** sends the deblocking filter intermediate data to ENC_CORE.
- Sending channel **TX channel 3** and **TX channel 4** send the reference picture (i.e., the previous deblocking filtered picture) to receiving channel **RX channel 5**.
- **RX channel 0** and **RX channel 1** send the result of the deblocking filter (i.e., reference picture) to external memory.
- **RX channel 2** sends the deblocking filter intermediate data to internal memory.
- **RX channel 3** sends the MV merge result to internal memory.
- **RX channel 4** sends the encoded bitstream to external memory.
- **RX channel 5** sends the reference picture (i.e., the previous deblocking filtered picture) to internal memory.

From the provided description, the H264_DMA channels can be divided into two types:

- Reading or writing internal memory (through AXI_M1 bus): TX channel 1/2, RX channel 2/3/5

- Reading or writing external memory (through AXI_M2 bus): TX channel 0/3/4, RX channel 0/1/4

39.5 Functional Description

39.5.1 Encoder Algorithm Core (ENC_CORE)

The function of the ENC_CORE is to compress and encode the video sequences according to the software configuration.

39.5.1.1 Architecture

The architecture of the encoder algorithm core is shown in Figure 39.5-1 *ENC_CORE Architecture*.

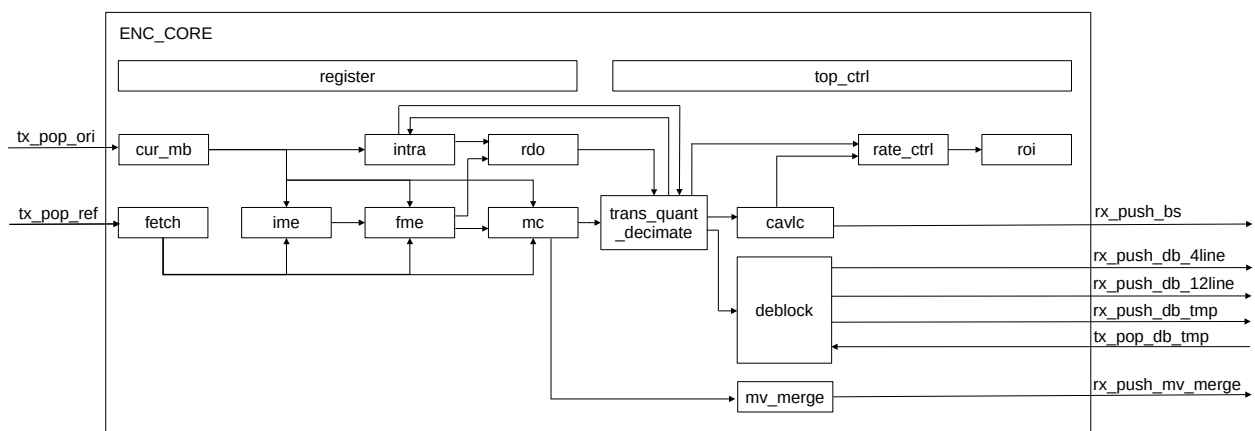


Figure 39.5-1. ENC_CORE Architecture

The functions of each module are as follows:

- register: Implements register configuration and selection.
- top_ctrl: Realizes the control and scheduling of each module of the encoder algorithm core.
- cur_mb: Reads the original picture MB from the TX channel 0 (tx_pop_ori) of H264_DMA, and distributes it to the intra, ime, fme, and mc modules.
- intra: Uses the reconstruction result calculated by the trans_quant_decimate module to perform intra-frame prediction of the MB, sends the intra-frame prediction and residual results to the trans_quant_decimate module, and sends the intra-frame prediction cost to the rdo module.
- fetch: Reads reference picture MBs within the search range from the TX channel 1 channel (tx_pop_ref) of H264_DMA, and distributes them to the ime, fme, and mc modules.
- ime: Receives the original picture MB from the cur_mb module, receives the reference picture MB from the fetch module, performs integer motion estimation within the search range, and sends the integer motion estimation result to the fme module.
- fme: Receives the original picture MB from the cur_mb module, receives the reference picture MB from the fetch module, receives the integer motion estimation result from the ime module, and performs

fractional motion estimation on the basis of the integer motion estimation. The fractional motion estimation result is sent to the mc module, and the inter prediction cost is sent to the rdo module.

- mc: Receives the original picture MB from the cur_mb module, receives the reference picture MB from the fetch module, receives the fractional motion estimation result from the fme module, performs motion compensation according to the fractional motion estimation result. The inter-frame prediction and the sum of the residual results are sent to the trans_quant_decimate module, and the motion vector is sent to the mv_merge module.
- rdo: Receives the intra-frame prediction cost from the intra module, receives the inter-frame prediction cost from the fme module, selects the final type of the current MB, implements the P slice to support the I MB function, and sends the final MB type to trans_quant_decimate module.
- trans_quant_decimate: Receives the final MB type from the rdo module, receives the intra prediction and residual results from the intra module, receives the inter prediction and residual results from the mc module. Then, it performs DCT, quantization, inverse quantization, and inverse DCT, and quantization result decimation, sends the residual information to the rate_ctrl module, sends the reduced quantization result to the cavlc module, and sends the reconstruction result to the deblock module.
- cavlc: Receives the quantization result from the trans_quant_decimate module and the prediction information of the current MB, performs context adaptive variable length coding (CAVLC), and writes the coded stream to the RX channel 4 of H264_DMA.
- deblock: Receives reconstruction result from the trans_quant_decimate module and the prediction information of the current MB, performs deblocking filtering, and writes the deblocking filter result (i.e., the reference picture) to the RX channel 0 (rx_push_db_12line) and RX channel 1 (rx_push_db_4line). This module reads and writes the deblocking filter intermediate data through the TX channel 2 channel (tx_pop_db_tmp) and RX channel 2 channel (rx_push_db_tmp) of H264_DMA.
- mv_merge: Receives the motion vector from the mc module, performs MV merging, and writes the merging result (MB motion information) to the RX channel 3 channel (rx_push_mv_merge) of H264_DMA.
- rate_ctrl: Receives the residual information from the trans_quant_decimate module, receives the encoding information from the cavlc module, receives the frame-level information configured by the register module, performs MB level rate control, and converts the calculated quantization parameter (QP) to the roi module.
- roi: Receives QPs from the rate_ctrl module, receives the ROI information configured by the register module, adjusts the QPs, and uses the adjusted QPs to process the current MB.

39.5.1.2 Quantization Result Decimation

The ENC_CORE calculates the score of each MB based on the magnitude of its final quantization result, and then compares the score with the configurable threshold register. If it is below the threshold, the quantization result will be set to 0. Otherwise, the quantization result will remain unchanged, so as to further improve the compression ratio. For each MB, its luma and chroma component carry out the above process independently. Different types of MBs have different comparison methods, as follows:

- luma component I MB with 16 x 16 partition:

$$l_score_mb_cmp + H264_x_I16\times16_DECI_SCORE_OFFSET < H264_x_L_DECI_SCORE$$

Among them, `l_score_mb_cmp` is the score of the luma component of the current MB calculated by `ENC_CORE`. `H264_x_I16×16_DECI_SCORE_OFFSET` is the configurable offset register. `H264_x_L_DECI_SCORE` is the configurable threshold register. `x` can be A or B, indicating the register corresponding to video sequence A or B. When the above requirement is met, the quantization result of the current luma component MB will be decimated to 0.

- luma component I MB with 4 x 4 partition: Current luma component MB quantization result will not be decimated to 0.
- chroma component I MB:

$$c_score_mb_cmp + H264_x_I_chroma_DECI_SCORE_OFFSET < H264_x_C_DECI_SCORE$$

Among them, `c_score_mb_cmp` is the score of current chroma component MB calculated by `ENC_CORE`. `H264_x_I_chroma_DECI_SCORE_OFFSET` is the configurable offset register. `H264_x_C_DECI_SCORE` is the configurable threshold register. `x` can be A or B, indicating the register corresponding to video sequence A or B. When the above requirement is met, the quantization result of the current chroma component MB will be decimated to 0.

- luma component P MB:

$$l_score_mb_cmp + H264_x_P16×16_DECI_SCORE_OFFSET < H264_x_L_DECI_SCORE$$

Among them, `l_score_mb_cmp` is the score of current luma component MB calculated by `ENC_CORE`. `H264_x_P16×16_DECI_SCORE_OFFSET` is the configurable offset register. `H264_x_L_DECI_SCORE` is the configurable threshold register. `x` can be A or B, indicating the register corresponding to video sequence A or B. When the above requirement is met, the quantization result of the current luma component MB will be decimated to 0.

- chroma component P MB:

$$c_score_mb_cmp + H264_x_P_chroma_DECI_SCORE_OFFSET < H264_x_C_DECI_SCORE$$

Among them, `c_score_mb_cmp` is the score of current chroma component MB calculated by `ENC_CORE`. `H264_x_P_chroma_DECI_SCORE_OFFSET` is the configurable offset register. `H264_x_C_DECI_SCORE` is the configurable threshold register. `x` can be A or B, indicating the register corresponding to video sequence A or B. When the above requirement is met, the quantization result of the current chroma component MB will be decimated to 0.

39.5.1.3 Motion Vector (MV) Merging

The `ENC_CORE` merges the MVs of the luma MB according to the configuration of the registers related to MV merging, and writes it out to the RX channel 3 channel (`rx_push_mv_merge`) of `H264_DMA`, and finally stores it in the internal memory for the upper layer application use, such as motion detection.

The registers related to MV merging are as follows:

- `H264_x_MV_MERGE_EN`: This register controls the enablement of MV merging. Setting this register to 1 means enabling the MV merging function of the corresponding video sequence. Setting this register to 0 means turning off the MV merging function of the corresponding video sequence. `x` can be A or B, indicating the register corresponding to video sequence A or B. Even if the MV merging function is enabled, the I MB will not output any data to the memory because the I MB has no MV.

- **H264_MV_MERGE_TYPE**: This register controls the type of MV merging. Configuring this register to 0 means only retaining the MV of the MB whose final type is P MB and the partition of luma component is 16 x 16. Configuring this register to 1 means only retaining the MV with the minimum modulus of the MB whose final type is P MB. Configuring this register to 2 means only retaining the MV with the maximum modulus of the MB whose final type is P MB. 3 is an invalid value. When the register **H264_x_MV_MERGE_EN** is set to 1, the configuration of this register will take effect.
- **H264_INT_MV_OUT_EN**: This register controls the type of merged MV ultimately stored in memory. Setting this register to 0 means that if the reserved merged MV is not 0, it will be stored in memory. Otherwise, it will be discarded. Setting this register to 1 means that if the integer part of the reserved merged MV is not 0, it will be stored in memory, otherwise discarded. When the register **H264_x_MV_MERGE_EN** is set to 1, the configuration of this register will take effect.
- **H264_MB_VALID_NUM**: This register represents the number of MV merging results stored in the memory in the current picture. One MB corresponds to one MV merging result. When the register **H264_x_MV_MERGE_EN** is set to 1, the value of this register is meaningful, otherwise the value of this register is meaningless and remains 0.

39.5.1.4 MB Level Rate Control

The ENC_CORE calculates the QP adjustment amount of each MB based on the configuration of the rate control related registers, the number of encoding bits of the current MB, and the mean absolute value of the residual (MAD). Then it uses the frame-level QP configured in the register and the QP adjustment amount to calculate the QP of each MB, thereby adjusting the number of coded bits.

The MB level rate control related registers are as follows:

- **H264_x_MB_RATE_CTRL_EN**: This register controls the enablement of MB level rate control. Setting this register to 1 means enabling the MB level rate control function of the corresponding video sequence. Setting this register to 0 means turning off the MB level rate control function of the corresponding video sequence. **x** can be A or B, indicating the register corresponding to video sequence A or B. Since the I frame uses a constant QP encoding method, the QP of each macro block of the I frame will not change even if the MB level rate control function is enabled.
- **H264_x_RATE_CTRL_U**: The value of this register is configured by the parameters calculated by the frame-level rate control in the software. ENC_CORE needs this value to calculate the QP adjustment amount of each macro block. **x** can be A or B, indicating the register corresponding to video sequence A or B. The calculation method of the configuration value of this register belongs to the category of the frame-level rate control algorithm in the software, which is beyond the scope of this chapter and will not be described.
- **H264_x_MAD_FRAME_PRED**: This register is the predicted value of frame-level MAD, which is configured by the frame-level rate control in the software. ENC_CORE needs this value to calculate the QP adjustment amount of each MB, **x** can be A or B, indicating the register corresponding to video sequence A or B. The calculation method of the configuration value of this register belongs to the category of the frame-level rate control algorithm in the software, which is beyond the scope of this chapter and will not be described.
- **H264_x_QP**: This register is the frame-level QP value, and the value range is 10 to 51. It is configured by the frame-level rate control in the software. ENC_CORE will adjust the QP value of each MB based on it, so as to realize the MB level rate control, **x** can be A or B, indicating the register corresponding to video

sequence A or B. The calculation method of the configuration value of this register belongs to the category of the frame-level rate control algorithm in the software, which is beyond the scope of this chapter and will not be described.

- **H264_x_QP_MAX**: This register is used to limit the maximum value of the QP finally calculated by the MB level rate control, and the value range is from 10 to 51. If the QP value finally calculated by the MB level rate control is greater than the configuration value of this register, it will be limited to the configured value of this register, otherwise the original value is retained. **x** can be A or B, indicating the register corresponding to video sequence A or B.
- **H264_x_QP_MIN**: This register is used to limit the minimum value of the QP finally calculated by the MB level rate control, and the value range is from 10 to 51. If the QP value finally calculated by the MB level rate control is smaller than the configuration value of this register, it will be limited to the configured value of this register, otherwise it remains the original value, **x** can be A or B, indicating the register corresponding to video sequence A or B.
- **H264_FRAME_MAD_SUM**: This register represents the cumulative sum of the MADs of all MBs in the current frame. The software needs to read the value of this register after the encoding of a frame of video is completed for calculation of frame-level rate control. The frame-level rate control algorithm in the software is beyond the scope of this chapter and will not be described.
- **H264_FRAME_ENC_BITS**: This register represents the sum of the number of encoding bits of all MBs in the current frame. The software needs to read the value of this register after completely encoding a frame of video to perform frame-level rate control. The frame-level rate control algorithm in the software is beyond the scope of this chapter and will not be described.
- **H264_FRAME_QP_SUM**: This register represents the sum of QP of all MBs in the current frame. The software needs to read the value of this register after completely encoding a frame of video to perform frame-level rate control. The frame-level rate control algorithm in the software is beyond the scope of this chapter and will not be described.

39.5.1.5 Region of Interest (ROI)

ENC_CORE can have up to 8 overlapping fixed-priority rectangular ROI in a frame of video according to the configuration of the relevant registers. The QP (calculated by MB level rate control) is adjusted, so that each ROI has different encoded picture quality.

The registers related to the ROI are as follows:

- **H264_x_ROI_EN**: This register is used to control the switch of the ROI function. Set it to 1 to enable the ROI function, and set it to 0 to disable the ROI function. **x** can be A or B, indicating the register corresponding to video sequence A or B.
- **H264_x_ROI_MODE**: This register is used to specify the working mode of ROI. If it is configured as 0, ROI will work in fixed QP mode. If it is configured as 1, ROI will work in QP offset mode. **x** can be A or B, indicating the register corresponding to video sequence A or B.
- **H264_x_ROI_REGION_n_REG**: This register is used to configure the enable and position information of each rectangular ROI. **x** can be A or B, indicating the register corresponding to video sequence A or B. **n** can be 0 to 7, indicating 8 ROI. From ROI 0 to ROI 7, the priority decreases in order. When there are overlapping areas, the configuration of the ROI with the highest priority takes effect. This register contains the following fields:

- `H264_x_ROI_REGIONn_EN`: Set to 1 to enable ROI *n*, set it to 0 to close the ROI *n*. When the register `H264_x_ROI_EN` set to 1, this register can control the ROI *n* switch, otherwise, this register has no effect.
- `H264_x_ROI_REGIONn_X`: Configure the horizontal starting MB coordinates of the ROI *n* in a frame of video.
- `H264_x_ROI_REGIONn_Y`: Configure the vertical starting MB coordinates of the ROI *n* in a frame of video.
- `H264_x_ROI_REGIONn_X_LEN`: Configure the number of horizontal MBs in the ROI *n* in a frame of video.
- `H264_x_ROI_REGIONn_Y_LEN`: Configure the number of vertical MBs in the ROI *n* in a frame of video.
- `H264_x_ROI_REGIONn_QP`: This register is used to configure the QP adjustment amount of the MBs in each rectangular ROI. *x* can be A or B, indicating the register corresponding to video sequence A or B. *n* can be 0 to 7, representing 8 ROI. The configuration method of this register is related to the configuration value of the `H264_x_ROI_MODE` register:
 - When `H264_x_ROI_MODE` is set to 0: It means that the ROI mode is fixed QP mode. At this time, the value of `H264_x_ROI_REGIONn_QP` register is directly As the QP value of the MB in the corresponding ROI, the value range is 10 to 51.
 - When `H264_x_ROI_MODE` is set to 1: It indicates that the ROI mode is QP offset mode. At this time, bit 0 to bit 5 of the `H264_x_ROI_REGIONn_QP` register indicates the QP offset, and the value range is 0 to 51. Bit 6 indicates the sign of the offset, where 0 indicates a positive offset, and 1 indicates a negative offset.
- `H264_x_NO_ROI_REGION_QP`: This register is used to configure the QP offset of the MB in the non-ROI. Bit 0 to bit 5 represent the QP offset, and the value range is 0 to 51. Bit 6 represents the sign of the offset, where 0 represents a positive offset, and is 1 indicates a negative offset. *x* can be A or B, indicating the register corresponding to video sequence A or B. When the ROI function is disabled, that is, when the `H264_x_ROI_EN` register is set to 0, this register will not take effect, and the QP of the MB in the non-ROI will not be changed.

Please note that the adjusted QP value of any ROI should not be less than 10.

Regardless of whether the MB is in the ROI or non-ROI, its final adjusted QP value will be limited to 0 to 51 to ensure that it does not exceed the value range in the H264 standard.

39.5.2 H264 Dedicated DMA

The H264 dedicated DMA controls the data transfers between encoder algorithm core and memory based on the software configuration.

39.5.2.1 Architecture

H264_DMA architecture is shown in the figure 39.5-2:

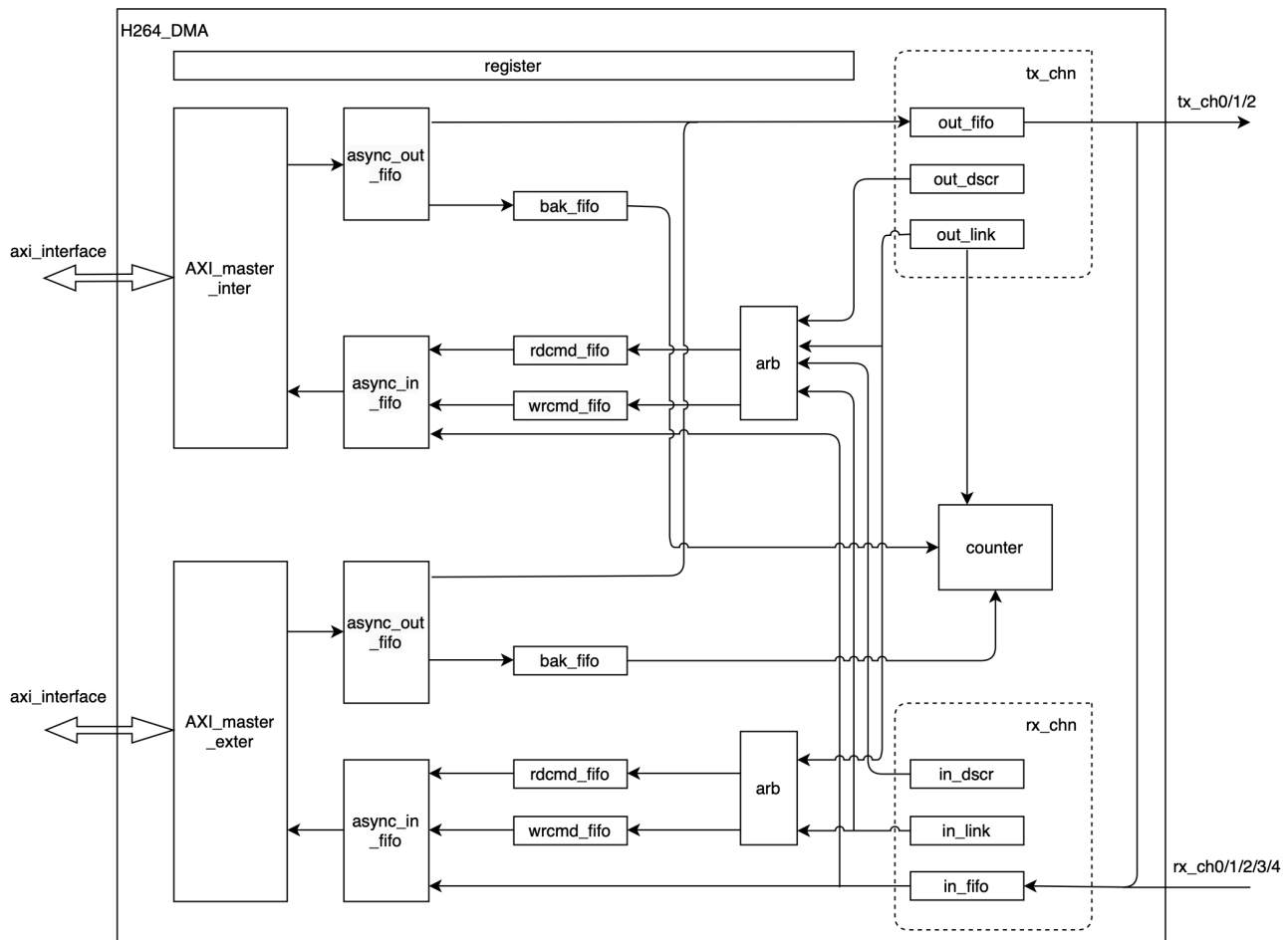


Figure 39.5-2. H264_DMA Architecture

The functions of each module are:

- **register:** This module implements register configuration and selection.
- **in_dscr:** This module initiates an AXI read request to internal memory, obtains the RX channel's linked list descriptor, and sends the descriptor to the in_link module of the corresponding RX channel.
- **in_link:** This module receives the descriptor of the corresponding channel in_dscr module and splits it into multiple AXI write data requests to guide the corresponding in_fifo module to write data to internal memory or external memory. RX channel 0/1/4 is designed for writing data to external memory, while RX channel 2/3/5 is for writing data to internal memory.
- **in_fifo:** Stores data transmitted by the h264 module or TX Channels 3/4, and waits for the AXI host to read the data from the corresponding channel. Specifically, the in_fifo of RX Channels 0/1/2/3/4 stores the data sent by the h264 module, while the in_fifo of RX Channel 5 stores the data transmitted by TX Channels 3/4.
- **out_dscr:** This module initiates an AXI read internal memory request, obtains the TX channel linked list descriptor, and sends the descriptor to the out_link module of corresponding TX channel.
- **out_link:** This module receives the descriptor of the corresponding channel out_dscr module and divides it into multiple AXI read data requests to obtain data from internal memory or external memory. TX channel 0/3/4 reads data from external memory and TX channel 1/2 reads data from internal memory.

- `out_fifo`: This module receives the data returned by the AXI read request and waits for h264 to read, or RX channel 5 to read. TX channel 0/1/2 is for h264 to read, and TX channel 3/4 is for RX channel 5 to read.
- `arb`: This module receives read/write requests for external or internal memory, and arbitrates the read/write requests. The requests are sent to the `rdcmd_fifo` and `wrcmd_fifo` modules of external memory, and the `rdcmd_fifo` and `wrcmd_fifo` modules of internal memory respectively.
- `wrcmd_fifo`: This module caches the AXI write request arbitrated by the `arb` module, and cuts the write request and data into multiple 8-byte data chunks before sending them to the `async_in_fifo` module.
- `rdcmd_fifo`: This module caches the AXI read request arbitrated by the `arb` module and sends the read request to the `async_in_fifo` module.
- `bak_fifo`: This module caches the read/write request information arbitrated by the `arb` module, including channel id, `AXI_id`, read/write request data length, read/write request source and read request starting address. The module finds the information corresponding to the request based on the `AXI_id` returned by the `async_out_fifo` module, and sends the completion information of the request and the read data in the `async_out_fifo` module to the corresponding channel.
- `async_in_fifo`: This module is used to receive AXI requests from the `wrcmd_fifo` module and `rdcmd_fifo` module, and synchronize the request from H264_DMA core clock domain to the AXI clock domain, waiting for the AXI master to read.
- `async_out_fifo`: This module is used to receive the read data returned by the `AXI_master_inter/AXI_master_exter` module and the completion of the read/write request, and synchronize it from the AXI clock domain to the H264_DMA core clock domain.
- `AXI_master_inter/AXI_master_exter`: This module reads the read/write request in the `async_in_fifo` module and converts it into an AXI bus protocol format output. The module receives the read data and read/write completion returned by the AXI slave and converts it into an internal private interface. The resulting format is written to the `async_out_fifo` module. The `AXI_master_inter` module is used to access the internal memory space, and the `AXI_master_exter` module is used to access the external memory space.

39.5.2.2 Linked List Descriptor

H264_DMA linked list descriptor is shown in the figure [39.5-3](#):

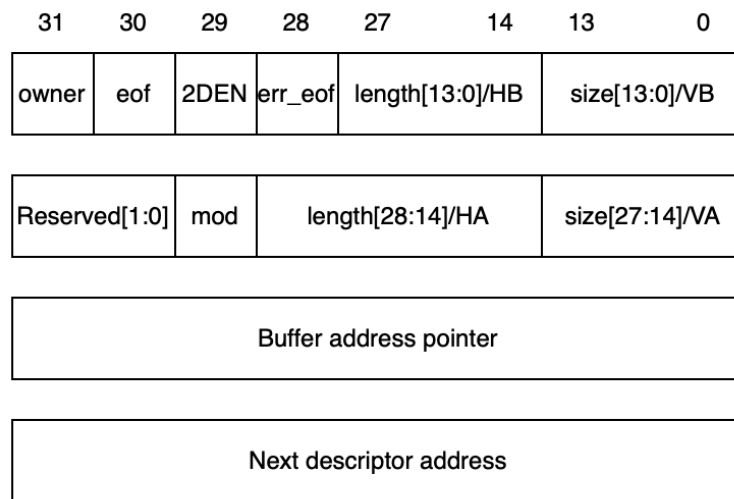


Figure 39.5-3. H264_DMA linked list descriptor

- **owner:** Indicates the operators allowed by the buffer corresponding to the current descriptor.
 - 0: Allow the operator to be CPU.
 - 1: The operator is allowed to be H264_DMA. H264_DMA will clear this bit after using the descriptor.
 - For transmit descriptors, since a descriptor will not be written back by default, the default owner will not be cleared. The user needs to configure H264_DMA_OUT_AUTO_WRBACK to enable writeback.
- **eof:** End flag.
 - 0: The current descriptor is not the last descriptor.
 - 1: The current descriptor is the last descriptor.
- **mod:** Data block read mode.
 - 0: H264_DMA read/write a picture block with a horizontal width of hb and a vertical height of vb only once.
 - 1: H264_DMA read/write multiple times the picture block with a horizontal width of hb and a vertical height of vb until the picture data with a horizontal width of HA and a vertical height of VA is read/written. Note: The unit of hb/vb/HA/VA is pixel.
 - In 1D mode, this field must be 0.
- **err_eof:** Receives end error flag.
 - For the receive descriptor, the hardware will set this bit to 1 after receiving a packet and detecting a received data error.
 - For transmit descriptors, this bit is fixed to 0.
- **2DEN:** 2D function enable signal.
 - 0: The function of H264_DMA is the same as the general DMA function. At this time, it can be used to transfer continuous data segments.
 - 1: The H264_DMA function is enabled and data is transferred according to the configured picture blocks.

- length[13:0]/HB:
 - When 2DEN = 0, this field represents the length of the data being transferred (least significant 14 bits). For data reception, this field is filled in after the hardware receives a frame of data. For data transmission, this field indicates the length of the data to be sent. The unit is byte.
 - When 2DEN = 1, this field represents the horizontal width of the picture block (HB). The unit is pixel.
- size[13:0]/VB:
 - When 2DEN = 0, this field represents the size of the memory space allocated for transferring data (least significant 14 bits). The unit is byte.
 - When 2DEN = 1, this field represents the vertical height of the picture block (VB). The unit is pixel.
- length[28:14]/HA:
 - When 2DEN = 0, this field represents the length of the transferred data (most significant 14 bits, only bit14–bit27 is used). For data reception, this field is filled after the hardware receives a frame of data. For data transmission, this field indicates the length of the data to be sent. The unit is byte.
 - When 2DEN = 1, this field represents the horizontal width of the picture corresponding to the descriptor (HA). The unit is pixel.
- size[27:14]/VA:
 - When 2DEN = 0, this field represents the size of the memory space for transferring data (most significant 14 bits). The unit is byte.
 - When 2DEN = 1, this field represents the vertical height of the picture corresponding to the descriptor (VA). The unit is pixel.
- Buffer address pointer:
 - When 2DEN = 0, it is the buffer address pointer.
 - When 2DEN = 1, it is the address pointer of the origin coordinates of the HA × VA picture block in the buffer.
- Next descriptor address: Next descriptor address pointer.

39.5.2.3 Transfer Initialization

H264_DMA will read the linked list from the internal memory to the local according to the configuration of the registers related to each channel linked list descriptor, so that it can be used by the corresponding channel to transfer data, thereby start the corresponding channel.

The registers associated with each linked list descriptor are as follows:

- H264_DMA_OUTLINK/INLINK_START_CH $\times$: This register controls whether the corresponding channel of H264_DMA starts to obtain descriptors. Setting this register to 1 means that the corresponding channel starts to obtain the descriptor. After obtaining the descriptor, the hardware automatically sets this register to 0, and then automatically obtain subsequent descriptors. $\times$ can be 0, 1, 2, 3, and 4, indicating the registers corresponding to channels 0, 1, 2, 3, and 4.
- H264_DMA_OUTLINK/INLINK_ADDR_CH $\times$: This register represents the memory address where the first descriptor is stored, and the address requires 8-byte alignment.

39.5.2.4 Transfer Reset

In order to ensure the safe reset of H264_DMA and prevent channel from being stuck, the reset of H264_DMA channel must be carried out according to the following procedure:

1. Write 1 to register H264_DMA_OUT/IN_CMD_DISABLE_CH_x of the corresponding channel.
2. Wait for H264_DMA_OUT/IN_RESET_AVAIL_CH_x of the corresponding channel to be set.
3. Write 1 to register H264_DMA_OUT/IN_RST_CH_x of the corresponding channel.
4. Write 0 to register H264_DMA_OUT/IN_RST_CH_x of the corresponding channel.

The registers related to channel reset are as follows:

- H264_DMA_OUT/IN_CMD_DISABLE_CH_x: This register controls whether the corresponding channel of H264_DMA stops initiating read/write requests to AXI. Setting this register to 1 means that the corresponding channel stops initiating read/write requests to AXI. Setting this register to 0 means that the corresponding channel can initiate read/write requests to AXI. _x can be 0, 1, and 2 in TX channel, indicating the registers corresponding to channels 0, 1, and 2. In RX channel, they can be 0, 1, 2, 3, 4, and 5, indicating the registers corresponding to channels 0, 1, 2, 3, 4, and 5.
- H264_DMA_OUT/IN_RESET_AVAIL_CH_x: This register indicates whether the channel corresponding to H264_DMA can be safely reset. If this register is 1, it means that the corresponding channel can be safely reset. If this register is 0, it means that the corresponding channel cannot be safely reset.
- H264_DMA_OUT/IN_RST_CH_x: This register controls whether the corresponding channel of H264_DMA performs soft reset. Setting this register to 1 means that the corresponding channel performs a soft reset. Setting this register to 0 means that the corresponding channel does not perform a soft reset.

Notes:

- Since TX channel 3, TX channel 4, and RX channel 5 are bound, these three channels only have one set of H264_DMA_OUT/IN_CMD_DISABLE_CH₅, H264_DMA_OUT/IN_RESET_AVAIL_CH₅, and H264_DMA_OUT/IN_RST_CH₅ registers. When resetting these three channels, you only need to operate this group of registers to realize the safe reset of the three H264_DMA channels.
- In order to ensure the completion of data transmission, before resetting RX channel 0, RX channel 1, RX channel 3, and RX channel 4, you need to check whether the corresponding channel linked list descriptor is written back successfully. Only after the linked list descriptor of the channel is written back successfully, which means all data transmission of the channel is completed, can the channel be reset. If the MV merging function is not enabled or the number of video frames in a GOP is 1, which means the register H264_GOP_NUM is configured as 1, there is no need to check whether the linked list descriptor of RX channel 3 is written back successfully.

39.5.2.5 Channel Configuration

Before the software starts H264_DMA, it needs to configure H264_DMA first. The configuration consists of register configuration and linked list descriptor configuration.

Linked list descriptor configuration: The channels that need to be configured with descriptors are TX channel 0 ~ TX channel 4 and RX channel 0 ~ RX channel 4. The linked list descriptors are all circular linked lists.

1. Configure the channel of reading the original picture (TX channel 0):

- mode: 2D, mode1.
 - length/size: HA is the number of horizontal pixels of the picture. VA is the number of vertical pixels of the picture, HB is 16 x 4, and VB is 16.
2. Configure the channel of reading reference picture (TX channel 1):
 - mode: 2D, mode1.
 - length/size: HA is NUM_H_MB x 16 x 16, VA is 3, HB is 16 x 16, and VB is 3.
 3. Configure the channel of reading DB intermediate data (TX channel 2):
 - mode: 1D.
 - length/size: The read space size is NUM_H_MB x (16 x 4 + 8 x 4 x 2).
 4. Configure the channel of reading first 12 lines of DB data (TX channel 3):
 - mode: 1D.
 - length/size: The read space size is (NUM_H_MB x (number of vertical macro block lines in the picture - 1) x (16 x 12 + 8 x 4 x 2)) + (NUM_H_MB x 16 x 16 x 1.5).
 5. Configure the channel of reading last 4 lines DB data (TX channel 4):
 - mode: 1D.
 - length/size: The read space size is (NUM_H_MB x (number of vertical MB lines in the picture - 1) x (16 x 4 + 8 x 4 x 2)).
 6. Configure the channel of writing first 12 lines of DB data (RX channel 0):
 - mode: 1D.
 - length/size: The size of the writing space is the same as the size of TX channel 3.
 7. Configure the channel of writing last 4 lines of DB data (RX channel 1):
 - mode: 1D.
 - length/size: The size of the writing space is the same as the size of TX channel 4.
 8. Configure the channel of writing DB intermediate data (RX channel 2):
 - mode: 1D.
 - length/size: The size of the writing space is the same as the size of TX channel 2.
 9. Configure the channel of writing MV merge data (RX channel 3):
 - mode: 1D.
 - length/size: The writing space size is NUM_H_MB x the number of vertical MB lines x 4.
 10. Configure the channel of writing bit stream data (RX channel 4):
 - mode: 1D.
 - length/size: The size of the writing space is the number of horizontal pixels of the picture x the number of vertical pixel lines x 1.5.

Register configuration:

- **H264_DMA_OUT/IN_MEM_BURST_LENGTH_CH x** : This register controls the AXI burst length of the corresponding channel.
 - 0: 8 bytes. 1: 16 bytes. 2: 32 bytes. 3: 64 bytes. 4: 128 bytes.
 - In TX channel, x can be 0 ~ 4. In RX channel, x can be 0 ~ 5.
 - The burst length of TX channel 2 and RX channel 2 must be configured the same.
- **H264_DMA_OUT_REORDER_EN_CHO**: This register controls the TX channel 0 reorder enable. When the linked list descriptor configuration HB is not 16 in TX channel 0, this register must be configured as 1.
- **H264_DMA_BLOCK_START_ADDR_CH5**: This register controls the data starting address of the RX channel 5. The register configuration must be consistent with the data buffer address in the linked list descriptor of the TX channel 1.
- **H264_DMA_BLOCK_ROW_LENGTH_4LINE_CH5**: This register specifies the number of bytes contained in a row block of 4 lines. The data size is $(NUM_H_MB \times (16 \times 4 + 8 \times 4 \times 2))$.
- **H264_DMA_BLOCK_ROW_LENGTH_12LINE_CH5**: This register specifies the number of bytes contained in a row block of 12 lines. The data size is $(NUM_H_MB \times (16 \times 12 + 8 \times 4 \times 2))$.
- **H264_DMA_BLOCK_LENGTH_4LINE_CH5**: This register specifies the number of bytes contained in a block of 4 lines. The data size is $(16 \times 4 + 8 \times 4 \times 2)$.
- **H264_DMA_BLOCK_LENGTH_12LINE_CH5**: This register specifies the number of bytes contained in a block of 12 lines. The data size is $(16 \times 12 + 8 \times 4 \times 2)$.

39.6 Interrupts

ESP32-P4's H264 Encoder can generate the following interrupt signals, which are then sent to the [Interrupt Matrix](#).

- H264_REG_INT
- H264_DMA_IN_CHO_INT
- H264_DMA_IN_CH1_INT
- H264_DMA_IN_CH2_INT
- H264_DMA_IN_CH3_INT
- H264_DMA_IN_CH4_INT
- H264_DMA_IN_CH5_INT
- H264_DMA_OUT_CHO_INT
- H264_DMA_OUT_CH1_INT
- H264_DMA_OUT_CH2_INT
- H264_DMA_OUT_CH3_INT
- H264_DMA_OUT_CH4_INT

There are several internal interrupt sources from H264 Encoder that can generate the above interrupt signals.

Interrupt signal H264_REG_INT can be generated by the following internal interrupt sources:

- H264_DMA_MOVE_2MB_LINE_DONE_INT: When H264_DMA completes moving the reference pictures of two MB rows from external memory to internal memory, the interrupt is triggered.
- H264_FRAME_DONE_INT: When ENC_CORE finishes compressing and encoding a frame of video, the interrupt is triggered.
- H264_REC_READY_INT: When ENC_CORE writes the deblocking filter result (i.e., the reference picture) of H264_x_REC_READY_TRIGGER_MB_LINES lines of MBs to the external memory, the interrupt is triggered. x can be A or B, indicating the register corresponds to video sequence A or B.
- H264_DB_TMP_READY_INT: When ENC_CORE writes the deblocking filter intermediate data of H264_x_DB_TMP_READY_TRIGGER_MB_NUM MBs to the internal memory, the interrupt is triggered. x can be A or B indicates the register corresponds to video sequence A or B.

The above interrupt sources can only be triggered when the interrupt enable register XXX_ENA is set to 1. Write 1 to XXX_CLR will clear the interrupt (XXX is the interrupt source name).

The H264_DMA_IN_CHx_INT (x is 0 ~ 5) signals are generated by the following interrupt sources:

- H264_DMA_IN_DONE_CHx_INT: Triggered when all data corresponding to a receive descriptor has been received via RX channel x (x is 0 ~ 5).
- H264_DMA_IN_SUC_EOF_CHx_INT: Triggered when the suc_eof bit in a receive descriptor is 1 and the data corresponding to this receive descriptor has been received (i.e. when the data frame or packet corresponding to an inlink has been received) via RX channel x (x is 0 ~ 5).
- H264_DMA_IN_ERR_EOF_CHx_INT: Triggered when an error is detected in the data frame or packet received via RX channel x (x is 0 ~ 4).
- H264_DMA_IN_DSCR_EMPTY_CHx_INT: Triggered when the size of the buffer pointed by receive descriptors is smaller than the length of data to be received via RX channel x (x is 0 ~ 4).
- H264_DMA_IN_DSCR_ERR_CHx_INT: Triggered when a receive descriptor on RX channel x (x is 0 ~ 4) fails any of the two descriptor checks.
- H264_DMA_INFIFO_OVF_L1_CHx_INT: Triggered when the L1 RX FIFO of RX channel x (x is 0 ~ 5) overflows.
- H264_DMA_INFIFO_OVF_L2_CHx_INT: Triggered when the L2 RX FIFO of RX channel x (x is 0 ~ 4) overflows.
- H264_DMA_INFIFO_UDF_L1_CHx_INT: Triggered when the L1 RX FIFO of RX channel x (x is 0 ~ 5) underflows.
- H264_DMA_INFIFO_UDF_L2_CHx_INT: Triggered when the L2 RX FIFO of RX channel x (x is 0 ~ 4) underflows.
- H264_DMA_IN_DSCR_TASK_OVF_CHx_INT: Triggered when descriptor ready task FIFO of RX channel x (x is 0 ~ 4) overflows.
- H264_DMA_FETCH_MB_COL_CNT_OVF_CHx_INT: Triggered when fetch MB columns cnt of RX channel x (x is 5) overflows.

The H264_DMA_OUT_CHx_INT (x is 0 ~ 4) signals are generated by the following interrupt sources:

- H264_DMA_OUT_DONE_CH x _INT: Triggered when all data corresponding to a transmit descriptor has been sent via TX channel x .
- H264_DMA_OUT_EOF_CH x _INT: Triggered when the suc_eof bit in a transmit descriptor is 1 and data corresponding to this descriptor has been sent via TX channel x . If H264_DMA_OUT_EOF_MODE_CH x is 0, this interrupt will be triggered when the last byte of data corresponding to this descriptor enters H264_DMA's TX channel; if H264_DMA_OUT_EOF_MODE_CH x is 1, this interrupt is triggered when the last byte of data is taken from H264_DMA's TX channel.
- H264_DMA_OUT_DSCR_ERR_CH x _INT: Triggered when a transmit descriptor on TX channel x fails any of the two descriptor checks.
- H264_DMA_OUT_TOTAL_EOF_CH x _INT: Triggered when all data corresponding to a linked list (including multiple descriptors) has been sent via TX channel x .
- H264_DMA_OUTFIFO_OVF_L1_CH x _INT: Triggered when the L1 TX FIFO of H264_DMA overflows.
- H264_DMA_OUTFIFO_OVF_L2_CH x _INT: Triggered when the L2 TX FIFO of H264_DMA overflows.
- H264_DMA_OUTFIFO_UDF_L1_CH x _INT: Triggered when the L1 TX FIFO of H264_DMA underflows.
- H264_DMA_OUTFIFO_UDF_L2_CH x _INT: Triggered when the L2 TX FIFO of H264_DMA underflows.
- H264_DMA_OUT_DSCR_TASK_OVF_CH x _INT: Triggered when descriptor ready task FIFO of TX channel x overflows.

39.7 Programming Procedures

The H264 encoder can work in two modes. In GOP mode, it periodically inserts an I frame into a specified number (configurable) of P frame intervals. In dual stream mode, it takes turns to process two video sequences, and the hardware automatically switches the ENC_CORE configurations.

39.7.1 GOP Mode

In GOP mode, a single video sequence A is encoded. Take a GOP containing 3 pictures as an example, the encoding sequence of pictures is shown in Figure 39.7-1:

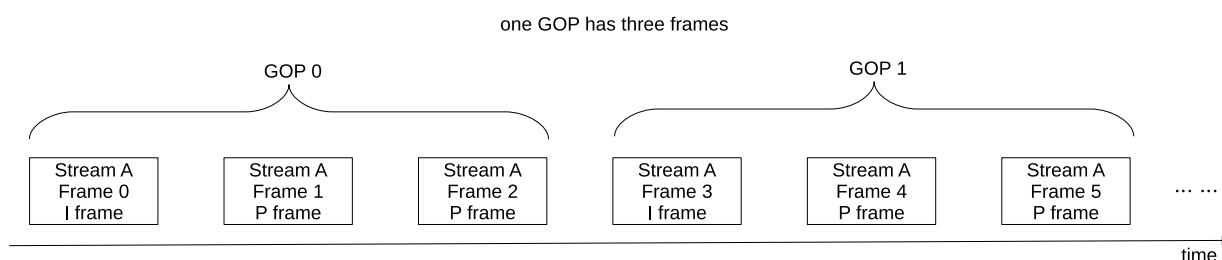


Figure 39.7-1. Coding order of pictures in GOP mode

In GOP mode, the register with the prefix H264_A_ is valid, and the register with the prefix H264_B_ is invalid.

The GOP mode configuration process is as follows:

1. Refer to the description in chapter 39.5.2.5 to configure each H264_DMA channel:

- (a) Configure H264_DMA TX channel 0.
- (b) Configure H264_DMA TX channel 1.
- (c) Configure H264_DMA TX channel 2.
- (d) Configure H264_DMA TX channel 3 and TX channel 4.
- (e) Configure H264_DMA RX channel 0 and RX channel 1.
- (f) Configure H264_DMA RX channel 2.
- (g) Configure H264_DMA RX channel 3.
- (h) Configure H264_DMA RX channel 4.
- (i) Configure H264_DMA RX channel 5.

2. Configure ENC_CORE according to the required functions:

- (a) Configure GOP mode: Set register [H264_FRAME_MODE](#) to 0 and register [H264_DUAL_STREAM_MODE](#) to 0.
- (b) Configure the number of pictures in a GOP: Configure the register [H264_GOP_NUM](#) as n, indicating that a GOP contains n pictures. When the register is configured as 0, the first frame of a GOP is an I frame, and the subsequent frames of a GOP are always P frame until the encoder is reset.
- (c) Configure the MB resolution of a picture: configure the register [H264_A_SYS_TOTAL_MB_X](#) as the number of MBs in the horizontal direction of the picture, and configure the register [H264_A_SYS_TOTAL_MB_Y](#) as the number of MBs in the vertical direction of the picture. The number of MBs in the horizontal direction is equal to the width of the video divided by 16 and rounded up. And the number of MBs in the vertical direction is equal to the height of the video divided by 16 and rounded up.
- (d) Configure the initial QP value of a picture: configure the register [H264_A_QP](#) as the initial QP value, ranging from 10 to 51.
- (e) Configure deblocking filter: Set register [H264_A_BYPASS_DB_FILTER](#) to 0 to enable the deblocking filter, set register [H264_A_BYPASS_DB_FILTER](#) to 1 to disable the deblocking filter.
- (f) Configure slice header information: According to the information required to be included in the slice header in the H264 standard, configure the relevant registers [H264_SLICE_REMAIN_BIT](#), [H264_SLICE_REMAIN_BITLENGTH](#), [H264_SLICE_BYTE_LENGTH](#), [H264_SLICE_BYTE_LSB](#), and [H264_SLICE_BYTE_MSB](#). For example, if the slice header is a variable with 41 bits and the highest significant bit is the slice header first bit, these registers should be set in the following ways.
 - [H264_SLICE_BYTE_MSB](#): From bit 41 to bit 10 of the slice header
 - [H264_SLICE_BYTE_LSB](#): From bit 9 to bit 2 of the slice header
 - [H264_SLICE_REMAIN_BIT](#): Bit 1 of the slice header
 - [H264_SLICE_BYTE_LENGTH](#): 5
 - [H264_SLICE_BYTE_REMAIN_BITLENGTH](#): 1
- (g) To use the quantization result decimation function, configure its related registers referring to the description in section [39.5.1.2](#).

- (h) To use the MB level rate control function, configure its related registers referring to the description in section [39.5.1.4](#).
 - (i) To use the ROI function, configure its related registers referring to the description in chapter [39.5.1.5](#).
 - (j) To use the MV merge function, configure its related registers referring to the description in section [39.5.1.3](#).
3. Perform the corresponding operations based on the type of video frame to be encoded (In a GOP, the first frame is an I frame, and subsequent frames are P frames):
- The video frame to be encoded is an I frame:
 - (a) Start H264_DMA TX channel 0: Refer to chapter [39.5.2.3](#) for the configuration.
 - (b) Start H264_DMA RX channel 0 and RX channel 1: Refer to chapter [39.5.2.3](#) for the configuration.
 - (c) Start H264_DMA RX channel 2: Refer to chapter [39.5.2.3](#) for the configuration.
 - (d) To use the MV merge function, start H264_DMA RX channel 3. Refer to chapter [39.5.2.3](#) for the configuration.
 - (e) Start H264_DMA RX channel 4: Refer to chapter [39.5.2.3](#) for the configuration.
 - (f) Clear H264 interrupts and enable interrupts.
 - (g) Start the encoding of a picture: Write 1 to register [H264_FRAME_START](#).
 - (h) Wait for the H264_DB_TMP_READY_INT interrupt. When the interrupt is triggered, start H264_DMA TX channel 2. Refer to chapter [39.5.2.3](#) for the configuration.
 - (i) Wait for the H264_REC_READY_INT interrupt. When the interrupt is triggered, start H264_DMA RX channel 5, H264_DMA TX channel 3, and TX channel 4. Refer to section [39.5.2.3](#) for the configuration.
 - (j) Start H264_DMA to move the reference picture from external memory to internal memory: Write 1 to register [H264_DMA_MOVE_START](#).
 - (k) Wait for the H264_DMA_MOVE_2MB_LINE_DONE_INT interrupt. When the interrupt is triggered, start H264_DMA TX channel 1. Refer to chapter [39.5.2.3](#) for the configuration.
 - (l) Wait for the H264_FRAME_DONE_INT interrupt. When the interrupt is triggered, it indicates that the picture encoding is completed.
 - (m) Reset H264_DMA TX channel 2 and RX channel 2: Refer to chapter [39.5.2.4](#) for the configuration.
 - (n) Reset the read and write time counter of the deblocking filter intermediate data in H264_DMA: Set register [H264_DMA_RX_CH2_INTER_COUNTER_RST](#) to 1, then set to 0.
 - The video frame to be encoded is P frame:
 - (a) Clear H264 interrupts and enable interrupts.
 - (b) Start H264_DMA RX channel 2: Refer to chapter [39.5.2.3](#) for the configuration.
 - (c) Start encoding a picture: Write 1 to register [H264_FRAME_START](#).

- (d) Wait for the H264_DB_TMP_READY_INT interrupt. When the interrupt is triggered, start H264_DMA TX channel 2. Refer to chapter 39.5.2.3 for the configuration.
 - (e) Wait for the H264_FRAME_DONE_INT interrupt. When the interrupt is triggered, it indicates that the picture encoding is completed.
 - (f) Reset H264_DMA TX channel 2 and RX channel 2: Refer to chapter 39.5.2.4 for the configuration.
 - (g) Reset the read and write time counter of deblocking filter intermediate data in H264_DMA: Set register H264_DMA_RX_CH2_INTER_COUNTER_RST to 1, then set to 0.
- After any frame ends, the following registers can be reconfigured:
 - Registers related to the quantization result decimate function. Refer to chapter 39.5.1.2 for details.
 - Registers related to the MB level rate control function. Refer to chapter 39.5.1.4 for details.
 - Deblocking filter enable register: H264_A_BYPASS_DB_FILTER.
 - ROI function related registers. Refer to chapter 39.5.1.5 for details.
 - Registers related to slice header information: H264_SLICE_RMEAIN_BIT, H264_SLICE_REMAIN_BITLENGTH, H264_SLICE_BYTE_LENGTH, H264_SLICE_BYTE_LSB, H264_SLICE_BYTE_MSB.
 - Registers related to the MV merge function. Refer to chapter 39.5.1.3 for details.
4. Repeat step 3 until all video frames in a GOP are encoded.
 5. Make sure the data is written completely. Refer to chapter 39.5.2.4 for the method.
 6. Reset each sending and receiving channel of H264_DMA referring to chapter 39.5.2.4.
 7. Reset the read and write times counter of deblocking filter result data, and the read and write times counter of reference picture data in H264_DMA. To be specific, set register H264_DMA_RX_CHO_EXTER_COUNTER_RST to 1, then set it to 0; Set register H264_DMA_RX_CH1_EXTER_COUNTER_RST to 1, then set it to 0; Set register H264_DMA_RX_CH5_INTER_COUNTER_RST to 1, then set it to 0.
 8. Reconfigure from step 2 and start encoding the next GOP.

39.7.2 Dual Stream Mode

In the dual stream mode, two video sequences A and B are encoded. The encoding sequence of pictures is shown in Figure 39.7-2 (a GOP contains 3 pictures).

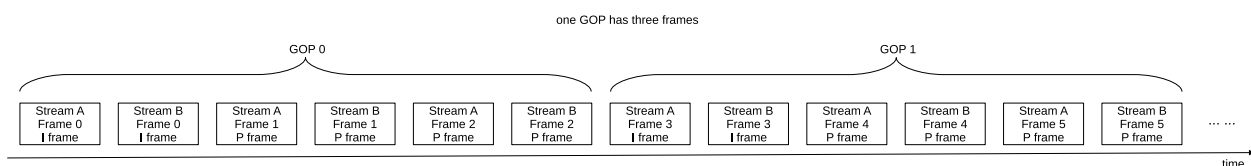


Figure 39.7-2. Coding order of pictures in dual stream mode

In dual stream mode, the frame order of H264 processing is fixed as A1, B1, A2, B2, A3, B3, etc. When processing frames in video sequence A, registers with the prefix H264_A_ will be automatically applied. When processing frames in video sequence B, registers with the prefix H264_B_ will be automatically applied. The registers with other prefixes are valid for both video sequence frames.

The configuration process of dual stream mode is as follows:

1. Configure ENC_CORE according to the required function:

- (a) Configure dual stream mode: Set register [H264_FRAME_MODE](#) to 1, and set register [H264_DUAL_STREAM_MODE](#) to 1.
- (b) Configure GOP (the GOP configuration of video sequence A and B must be the same): Configure the register [H264_GOP_NUM](#) as n, indicating that a GOP contains n pictures. When the register is configured as 0, the first frame of a GOP is an I frame, and the subsequent frames of a GOP are always P frames until the encoder is reset.
- (c) Configure the MB resolution of a picture: Configure the register [H264_A_SYS_TOTAL_MB_X](#) as the number of MBs in the horizontal direction of the video sequence A, and configure the register [H264_A_SYS_TOTAL_MB_Y](#) as the number of MBs in the vertical direction of the video sequence A. Configure the register [H264_B_SYS_TOTAL_MB_X](#) as the number of MBs in the horizontal direction of the video sequence B, and configure the register [H264_B_SYS_TOTAL_MB_Y](#) as the number of MBs in the vertical direction of the video sequence B. The number of MBs in the horizontal direction is equal to the width of the video divided by 16 and rounded up. And the number of MBs in the vertical direction is equal to the height of the video divided by 16 and rounded up.
- (d) Configure the initial QP of a picture: Configure the register [H264_A_QP](#) as the initial QP of video sequence A, ranging from 10 to 51. Configure the register [H264_B_QP](#) as the initial QP of video sequence B, ranging from 10 to 51.
- (e) Configure the deblocking filter: Set the register [H264_A_BYPASS_DB_FILTER](#) to 0 to enable the deblocking filter of video sequence A. Set the register [H264_A_BYPASS_DB_FILTER](#) to 1 to disable the deblocking filter of video sequence A. Set the register [H264_B_BYPASS_DB_FILTER](#) to 0 to enable the deblocking filter of video sequence B. Set the register [H264_B_BYPASS_DB_FILTER](#) to 1 to disable the deblocking filter of video sequence B.
- (f) Configure slice header information: According to the information required to be included in the slice header in the H264 standard, configure the relevant registers [H264_SLICE_RMAIN_BIT](#), [H264_SLICE_REMAIN_BITLENGTH](#), [H264_SLICE_BYTE_LENGTH](#), [H264_SLICE_BYTE_LSB](#), and [H264_SLICE_BYTE_MSB](#). For example, if the slice header is a variable with 41 bits and the highest significant bit is the slice header first bit, these registers should be set in the following ways.
 - [H264_SLICE_BYTE_MSB](#): From bit 41 to bit 10 of the slice header
 - [H264_SLICE_BYTE_LSB](#): From bit 9 to bit 2 of the slice header
 - [H264_SLICE_RMAIN_BIT](#): Bit 1 of the slice header
 - [H264_SLICE_BYTE_LENGTH](#): 5
 - [H264_SLICE_BYTE_REMAIN_BITLENGTH](#): 1
- (g) To use the quantization result decimate function, configure its related registers referring to section [39.5.1.2](#).

- (h) To use the MB level rate control function, configure its related registers referring to chapter [39.5.1.4](#).
 - (i) To use the ROI function, configure its related registers referring to chapter [39.5.1.5](#).
 - (j) To use the MV merge function, configure its related registers referring to chapter [39.5.1.3](#).
2. According to whether the frame to be encoded belongs to video sequence A or B, use the information corresponding to the video sequence and configure each H264_DMA channel referring to section [39.5.2.5](#):
- (a) Configure H264_DMA TX channel 0.
 - (b) Configure H264_DMA TX channel 1.
 - (c) Configure H264_DMA TX channel 2.
 - (d) Configure H264_DMA TX channel 3 and TX channel 4.
 - (e) Configure H264_DMA RX channel 0 and RX channel 1.
 - (f) Configure H264_DMA RX channel 2.
 - (g) Configure H264_DMA RX channel 3.
 - (h) Configure H264_DMA RX channel 4.
 - (i) Configure H264_DMA RX channel 5.
3. Perform the corresponding operations based on the type of video frame to be encoded. Video sequence A and B are encoded alternately, i.e., first a picture in video sequence A is encoded, and then a picture in video sequence B, etc. In a GOP, the first frame is an I frame, and the subsequent frames are P frames for any video sequence.
- The video frame to be encoded is an I frame:
 - (a) Start H264_DMA TX channel 0: Refer to section [39.5.2.3](#) for the configuration.
 - (b) Start H264_DMA RX channel 0 and RX channel 1: Refer to section [39.5.2.3](#) for the configuration.
 - (c) Start H264_DMA RX channel 2: Refer to section [39.5.2.3](#) for the configuration.
 - (d) Start H264_DMA RX channel 4: Refer to section [39.5.2.3](#) for the configuration.
 - (e) Clear H264 interrupts and enable interrupts.
 - (f) Start encoding a picture: Write 1 to register [H264_FRAME_START](#).
 - (g) Wait for the H264_DB_TMP_READY_INT interrupt. When the interrupt is triggered, start H264_DMA TX channel 2. Refer to section [39.5.2.3](#) for the configuration.
 - (h) Wait for the H264_FRAME_DONE_INT interrupt. When the interrupt is triggered, it means that the picture encoding is completed.
 - (i) Make sure the data is written completely. Refer to section [39.5.2.4](#) for the method.
 - (j) Reset each sending and receiving channel of H264_DMA referring to section [39.5.2.4](#).
 - (k) Reset the read and write counter of the deblocking filter intermediate data, and the read and write counter of the reference picture data in H264_DMA. To be specific, set register [H264_DMA_RX_CH2_INTER_COUNTER_RST](#) to 1, and then set it to 0. Set the register [H264_DMA_RX_CH5_INTER_COUNTER_RST](#) is set to 1 and then set it to 0.

- The video frame to be encoded is P frame:
 - (a) Clear H264 interrupts and enable interrupts.
 - (b) Start H264_DMA TX channel 0: Refer to section [39.5.2.3](#) for the configuration.
 - (c) Start H264_DMA RX channel 0 and RX channel 1: Refer to section [39.5.2.3](#) for the configuration.
 - (d) Start H264_DMA RX channel 2: Refer to section [39.5.2.3](#) for the configuration.
 - (e) Start H264_DMA RX channel 4: Refer to section [39.5.2.3](#) for the configuration.
 - (f) Start H264_DMA RX channel 5: Refer to section [39.5.2.3](#) for the configuration.
 - (g) Start H264_DMA TX channel 3 and TX channel 4: Refer to section [39.5.2.3](#) for the configuration.
 - (h) To use the MV merge function, start H264_DMA RX channel 3: Refer to section [39.5.2.3](#) for the configuration.
 - (i) Start H264_DMA to move the reference picture from external memory to internal memory: Write 1 to register [H264_DMA_MOVE_START](#).
 - (j) Wait for H264_DMA_MOVE_2MB_LINE_DONE_INT interrupt. When the interrupt is triggered, start H264_DMA TX channel 1. Refer to section [39.5.2.3](#) for the configuration.
 - (k) Start encoding of a picture: Write 1 to register [H264_FRAME_START](#).
 - (l) Wait for the H264_DB_TMP_READY_INT interrupt. When the interrupt is triggered, start H264_DMA TX channel 2. Refer to section [39.5.2.3](#) for the configuration.
 - (m) Wait for the H264_FRAME_DONE_INT interrupt. When the interrupt is triggered, it indicates that the picture encoding is completed.
 - (n) Make sure the data is written completely referring to section [39.5.2.4](#).
 - (o) Reset each sending and receiving channel of H264_DMA referring to section [39.5.2.4](#).
 - (p) Reset the read and write counter of the deblocking filter intermediate data and the read and write counter of the reference picture data in H264_DMA. To be specific, set the register [H264_DMA_RX_CH2_INTER_COUNTER_RST](#) to 1, and then set it to 0; Set the register [H264_DMA_RX_CH5_INTER_COUNTER_RST](#) to 1 and then set it to 0.
- After any frame ends, the following registers corresponding to the currently encoded video sequence can be reconfigured:
 - Registers related to the quantization result decimate function, refer to the description in chapter [39.5.1.2](#).
 - Registers related to the MB level rate control function, refer to the description in chapter [39.5.1.4](#).
 - Deblocking filter enable register: [H264_A_BYPASS_DB_FILTER](#).
 - ROI function related registers, refer to the description in chapter [39.5.1.5](#).
 - Registers related to the slice header information: [H264_SLICE_RMEAIN_BIT](#), [H264_SLICE_REMAIN_BITLENGTH](#), [H264_SLICE_BYTE_LENGTH](#), [H264_SLICE_BYTE_LSB](#), and [H264_SLICE_BYTE_MSB](#).

- Registers related to the MV merge function, refer to the description in chapter [39.5.1.3](#).
- 4. Repeat steps [2](#) and [3](#) until one GOP of each of the 2 video sequences is encoded.
- 5. Reset the read and write times counter of the deblocking filter result data in H264_DMA. To be specific, set the register [H264_DMA_RX_CHO_EXTER_COUNTER_RST](#) to 1, then set to 0. Set the register [H264_DMA_RX_CH1_EXTER_COUNTER_RST](#) to 1, then set to 0.
- 6. Reconfigure from step [1](#) to start the encoding of the next GOP.

39.7.3 Soft Reset

No matter what working mode H264 is in, reset it according to the following process can stop encoding at any time.

1. Reset ENC_CORE: Write 1 to register [H264_SYS_RST_PULSE](#).
2. Reset each sending and receiving channel of H264_DMA referring to section [39.5.2.4](#).
3. Reset each read and write time counter in H264_DMA. To be specific, set register [H264_DMA_RX_CHO_EXTER_COUNTER_RST](#) to 1, then set it to 0; Set register [H264_DMA_RX_CH1_EXTER_COUNTER_RST](#) to 1, then set it to 0; Set the register [H264_DMA_RX_CH2_INTER_COUNTER_RST](#) to 1, and then set it to 0; Set the register [H264_DMA_RX_CH5_INTER_COUNTER_RST](#) to 1, and then set it to 0.
4. H264 returns to the initial state and can be reconfigured as needed.

39.8 Register Summary

The addresses in this section are relative to **H264 Encoder** base address and **H264 DMA** base address. These base addresses are provided in Table [7.3-2 Module/Peripheral Address Mapping](#) in Chapter [7 System and Memory](#).

The abbreviations given in Column **Access** are explained in Section [Access Types for Registers](#).

39.8.1 H264 Encoder Register Summary

Name	Description	Address	Access
Configuration Registers			
H264_SYS_CTRL_REG	H264 system level control register	0x0000	varies
H264_GOP_CONF_REG	GOP related configuration register	0x0004	R/W
H264_SLICE_HEADER_REMAIN_REG	Frame slice header remain bit register	0x00AC	R/W
H264_SLICE_HEADER_BYTE_LENGTH_REG	Frame slice header byte length register	0x00B0	R/W
H264_BS_THRESHOLD_REG	Bitstream buffer overflow threshold register	0x00B4	R/W
H264_SLICE_HEADER_BYTE0_REG	Frame slice header byte low 32 bit register	0x00B8	R/W
H264_SLICE_HEADER_BYTE1_REG	Frame slice header byte high 32 bit register	0x00BC	R/W
H264_CONF_REG	General configuration register	0x00D0	R/W
H264_MV_MERGE_CONFIG_REG	MV merge configuration register	0x00D4	varies
Video Sequence A Configuration Registers			
H264_A_SYS_MB_RES_REG	Video sequence A horizontal and vertical MB resolution register	0x0008	R/W
H264_A_SYS_CONF_REG	Video sequence A system level configuration register	0x000C	R/W
H264_A_DECI_SCORE_REG	Video sequence A luma and chroma MB decimate score register	0x0010	R/W
H264_A_DECI_SCORE_OFFSET_REG	Video sequence A luma and chroma MB decimate score offset register	0x0014	R/W
H264_A_RC_CONFO_REG	Video sequence A rate control configuration register0	0x0018	R/W
H264_A_RC_CONF1_REG	Video sequence A rate control configuration register1	0x001C	R/W
H264_A_DB_BYPASS_REG	Video sequence A Deblocking bypass register	0x0020	R/W
H264_A_ROI_REGION0_REG	Video sequence A H264 ROI0 range configure register	0x0024	R/W
H264_A_ROI_REGION1_REG	Video sequence A H264 ROI1 range configure register	0x0028	R/W
H264_A_ROI_REGION2_REG	Video sequence A H264 ROI2 range configure register	0x002C	R/W

Name	Description	Address	Access
H264_A_ROI_REGION3_REG	Video sequence A H264 ROI3 range configure register	0x0030	R/W
H264_A_ROI_REGION4_REG	Video sequence A H264 ROI4 range configure register	0x0034	R/W
H264_A_ROI_REGION5_REG	Video sequence A H264 ROI5 range configure register	0x0038	R/W
H264_A_ROI_REGION6_REG	Video sequence A H264 ROI6 range configure register	0x003C	R/W
H264_A_ROI_REGION7_REG	Video sequence A H264 ROI7 range configure register	0x0040	R/W
H264_A_ROI_REGION0_3_QP_REG	Video sequence A H264 ROI0-3 QP register	0x0044	R/W
H264_A_ROI_REGION4_7_QP_REG	Video sequence A H264 ROI4-7 QP register	0x0048	R/W
H264_A_NO_ROI_REGION_QP_OFFSET_REG	Video sequence A H264 no ROI QP register	0x004C	R/W
H264_A_ROI_CONFIG_REG	Video sequence A H264 ROI configure register	0x0050	R/W
Video Sequence B Configuration Registers			
H264_B_SYS_MB_RES_REG	Video sequence B horizontal and vertical MB resolution register	0x0054	R/W
H264_B_SYS_CONF_REG	Video sequence B system level configuration register	0x0058	R/W
H264_B_DECI_SCORE_REG	Video sequence B luma and chroma MB decimate score register	0x005C	R/W
H264_B_DECI_SCORE_OFFSET_REG	Video sequence B luma and chroma MB decimate score offset register	0x0060	R/W
H264_B_RC_CONFO_REG	Video sequence B rate control configuration register0.	0x0064	R/W
H264_B_RC_CONF1_REG	Video sequence B rate control configuration register1.	0x0068	R/W
H264_B_DB_BYPASS_REG	Video sequence B Deblocking bypass register	0x006C	R/W
H264_B_ROI_REGION0_REG	Video sequence B H264 ROI0 range configure register	0x0070	R/W
H264_B_ROI_REGION1_REG	Video sequence B H264 ROI1 range configure register	0x0074	R/W
H264_B_ROI_REGION2_REG	Video sequence B H264 ROI2 range configure register	0x0078	R/W
H264_B_ROI_REGION3_REG	Video sequence B H264 ROI3 range configure register	0x007C	R/W
H264_B_ROI_REGION4_REG	Video sequence B H264 ROI4 range configure register	0x0080	R/W

Name	Description	Address	Access
H264_B_ROI_REGION5_REG	Video sequence B H264 ROI5 range configure register	0x0084	R/W
H264_B_ROI_REGION6_REG	Video sequence B H264 ROI6 range configure register	0x0088	R/W
H264_B_ROI_REGION7_REG	Video sequence B H264 ROI7 range configure register	0x008C	R/W
H264_B_ROI_REGION0_3_QP_REG	Video sequence B H264 ROI0-3 QP register	0x0090	R/W
H264_B_ROI_REGION4_7_QP_REG	Video sequence B H264 ROI4-7 QP register	0x0094	R/W
H264_B_NO_ROI_REGION_QP_OFFSET_REG	Video sequence B H264 no ROI QP register	0x0098	R/W
H264_B_ROI_CONFIG_REG	Video sequence B H264 ROI configure register	0x009C	R/W
Status Registers			
H264_RC_STATUS0_REG	Rate control status register0	0x00A0	RO
H264_RC_STATUS1_REG	Rate control status register1	0x00A4	RO
H264_RC_STATUS2_REG	Rate control status register2	0x00A8	RO
H264_SYS_STATUS_REG	System status register	0x00DC	RO
H264_FRAME_CODE_LENGTH_REG	Frame code byte length register	0x00E0	RO
Interrupt Registers			
H264_INT_RAW_REG	Interrupt raw status register	0x00C0	R/WTC/SS
H264_INT_ST_REG	Interrupt masked status register	0x00C4	RO
H264_INT_ENA_REG	Interrupt enable register	0x00C8	R/W
H264_INT_CLR_REG	Interrupt clear register	0x00CC	WT
Version Register			
H264_DATE_REG	Version control register	0x00F0	R/W

39.8.2 H264 DMA Register Summary

Name	Description	Address	Access
Configuration Registers			
TX Configuration Registers			
H264_DMA_OUT_CONFO_CHO_REG	TX CHO config0 register	0x0000	R/W
H264_DMA_OUT_PUSH_CHO_REG	TX CHO outfifo push register	0x0018	varies
H264_DMA_OUT_LINK_CONF_CHO_REG	TX CHO out_link dscr ctrl register	0x001C	varies
H264_DMA_OUT_LINK_ADDR_CHO_REG	TX CHO out_link dscr addr register	0x0020	R/W
H264_DMA_OUT_ARB_CHO_REG	TX CHO arb register	0x003C	R/W
H264_DMA_OUT_RO_PD_CONF_CHO_REG	TX CHO reorder power config register	0x0044	R/W
H264_DMA_OUT_MODE_ENABLE_CHO_REG	TX CHO mode enable register	0x0050	R/W
H264_DMA_OUT_MODE_YUV_CHO_REG	TX CHO test mode yuv value register	0x0054	R/W
H264_DMA_OUT_CONFO_CH1_REG	TX CH1 config0 register	0x0100	R/W
H264_DMA_OUT_PUSH_CH1_REG	TX CH1 outfifo push register	0x0118	varies

Name	Description	Address	Access
H264_DMA_OUT_LINK_CONF_CH1_REG	TX CH1 out_link dscr ctrl register	0x011C	varies
H264_DMA_OUT_LINK_ADDR_CH1_REG	TX CH1 out_link dscr addr register	0x0120	R/W
H264_DMA_OUT_ARB_CH1_REG	TX CH1 arb register	0x013C	R/W
H264_DMA_OUT_CONFO_CH2_REG	TX CH2 config0 register	0x0200	R/W
H264_DMA_OUT_PUSH_CH2_REG	TX CH2 outfifo push register	0x0218	varies
H264_DMA_OUT_LINK_CONF_CH2_REG	TX CH2 out_link dscr ctrl register	0x021C	varies
H264_DMA_OUT_LINK_ADDR_CH2_REG	TX CH2 out_link dscr addr register	0x0220	R/W
H264_DMA_OUT_ARB_CH2_REG	TX CH2 arb register	0x023C	R/W
H264_DMA_OUT_CONFO_CH3_REG	TX CH3 config0 register	0x0300	R/W
H264_DMA_OUT_PUSH_CH3_REG	TX CH3 outfifo push register	0x0318	varies
H264_DMA_OUT_LINK_CONF_CH3_REG	TX CH3 out_link dscr ctrl register	0x031C	varies
H264_DMA_OUT_LINK_ADDR_CH3_REG	TX CH3 out_link dscr addr register	0x0320	R/W
H264_DMA_OUT_ARB_CH3_REG	TX CH3 arb register	0x033C	R/W
H264_DMA_OUT_CONFO_CH4_REG	TX CH4 config0 register	0x0400	R/W
H264_DMA_OUT_PUSH_CH4_REG	TX CH4 outfifo push register	0x0418	varies
H264_DMA_OUT_LINK_CONF_CH4_REG	TX CH4 out_link dscr ctrl register	0x041C	varies
H264_DMA_OUT_LINK_ADDR_CH4_REG	TX CH4 out_link dscr addr register	0x0420	R/W
H264_DMA_OUT_ARB_CH4_REG	TX CH4 arb register	0x043C	R/W
RX Configuration Registers			
H264_DMA_IN_CONFO_CHO_REG	RX CH0 config0 register	0x0500	R/W
H264_DMA_IN_POP_CHO_REG	RX CH0 INFIFO pop register	0x0518	varies
H264_DMA_IN_LINK_CONF_CHO_REG	RX CH0 in_link dscr ctrl register	0x051C	varies
H264_DMA_IN_LINK_ADDR_CHO_REG	RX CH0 in_link dscr addr register	0x0520	R/W
H264_DMA_IN_ARB_CHO_REG	RX CH0 arb register	0x0540	R/W
H264_DMA_IN_RO_PD_CONF_CHO_REG	RX CH0 reorder power config register	0x0548	R/W
H264_DMA_IN_CONFO_CH1_REG	RX CH1 config0 register	0x0600	R/W
H264_DMA_IN_POP_CH1_REG	RX CH1 INFIFO pop register	0x0618	varies
H264_DMA_IN_LINK_CONF_CH1_REG	RX CH1 in_link dscr ctrl register	0x061C	varies
H264_DMA_IN_LINK_ADDR_CH1_REG	RX CH1 in_link dscr addr register	0x0620	R/W
H264_DMA_IN_ARB_CH1_REG	RX CH1 arb register	0x0640	R/W
H264_DMA_IN_CONFO_CH2_REG	RX CH2 config0 register	0x0700	R/W
H264_DMA_IN_POP_CH2_REG	RX CH2 INFIFO pop register	0x0718	varies
H264_DMA_IN_LINK_CONF_CH2_REG	RX CH2 in_link dscr ctrl register	0x071C	varies
H264_DMA_IN_LINK_ADDR_CH2_REG	RX CH2 in_link dscr addr register	0x0720	R/W
H264_DMA_IN_ARB_CH2_REG	RX CH2 arb register	0x0740	R/W
H264_DMA_IN_CONFO_CH3_REG	RX CH3 config0 register	0x0800	R/W
H264_DMA_IN_POP_CH3_REG	RX CH3 INFIFO pop register	0x0818	varies
H264_DMA_IN_LINK_CONF_CH3_REG	RX CH3 in_link dscr ctrl register	0x081C	varies
H264_DMA_IN_LINK_ADDR_CH3_REG	RX CH3 in_link dscr addr register	0x0820	R/W
H264_DMA_IN_ARB_CH3_REG	RX CH3 arb register	0x0840	R/W
H264_DMA_IN_CONFO_CH4_REG	RX CH4 config0 register	0x0900	R/W
H264_DMA_IN_POP_CH4_REG	RX CH4 INFIFO pop register	0x0918	varies
H264_DMA_IN_LINK_CONF_CH4_REG	RX CH4 in_link dscr ctrl register	0x091C	varies

Name	Description	Address	Access
H264_DMA_IN_LINK_ADDR_CH4_REG	RX CH4 in_link dscr addr register	0x0920	R/W
H264_DMA_IN_ARB_CH4_REG	RX CH4 arb register	0x0940	R/W
H264_DMA_IN_CONFO_CH5_REG	RX CH5 config0 register	0x0A00	R/W
H264_DMA_IN_CONF1_CH5_REG	RX CH5 config1 register	0x0A04	R/W
H264_DMA_IN_CONF2_CH5_REG	RX CH5 config2 register	0x0A08	R/W
H264_DMA_IN_CONF3_CH5_REG	RX CH5 config3 register	0x0A0C	R/W
H264_DMA_IN_POP_CH5_REG	RX CH5 INFIPO pop register	0x0A24	varies
H264_DMA_IN_ARB_CH5_REG	RX CH5 arb register	0x0A40	R/W
Other Configuration Registers			
H264_DMA_RST_CONF_REG	AXI reset config register	0x0B08	R/W
H264_DMA_INTER_MEM_START_ADDRO_REG	Start address of internal memory range0 register	0x0B0C	R/W
H264_DMA_INTER_MEM_END_ADDRO_REG	end address of internal memory range0 register	0x0B10	R/W
H264_DMA_INTER_MEM_START_ADDR1_REG	Start address of internal memory range1 register	0x0B14	R/W
H264_DMA_INTER_MEM_END_ADDR1_REG	end address of internal memory range1 register	0x0B18	R/W
H264_DMA_EXTER_MEM_START_ADDRO_REG	Start address of external memory range0 register	0x0B20	R/W
H264_DMA_EXTER_MEM_END_ADDRO_REG	end address of external memory range0 register	0x0B24	R/W
H264_DMA_EXTER_MEM_START_ADDR1_REG	Start address of external memory range1 register	0x0B28	R/W
H264_DMA_EXTER_MEM_END_ADDR1_REG	end address of external memory range1 register	0x0B2C	R/W
H264_DMA_OUT_ARB_CONFIG_REG	reserved	0x0B30	R/W
H264_DMA_IN_ARB_CONFIG_REG	reserved	0x0B34	R/W
H264_DMA_COUNTER_RST_REG	counter reset register	0x0B50	R/W
Interrupt Registers			
TX Interrupt Registers			
H264_DMA_OUT_INT_RAW_CHO_REG	TX CH0 interrupt raw register	0x0004	R/WTC/SS
H264_DMA_OUT_INT_ENA_CHO_REG	TX CH0 interrupt ena register	0x0008	R/W
H264_DMA_OUT_INT_ST_CHO_REG	TX CH0 interrupt st register	0x000C	RO
H264_DMA_OUT_INT_CLR_CHO_REG	TX CH0 interrupt clr register	0x0010	WT
H264_DMA_OUT_INT_RAW_CH1_REG	TX CH1 interrupt raw register	0x0104	R/WTC/SS
H264_DMA_OUT_INT_ENA_CH1_REG	TX CH1 interrupt ena register	0x0108	R/W
H264_DMA_OUT_INT_ST_CH1_REG	TX CH1 interrupt st register	0x010C	RO
H264_DMA_OUT_INT_CLR_CH1_REG	TX CH1 interrupt clr register	0x0110	WT
H264_DMA_OUT_INT_RAW_CH2_REG	TX CH2 interrupt raw register	0x0204	R/WTC/SS
H264_DMA_OUT_INT_ENA_CH2_REG	TX CH2 interrupt ena register	0x0208	R/W
H264_DMA_OUT_INT_ST_CH2_REG	TX CH2 interrupt st register	0x020C	RO
H264_DMA_OUT_INT_CLR_CH2_REG	TX CH2 interrupt clr register	0x0210	WT

Name	Description	Address	Access
H264_DMA_OUT_INT_RAW_CH3_REG	TX CH3 interrupt raw register	0x0304	R/WTC/SS
H264_DMA_OUT_INT_ENA_CH3_REG	TX CH3 interrupt ena register	0x0308	R/W
H264_DMA_OUT_INT_ST_CH3_REG	TX CH3 interrupt st register	0x030C	RO
H264_DMA_OUT_INT_CLR_CH3_REG	TX CH3 interrupt clr register	0x0310	WT
H264_DMA_OUT_INT_RAW_CH4_REG	TX CH4 interrupt raw register	0x0404	R/WTC/SS
H264_DMA_OUT_INT_ENA_CH4_REG	TX CH4 interrupt ena register	0x0408	R/W
H264_DMA_OUT_INT_ST_CH4_REG	TX CH4 interrupt st register	0x040C	RO
H264_DMA_OUT_INT_CLR_CH4_REG	TX CH4 interrupt clr register	0x0410	WT
RX Interrupt Registers			
H264_DMA_IN_INT_RAW_CHO_REG	RX CH0 interrupt raw register	0x0504	R/WTC/SS
H264_DMA_IN_INT_ENA_CHO_REG	RX CH0 interrupt ena register	0x0508	R/W
H264_DMA_IN_INT_ST_CHO_REG	RX CH0 interrupt st register	0x050C	RO
H264_DMA_IN_INT_CLR_CHO_REG	RX CH0 interrupt clr register	0x0510	WT
H264_DMA_IN_INT_RAW_CH1_REG	RX CH1 interrupt raw register	0x0604	R/WTC/SS
H264_DMA_IN_INT_ENA_CH1_REG	RX CH1 interrupt ena register	0x0608	R/W
H264_DMA_IN_INT_ST_CH1_REG	RX CH1 interrupt st register	0x060C	RO
H264_DMA_IN_INT_CLR_CH1_REG	RX CH1 interrupt clr register	0x0610	WT
H264_DMA_IN_INT_RAW_CH2_REG	RX CH2 interrupt raw register	0x0704	R/WTC/SS
H264_DMA_IN_INT_ENA_CH2_REG	RX CH2 interrupt ena register	0x0708	R/W
H264_DMA_IN_INT_ST_CH2_REG	RX CH2 interrupt st register	0x070C	RO
H264_DMA_IN_INT_CLR_CH2_REG	RX CH2 interrupt clr register	0x0710	WT
H264_DMA_IN_INT_RAW_CH3_REG	RX CH3 interrupt raw register	0x0804	R/WTC/SS
H264_DMA_IN_INT_ENA_CH3_REG	RX CH3 interrupt ena register	0x0808	R/W
H264_DMA_IN_INT_ST_CH3_REG	RX CH3 interrupt st register	0x080C	RO
H264_DMA_IN_INT_CLR_CH3_REG	RX CH3 interrupt clr register	0x0810	WT
H264_DMA_IN_INT_RAW_CH4_REG	RX CH4 interrupt raw register	0x0904	R/WTC/SS
H264_DMA_IN_INT_ENA_CH4_REG	RX CH4 interrupt ena register	0x0908	R/W
H264_DMA_IN_INT_ST_CH4_REG	RX CH4 interrupt st register	0x090C	RO
H264_DMA_IN_INT_CLR_CH4_REG	RX CH4 interrupt clr register	0x0910	WT
H264_DMA_IN_INT_RAW_CH5_REG	RX CH5 interrupt raw register	0x0A10	R/WTC/SS
H264_DMA_IN_INT_ENA_CH5_REG	RX CH5 interrupt ena register	0x0A14	R/W
H264_DMA_IN_INT_ST_CH5_REG	RX CH5 interrupt st register	0x0A18	RO
H264_DMA_IN_INT_CLR_CH5_REG	RX CH5 interrupt clr register	0x0A1C	WT
Status Registers			
TX Status Registers			
H264_DMA_OUTFIFO_STATUS_CHO_REG	TX CH0 outfifo status register	0x0014	RO
H264_DMA_OUT_STATE_CHO_REG	TX CH0 state register	0x0024	RO
H264_DMA_OUT_EOF_DES_ADDR_CHO_REG	TX CH0 eof des addr register	0x0028	RO
H264_DMA_OUT_DSCR_CHO_REG	TX CH0 next dscr addr register	0x002C	RO
H264_DMA_OUT_DSCR_BF0_CHO_REG	TX CH0 last dscr addr register	0x0030	RO
H264_DMA_OUT_DSCR_BF1_CHO_REG	TX CH0 second-to-last dscr addr register	0x0034	RO
H264_DMA_OUT_RO_STATUS_CHO_REG	TX CH0 reorder status register	0x0040	RO

Name	Description	Address	Access
H264_DMA_OUT_BUF_LEN_CH0_REG	TX CH0 buf len register	0x0070	RO
H264_DMA_OUT_FIFO_BCNT_CH0_REG	TX CH0 fifo byte cnt register	0x0074	RO
H264_DMA_OUT_PUSH_BYTECNT_CH0_REG	TX CH0 push byte cnt register	0x0078	RO
H264_DMA_OUT_XADDR_CH0_REG	TX CH0 xaddr register	0x007C	RO
H264_DMA_OUTFIFO_STATUS_CH1_REG	TX CH1 outfifo status register	0x0114	RO
H264_DMA_OUT_STATE_CH1_REG	TX CH1 state register	0x0124	RO
H264_DMA_OUT_EOF_DES_ADDR_CH1_REG	TX CH1 eof des addr register	0x0128	RO
H264_DMA_OUT_DSCR_CH1_REG	TX CH1 next dscr addr register	0x012C	RO
H264_DMA_OUT_DSCR_BF0_CH1_REG	TX CH1 last dscr addr register	0x0130	RO
H264_DMA_OUT_DSCR_BF1_CH1_REG	TX CH1 second-to-last dscr addr register	0x0134	RO
H264_DMA_OUT_BUF_LEN_CH1_REG	TX CH1 buf len register	0x0170	RO
H264_DMA_OUT_FIFO_BCNT_CH1_REG	TX CH1 fifo byte cnt register	0x0174	RO
H264_DMA_OUT_PUSH_BYTECNT_CH1_REG	TX CH1 push byte cnt register	0x0178	RO
H264_DMA_OUT_XADDR_CH1_REG	TX CH1 xaddr register	0x017C	RO
H264_DMA_OUTFIFO_STATUS_CH2_REG	TX CH2 outfifo status register	0x0214	RO
H264_DMA_OUT_STATE_CH2_REG	TX CH2 state register	0x0224	RO
H264_DMA_OUT_EOF_DES_ADDR_CH2_REG	TX CH2 eof des addr register	0x0228	RO
H264_DMA_OUT_DSCR_CH2_REG	TX CH2 next dscr addr register	0x022C	RO
H264_DMA_OUT_DSCR_BF0_CH2_REG	TX CH2 last dscr addr register	0x0230	RO
H264_DMA_OUT_DSCR_BF1_CH2_REG	TX CH2 second-to-last dscr addr register	0x0234	RO
H264_DMA_OUT_BUF_LEN_CH2_REG	TX CH2 buf len register	0x0270	RO
H264_DMA_OUT_FIFO_BCNT_CH2_REG	TX CH2 fifo byte cnt register	0x0274	RO
H264_DMA_OUT_PUSH_BYTECNT_CH2_REG	TX CH2 push byte cnt register	0x0278	RO
H264_DMA_OUT_XADDR_CH2_REG	TX CH2 xaddr register	0x027C	RO
H264_DMA_OUTFIFO_STATUS_CH3_REG	TX CH3 outfifo status register	0x0314	RO
H264_DMA_OUT_STATE_CH3_REG	TX CH3 state register	0x0324	RO
H264_DMA_OUT_EOF_DES_ADDR_CH3_REG	TX CH3 eof des addr register	0x0328	RO
H264_DMA_OUT_DSCR_CH3_REG	TX CH3 next dscr addr register	0x032C	RO
H264_DMA_OUT_DSCR_BF0_CH3_REG	TX CH3 last dscr addr register	0x0330	RO
H264_DMA_OUT_DSCR_BF1_CH3_REG	TX CH3 second-to-last dscr addr register	0x0334	RO
H264_DMA_OUT_BUF_LEN_CH3_REG	TX CH3 buf len register	0x0370	RO
H264_DMA_OUT_FIFO_BCNT_CH3_REG	TX CH3 fifo byte cnt register	0x0374	RO
H264_DMA_OUT_PUSH_BYTECNT_CH3_REG	TX CH3 push byte cnt register	0x0378	RO
H264_DMA_OUT_XADDR_CH3_REG	TX CH3 xaddr register	0x037C	RO
H264_DMA_OUT_BLOCK_BUF_LEN_CH3_REG	TX CH3 block buf len register	0x0380	RO
H264_DMA_OUTFIFO_STATUS_CH4_REG	TX CH4 outfifo status register	0x0414	RO
H264_DMA_OUT_STATE_CH4_REG	TX CH4 state register	0x0424	RO
H264_DMA_OUT_EOF_DES_ADDR_CH4_REG	TX CH4 eof des addr register	0x0428	RO
H264_DMA_OUT_DSCR_CH4_REG	TX CH4 next dscr addr register	0x042C	RO
H264_DMA_OUT_DSCR_BF0_CH4_REG	TX CH4 last dscr addr register	0x0430	RO

Name	Description	Address	Access
H264_DMA_OUT_DSCR_BF1_CH4_REG	TX CH4 second-to-last dscr addr register	0x0434	RO
H264_DMA_OUT_BUF_LEN_CH4_REG	TX CH4 buf len register	0x0470	RO
H264_DMA_OUT_FIFO_BCNT_CH4_REG	TX CH4 fifo byte cnt register	0x0474	RO
H264_DMA_OUT_PUSH_BYTECNT_CH4_REG	TX CH4 push byte cnt register	0x0478	RO
H264_DMA_OUT_XADDR_CH4_REG	TX CH4 xaddr register	0x047C	RO
H264_DMA_OUT_BLOCK_BUF_LEN_CH4_REG	TX CH4 block buf len register	0x0480	RO
RX Status Registers			
H264_DMA_INFIFO_STATUS_CHO_REG	RX CH0 INFIFO status register	0x0514	RO
H264_DMA_IN_STATE_CHO_REG	RX CH0 state register	0x0524	RO
H264_DMA_IN_SUC_EOF_DES_ADDR_CHO_REG	RX CH0 eof des addr register	0x0528	RO
H264_DMA_IN_ERR_EOF_DES_ADDR_CHO_REG	RX CH0 err eof des addr register	0x052C	RO
H264_DMA_IN_DSCR_CHO_REG	RX CH0 next dscr addr register	0x0530	RO
H264_DMA_IN_DSCR_BFO_CHO_REG	RX CH0 last dscr addr register	0x0534	RO
H264_DMA_IN_DSCR_BF1_CHO_REG	RX CH0 second-to-last dscr addr register	0x0538	RO
H264_DMA_IN_FIFO_CNT_CHO_REG	RX CH0 fifo cnt register	0x0580	RO
H264_DMA_IN_POP_DATA_CNT_CHO_REG	RX CH0 pop data cnt register	0x0584	RO
H264_DMA_IN_XADDR_CHO_REG	RX CH0 xaddr register	0x0588	RO
H264_DMA_IN_BUF_HB_RCV_CHO_REG	RX CH0 buf len hb rcv register	0x058C	RO
H264_DMA_INFIFO_STATUS_CH1_REG	RX CH1 INFIFO status register	0x0614	RO
H264_DMA_IN_STATE_CH1_REG	RX CH1 state register	0x0624	RO
H264_DMA_IN_SUC_EOF_DES_ADDR_CH1_REG	RX CH1 eof des addr register	0x0628	RO
H264_DMA_IN_ERR_EOF_DES_ADDR_CH1_REG	RX CH1 err eof des addr register	0x062C	RO
H264_DMA_IN_DSCR_CH1_REG	RX CH1 next dscr addr register	0x0630	RO
H264_DMA_IN_DSCR_BFO_CH1_REG	RX CH1 last dscr addr register	0x0634	RO
H264_DMA_IN_DSCR_BF1_CH1_REG	RX CH1 second-to-last dscr addr register	0x0638	RO
H264_DMA_IN_FIFO_CNT_CH1_REG	RX CH1 fifo cnt register	0x0680	RO
H264_DMA_IN_POP_DATA_CNT_CH1_REG	RX CH1 pop data cnt register	0x0684	RO
H264_DMA_IN_XADDR_CH1_REG	RX CH1 xaddr register	0x0688	RO
H264_DMA_IN_BUF_HB_RCV_CH1_REG	RX CH1 buf len hb rcv register	0x068C	RO
H264_DMA_INFIFO_STATUS_CH2_REG	RX CH2 INFIFO status register	0x0714	RO
H264_DMA_IN_STATE_CH2_REG	RX CH2 state register	0x0724	RO
H264_DMA_IN_SUC_EOF_DES_ADDR_CH2_REG	RX CH2 eof des addr register	0x0728	RO
H264_DMA_IN_ERR_EOF_DES_ADDR_CH2_REG	RX CH2 err eof des addr register	0x072C	RO
H264_DMA_IN_DSCR_CH2_REG	RX CH2 next dscr addr register	0x0730	RO
H264_DMA_IN_DSCR_BFO_CH2_REG	RX CH2 last dscr addr register	0x0734	RO
H264_DMA_IN_DSCR_BF1_CH2_REG	RX CH2 second-to-last dscr addr register	0x0738	RO
H264_DMA_IN_FIFO_CNT_CH2_REG	RX CH2 fifo cnt register	0x0780	RO
H264_DMA_IN_POP_DATA_CNT_CH2_REG	RX CH2 pop data cnt register	0x0784	RO
H264_DMA_IN_XADDR_CH2_REG	RX CH2 xaddr register	0x0788	RO

Name	Description	Address	Access
H264_DMA_IN_BUF_HB_RCV_CH2_REG	RX CH2 buf len hb rcv register	0x078C	RO
H264_DMA_INFIFO_STATUS_CH3_REG	RX CH3 INFIFO status register	0x0814	RO
H264_DMA_IN_STATE_CH3_REG	RX CH3 state register	0x0824	RO
H264_DMA_IN_SUC_EOF_DES_ADDR_CH3_REG	RX CH3 eof des addr register	0x0828	RO
H264_DMA_IN_ERR_EOF_DES_ADDR_CH3_REG	RX CH3 err eof des addr register	0x082C	RO
H264_DMA_IN_DSCR_CH3_REG	RX CH3 next dscr addr register	0x0830	RO
H264_DMA_IN_DSCR_BFO_CH3_REG	RX CH3 last dscr addr register	0x0834	RO
H264_DMA_IN_DSCR_BF1_CH3_REG	RX CH3 second-to-last dscr addr register	0x0838	RO
H264_DMA_IN_FIFO_CNT_CH3_REG	RX CH3 fifo cnt register	0x0880	RO
H264_DMA_IN_POP_DATA_CNT_CH3_REG	RX CH3 pop data cnt register	0x0884	RO
H264_DMA_IN_XADDR_CH3_REG	RX CH3 xaddr register	0x0888	RO
H264_DMA_IN_BUF_HB_RCV_CH3_REG	RX CH3 buf len hb rcv register	0x088C	RO
H264_DMA_INFIFO_STATUS_CH4_REG	RX CH4 INFIFO status register	0x0914	RO
H264_DMA_IN_STATE_CH4_REG	RX CH4 state register	0x0924	RO
H264_DMA_IN_SUC_EOF_DES_ADDR_CH4_REG	RX CH4 eof des addr register	0x0928	RO
H264_DMA_IN_ERR_EOF_DES_ADDR_CH4_REG	RX CH4 err eof des addr register	0x092C	RO
H264_DMA_IN_DSCR_CH4_REG	RX CH4 next dscr addr register	0x0930	RO
H264_DMA_IN_DSCR_BFO_CH4_REG	RX CH4 last dscr addr register	0x0934	RO
H264_DMA_IN_DSCR_BF1_CH4_REG	RX CH4 second-to-last dscr addr register	0x0938	RO
H264_DMA_IN_FIFO_CNT_CH4_REG	RX CH4 fifo cnt register	0x0980	RO
H264_DMA_IN_POP_DATA_CNT_CH4_REG	RX CH4 pop data cnt register	0x0984	RO
H264_DMA_IN_XADDR_CH4_REG	RX CH4 xaddr register	0x0988	RO
H264_DMA_IN_BUF_HB_RCV_CH4_REG	RX CH4 buf len hb rcv register	0x098C	RO
H264_DMA_INFIFO_STATUS_CH5_REG	RX CH5 INFIFO status register	0x0A20	RO
H264_DMA_IN_STATE_CH5_REG	RX CH5 state register	0x0A28	RO
H264_DMA_IN_FIFO_CNT_CH5_REG	RX CH5 fifo cnt register	0x0A80	RO
H264_DMA_IN_POP_DATA_CNT_CH5_REG	RX CH5 pop data cnt register	0x0A84	RO
H264_DMA_IN_XADDR_CH5_REG	RX CH5 xaddr register	0x0A88	RO
H264_DMA_IN_BUF_HB_RCV_CH5_REG	RX CH5 buf len hb rcv register	0x0A8C	RO
H264_DMA_INTER_AXI_ERR_REG	internal memory AXI err register	0x0B00	RO
H264_DMA_EXTER_AXI_ERR_REG	external memory AXI err register	0x0B04	RO
H264_DMA_RX_CH0_COUNTER_REG	RX ch0 counter register	0x0B54	RO
H264_DMA_RX_CH1_COUNTER_REG	RX ch1 counter register	0x0B58	RO
H264_DMA_RX_CH2_COUNTER_REG	RX ch2 counter register	0x0B5C	RO
H264_DMA_RX_CH5_COUNTER_REG	RX ch5 counter register	0x0B60	RO
Version Register			
H264_DMA_DATE_REG	Version control register	0x0B3C	R/W

39.9 Registers

The addresses in this section are relative to **H264 Encoder** base address and **H264 DMA** base address. These base addresses are provided in Table 7.3-2 in Chapter 7 *System and Memory*.

For how to program reserved fields, please refer to Section *Programming Reserved Register Field*.

39.9.1 H264 Encoder Registers

Register 39.1. H264_SYS_CTRL_REG (0x0000)

(reserved)																																H264_SYS_RST_PULSE H264_FRAME_MODE H264_DMA_MOVE_START H264_FRAME_START																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																												
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H264_FRAME_START Configures whether to start encoding of a picture.
0: Invalid
1: Start encoding of a picture
(WT)

H264_DMA_MOVE_START Configures whether to start moving reference data from external mem.
0: Invalid
1: H264 start moving two MB lines of reference frame from external mem to internal mem
(WT)

H264_FRAME_MODE Configures H264 running mode. When field H264_DUAL_STREAM_MODE is set to 1, this field must be set to 1 too.
0: GOP mode. Before the start of every GOP, the H264_DMA needs to be reconfigured
1: Frame mode. Before the start of every frame, the H264_DMA needs to be reconfigured
(R/W)

H264_SYS_RST_PULSE Configures whether to reset H264 encoder.
0: Invalid
1: Reset H264 encoder
(WT)

Register 39.2. H264_GOP_CONF_REG (0x0004)

(reserved)																								H264_GOP_NUM								H264_DUAL_STREAM_MODE							
31																								9	8	1								0					
0 0																								0								0	Reset						

H264_DUAL_STREAM_MODE Configures whether to enable dual stream mode. When this field is set to 1, H264_FRAME_MODE field must be set to 1 too.

0: Normal mode

1: Dual stream mode

(R/W)

H264_GOP_NUM Configures the frame number of one GOP.

0: The frame number of one GOP is infinite

Others: Actual frame number of one GOP

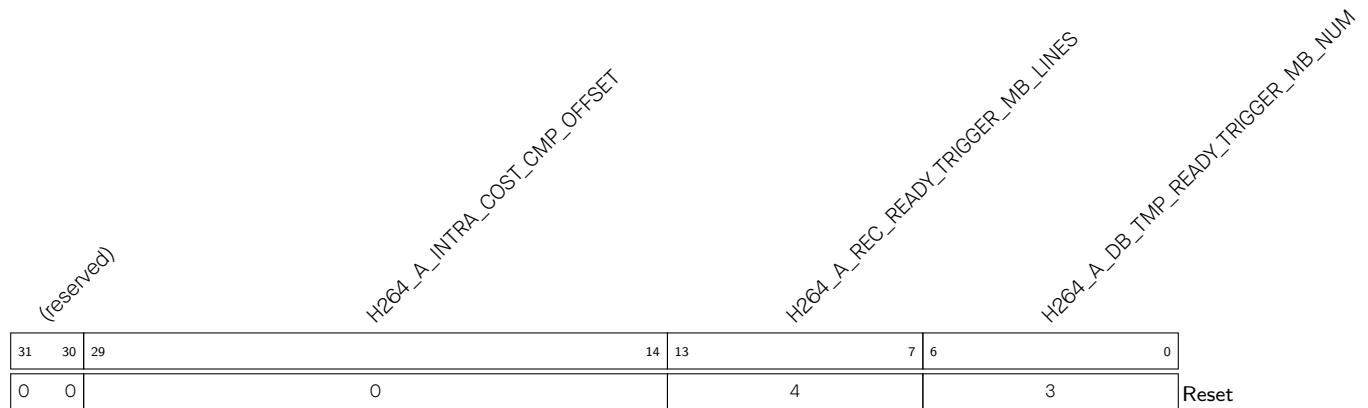
(R/W)

Register 39.3. H264_A_SYS_MB_RES_REG (0x0008)

(reserved)																H264_A_SYS_TOTAL_MB_X								H264_A_SYS_TOTAL_MB_Y																															
31																14								13								7								6								0							
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0																0								0								Reset																							

H264_A_SYS_TOTAL_MB_Y Configures video sequence A vertical MB resolution. (R/W)

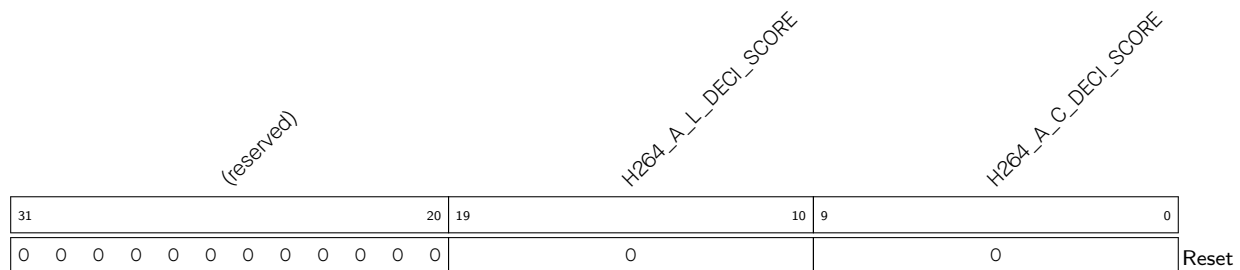
H264_A_SYS_TOTAL_MB_X	Configures video sequence A horizontal MB resolution. (R/W)
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Register 39.4. H264_A_SYS_CONF_REG (0x000C)

H264_A_DB_TMP_READY_TRIGGER_MB_NUM Configures when to trigger video sequence A H264_DB_TMP_READY_INT. When the (MB number of written db temp+1) is greater than this filed in first MB line, trigger H264_DB_TMP_READY_INT. Min is 3. (R/W)

H264_A_REC_READY_TRIGGER_MB_LINES Configures when to trigger video sequence A H264_REC_READY_INT. When the MB line number of generated reconstruct pixel is greater than this filed, trigger H264_REC_READY_INT. Min is 4. (R/W)

H264_A_INTRA_COST_CMP_OFFSET Configures video sequence A intra cost offset when I MB compared with P MB. (R/W)

Register 39.5. H264_A_DECI_SCORE_REG (0x0010)

H264_A_C_DECI_SCORE Configures video sequence A chroma MB decimate score. When chroma score is smaller than it, chroma decimate will be enabled. (R/W)

H264_A_L_DECI_SCORE Configures video sequence A luma MB decimate score. When luma score is smaller than it, luma decimate will be enabled. (R/W)

Register 39.6. H264_A_DECI_SCORE_OFFSET_REG (0x0014)

(reserved)								H264_A_P_CHROMA_DECI_SCORE_OFFSET								H264_A_P16x16_DECI_SCORE_OFFSET								H264_A_I_CHROMA_DECI_SCORE_OFFSET								H264_A_I16x16_DECI_SCORE_OFFSET							
31								24	23					18	17					12	11					6	5					0							
0	0	0	0	0	0	0	0	0	0				0				0				0				0				Reset										

H264_A_I16×16_DECI_SCORE_OFFSET Configures video sequence A i16x16 MB decimate score offset. This offset will be added to i16x16 MB score. (R/W)

H264_A_I_CHROMA_DECI_SCORE_OFFSET Configures video sequence A I chroma MB decimate score offset. This offset will be added to I chroma MB score. (R/W)

H264_A_P16×16_DECI_SCORE_OFFSET Configures video sequence A p16x16 MB decimate score offset. This offset will be added to p16x16 MB score. (R/W)

H264_A_P_CHROMA_DECI_SCORE_OFFSET Configures video sequence A p chroma MB decimate score offset. This offset will be added to p chroma MB score. (R/W)

Register 39.7. H264_A_RC_CONFO_REG (0x0018)

(reserved)										H264_A_MB_RATE_CTRL_EN												H264_A_RATE_CTRL_U												H264_A_QP									
31										23		22		21																		6		5		0							
0 0 0 0 0 0 0 0 0 0										0		0																				0		Reset									

H264_A_QP Configures video sequence A frame level initial luma QP value. (R/W)

H264_A_RATE_CTRL_U Configures video sequence A parameter U value. $U = \text{int}((\text{float}) u \ll 8)$. (R/W)

H264_A_MB_RATE_CTRL_EN Configures video sequence A whether to enable the MB level rate control.
 0: Disable the MB level rate control
 1: Enable the MB level rate control
 (R/W)

Register 39.8. H264_A_RC_CONF1_REG (0x001C)

(reserved)		H264_A_MAD_FRAME_PRED																	H264_A_QP_MAX					H264_A_QP_MIN		H264_A_CHROMA_QP_DELTA			H264_A_CHROMA_DC_QP_DELTA							
31	30																	19	18						13	12				7	6			3	2	0
0	0																	0		0					0		0			0		0		Reset		

Reset

H264_A_CHROMA_DC_QP_DELTA Configures video sequence A chroma DC QP offset based on Chroma QP. Chroma DC QP = Chroma QP(after map) + reg_chroma_dc_qp_delta. (R/W)

H264_A_CHROMA_QP_DELTA Configures video sequence A chroma QP offset based on luma QP. Chroma QP(before map) = Luma QP + reg_chroma_qp_delta. (R/W)

H264_A_QP_MIN Configures video sequence A allowed luma QP min value. (R/W)

H264_A_QP_MAX Configures video sequence A allowed luma QP max value. (R/W)

H264_A_MAD_FRAME_PRED Configures video sequence A frame level predicted MB MAD value. (R/W)

Register 39.9. H264_A_DB_BYPASS_REG (0x0020)

(reserved)																												H264_A_BYPASS_DB_FILTER	
31																												1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Reset

H264_A_BYPASS_DB_FILTER Configures whether to bypass video sequence A deblocking filter.

0: Do not bypass the deblocking filter

1: Bypass the deblocking filter

(R/W)

Register 39.10. H264_A_ROI_REGION0_REG (0x0024)

(reserved)				H264_A_ROI_REGION0_EN				H264_A_ROI_REGION0_Y_LEN				H264_A_ROI_REGION0_X_LEN				H264_A_ROI_REGION0_Y				H264_A_ROI_REGION0_X								
31	29	28	27					21	20					14	13					7	6					0		
0	0	0	0	0								0								0								Reset

H264_A_ROI_REGION0_X Configures the horizontal start MBs of ROI 0 in Video sequence A. (R/W)

H264_A_ROI_REGION0_Y Configures the vertical start MBs of ROI 0 in Video sequence A. (R/W)

H264_A_ROI_REGION0_X_LEN Configures the number of MBs in horizontal direction of the ROI 0 in Video sequence A. (R/W)

H264_A_ROI_REGION0_Y_LEN Configures the number of MBs in vertical direction of the ROI 0 in Video sequence A. (R/W)

H264_A_ROI_REGION0_EN Configures whether to enable Video sequence A ROI 0.

0: Disable ROI

1: Enable ROI

(R/W)

Register 39.11. H264_A_ROI_REGION1_REG (0x0028)

(reserved)				H264_A_ROI_REGION1_EN				H264_A_ROI_REGION1_Y_LEN				H264_A_ROI_REGION1_X_LEN				H264_A_ROI_REGION1_Y				H264_A_ROI_REGION1_X			
31	29	28	27	21	20	14	13	7	6	0													
0	0	0	0	0	0	0	0	0	0	0	Reset												

H264_A_ROI_REGION1_X Configures the horizontal start MBs of ROI 1 in Video sequence A. (R/W)

H264_A_ROI_REGION1_Y Configures the vertical start MBs of ROI 1 in Video sequence A. (R/W)

H264_A_ROI_REGION1_X_LEN Configures the number of MBs in horizontal direction of the ROI 1 in Video sequence A. (R/W)

H264_A_ROI_REGION1_Y_LEN Configures the number of MBs in vertical direction of the ROI 1 in Video sequence A. (R/W)

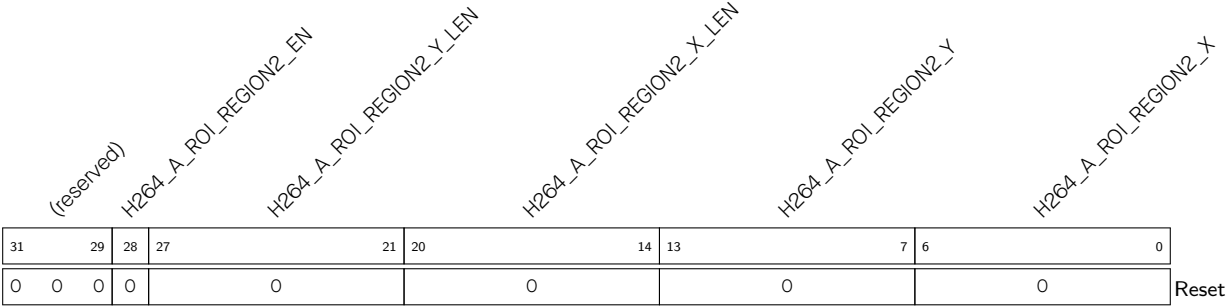
H264_A_ROI_REGION1_EN Configures whether to enable Video sequence A ROI 1.

0: Disable ROI

1: Enable ROI

(R/W)

Register 39.12. H264_A_ROI_REGION2_REG (0x002C)



H264_A_ROI_REGION2_X Configures the horizontal start MBs of ROI 2 in Video sequence A. (R/W)

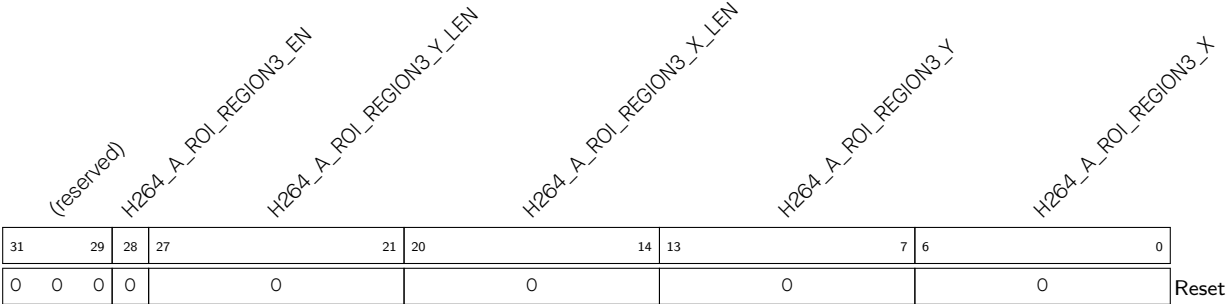
H264_A_ROI_REGION2_Y Configures the vertical start MBs of ROI 2 in Video sequence A. (R/W)

H264_A_ROI_REGION2_X_LEN Configures the number of MBs in horizontal direction of the ROI 2 in Video sequence A. (R/W)

H264_A_ROI_REGION2_Y_LEN Configures the number of MBs in vertical direction of the ROI 2 in Video sequence A. (R/W)

H264_A_ROI_REGION2_EN Configures whether to enable Video sequence A ROI 2.
0: Disable ROI
1: Enable ROI
(R/W)

Register 39.13. H264_A_ROI_REGION3_REG (0x0030)



H264_A_ROI_REGION3_X Configures the horizontal start MBs of ROI 3 in Video sequence A. (R/W)

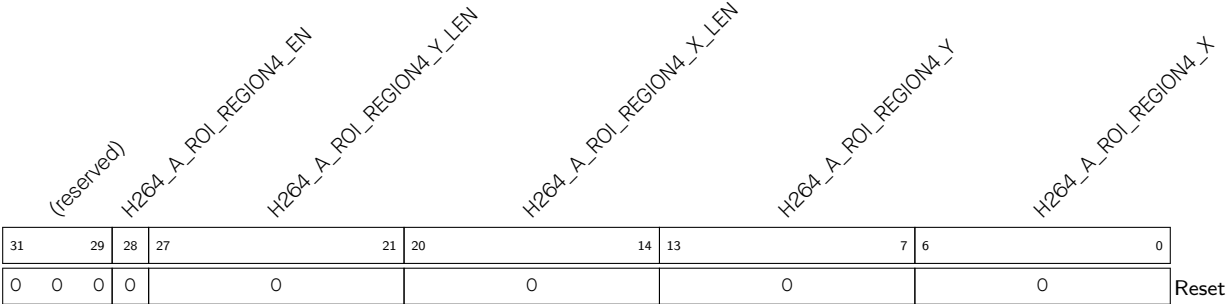
H264_A_ROI_REGION3_Y Configures the vertical start MBs of ROI 3 in Video sequence A. (R/W)

H264_A_ROI_REGION3_X_LEN Configures the number of MBs in horizontal direction of the ROI 3 in video sequence A. (R/W)

H264_A_ROI_REGION3_Y_LEN Configures the number of MBs in vertical direction of the ROI 3 in video sequence A. (R/W)

H264_A_ROI_REGION3_EN Configures whether to enable Video sequence A ROI 3.
0: Disable ROI
1: Enable ROI
(R/W)

Register 39.14. H264_A_ROI_REGION4_REG (0x0034)



H264_A_ROI_REGION4_X Configures the horizontal start MBs of ROI 4 in Video sequence A. (R/W)

H264_A_ROI_REGION4_Y Configures the vertical start MBs of ROI 4 in Video sequence A. (R/W)

H264_A_ROI_REGION4_X_LEN Configures the number of MBs in horizontal direction of the ROI 4 in video sequence A. (R/W)

H264_A_ROI_REGION4_Y_LEN Configures the number of MBs in vertical direction of the ROI 4 in video sequence A. (R/W)

H264_A_ROI_REGION4_EN Configures whether to enable Video sequence A ROI 4.
0: Disable ROI
1: Enable ROI
(R/W)

Register 39.15. H264_A_ROI_REGION5_REG (0x0038)

(reserved)				H264_A_ROI_REGION5_EN				H264_A_ROI_REGION5_Y_LEN				H264_A_ROI_REGION5_X_LEN				H264_A_ROI_REGION5_Y				H264_A_ROI_REGION5_X			
31	29	28	27	21				20	14				13	7				6	0				
0	0	0	0	0				0				0				0				Reset			

Reset

H264_A_ROI_REGION5_X Configures the horizontal start MBs of ROI 5 video sequence A. (R/W)

H264_A_ROI_REGION5_Y Configures the vertical start MBs of ROI 5 video sequence A. (R/W)

H264_A_ROI_REGION5_X_LEN Configures the number of MBs in horizontal direction of the ROI 5 video sequence A. (R/W)

H264_A_ROI_REGION5_Y_LEN Configures the number of MBs in vertical direction of the ROI 5 in video sequence A. (R/W)

H264_A_ROI_REGION5_EN Configures whether to enable Video sequence A ROI 5.

0: Disable ROI

1: Enable ROI

(R/W)

Register 39.16. H264_A_ROI_REGION6_REG (0x003C)

(reserved)				H264_A_ROI_REGION6_EN				H264_A_ROI_REGION6_Y_LEN				H264_A_ROI_REGION6_X_LEN				H264_A_ROI_REGION6_Y				H264_A_ROI_REGION6_X						
31	29	28	27					21	20					14	13					7	6					0
0	0	0	0	0				0				0				0				0				Reset		

Reset

H264_A_ROI_REGION6_X Configures the horizontal start MBs of ROI 6 video sequence A. (R/W)

H264_A_ROI_REGION6_Y Configures the vertical start MBs of ROI 6 in video sequence A. (R/W)

H264_A_ROI_REGION6_X_LEN Configures the number of MBs in horizontal direction of the ROI 6 in video sequence A. (R/W)

H264_A_ROI_REGION6_Y_LEN Configures the number of MBs in vertical direction of the ROI 6 in video sequence A. (R/W)

H264_A_ROI_REGION6_EN Configures whether to enable Video sequence A ROI 6.

0: Disable ROI

1: Enable ROI

(R/W)

Register 39.17. H264_A_ROI_REGION7_REG (0x0040)

(reserved)				H264_A_ROI_REGION7_EN				H264_A_ROI_REGION7_Y_LEN				H264_A_ROI_REGION7_X_LEN				H264_A_ROI_REGION7_Y				H264_A_ROI_REGION7_X			
31	29	28	27	21				20	14				13	7				6	0				
0	0	0	0	0				0				0				0				Reset			

Reset

H264_A_ROI_REGION7_X Configures the horizontal start MBs of ROI 7 in video sequence A. (R/W)

H264_A_ROI_REGION7_Y Configures the vertical start MBs of ROI 7 in video sequence A. (R/W)

H264_A_ROI_REGION7_X_LEN Configures the number of MBs in horizontal direction of the ROI 7 in video sequence A. (R/W)

H264_A_ROI_REGION7_Y_LEN Configures the number of MBs in vertical direction of the ROI 7 in video sequence A. (R/W)

H264_A_ROI_REGION7_EN Configures whether to enable Video sequence A ROI 7.

0: Disable ROI

1: Enable ROI

(R/W)

Register 39.18. H264_A_ROI_REGION0_3_QP_REG (0x0044)

(reserved)				H264_A_ROI_REGION3_QP				H264_A_ROI_REGION2_QP				H264_A_ROI_REGION1_QP				H264_A_ROI_REGION0_QP			
31	28	27	21	20	14	13	7	6	0										
0	0	0	0	0	0	0	0	0	0	Reset									

Reset

H264_A_ROI_REGION0_QP Configures H264 ROI 0 QP adjustment amount in video sequence A. (R/W)

H264_A_ROI_REGION1_QP Configures H264 ROI 1 QP adjustment amount in video sequence A. (R/W)

H264_A_ROI_REGION2_QP Configures H264 ROI 2 QP adjustment amount in video sequence A. (R/W)

H264_A_ROI_REGION3_QP Configures H264 ROI 3 QP adjustment amount in video sequence A. (R/W)

Register 39.19. H264_A_ROI_REGION4_7_QP_REG (0x0048)

(reserved)				H264_A_ROI_REGION7_QP				H264_A_ROI_REGION6_QP				H264_A_ROI_REGION5_QP				H264_A_ROI_REGION4_QP			
31	28	27		21	20		14	13		7	6		0						
0	0	0	0	0	0	0	0	0	0	0	0	0	0						
													Reset						

Reset

H264_A_ROI_REGION4_QP Configures H264 ROI 4 QP adjustment amount in video sequence A.
(R/W)

H264_A_ROI_REGION5_QP Configures H264 ROI 5 QP adjustment amount in video sequence A.
(R/W)

H264_A_ROI_REGION6_QP Configures H264 ROI 6 QP adjustment amount in video sequence A.
(R/W)

H264_A_ROI_REGION7_QP Configures H264 ROI 7 QP adjustment amount in video sequence A.
(R/W)

Register 39.20. H264_A_NO_ROI_REGION_QP_OFFSET_REG (0x004C)

(reserved)																								H264_A_NO_ROI_REGION_QP													
31																								7	6	0											
0 0																								0												Reset	

Reset

H264_A_NO_ROI_REGION_QP Configures H264 non-ROI QP in video sequence A, delta QP. (R/W)

Register 39.21. H264_A_ROI_CONFIG_REG (0x0050)

(reserved)																												H264_A_ROI_MODE H264_A_ROI_EN	
31																											2	1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Reset

H264_A_ROI_EN Configures whether to enable ROI in video sequence A.

0: Disable ROI

1: Enable ROI

(R/W)

H264_A_ROI_MODE Configures the mode of ROI in video sequence A.

0: Fixed QP

1: Delta QP

(R/W)

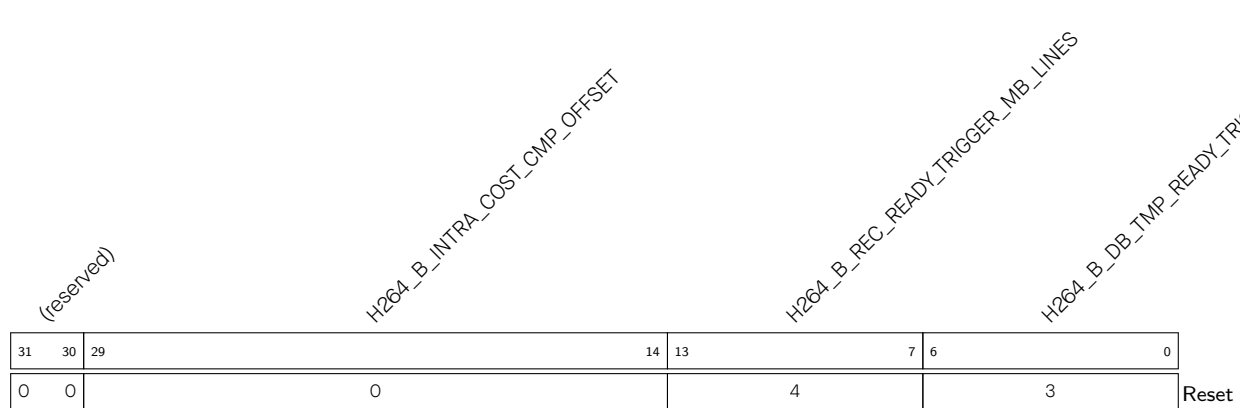
Register 39.22. H264_B_SYS_MB_RES_REG (0x0054)

(reserved)														H264_B_SYS_TOTAL_MB_X							H264_B_SYS_TOTAL_MB_Y																											
31														14							13							7							6							0						
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0														0							0							Reset																				

Reset

H264_B_SYS_TOTAL_MB_Y Configures video sequence B vertical MB resolution. (R/W)**H264_B_SYS_TOTAL_MB_X** Configures video sequence B horizontal MB resolution. (R/W)

Register 39.23. H264_B_SYS_CONF_REG (0x0058)

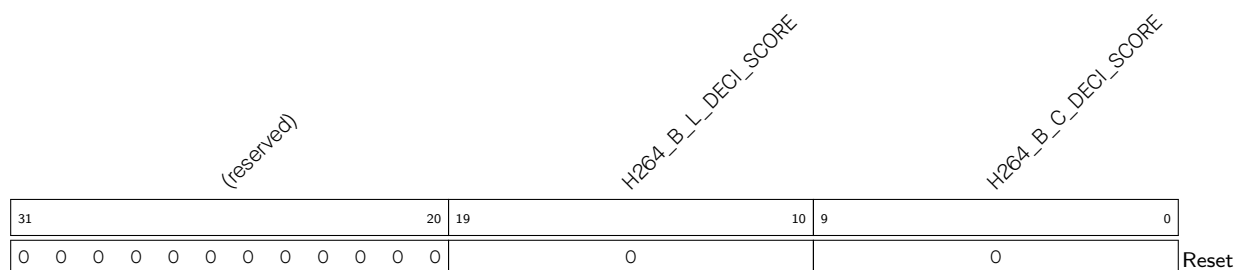


H264_B_DB_TMP_READY_TRIGGER_MB_NUM Configures when to trigger video sequence B H264_DB_TMP_READY_INT. When the (MB number of written db temp+1) is greater than this filed in first MB line, trigger H264_DB_TMP_READY_INT. Min is 3. (R/W)

H264_B_REC_READY_TRIGGER_MB_LINES Configures when to trigger video sequence B H264_REC_READY_INT. When the MB line number of generated reconstruct pixel is greater than this filed, trigger H264_REC_READY_INT. Min is 4. (R/W)

H264_B_INTRA_COST_CMP_OFFSET Configures video sequence B intra cost offset when I MB compared with P MB. (R/W)

Register 39.24. H264_B_DECI_SCORE_REG (0x005C)



H264_B_C_DECI_SCORE Configures video sequence B chroma MB decimate score. When chroma score is smaller than it, chroma decimate will be enabled. (R/W)

H264_B_L_DECI_SCORE Configures video sequence B luma MB decimate score. When luma score is smaller than it, luma decimate will be enabled. (R/W)

Register 39.25. H264_B_DECI_SCORE_OFFSET_REG (0x0060)

(reserved)								H264_B_P_CHROMA_DEC1_SCORE_OFFSET								H264_B_P16x16_DEC1_SCORE_OFFSET								H264_B_I_CHROMA_DEC1_SCORE_OFFSET								H264_B_I16x16_DEC1_SCORE_OFFSET																																															
31								24								23								18								17								12								11								6								5								0							
0 0 0 0 0 0 0 0								0								0								0								0								Reset																																							

H264_B_I16x16_DECI_SCORE_OFFSET Configures video sequence B i16 x 16 MB decimate score offset. This offset will be added to i16 x 16 MB score. (R/W)

H264_B_I_CHROMA_DECI_SCORE_OFFSET Configures video sequence B I chroma MB decimate score offset. This offset will be added to I chroma MB score. (R/W)

H264_B_P16x16_DECI_SCORE_OFFSET	Configures video sequence B p16 x 16 MB decimate score offset. This offset will be added to p16 x 16 MB score. (R/W)
--	--

H264_B_P_CHROMA_DECI_SCORE_OFFSET	Configures video sequence B p chroma MB decimate score offset. This offset will be added to p chroma MB score. (R/W)
--	--

Register 39.26. H264_B_RC_CONFO_REG (0x0064)

[illegible]

H264_B_QP Configures video sequence B frame level initial luma QP value. (R/W)

H264_B_RATE_CTRL_U Configures video sequence B parameter U value. $U = \text{int}((\text{float } u \ll 8)$.
(R/W)

H264_B_MB_RATE_CTRL_EN Configures video sequence B whether to enable MB level rate control.

0: Disable the MB level rate control

1: Enable the MB level rate control

(R/W)

Register 39.27. H264_B_RC_CONF1_REG (0x0068)

(reserved)		H264_B_MAD_FRAME_PRED																H264_B_QP_MAX				H264_B_QP_MIN		H264_B_CHROMA_QP_DELTA				H264_B_CHROMA_DC_QP_DELTA				
31	30	19																18	13				12	7		6	3				2	0
0	0																0				0				0		0		Reset			

H264_B_CHROMA_DC_QP_DELTA Configures video sequence B chroma DC QP offset based on Chroma QP. Chroma DC QP = Chroma QP(after ping) + reg_chroma_dc_QP_delta. (R/W)

H264_B_CHROMA_QP_DELTA Configures video sequence B chroma QP offset based on luma QP. Chroma QP(before mapping) = Luma QP + reg_chroma_QP_delta. (R/W)

H264_B_QP_MIN Configures the minimum allowed luma QP value for video sequence B. (R/W)

H264_B_QP_MAX Configures the maximum allowed luma QP value for video sequence B. (R/W)

H264_B_MAD_FRAME_PRED Configures the MAD value for frame-level predicted MB in video sequence B. (R/W)

Register 39.28. H264_B_DB_BYPASS_REG (0x006C)

(reserved)																																H264_B_BYPASS_DB_FILTER			
31																															1	0			
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset

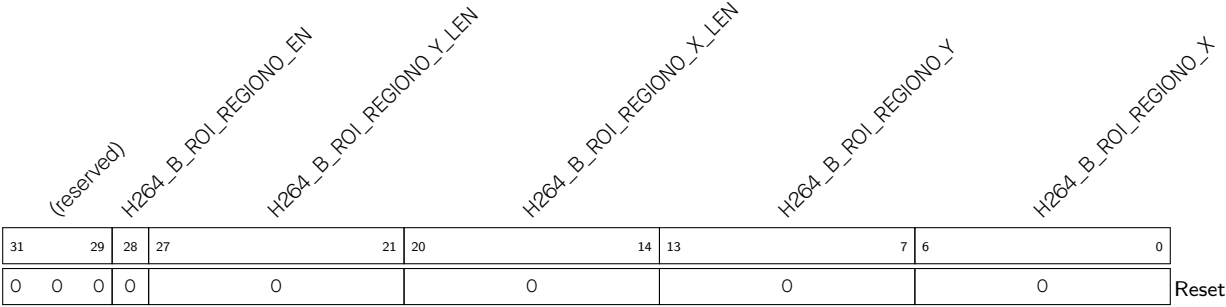
H264_B_BYPASS_DB_FILTER Configures whether to bypass video sequence B deblocking filter.

0: Not bypass the deblocking filter

1: Bypass the deblocking filter

(R/W)

Register 39.29. H264_B_ROI_REGION0_REG (0x0070)



H264_B_ROI_REGION0_X Configures the horizontal start MBs of ROI 0 in Video sequence B. (R/W)

H264_B_ROI_REGION0_Y Configures the vertical start MBs of ROI 0 in Video sequence B. (R/W)

H264_B_ROI_REGION0_X_LEN Configures the number of MBs in horizontal direction of the ROI 0 in Video sequence B. (R/W)

H264_B_ROI_REGION0_Y_LEN Configures the number of MBs in vertical direction of the ROI 0 in Video sequence B. (R/W)

H264_B_ROI_REGION0_EN Configures whether to enable Video sequence B ROI 0.

- 0: Disable ROI
- 1: Enable ROI

(R/W)

Register 39.30. H264_B_ROI_REGION1_REG (0x0074)

(reserved)				H264_B_ROI_REGION1_EN				H264_B_ROI_REGION1_Y_LEN				H264_B_ROI_REGION1_X_LEN				H264_B_ROI_REGION1_Y				H264_B_ROI_REGION1_X			
31	29	28	27	21				20	14				13	7				6	0				
0	0	0	0	0				0				0				0				Reset			

Reset

H264_B_ROI_REGION1_X Configures the horizontal start MBs of ROI 1 in Video sequence B. (R/W)

H264_B_ROI_REGION1_Y Configures the vertical start MBs of ROI 1 in Video sequence B. (R/W)

H264_B_ROI_REGION1_X_LEN Configures the number of MBs in horizontal direction of the ROI 1 in Video sequence B. (R/W)

H264_B_ROI_REGION1_Y_LEN Configures the number of MBs in vertical direction of the ROI 1 in Video sequence B. (R/W)

H264_B_ROI_REGION1_EN Configures whether to enable Video sequence B ROI 1.

0: Disable ROI

1: Enable ROI

(R/W)

Register 39.31. H264_B_ROI_REGION2_REG (0x0078)

(reserved)				H264_B_ROI_REGION2_EN				H264_B_ROI_REGION2_Y_LEN				H264_B_ROI_REGION2_X_LEN				H264_B_ROI_REGION2_Y				H264_B_ROI_REGION2_X			
31	29	28	27	21	20	14	13	7	6	0													
0	0	0	0	0	0	0	0	0	0	Reset													

H264_B_ROI_REGION2_X Configures the horizontal start MBs of ROI 2 in Video sequence B. (R/W)

H264_B_ROI_REGION2_Y Configures the vertical start MBs of ROI 2 in Video sequence B. (R/W)

H264_B_ROI_REGION2_X_LEN Configures the number of MBs in horizontal direction of the ROI 2 in Video sequence B. (R/W)

H264_B_ROI_REGION2_Y_LEN Configures the number of MBs in vertical direction of the ROI 2 in Video sequence B. (R/W)

H264_B_ROI_REGION2_EN Configures whether to enable Video sequence B ROI 2.

- 0: Disable ROI
- 1: Enable ROI

(R/W)

Register 39.32. H264_B_ROI_REGION3_REG (0x007C)

(reserved)			H264_B_ROI_REGION3_EN			H264_B_ROI_REGION3_Y_LEN			H264_B_ROI_REGION3_X_LEN			H264_B_ROI_REGION3_Y			H264_B_ROI_REGION3_X		
31	29	28	27	21	20	14	13	7	6	0							
0	0	0	0	0		0		0		0		0		Reset			

H264_B_ROI_REGION3_X Configures the horizontal start MBs of ROI 3 in Video sequence B. (R/W)

H264_B_ROI_REGION3_Y Configures the vertical start MBs of ROI 3 in Video sequence B. (R/W)

H264_B_ROI_REGION3_X_LEN Configures the number of MBs in horizontal direction of the ROI 3 in video sequence B. (R/W)

H264_B_ROI_REGION3_Y_LEN Configures the number of MBs in vertical direction of the ROI 3 in video sequence B. (R/W)

H264_B_ROI_REGION3_EN Configures whether to enable Video sequence B ROI 3.

0: Disable ROI

1: Enable ROI

(R/W)

Register 39.33. H264_B_ROI_REGION4_REG (0x0080)

(reserved)				H264_B_ROI_REGION4_EN				H264_B_ROI_REGION4_Y_LEN				H264_B_ROI_REGION4_X_LEN				H264_B_ROI_REGION4_Y				H264_B_ROI_REGION4_X			
31	29	28	27	21	20	14	13	7	6	0													
0	0	0	0	0				0				0				0				Reset			

H264_B_ROI_REGION4_X Configures the horizontal start MBs of ROI 4 in Video sequence B. (R/W)

H264_B_ROI_REGION4_Y Configures the vertical start MBs of ROI 4 in Video sequence B. (R/W)

H264_B_ROI_REGION4_X_LEN Configures the number of MBs in horizontal direction of the ROI 4 in video sequence B. (R/W)

H264_B_ROI_REGION4_Y_LEN Configures the number of MBs in vertical direction of the ROI 4 in video sequence B. (R/W)

H264_B_ROI_REGION4_EN Configures whether to enable Video sequence B ROI 4.

0: Disable ROI

1: Enable ROI

(R/W)

Register 39.34. H264_B_ROI_REGION5_REG (0x0084)

(reserved)				H264_B_ROI_REGION5_EN				H264_B_ROI_REGION5_Y_LEN				H264_B_ROI_REGION5_X_LEN				H264_B_ROI_REGION5_Y				H264_B_ROI_REGION5_X			
31	29	28	27	21				20	14				13	7				6	0				
0	0	0	0	0				0				0				0				Reset			

Reset

H264_B_ROI_REGION5_X Configures the horizontal start MBs of ROI 5 video sequence B. (R/W)

H264_B_ROI_REGION5_Y Configures the vertical start MBs of ROI 5 video sequence B. (R/W)

H264_B_ROI_REGION5_X_LEN Configures the number of MBs in horizontal direction of the ROI 5 video sequence B. (R/W)

H264_B_ROI_REGION5_Y_LEN Configures the number of MBs in vertical direction of the ROI 5 in video sequence B. (R/W)

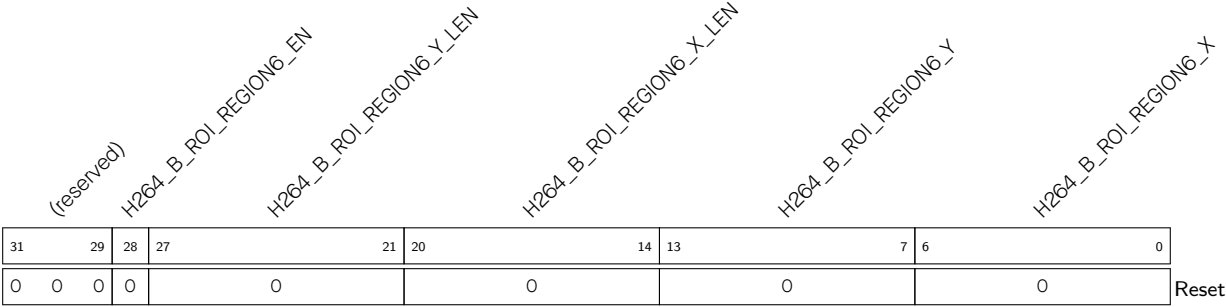
H264_B_ROI_REGION5_EN Configures whether to enable Video sequence B ROI 5.

0: Disable ROI

1: Enable ROI

(R/W)

Register 39.35. H264_B_ROI_REGION6_REG (0x0088)



H264_B_ROI_REGION6_X Configures the horizontal start MBs of ROI 6 video sequence B. (R/W)

H264_B_ROI_REGION6_Y Configures the vertical start MBs of ROI 6 in video sequence B. (R/W)

H264_B_ROI_REGION6_X_LEN Configures the number of MBs in horizontal direction of the ROI 6 in video sequence B. (R/W)

H264_B_ROI_REGION6_Y_LEN Configures the number of MBs in vertical direction of the ROI 6 in video sequence B. (R/W)

H264_B_ROI_REGION6_EN Configures whether to enable Video sequence B ROI 6.

- 0: Disable ROI
- 1: Enable ROI

(R/W)

Register 39.36. H264_B_ROI_REGION7_REG (0x008C)

(reserved)				H264_B_ROI_REGION7_EN				H264_B_ROI_REGION7_Y_LEN				H264_B_ROI_REGION7_X_LEN				H264_B_ROI_REGION7_Y				H264_B_ROI_REGION7_X			
31	29	28	27	21				20	14				13	7				6	0				
0	0	0	0	0				0				0				0				Reset			

Reset

H264_B_ROI_REGION7_X Configures the horizontal start MBs of ROI 7 in video sequence B. (R/W)

H264_B_ROI_REGION7_Y Configures the vertical start MBs of ROI 7 in video sequence B. (R/W)

H264_B_ROI_REGION7_X_LEN Configures the number of MBs in horizontal direction of the ROI 7 in video sequence B. (R/W)

H264_B_ROI_REGION7_Y_LEN Configures the number of MBs in vertical direction of the ROI 7 in video sequence B. (R/W)

H264_B_ROI_REGION7_EN Configures whether to enable Video sequence B ROI 7.

0: Disable ROI

1: Enable ROI

(R/W)

Register 39.37. H264_B_ROI_REGION0_3_QP_REG (0x0090)

(reserved)				H264_B_ROI_REGION3_QP				H264_B_ROI_REGION2_QP				H264_B_ROI_REGION1_QP				H264_B_ROI_REGION0_QP			
31	28	27	21	20	14	13	7	6	0										
0	0	0	0	0	0	0	0	0	0	Reset									

Reset

H264_B_ROI_REGION0_QP Configures H264 ROI 0 QP adjustment amount in video sequence B. (R/W)

H264_B_ROI_REGION1_QP Configures H264 ROI 1 QP adjustment amount in video sequence B. (R/W)

H264_B_ROI_REGION2_QP Configures H264 ROI 2 QP adjustment amount in video sequence B. (R/W)

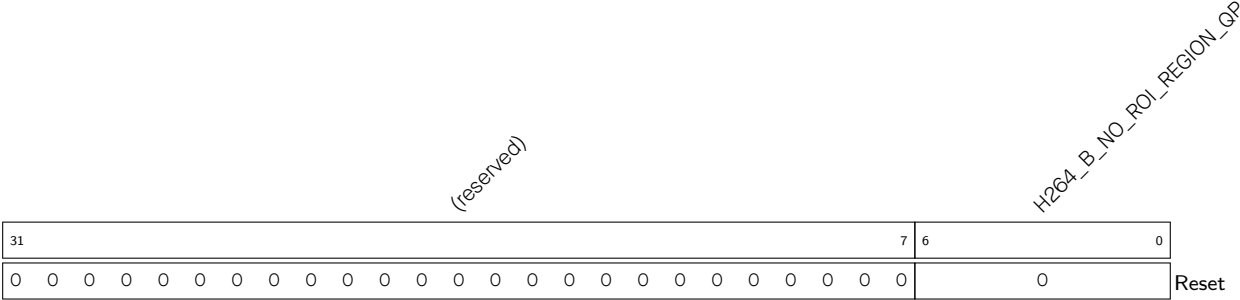
H264_B_ROI_REGION3_QP Configures H264 ROI 3 QP adjustment amount in video sequence B. (R/W)

Register 39.38. H264_B_ROI_REGION4_7_QP_REG (0x0094)



- H264_B_ROI_REGION4_QP Configures H264 ROI 4 QP adjustment amount in video sequence B.
(R/W)
- H264_B_ROI_REGION5_QP Configures H264 ROI 5 QP adjustment amount in video sequence B.
(R/W)
- H264_B_ROI_REGION6_QP Configures H264 ROI 6 QP adjustment amount in video sequence B.
(R/W)
- H264_B_ROI_REGION7_QP Configures H264 ROI 7 QP adjustment amount in video sequence B.
(R/W)

Register 39.39. H264_B_NO_ROI_REGION_QP_OFFSET_REG (0x0098)



- H264_B_NO_ROI_REGION_QP Configures H264 non-ROI QP in video sequence B, delta QP. (R/W)

Register 39.40. H264_B_ROI_CONFIG_REG (0x009C)

(reserved)																												H264_B_ROI_MODE H264_B_ROI_EN	
31																											2	1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Reset

H264_B_ROI_EN Configures whether to enable ROI in video sequence B.

0: Disable ROI

1: Enable ROI

(R/W)

H264_B_ROI_MODE Configures the mode of ROI in video sequence B.

0: Fixed QP

1: Delta QP

(R/W)

Register 39.41. H264_SLICE_HEADER_REMAIN_REG (0x00AC)

(reserved)																				H264_SLICE_RMEAIN_BIT												H264_SLICE_REMAIN_BITLENGTH																						
31																				11											10											3	2	0										
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0																				0											0											Reset												

H264_SLICE_REMAIN_BITLENGTH Configures slice header remain bit number. (R/W)

H264_SLICE_REMAIN_BIT Configures slice header remain bit. (R/W)

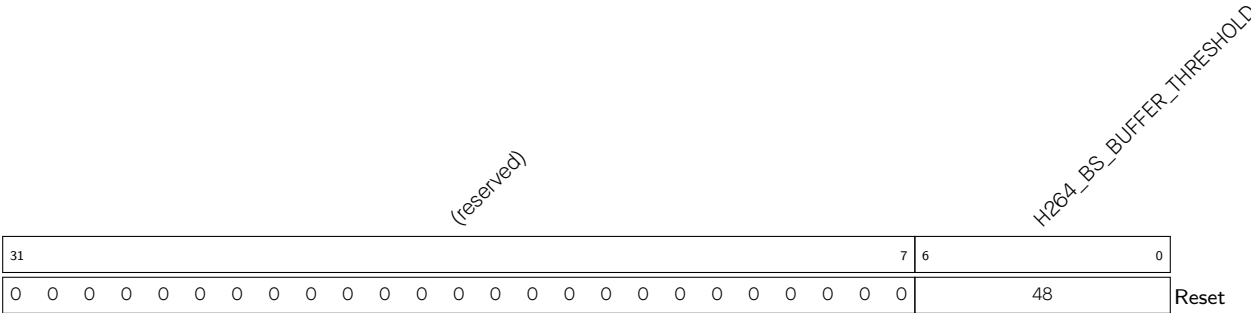
Register 39.42. H264_SLICE_HEADER_BYTE_LENGTH_REG (0x00B0)

(reserved)																						H264_SLICE_BYTE_LENGTH			
31																					4	3			0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Reset

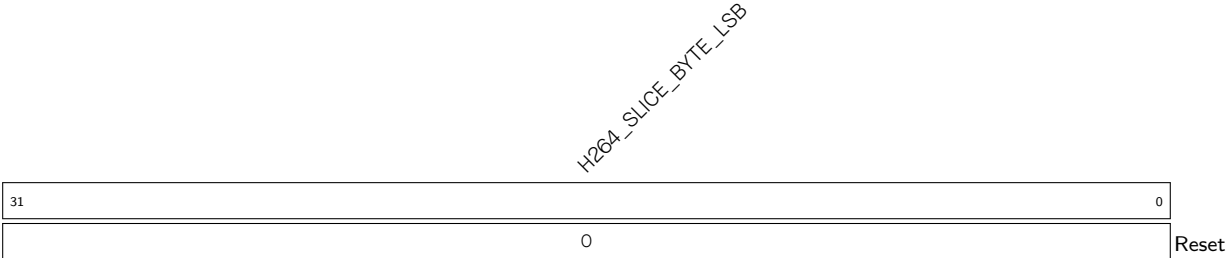
H264_SLICE_BYTE_LENGTH Configures slice header byte number. (R/W)

Register 39.43. H264_BS_THRESHOLD_REG (0x00B4)



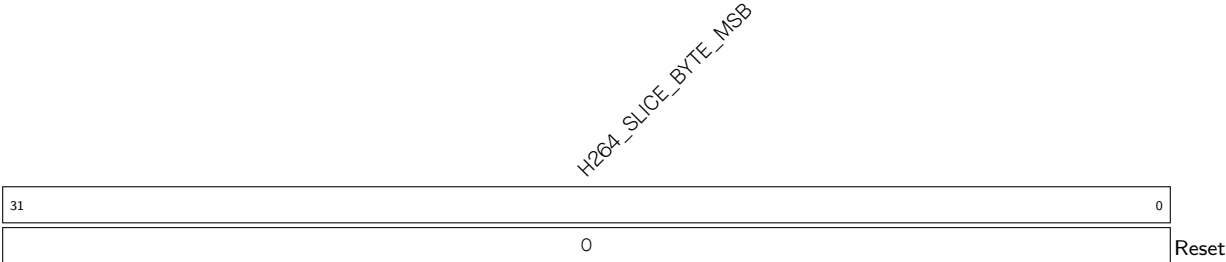
H264_BS_BUFFER_THRESHOLD Configures bitstream buffer overflow threshold. This value should be bigger than the encoded bytes of one 4x4 sub MB. (R/W)

Register 39.44. H264_SLICE_HEADER_BYTE0_REG (0x00B8)



H264_SLICE_BYTE_LSB Configures slice header low 32 bit. (R/W)

Register 39.45. H264_SLICE_HEADER_BYTE1_REG (0x00BC)



H264_SLICE_BYTE_MSB Configures slice header high 32 bit. (R/W)

Register 39.46. H264_CONF_REG (0x00D0)

(reserved)																H264_CLK_EN			
31																		1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
																			Reset

H264_CLK_EN Configures whether to enable the register clock gate.

0: Enable the clock gate only when application writes registers

1: Force enable the clock gate for register

(R/W)

Register 39.47. H264_MV_MERGE_CONFIG_REG (0x00D4)

(reserved)														H264_MB_VALID_NUM												H264_B_MV_MERGE_EN					H264_A_MV_MERGE_EN					H264_INT_MV_OUT_EN					H264_MV_MERGE_TYPE												
31														18														17												5					4	3	2	1	0				
0														0														0					0	0	0	0	Reset																

H264_MV_MERGE_TYPE Configures MV merge type.

0: Merge p16x16 mv

1: Merge min mv

2: Merge max mv

3: Not valid

(R/W)

H264_INT_MV_OUT_EN Configures the output type of the merged MV.

0: Output the merged MV where any part (integer and fractional) is not zero

1: Output the merged MV where the integer part is not zero

(R/W)

H264_A_MV_MERGE_EN Configures whether to enable MV merge for video sequence A.

0: Disable

1: Enable

(R/W)

H264_B_MV_MERGE_EN Configures whether to enable MV merge for video sequence B.

0: Disable

1: Enable

(R/W)

H264_MB_VALID_NUM Represents the number of MBs in the output that contains valid merged MVs. (RO)

Register 39.48. H264_RC_STATUS0_REG (0x00A0)

[illegible]

H264_FRAME_MAD_SUM Represents the sum of all MB actual MAD for one frame. (RO)

Register 39.49. H264_RC_STATUS1_REG (0x00A4)

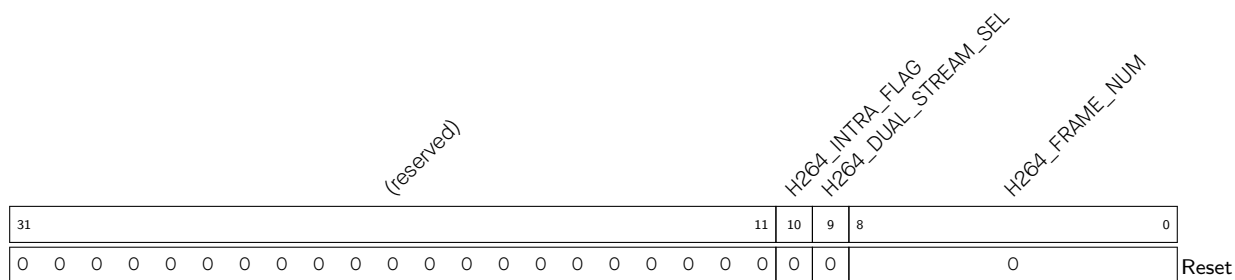
Diagram illustrating the H264_FRAME_ENC register structure. The register is 32 bits wide. The H264_FRAME_ENC_BITS field is 26 bits wide, starting at bit 26 and ending at bit 31. The field is divided into a 5-bit sub-field (bits 26-30) and a 21-bit sub-field (bits 31-31). The 5-bit sub-field is labeled "(reserved)" and contains the value 00000. The 21-bit sub-field is labeled "H264_FRAME_ENC_BITS" and contains the value 0. The register is labeled "Reset".

H264_FRAME_ENC_BITS Represents the sum of all MB actual encoding bits for one frame. (RO)

Register 39.50. H264_RC_STATUS2_REG (0x00A8)

(reserved)														H264_FRAME_QP_SUM																		
31														0																		
19														18																		
0 0 0 0 0 0 0 0 0 0 0 0 0 0														0																		
																																Reset

H264_FRAME_QP_SUM Represents all MB actual luma QP sum value of one frame. (RO)

Register 39.51. H264_SYS_STATUS_REG (0x00DC)

H264_FRAME_NUM Represents the current frame number. (RO)

H264_DUAL_STREAM_SEL Represents which register group is used for the current frame.

0: Register group A is used

1: Register group B is used

(RO)

H264_INTRA_FLAG Represents the type of the current encoding frame.

0: P frame

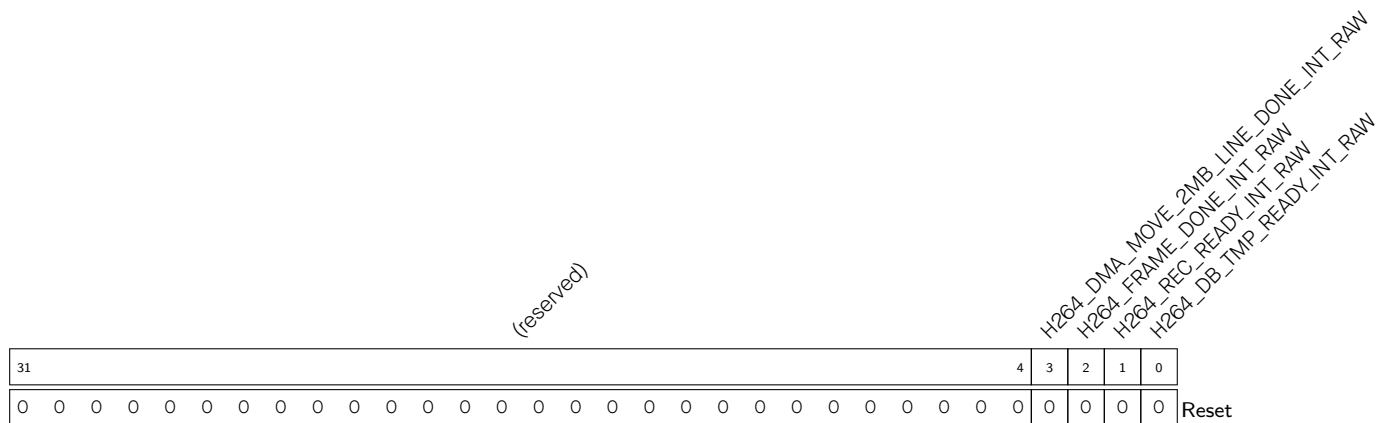
1: I frame

(RO)

Register 39.52. H264_FRAME_CODE_LENGTH_REG (0x00E0)

H264_FRAME_CODE_LENGTH Represents the byte length of the current frame code. (RO)

Register 39.53. H264_INT_RAW_REG (0x00C0)



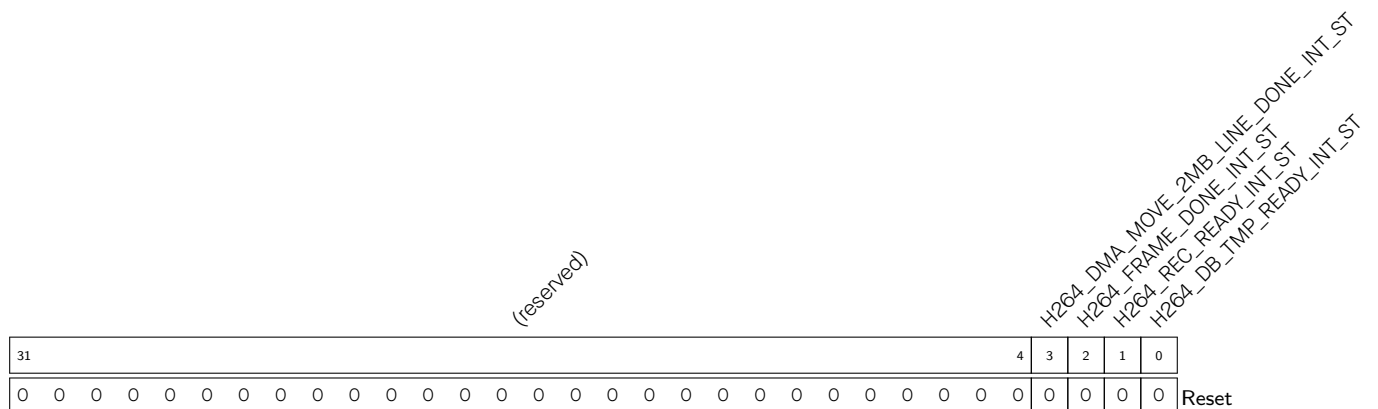
H264_DB_TMP_READY_INT_RAW The raw interrupt status of H264_DB_TMP_READY_INT. (R/WTC/SS)

H264_REC_READY_INT_RAW The raw interrupt status of H264_REC_READY_INT. (R/WTC/SS)

H264_FRAME_DONE_INT_RAW The raw interrupt status of H264_FRAME_DONE_INT. (R/WTC/SS)

H264_DMA_MOVE_2MB_LINE_DONE_INT_RAW The raw interrupt status of H264_DMA_MOVE_2MB_LINE_DONE_INT. (R/WTC/SS)

Register 39.54. H264_INT_ST_REG (0x00C4)



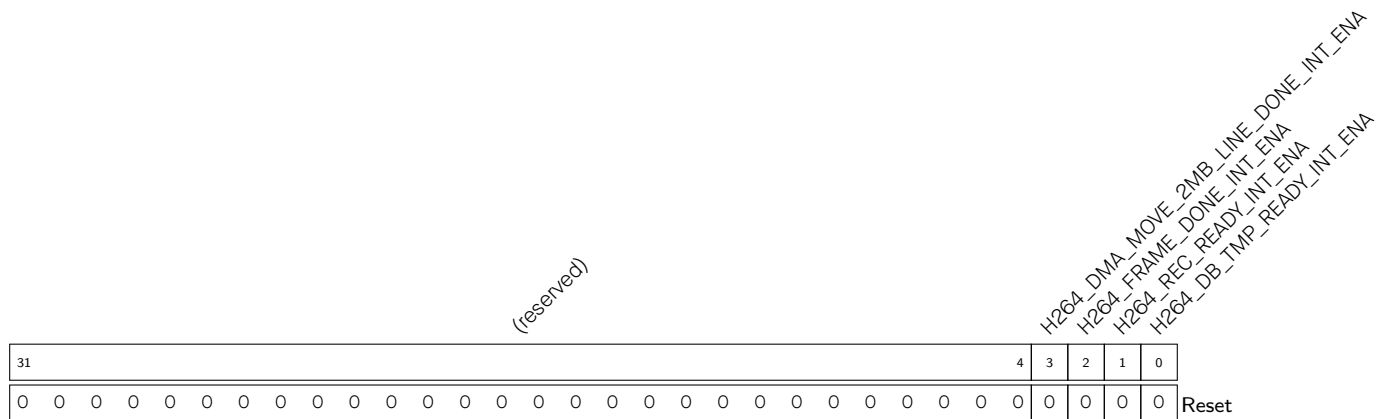
H264_DB_TMP_READY_INT_ST The masked interrupt status of H264_DB_TMP_READY_INT. (RO)

H264_REC_READY_INT_ST The masked interrupt status of H264_REC_READY_INT. (RO)

H264_FRAME_DONE_INT_ST The masked interrupt status of H264_FRAME_DONE_INT. (RO)

H264_DMA_MOVE_2MB_LINE_DONE_INT_ST The masked interrupt status of H264_DMA_MOVE_2MB_LINE_DONE_INT. (RO)

Register 39.55. H264_INT_ENA_REG (0x00C8)



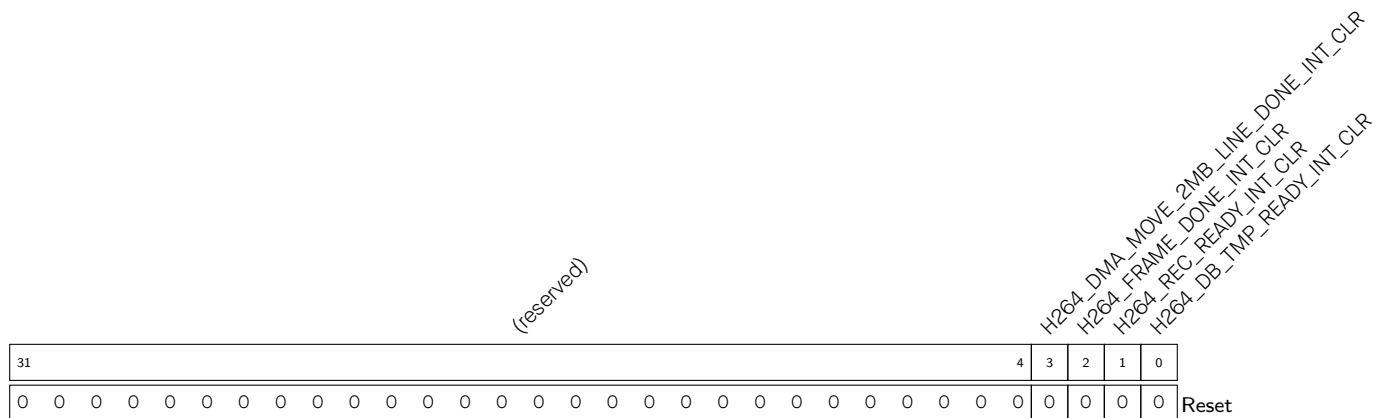
H264_DB_TMP_READY_INT_ENA Write 1 to enable H264_DB_TMP_READY_INT. (R/W)

H264_REC_READY_INT_ENA Write 1 to enable H264_REC_READY_INT. (R/W)

H264_FRAME_DONE_INT_ENA Write 1 to enable H264_FRAME_DONE_INT. (R/W)

H264_DMA_MOVE_2MB_LINE_DONE_INT_ENA Write 1 to enable H264_DMA_MOVE_2MB_LINE_DONE_INT. (R/W)

Register 39.56. H264_INT_CLR_REG (0x00CC)



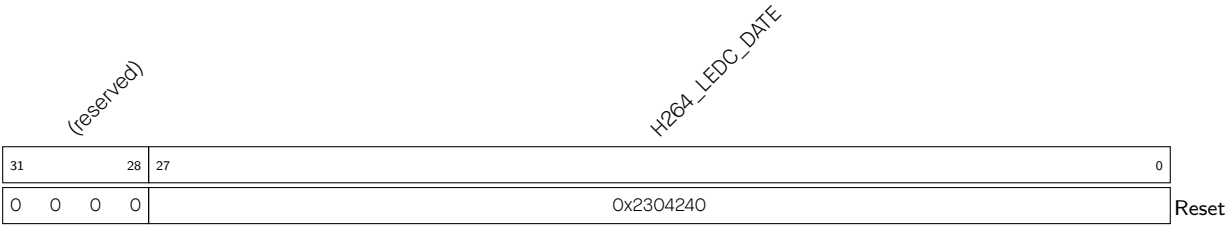
H264_DB_TMP_READY_INT_CLR Write 1 to clear H264_DB_TMP_READY_INT. (WT)

H264_REC_READY_INT_CLR Write 1 to clear H264_REC_READY_INT. (WT)

H264_FRAME_DONE_INT_CLR Write 1 to clear H264_FRAME_DONE_INT. (WT)

H264_DMA_MOVE_2MB_LINE_DONE_INT_CLR Write 1 to clear H264_DMA_MOVE_2MB_LINE_DONE_INT. (WT)

Register 39.57. H264_DATE_REG (0x00F0)



H264_LEDC_DATE Version control register. (R/W)

39.9.2 H264 DMA Registers

Register 39.58. H264_DMA_OUT_CONFO_CHO_REG (0x0000)

[illegible]

H264_DMA_OUT_AUTO_WRBACK_CHO Configures whether to enable automatic outlink-writeback when all the data pointed by outlink descriptor has been received.

0: Disable

1: Enable

(R/W)

H264_DMA_OUT_EOF_MODE_CHO Configures EOF flag generation mode for TX channel 0.

0: EOF flag is generated when last data has been pushed into FIFO in DMA

1: EOF flag is generated when last data has been popped from FIFO in DMA

(R/W)

H264_DMA_OUTDSCR_BURST_EN_CHO Configures whether to enable INCR burst transfer for TX channel 0 to read descriptors.

0: Disable

1: Enable

(R/W)

H264_DMA_OUT_ECC_AES_EN_CHO Configures whether DMA can access external memory space for ECC and AES via TX channel 0.

0: Disable access

1: Enable access to the ECC/AES area. In this case, the start address of the memory region should be 16-byte aligned. The product of the width and the bytes per pixel must also be 16-byte aligned.

(R/W)

H264_DMA_OUT_CHECK_OWNER_CHO Configures whether to enable owner bit check for TX channel 0.

0: Disable

1: Enable

(R/W)

Continued on the next page...

Register 39.58. H264_DMA_OUT_CONFO_CHO_REG (0x0000)

Continued from the previous page...

H264_DMA_OUT_MEM_BURST_LENGTH_CHO Configures the burst length for TX channel 0.

- 0: 8 bytes
 - 1: 16 bytes
 - 2: 32 bytes
 - 3: 64 bytes
 - 4: 128 bytes
 - 5 ~ 7: Invalid
- (R/W)

H264_DMA_OUT_PAGE_BOUND_EN_CHO Configures whether to make sure AXI read data doesn't cross the address boundary which define by mem_burst_length.

- 0: AXI read data can cross the address boundary
 - 1: AXI read data doesn't cross the address boundary
- (R/W)

H264_DMA_OUT_REORDER_EN_CHO Configures whether to enable TX channel 0 macro block re-order.

- 0: Disable
 - 1: Enable
- (R/W)

H264_DMA_OUT_RST_CHO Write 1 then write 0 to reset TX channel 0. (R/W)

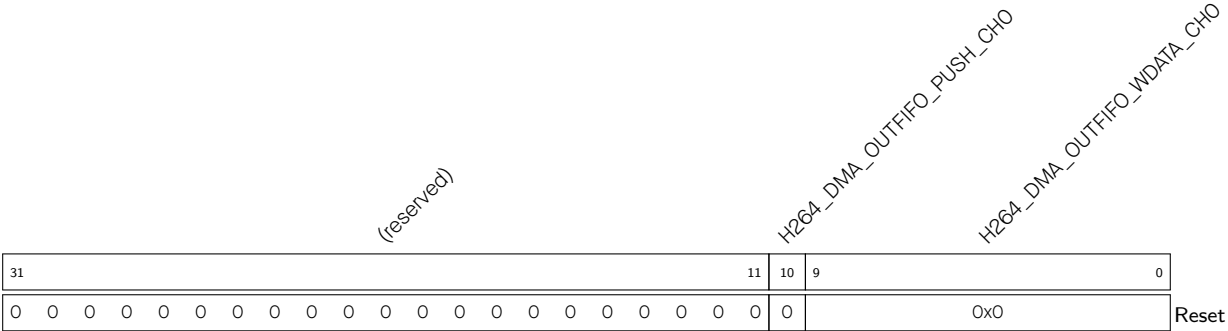
H264_DMA_OUT_CMD_DISABLE_CHO Configures whether to disable command on TX channel 0.

- 0: Disable
 - 1: Enable
- (R/W)

H264_DMA_OUT_ARB_WEIGHT_OPT_DIS_CHO Configures whether to disable weight arbitration for TX channel 0.

- 0: Enable
 - 1: Disable
- (R/W)

Register 39.59. H264_DMA_OUT_PUSH_CHO_REG (0x0018)



- H264_DMA_OUTFIFO_WDATA_CHO** Represents the data that needs to be pushed into DMA TX FIFO. (R/W)
- H264_DMA_OUTFIFO_PUSH_CHO** Configures whether to push data into DMA TX FIFO.
0: No effect
1: Push
(R/W/SC)

Register 39.60. H264_DMA_OUT_LINK_CONF_CHO_REG (0x001C)

(reserved)								H264_DMA_OUTLINK_PARK_CHO H264_DMA_OUTLINK_RESTART_CHO H264_DMA_OUTLINK_START_CHO H264_DMA_OUTLINK_STOP_CHO																(reserved)							
31								24	23	22	21	20	19																	0	
0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset

H264_DMA_OUTLINK_STOP_CHO Configures whether to stop processing the outlink descriptors.

0: No effect

1: Stop

(R/W/SC)

H264_DMA_OUTLINK_START_CHO Configures whether to start processing the outlink descriptors.

0: No effect

1: Start

(R/W/SC)

H264_DMA_OUTLINK_RESTART_CHO Configures whether to restart a new outlink from the last address.

0: No effect

1: Restart

(R/W/SC)

H264_DMA_OUTLINK_PARK_CHO Represents the status of the transmit descriptor's FSM.

0: The outlink descriptor's FSM is working

1: The outlink descriptor's FSM is in idle state

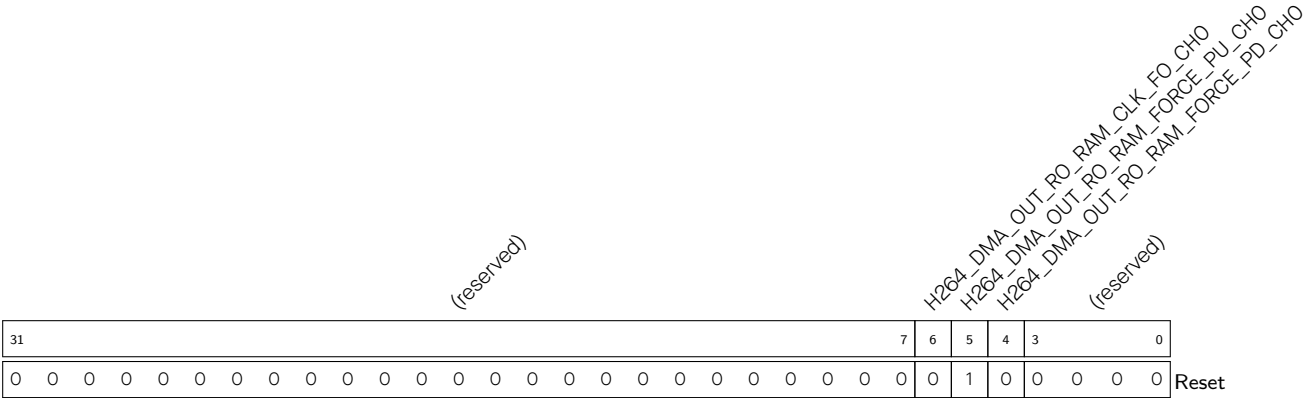
(RO)

Register 39.61. H264_DMA_OUT_LINK_ADDR_CHO_REG (0x0020)

H264_DMA_OUTLINK_ADDR_CHO																															
31																															0
0x000000																															
Reset																															

H264_DMA_OUTLINK_ADDR_CHO Configures the first outlink descriptor's address. (R/W)

Register 39.63. H264_DMA_OUT_RO_PD_CONF_CHO_REG (0x0044)



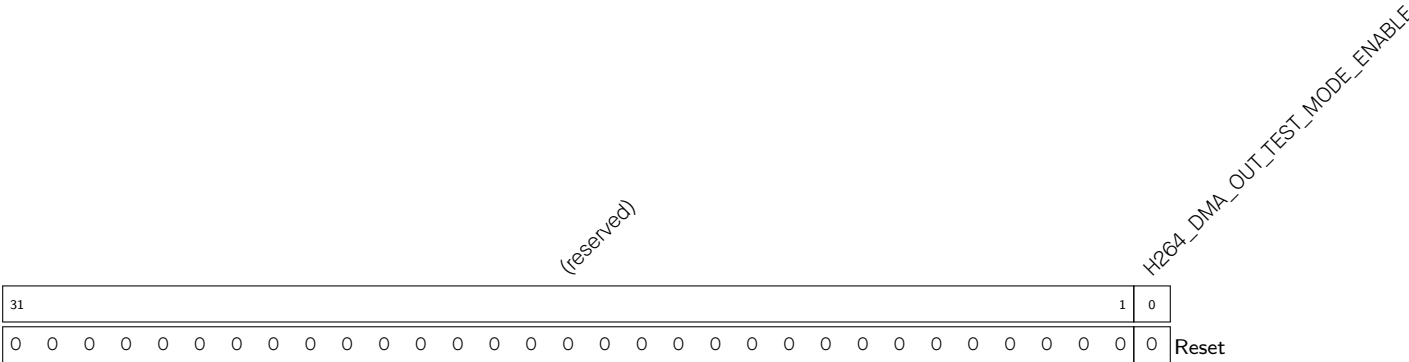
- H264_DMA_OUT_RO_RAM_FORCE_PD_CHO

Configures whether to force power down for DMA reorder RAM.
0: Not force power down
1: Force power down
(R/W)
- H264_DMA_OUT_RO_RAM_FORCE_PU_CHO

Configures whether to force power up for DMA reorder RAM.
0: Not force power up
1: Force power up
(R/W)
- H264_DMA_OUT_RO_RAM_CLK_FO_CHO

Configures whether to force enable clock gating for DMA TX reorder RAM.
0: A clock gating will be used when accessing the RAM in DMA
1: Force to open the clock and bypass the clock gating when accessing the RAM in DMA
(R/W)

Register 39.64. H264_DMA_OUT_MODE_ENABLE_CHO_REG (0x0050)



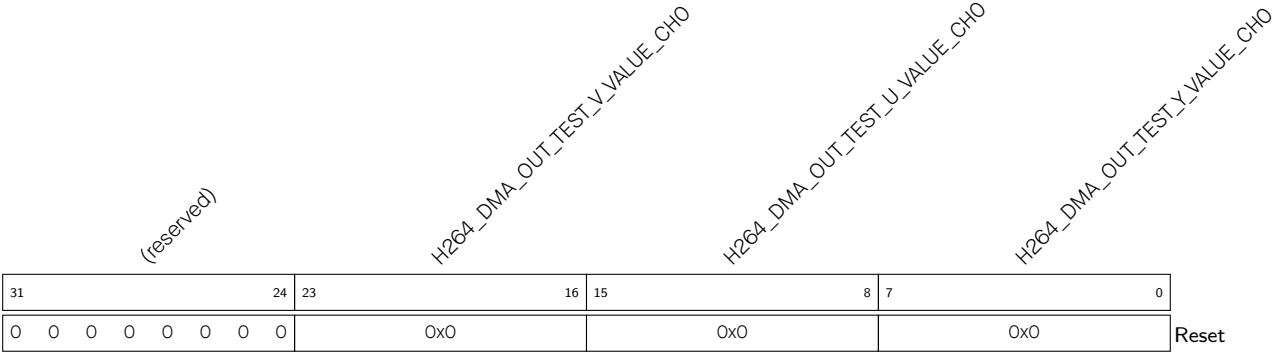
H264_DMA_OUT_TEST_MODE_ENABLE_CHO Configures whether to enable test mode of TX channel 0.

0: TX channel 0 work in normal mode

1: TX channel 0 work in test mode

(R/W)

Register 39.65. H264_DMA_OUT_MODE_YUV_CHO_REG (0x0054)



H264_DMA_OUT_TEST_Y_VALUE_CHO Configures y value in test mode. (R/W)

H264_DMA_OUT_TEST_U_VALUE_CHO Configures u value in test mode. (R/W)

H264_DMA_OUT_TEST_V_VALUE_CHO Configures v value in test mode. (R/W)

Register 39.66. H264_DMA_OUT_CONFO_CH1_REG (0x0100)

(reserved)						H264_DMA_OUT_ARB_WEIGHT_OPT_DIS_CH1												(reserved)												H264_DMA_OUT_PAGE_BOUND_EN_CH1				(reserved)				H264_DMA_OUT_MEM_BURST_LENGTH_CH1				H264_DMA_OUT_CHECK_OWNER_CH1				H264_DMA_OUT_ECC_AES_EN_CH1				H264_DMA_OUT_EOF_MODE_CH1				H264_DMA_OUT_AUTO_WRRBACK_CH1			
31					27		26	25	24	23													13		12	11	9		8	6		5	4	3	2	1	0																				
0					0		0	0	0	0													0		0	0	0		0x0	0		0	0	0	0	1	0	Reset																			

H264_DMA_OUT_AUTO_WRBACK_CH1 Configures whether to enable automatic outlink-writeback when all the data pointed by outlink descriptor has been received.

0: Disable

1: Enable

(R/W)

H264_DMA_OUT_EOF_MODE_CH1 Configures EOF flag generation mode for TX channel 1.

0: EOF flag is generated when last data has been pushed into FIFO in DMA

1: EOF flag is generated when last data has been popped from FIFO in DMA

(R/W)

H264_DMA_OUTDSCR_BURST_EN_CH1 Configures whether to enable INCR burst transfer for TX channel 1 to read descriptors.

0: Disable

1: Enable

(R/W)

H264_DMA_OUT_ECC_AES_EN_CH1	Configures whether DMA can access external memory space for ECC and AES via TX channel 1.
------------------------------------	---

0: Not access

1: Access ECC/AES area. In this case, the start address of the memory region should be 16-byte aligned. The product of the width and the bytes per pixel must also be 16-byte aligned.

(R/W)

H264_DMA_OUT_CHECK_OWNER_CH1	Configures whether to enable owner bit check for TX channel 1.
-------------------------------------	--

0: Disable

1: Enable

(R/W)

Continued on the next page...

Register 39.66. H264_DMA_OUT_CONFO_CH1_REG (0x0100)

Continued from the previous page...

H264_DMA_OUT_MEM_BURST_LENGTH_CH1 Configures the burst length for TX channel 1.

- 0: 8 bytes
 - 1: 16 bytes
 - 2: 32 bytes
 - 3: 64 bytes
 - 4: 128 bytes
 - 5 ~ 7: Invalid
- (R/W)

H264_DMA_OUT_PAGE_BOUND_EN_CH1 Configures whether to make sure AXI read data doesn't cross the address boundary which define by mem_burst_length.

- 0: AXI read data can cross the address boundary
 - 1: AXI read data doesn't cross the address boundary
- (R/W)

H264_DMA_OUT_RST_CH1 Write 1 then write 0 to reset TX channel 1. (R/W)

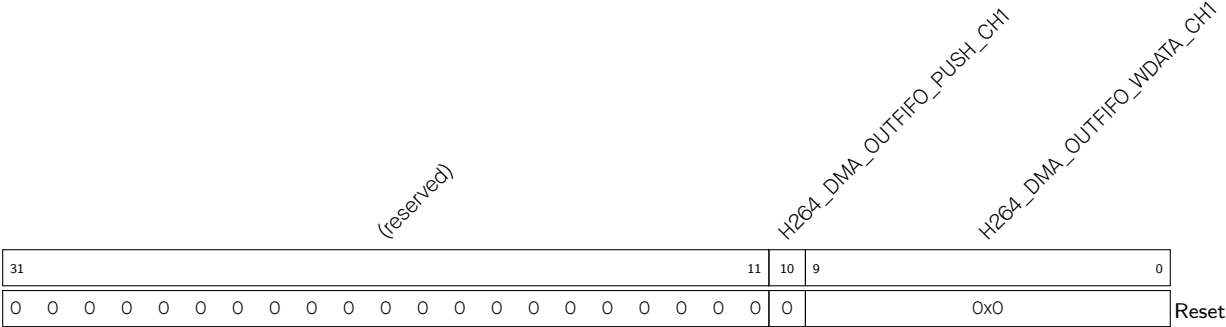
H264_DMA_OUT_CMD_DISABLE_CH1 Configures whether to disable command on TX channel 1.

- 0: Disable
 - 1: Enable
- (R/W)

H264_DMA_OUT_ARB_WEIGHT_OPT_DIS_CH1 Configures whether to disable weight arbitration for TX channel 1.

- 0: Enable
 - 1: Disable
- (R/W)

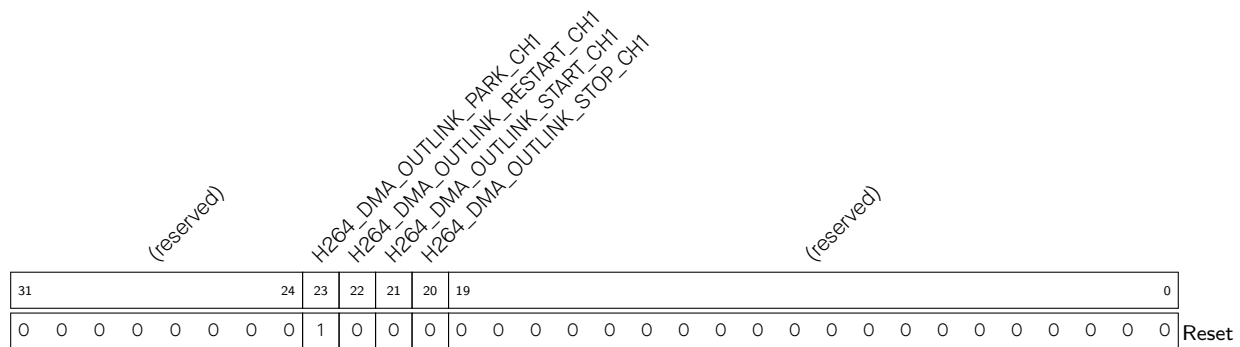
Register 39.67. H264_DMA_OUT_PUSH_CH1_REG (0x0118)



H264_DMA_OUTFIFO_WDATA_CH1 Represents the data that needs to be pushed into DMA TX FIFO.
(R/W)

H264_DMA_OUTFIFO_PUSH_CH1 Configures whether to push data into DMA TX FIFO.
0: No effect
1: Push
(R/W/SC)

Register 39.68. H264_DMA_OUT_LINK_CONF_CH1_REG (0x011C)



H264_DMA_OUTLINK_STOP_CH1 Configures whether to stop processing the outlink descriptors.

0: No effect

1: Stop

(R/W/SC)

H264_DMA_OUTLINK_START_CH1 Configures whether to start processing the outlink descriptors.

0: No effect

1: Start

(R/W/SC)

H264_DMA_OUTLINK_RESTART_CH1 Configures whether to restart a new outlink from the last address.

0: No effect

1: Restart

(R/W/SC)

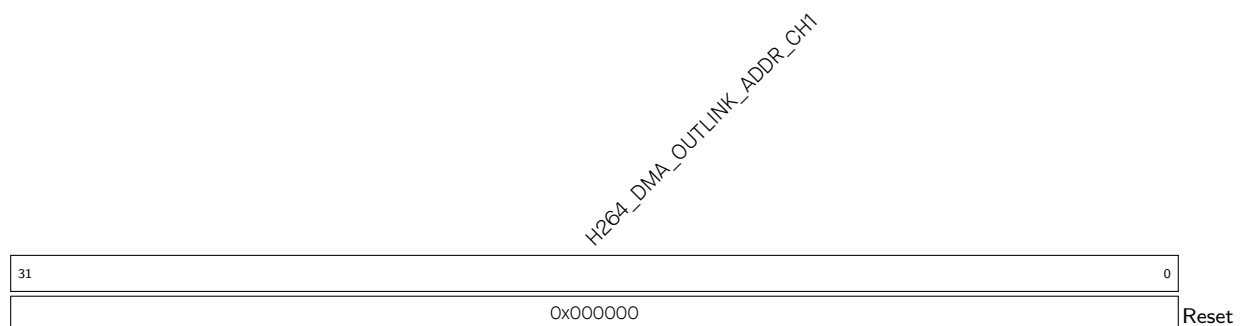
H264_DMA_OUTLINK_PARK_CH1 Represents the status of the transmit descriptor's FSM.

0: The outlink descriptor's FSM is working

1: The outlink descriptor's FSM is in idle state

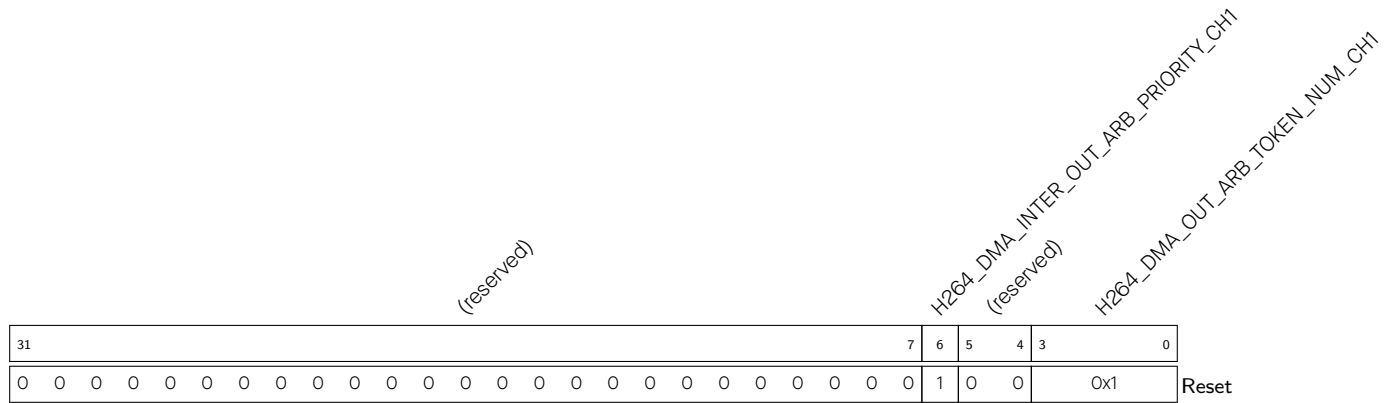
(RO)

Register 39.69. H264_DMA_OUT_LINK_ADDR_CH1_REG (0x0120)



H264_DMA_OUTLINK_ADDR_CH1 Configures the first outlink descriptor's address. (R/W)

Register 39.70. H264_DMA_OUT_ARB_CH1_REG (0x013C)



H264_DMA_OUT_ARB_TOKEN_NUM_CH1 Configures the maximum token count for the arbiter configuration. (R/W)

H264_DMA_INTER_OUT_ARB_PRIORITY_CH1	Configures the priority of internal memory access for TX channel 1. (R/W)
--	---

Register 39.71. H264_DMA_OUT_CONFO_CH2_REG (0x0200)

[illegible]

H264_DMA_OUT_AUTO_WRBACK_CH2 Configures whether to enable automatic outlink-writeback when all the data pointed by outlink descriptor has been received.

0: Disable

1: Enable

(R/W)

H264_DMA_OUT_EOF_MODE_CH2 Configures EOF flag generation mode for TX channel 2.

0: EOF flag is generated when last data has been pushed into FIFO in DMA

1: EOF flag is generated when last data has been popped from FIFO in DMA

(R/W)

H264_DMA_OUTDESCR_BURST_EN_CH2 Configures whether to enable INCR burst transfer for TX channel 2 to read descriptors.

0: Disable

1: Enable

(R/W)

H264_DMA_OUT_ECC_AES_EN_CH2 Configures whether DMA can access external memory space for ECC and AES via TX channel 2.

0: Not access

1: Access ECC/AES area. In this case, the start address of the memory region should be 16-byte aligned. The product of the width and the bytes per pixel must also be 16-byte aligned.

(R/W)

H264_DMA_OUT_CHECK_OWNER_CH2 Configures whether to enable owner bit check for TX channel 2.

0: Disable

1: Enable

(R/W)

Continued on the next page...

Register 39.71. H264_DMA_OUT_CONFO_CH2_REG (0x0200)

Continued from the previous page...

H264_DMA_OUT_MEM_BURST_LENGTH_CH2 Configures the burst length for TX channel 2.

- 0: 8 bytes
 - 1: 16 bytes
 - 2: 32 bytes
 - 3: 64 bytes
 - 4: 128 bytes
 - 5 ~ 7: Invalid
- (R/W)

H264_DMA_OUT_PAGE_BOUND_EN_CH2 Configures whether to make sure AXI read data doesn't cross the address boundary which define by mem_burst_length.

- 0: AXI read data can cross the address boundary
 - 1: AXI read data doesn't cross the address boundary
- (R/W)

H264_DMA_OUT_RST_CH2 Write 1 then write 0 to reset TX channel 2. (R/W)

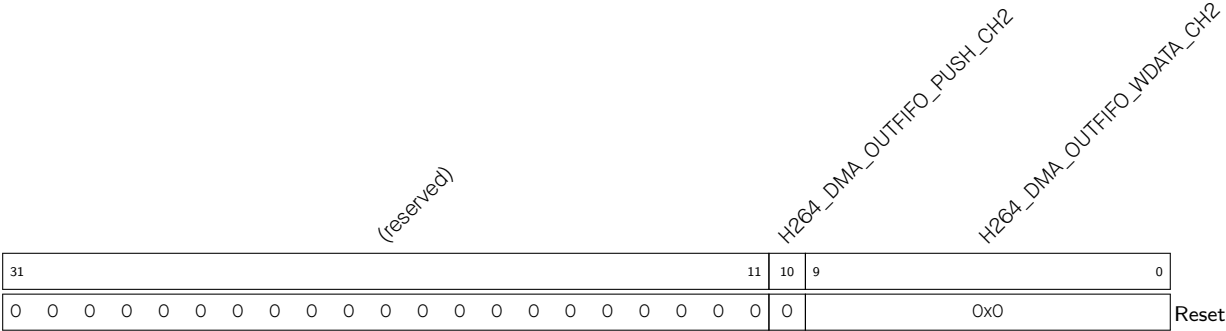
H264_DMA_OUT_CMD_DISABLE_CH2 Configures whether to disable command on TX channel 2.

- 0: Disable
 - 1: Enable
- (R/W)

H264_DMA_OUT_ARB_WEIGHT_OPT_DIS_CH2 Configures whether to disable weight arbitration for TX channel 2.

- 0: Enable
 - 1: Disable
- (R/W)

Register 39.72. H264_DMA_OUT_PUSH_CH2_REG (0x0218)



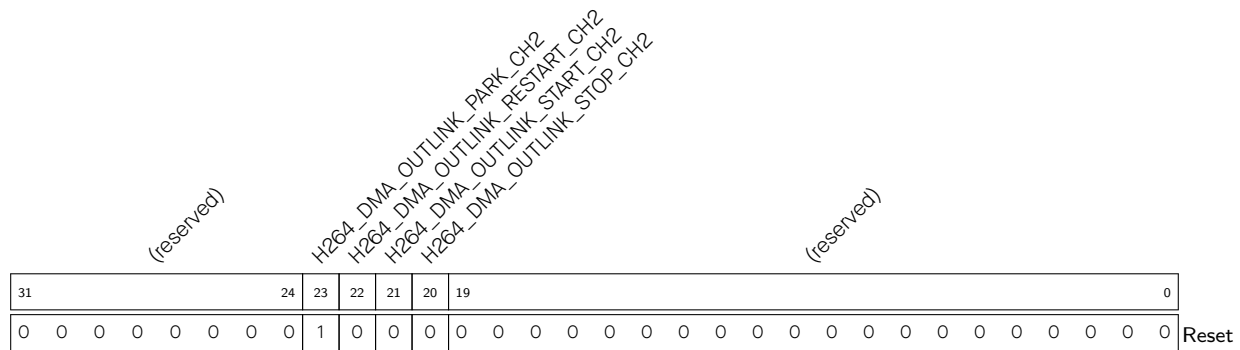
H264_DMA_OUTFIFO_WDATA_CH2 Represents the data that needs to be pushed into DMA TX FIFO. (R/W)

H264_DMA_OUTFIFO_PUSH_CH2 Configures whether to push data into DMA TX FIFO.

- 0: No effect
- 1: Push

(R/W/SC)

Register 39.73. H264_DMA_OUT_LINK_CONF_CH2_REG (0x021C)



H264_DMA_OUTLINK_STOP_CH2 Configures whether to stop processing outlink descriptors.

0: No effect

1: Stop

(R/W/SC)

H264_DMA_OUTLINK_START_CH2 Configures whether to start processing outlink descriptors.

0: No effect

1: Start

(R/W/SC)

H264_DMA_OUTLINK_RESTART_CH2 Configures whether to restart a new outlink from the last address.

0: No effect

1: Restart

(R/W/SC)

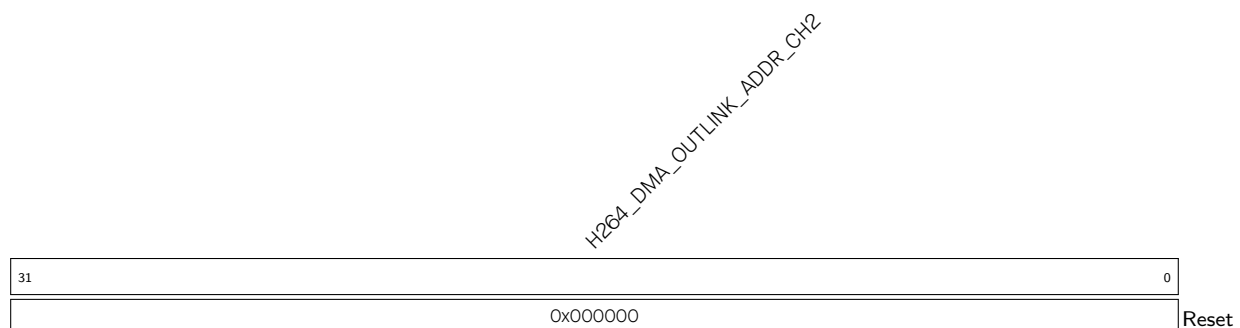
H264_DMA_OUTLINK_PARK_CH2 Represents the status of the transmit descriptor's FSM.

0: The outlink descriptor's FSM is working

1: The outlink descriptor's FSM is in idle state

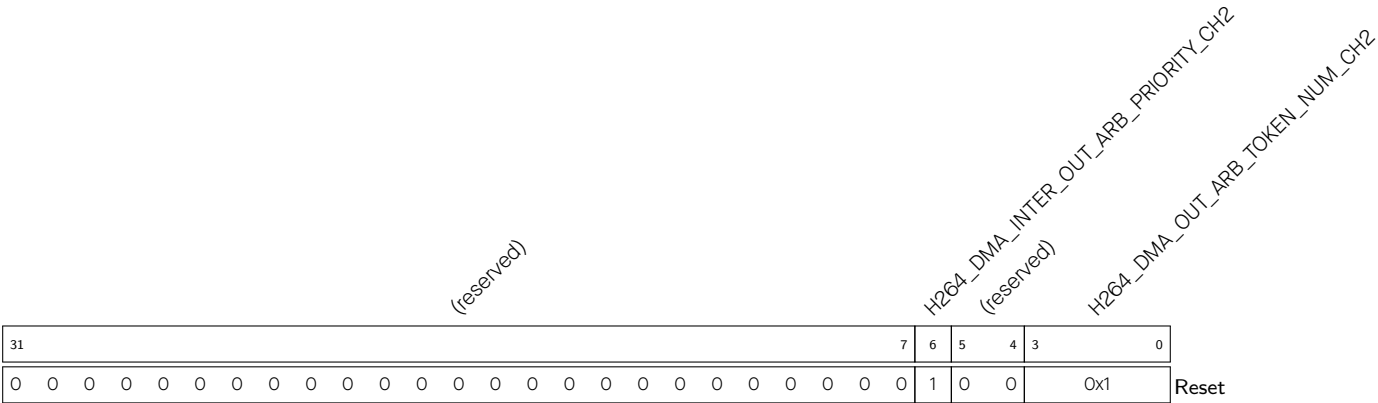
(RO)

Register 39.74. H264_DMA_OUT_LINK_ADDR_CH2_REG (0x0220)



H264_DMA_OUTLINK_ADDR_CH2 Configures the first outlink descriptor's address. (R/W)

Register 39.75. H264_DMA_OUT_ARB_CH2_REG (0x023C)



- H264_DMA_OUT_ARB_TOKEN_NUM_CH2

Configures the maximum token count for the arbiter configuration. (R/W)
- H264_DMA_INTER_OUT_ARB_PRIORITY_CH2

Configures the priority of internal memory access for TX channel 2.

Register 39.76. H264_DMA_OUT_CONF0_CH3_REG (0x0300)

H264_DMA_OUT_AUTO_WRBACK_CH3 Configures whether to enable automatic outlink-writeback when all the data pointed by outlink descriptor has been received.

0: Disable

1: Enable

(R/W)

H264_DMA_OUT_EOF_MODE_CH3 Configures EOF flag generation mode for TX channel 3.

0: EOF flag is generated when last data has been pushed into FIFO in DMA

1: EOF flag is generated when last data has been popped from FIFO in DMA

(R/W)

H264_DMA_OUTDSCR_BURST_EN_CH3 Configures whether to enable INCR burst transfer for TX channel 3 to read descriptors.

0: Disable

1: Enable

(R/W)

H264_DMA_OUT_ECC_AES_EN_CH3 Configures whether DMA can access external memory space for ECC and AES via TX channel 3.

0: Not access

1: Access ECC/AES area. In this case, the start address of the memory region should be 16-byte aligned. The product of the width and the bytes per pixel must also be 16-byte aligned.

(R/W)

H264_DMA_OUT_CHECK_OWNER_CH3 Configures whether to enable owner bit check for TX channel 3.

0: Disable

1: Enable

(R/W)

Continued on the next page...

Register 39.76. H264_DMA_OUT_CONFO_CH3_REG (0x0300)

Continued from the previous page...

H264_DMA_OUT_MEM_BURST_LENGTH_CH3 Configures the burst length for TX channel 3.

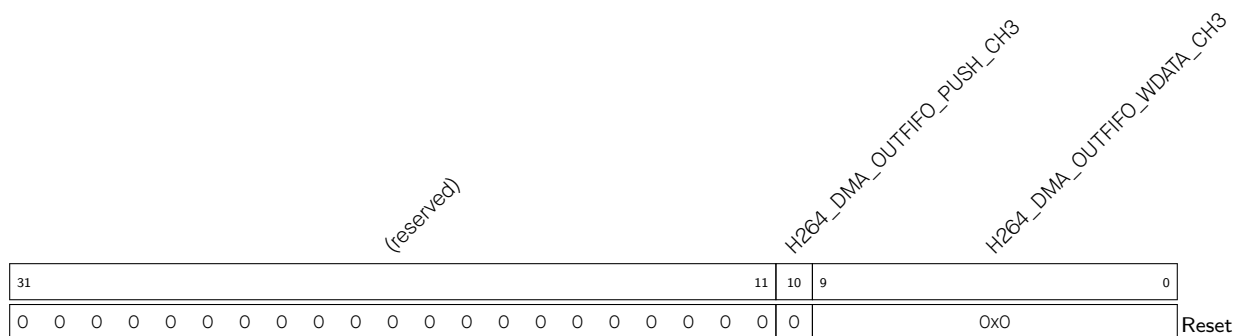
- 0: 8 bytes
 - 1: 16 bytes
 - 2: 32 bytes
 - 3: 64 bytes
 - 4: 128 bytes
 - 5 ~ 7: Invalid
- (R/W)

H264_DMA_OUT_PAGE_BOUND_EN_CH3 Configures whether to make sure AXI read data doesn't cross the address boundary which define by mem_burst_length.

- 0: AXI read data can cross the address boundary
 - 1: AXI read data doesn't cross the address boundary
- (R/W)

H264_DMA_OUT_ARB_WEIGHT_OPT_DIS_CH3 Configures whether to disable weight arbitration for TX channel 3.

- 0: Enable
 - 1: Disable
- (R/W)

Register 39.77. H264_DMA_OUT_PUSH_CH3_REG (0x0318)**H264_DMA_OUTFIFO_WDATA_CH3** Represents the data that needs to be pushed into DMA TX FIFO. (R/W)**H264_DMA_OUTFIFO_PUSH_CH3** Configures whether to push data into DMA TX FIFO.

- 0: No effect
 - 1: Push
- (R/W/SC)

Register 39.78. H264_DMA_OUT_LINK_CONF_CH3_REG (0x031C)

[illegible]

H264_DMA_OUTLINK_STOP_CH3 Configures whether to stop processing the outlink descriptors.

0: No effect

1: Stop

(R/W/SC)

H264_DMA_OUTLINK_START_CH3 Configures whether to start processing the outlink descriptors.

0: No effect

1: Start

(R/W/SC)

H264_DMA_OUTLINK_RESTART_CH3 Configures whether to restart a new outlink from the last address.

0: No effect

1: Restart

(R/W/SC)

H264_DMA_OUTLINK_PARK_CH3 Represents the status of the transmit descriptor's FSM.

0: The outlink descriptor's FSM is working

1: The outlink descriptor's FSM is in idle state

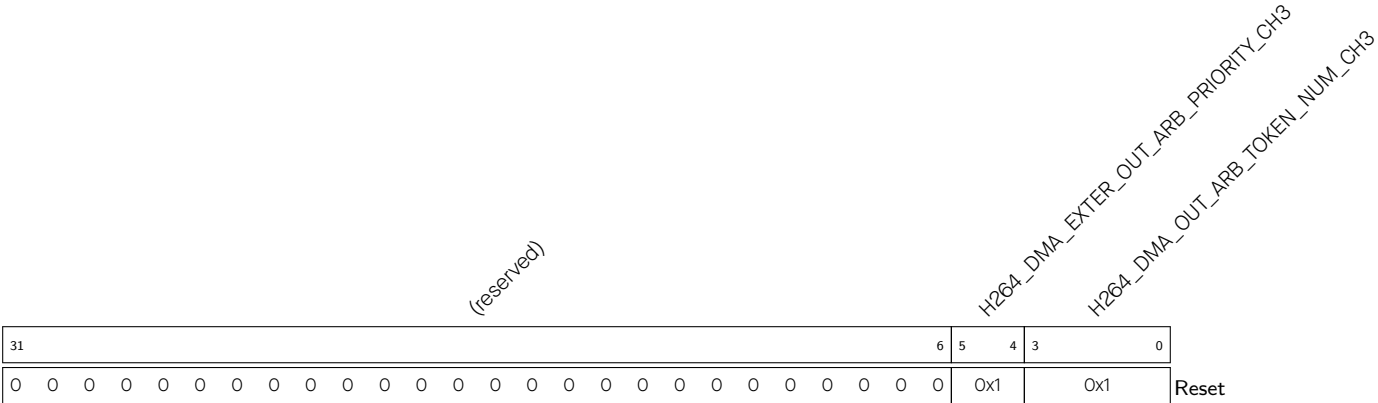
(RO)

Register 39.79. H264_DMA_OUT_LINK_ADDR_CH3_REG (0x0320)

Diagram of the H264_DMA_OUTLINK_ADDR_CH3 register. The register is 32 bits wide, with bit positions 31 down to 0 labeled. The value 0x00000000 is shown, and the label 'Reset' is at the bottom right.

H264_DMA_OUTLINK_ADDR_CH3 Configures the first outlink descriptor's address. (R/W)

Register 39.80. H264_DMA_OUT_ARB_CH3_REG (0x033C)



- H264_DMA_OUT_ARB_TOKEN_NUM_CH3 Configures the maximum token count for the arbiter configuration. (R/W)
- H264_DMA_EXTER_OUT_ARB_PRIORITY_CH3 Configures the priority of external memory access for TX channel 3. (R/W)

Register 39.81. H264_DMA_OUT_CONFO_CH4_REG (0x0400)

(reserved)						H264_DMA_OUT_ARB_WEIGHT_OPT_DIS_CH4																		(reserved)						H264_DMA_OUT_PAGE_BOUND_EN_CH4						(reserved)						H264_DMA_OUT_MEM_BURST_LENGTH_CH4						H264_DMA_OUT_CHECK_OWNER_CH4						H264_DMA_OUT_ECC_AES_EN_CH4						H264_DMA_OUT_OUTDSCR_BURST_EN_CH4						H264_DMA_OUT_EOF_MODE_CH4						H264_DMA_OUT_AUTO_WBACK_CH4					
31						27		26		25												13		12		11				9		8				6		5		4		3		2		1		0																													
0		0		0		0		0		0		0		0		0		0		0		0		0		0		0		0		0		0		0		0		1		0		Reset																																	

H264_DMA_OUT_AUTO_WRBACK_CH4 Configures whether to enable automatic outlink-writeback when all the data pointed by outlink descriptor has been received.

0: Disable

1: Enable

(R/W)

H264_DMA_OUT_EOF_MODE_CH4 Configures EOF flag generation mode for TX channel 4.

0: EOF flag is generated when last data has been pushed into FIFO in DMA

1: EOF flag is generated when last data has been popped from FIFO in DMA

(R/W)

H264_DMA_OUTDSCR_BURST_EN_CH4 Configures whether to enable INCR burst transfer for TX channel 4 to read descriptors.

0: Disable

1: Enable

(R/W)

H264_DMA_OUT_ECC_AES_EN_CH4 Configures whether DMA can access external memory space for ECC and AES via TX channel 4.

0: Not access

1: Access ECC/AES area. In this case, the start address of the memory region should be 16-byte aligned. The product of the width and the bytes per pixel must also be 16-byte aligned.

(R/W)

H264_DMA_OUT_CHECK_OWNER_CH4 Configures whether to enable owner bit check for TX channel 4.

0: Disable

1: Enable

(R/W)

Continued on the next page...

Register 39.81. H264_DMA_OUT_CONFO_CH4_REG (0x0400)

Continued from the previous page...

H264_DMA_OUT_MEM_BURST_LENGTH_CH4 Configures the burst length for TX channel 4.

- 0: 8 bytes
- 1: 16 bytes
- 2: 32 bytes
- 3: 64 bytes
- 4: 128 bytes
- 5 ~ 7: Invalid

(R/W)

H264_DMA_OUT_PAGE_BOUND_EN_CH4 Configures whether to make sure AXI read data doesn't cross the address boundary which define by mem_burst_length.

- 0: AXI read data can cross the address boundary
- 1: AXI read data doesn't cross the address boundary

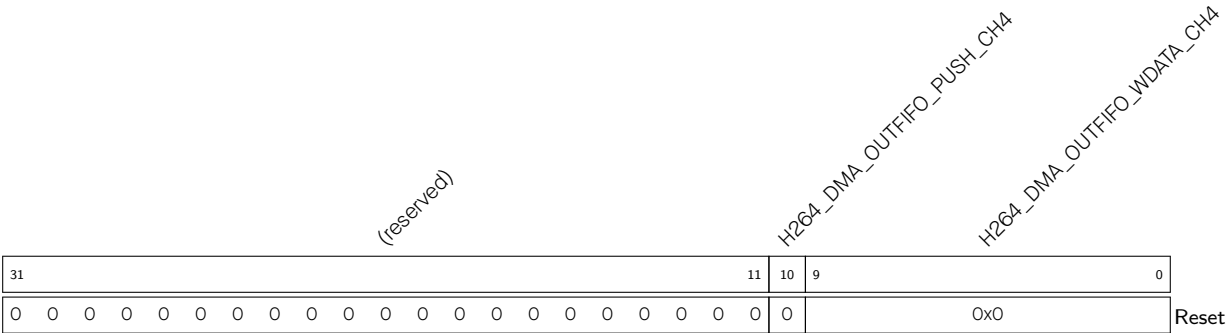
(R/W)

H264_DMA_OUT_ARB_WEIGHT_OPT_DIS_CH4 Configures whether to disable weight arbitration for TX channel 4.

- 0: Enable
- 1: Disable

(R/W)

Register 39.82. H264_DMA_OUT_PUSH_CH4_REG (0x0418)

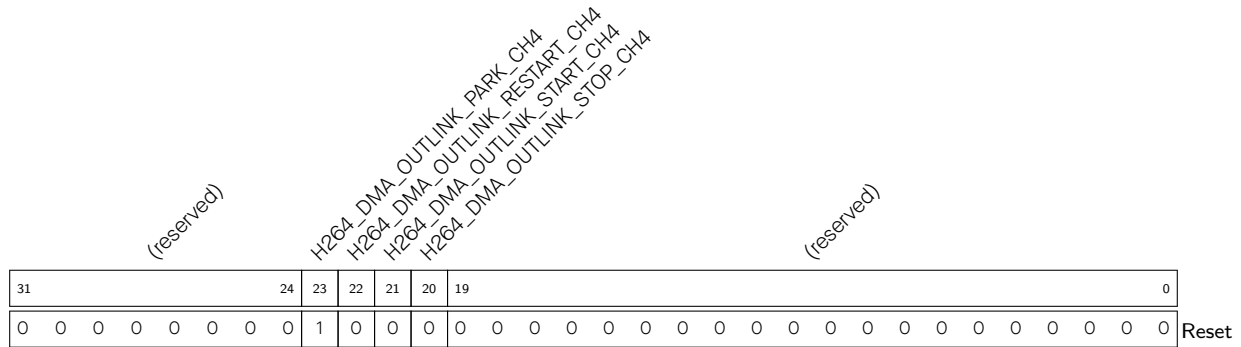


H264_DMA_OUTFIFO_WDATA_CH4 Represents the data that needs to be pushed into DMA TX FIFO. (R/W)

H264_DMA_OUTFIFO_PUSH_CH4 Configures whether to push data into DMA TX FIFO.

- 0: No effect
- 1: Push

(R/W/SC)

Register 39.83. H264_DMA_OUT_LINK_CONF_CH4_REG (0x041C)

H264_DMA_OUTLINK_STOP_CH4 Configures whether to stop processing the outlink descriptors.

- 0: No effect
 - 1: Stop
- (R/W/SC)

H264_DMA_OUTLINK_START_CH4 Configures whether to start processing the outlink descriptors.

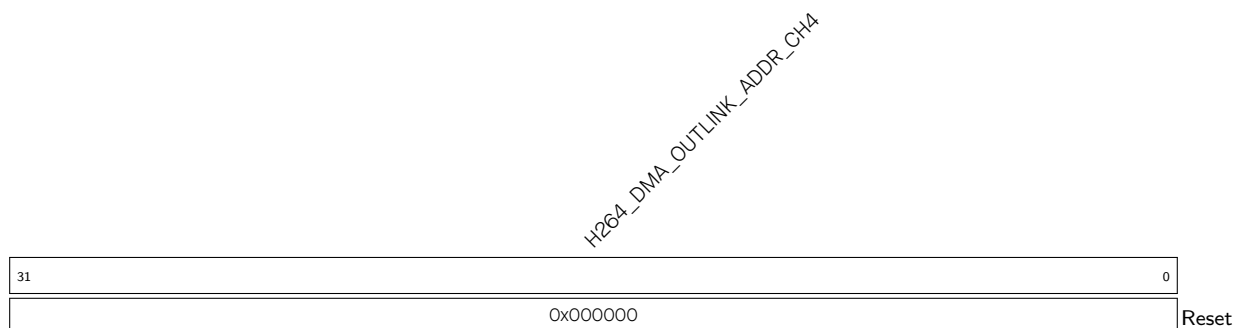
- 0: No effect
 - 1: Start
- (R/W/SC)

H264_DMA_OUTLINK_RESTART_CH4 Configures whether to restart a new outlink from the last address.

- 0: No effect
 - 1: Restart
- (R/W/SC)

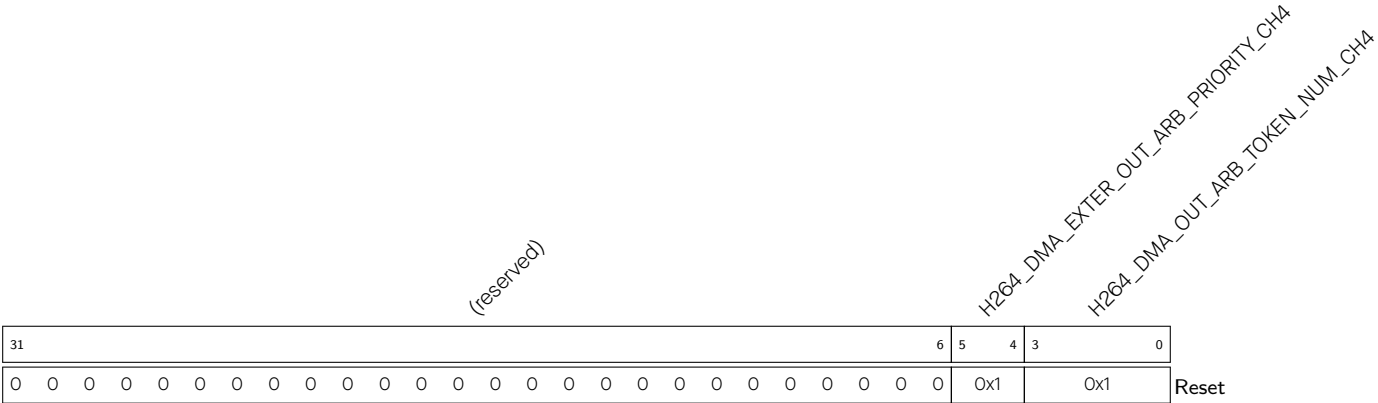
H264_DMA_OUTLINK_PARK_CH4 Represents the status of the transmit descriptor's FSM.

- 0: The outlink descriptor's FSM is working
 - 1: The outlink descriptor's FSM is in idle state
- (RO)

Register 39.84. H264_DMA_OUT_LINK_ADDR_CH4_REG (0x0420)

H264_DMA_OUTLINK_ADDR_CH4 Configures the first outlink descriptor's address. (R/W)

Register 39.85. H264_DMA_OUT_ARB_CH4_REG (0x043C)



- H264_DMA_OUT_ARB_TOKEN_NUM_CH4 Configures the maximum token count for the arbiter configuration. (R/W)
- H264_DMA_EXTER_OUT_ARB_PRIORITY_CH4 Configures the priority of external memory access for TX channel 4. (R/W)

Register 39.86. H264_DMA_IN_CONF0_CHO_REG (0x0500)

[illegible]

H264_DMA_INDSR_BURST_EN_CHO Configures whether to enable INCR burst transfer for RX channel 0 to read descriptors.

0: Disable

1: Enable

(R/W)

H264_DMA_IN_ECC_AES_EN_CHO Configures whether DMA can access external memory space for ECC and AES via RX channel 0.

0: Not access

1: Access. In this case, the start address of the memory region should be 16-byte aligned. The product of the width and the bytes per pixel must also be 16-byte aligned.

(R/W)

H264_DMA_IN_CHECK_OWNER_CHO	Configures whether to enable owner bit check for RX channel 0.
------------------------------------	--

0: Disable

1: Enable

(R/W)

H264_DMA_IN_MEM_BURST_LENGTH_CHO Configures the burst length for RX channel 0.

0: 8 bytes

1: 16 bytes

2: 32 bytes

3: 64 bytes

4: 128 bytes

5 ~ 7: Invalid

(R/W)

H264_DMA_IN_PAGE_BOUND_EN_CHO Configures whether to make sure AXI read data doesn't cross the address boundary which define by mem_burst_length.

0: AXI read data can cross the address boundary

1: AXI read data doesn't cross the address boundary

(R/W)

H264_DMA_IN_RST_CHO Write 1 then write 0 to reset RX channel 0. (R/W)

Continued on the next page...

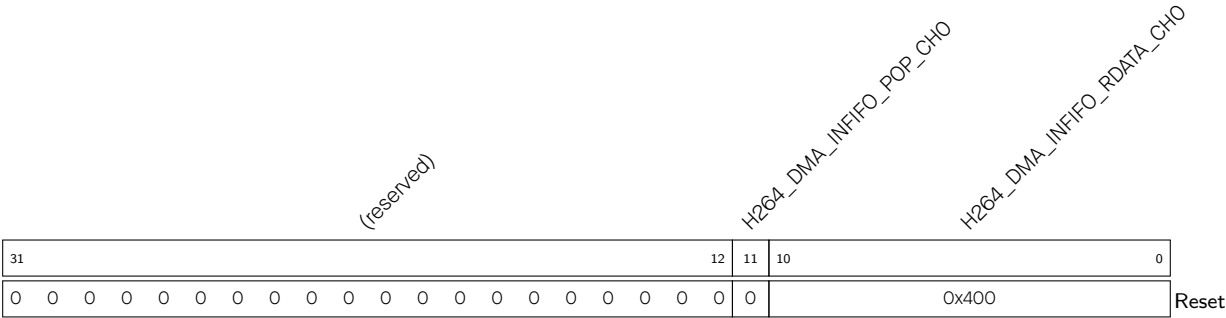
Register 39.86. H264_DMA_IN_CONFO_CHO_REG (0x0500)

Continued from the previous page...

H264_DMA_IN_CMD_DISABLE_CHO Configures whether to disable command on RX channel 0.
0: Disable
1: Enable
(R/W)

H264_DMA_IN_ARB_WEIGHT_OPT_DIS_CHO Configures whether to disable weight arbitration for RX channel 0.
0: Enable
1: Disable
(R/W)

Register 39.87. H264_DMA_IN_POP_CHO_REG (0x0518)



H264_DMA_INFIFO_RDATA_CHO Represents the data popped from DMA RX FIFO. (RO)

H264_DMA_INFIFO_POP_CHO Configures whether to pop data from DMA RX FIFO.
0: No effect
1: Pop
(R/W/SC)

Register 39.88. H264_DMA_IN_LINK_CONF_CHO_REG (0x051C)

H264_DMA_INLINK_AUTO_RET_CHO Configures whether to return to the address of the current receive descriptor when there are errors during data reception.

0: Not return

1: Return

(R/W)

H264_DMA_INLINK_STOP_CHO Configures whether to stop processing the inlink descriptors.

0: No effect

1: Stop

(R/W/SC)

H264_DMA_INLINK_START_CHO Configures whether to start processing the inlink descriptors.

0: No effect

1: Start

(R/W/SC)

H264_DMA_INLINK_RESTART_CHO Configures whether to restart a new inlink from the last address.

0: No effect

1: Restart

(R/W/SC)

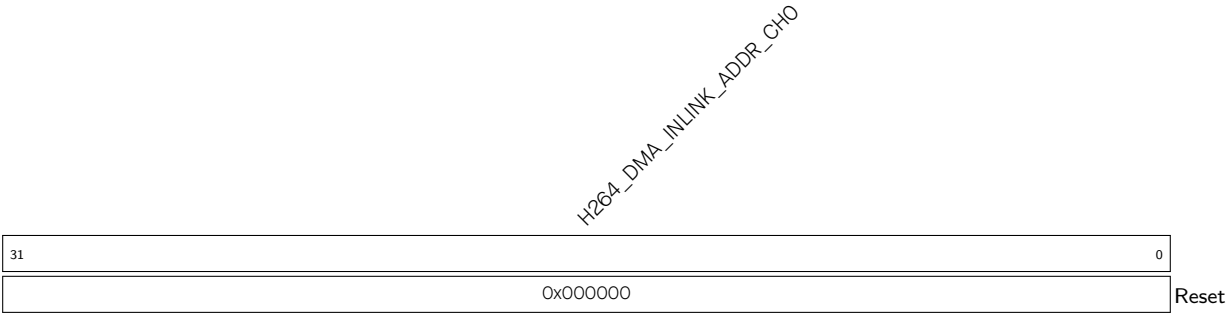
H264_DMA_INLINK_PARK_CHO Represents the status of the inlink descriptor's FSM.

0: The inlink descriptor's FSM is working

1: The inlink descriptor's FSM is in idle state

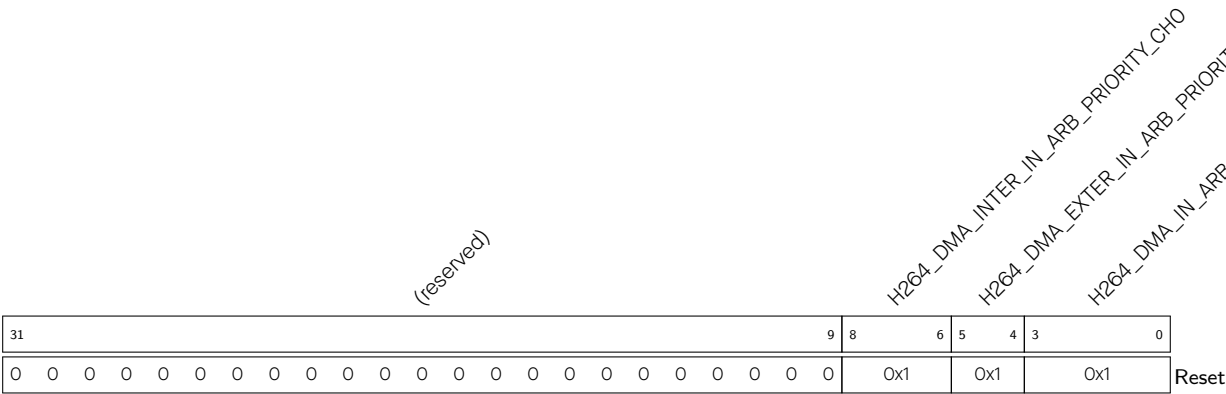
(RO)

Register 39.89. H264_DMA_IN_LINK_ADDR_CHO_REG (0x0520)



H264_DMA_INLINK_ADDR_CHO Configures the first inlink descriptor’s address. (R/W)

Register 39.90. H264_DMA_IN_ARB_CHO_REG (0x0540)

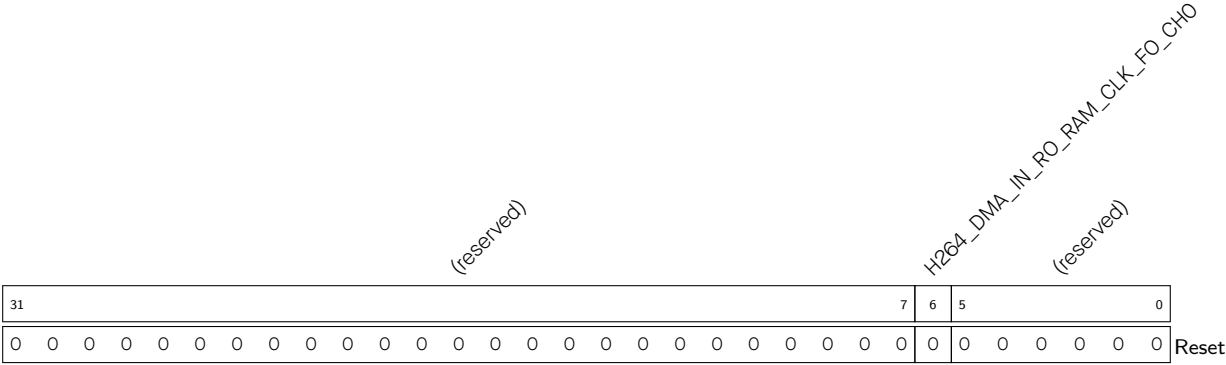


H264_DMA_IN_ARB_TOKEN_NUM_CHO Configures the maximum token count for the arbiter configuration. (R/W)

H264_DMA_EXTER_IN_ARB_PRIORITY_CHO Configures the priority of external memory access for RX channel 0. (R/W)

H264_DMA_INTER_IN_ARB_PRIORITY_CHO Configures the priority of internal memory access for RX channel 0. (R/W)

Register 39.91. H264_DMA_IN_RO_PD_CONF_CHO_REG (0x0548)



H264_DMA_IN_RO_RAM_CLK_FO_CHO Configures whether to force enable clock gating for DMA RX reorder RAM.

0: A clock gating will be used when accessing the RAM in DMA

1: Force to open the clock and bypass the clock gating when accessing the RAM in DMA

(R/W)

Register 39.92. H264_DMA_IN_CONFO_CH1_REG (0x0600)

[illegible]

H264_DMA_INDSR_BURST_EN_CH1 Configures whether to enable INCR burst transfer for RX channel 1 to read descriptors.

0: Disable

1: Enable

(R/W)

H264_DMA_IN_ECC_AES_EN_CH1 Configures whether DMA can access external memory space for ECC and AES via RX channel 1.

0: Not access

1: Access. In this case, the start address of the memory region should be 16-byte aligned. The product of the width and the bytes per pixel must also be 16-byte aligned.

(R/W)

H264_DMA_IN_CHECK_OWNER_CH1	Configures whether to enable owner bit check for RX channel 1.
------------------------------------	--

0: Disable

1: Enable

(R/W)

H264_DMA_IN_MEM_BURST_LENGTH_CH1 Configures the burst length for RX channel 1.

0: 8 bytes

1: 16 bytes

2: 32 bytes

3: 64 bytes

4: 128 bytes

5 ~ 7: Invalid

(R/W)

H264_DMA_IN_PAGE_BOUND_EN_CH1 Configures whether to make sure AXI read data doesn't cross the address boundary which define by mem_burst_length.

0: AXI read data can cross the address boundary

1: AXI read data doesn't cross the address boundary

(R/W)

H264_DMA_IN_RST_CH1 Write 1 then write 0 to reset RX channel 1. (R/W)

Continued on the next page...

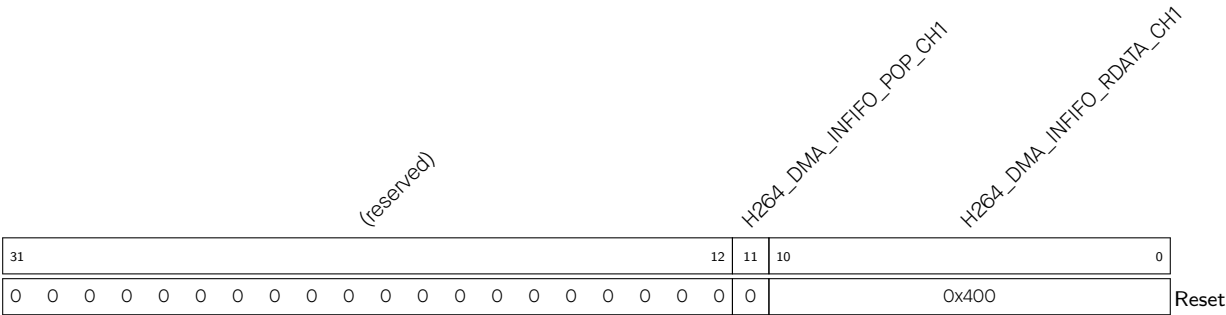
Register 39.92. H264_DMA_IN_CONFO_CH1_REG (0x0600)

Continued from the previous page...

H264_DMA_IN_CMD_DISABLE_CH1 Configures whether to disable command on RX channel 1.
0: Disable
1: Enable
(R/W)

H264_DMA_IN_ARB_WEIGHT_OPT_DIS_CH1 Configures whether to disable weight arbitration for RX channel 1.
0: Enable
1: Disable
(R/W)

Register 39.93. H264_DMA_IN_POP_CH1_REG (0x0618)



H264_DMA_INFIFO_RDATA_CH1 Represents the data popped from DMA RX FIFO. (RO)

H264_DMA_INFIFO_POP_CH1 Configures whether to pop data from DMA RX FIFO.
0: No effect
1: Pop
(R/W/SC)

Register 39.94. H264_DMA_IN_LINK_CONF_CH1_REG (0x061C)

(reserved)								H264_DMA_INLINK_PARK_CH1 H264_DMA_INLINK_RESTART_CH1 H264_DMA_INLINK_START_CH1 H264_DMA_INLINK_STOP_CH1 H264_DMA_INLINK_AUTO_RET_CH1																(reserved)											
31	25				24	23	22	21	20	19																									0
0	0	0	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset	

H264_DMA_INLINK_AUTO_RET_CH1 Configures whether to return to the address of the current receive descriptor when there are errors during data reception.

0: Not return

1: Return

(R/W)

H264_DMA_INLINK_STOP_CH1 Configures whether to stop processing the inlink descriptors.

0: No effect

1: Stop

(R/W/SC)

H264_DMA_INLINK_START_CH1 Configures whether to start processing the inlink descriptors.

0: No effect

1: Start

(R/W/SC)

H264_DMA_INLINK_RESTART_CH1 Configures whether to restart a new inlink from the last address.

0: No effect

1: Restart

(R/W/SC)

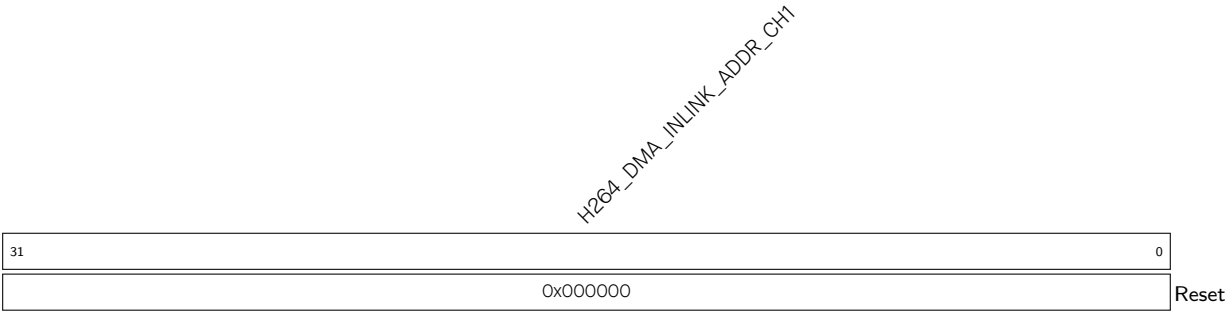
H264_DMA_INLINK_PARK_CH1 Represents the status of the inlink descriptor's FSM.

0: The inlink descriptor's FSM is working

1: The inlink descriptor's FSM is in idle state

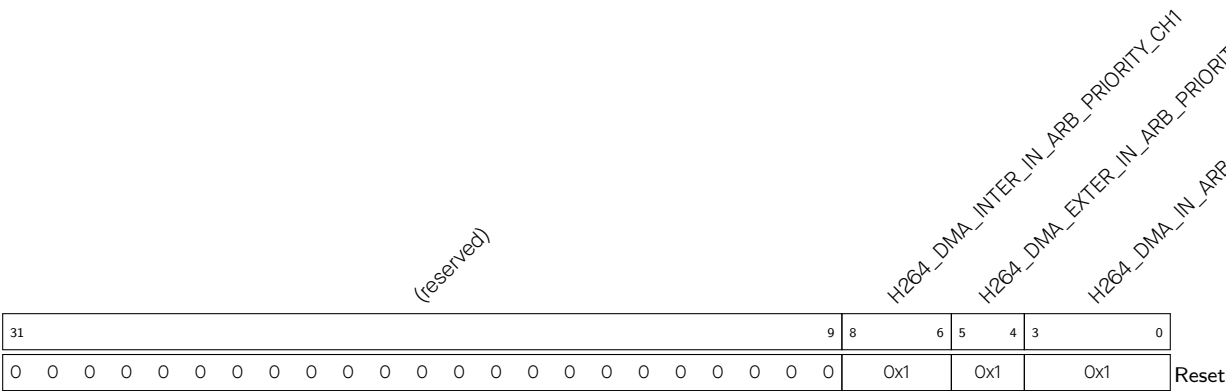
(RO)

Register 39.95. H264_DMA_IN_LINK_ADDR_CH1_REG (0x0620)



H264_DMA_INLINK_ADDR_CH1 Configures the first inlink descriptor’s address. (R/W)

Register 39.96. H264_DMA_IN_ARB_CH1_REG (0x0640)



- H264_DMA_IN_ARB_TOKEN_NUM_CH1** Configures the maximum token count for the arbiter configuration. (R/W)
- H264_DMA_EXTER_IN_ARB_PRIORITY_CH1** Configures the priority of external memory access for RX channel 1. (R/W)
- H264_DMA_INTER_IN_ARB_PRIORITY_CH1** Configures the priority of internal memory access for RX channel 1. (R/W)

Register 39.97. H264_DMA_IN_CONFO_CH2_REG (0x0700)

(reserved)						H264_DMA_IN_ARB_WEIGHT_OPT_DIS_CH2	H264_DMA_IN_CMD_DISABLE_CH2	H264_DMA_IN_RST_CH2	(reserved)							H264_DMA_IN_PAGE_BOUND_EN_CH2	(reserved)	H264_DMA_IN_MEM_BURST_LENGTH_CH2	H264_DMA_IN_CHECK_OWNER_CH2	H264_DMA_IN_ECC_AES_EN_CH2	H264_DMA_INDSCR_BURST_EN_CH2	(reserved)	
31		27	26	25	24	23			13	12	11		9	8		6	5	4	3	2	1	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	OxO	0	0	0	0	0	0	0	Reset

H264_DMA_INDSR_BURST_EN_CH2 Configures whether to enable INCR burst transfer for RX channel 2 to read descriptors.

0: Disable

1: Enable

(R/W)

H264_DMA_IN_ECC_AES_EN_CH2 Configures whether DMA can access external memory space for ECC and AES via RX channel 2.

0: Not access

1: Access. In this case, the start address of the memory region should be 16-byte aligned. The product of the width and the bytes per pixel must also be 16-byte aligned.

(R/W)

H264_DMA_IN_CHECK_OWNER_CH2 Configures whether to enable owner bit check for RX channel 2.

0: Disable

1: Enable

(R/W)

H264_DMA_IN_MEM_BURST_LENGTH_CH2 Configures the burst length for RX channel 2.

0: 8 bytes

1: 16 bytes

2: 32 bytes

3: 64 bytes

4: 128 bytes

5 ~ 7: Invalid

(R/W)

H264_DMA_IN_PAGE_BOUND_EN_CH2 Configures whether to make sure AXI read data doesn't cross the address boundary which define by `mem_burst_length`.

0: AXI read data can cross the address boundary

1: AXI read data doesn't cross the address boundary

(R/W)

H264_DMA_IN_RST_CH2 Write 1 then write 0 to reset RX channel 2. (R/W)

Continued on the next page...

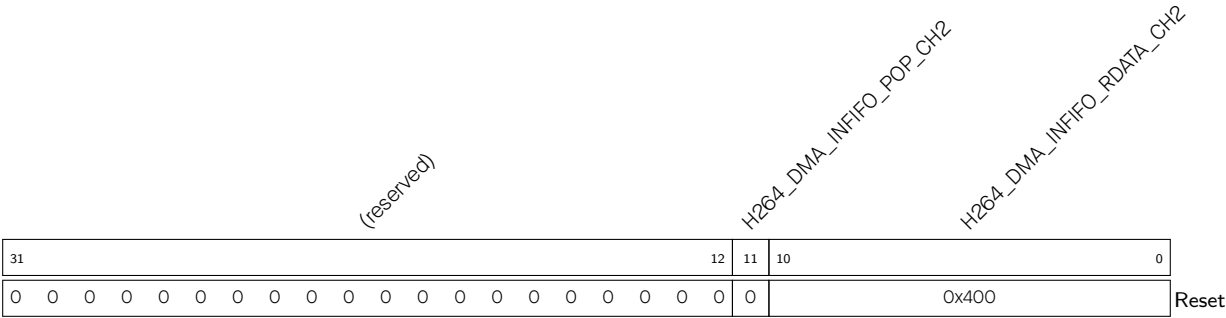
Register 39.97. H264_DMA_IN_CONFO_CH2_REG (0x0700)

Continued from the previous page...

H264_DMA_IN_CMD_DISABLE_CH2 Configures whether to disable command on RX channel 2.
0: Disable
1: Enable
(R/W)

H264_DMA_IN_ARB_WEIGHT_OPT_DIS_CH2 Configures whether to disable weight arbitration for RX channel 2.
0: Enable
1: Disable
(R/W)

Register 39.98. H264_DMA_IN_POP_CH2_REG (0x0718)



H264_DMA_INFIFO_RDATA_CH2 Represents the data popped from DMA RX FIFO. (RO)

H264_DMA_INFIFO_POP_CH2 Configures whether to pop data from DMA RX FIFO.
0: No effect
1: Pop
(R/W/SC)

Register 39.99. H264_DMA_IN_LINK_CONF_CH2_REG (0x071C)

(reserved)							H264_DMA_INLINK_PARK_CH2 H264_DMA_INLINK_RESTART_CH2 H264_DMA_INLINK_START_CH2 H264_DMA_INLINK_STOP_CH2 H264_DMA_INLINK_AUTO_RET_CH2																									(reserved)						
31						25	24	23	22	21	20	19																									0	
0	0	0	0	0	0	0	0	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset		

H264_DMA_INLINK_AUTO_RET_CH2 Configures whether to return to the address of the current receive descriptor when there are errors during data reception.

0: Not return

1: Return

(R/W)

H264_DMA_INLINK_STOP_CH2 Configures whether to stop processing the inlink descriptors.

0: No effect

1: Stop

(R/W/SC)

H264_DMA_INLINK_START_CH2 Configures whether to start processing the inlink descriptors.

0: No effect

1: Start

(R/W/SC)

H264_DMA_INLINK_RESTART_CH2 Configures whether to restart a new inlink from the last address.

0: No effect

1: Restart

(R/W/SC)

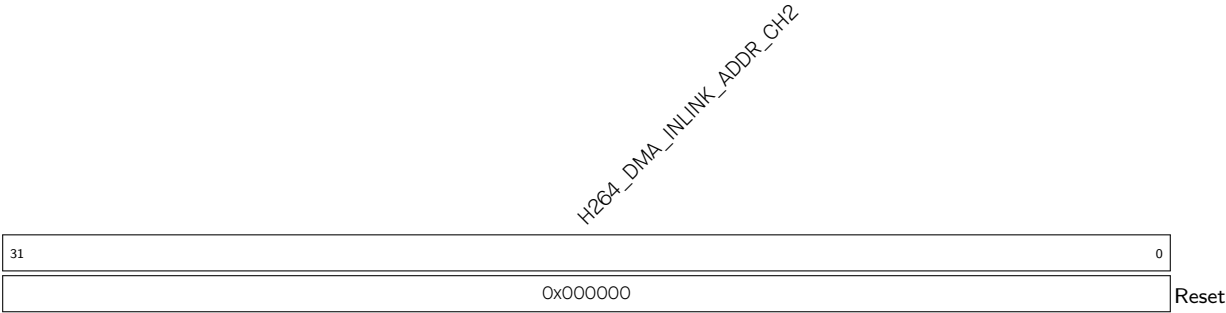
H264_DMA_INLINK_PARK_CH2 Represents the status of the inlink descriptor's FSM.

0: The inlink descriptor's FSM is working

1: The inlink descriptor's FSM is in idle state

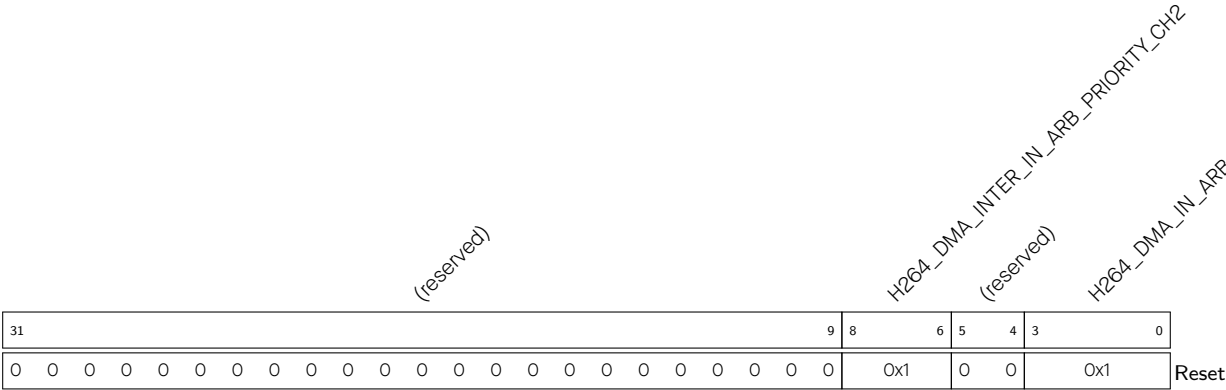
(RO)

Register 39.100. H264_DMA_IN_LINK_ADDR_CH2_REG (0x0720)



H264_DMA_INLINK_ADDR_CH2 Configures the first inlink descriptor’s address. (R/W)

Register 39.101. H264_DMA_IN_ARB_CH2_REG (0x0740)



- H264_DMA_IN_ARB_TOKEN_NUM_CH2 Configures the maximum token count for the arbiter configuration. (R/W)
- H264_DMA_INTER_IN_ARB_PRIORITY_CH2 Configures the priority of internal memory access for RX channel 2. (R/W)

Register 39.102. H264_DMA_IN_CONFO_CH3_REG (0x0800)

(reserved) H264_DMA_IN_ARB_WEIGHT_OPT_DIS_CH3 H264_DMA_IN_CMD_DISABLE_CH3 H264_DMA_IN_RST_CH3 (reserved) H264_DMA_IN_PAGE_BOUND_EN_CH3 (reserved) H264_DMA_IN_MEM_BURST_LENGTH_CH3 H264_DMA_IN_CHECK_OWNER_CH3 H264_DMA_IN_ECC_AES_EN_CH3 H264_DMA_INDSR_BURST_EN_CH3 (reserved)																														
31					27	26	25	24	23								13	12	11	9		8	6		5	4	3	2	1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset	

H264_DMA_INDSR_BURST_EN_CH3 Configures whether to enable INCR burst transfer for RX channel 3 to read descriptors.

0: Disable

1: Enable

(R/W)

H264_DMA_IN_ECC_AES_EN_CH3 Configures whether DMA can access external memory space for ECC and AES via RX channel 3.

0: Not access

1: Access. In this case, the start address of the memory region should be 16-byte aligned. The product of the width and the bytes per pixel must also be 16-byte aligned.

(R/W)

H264_DMA_IN_CHECK_OWNER_CH3 Configures whether to enable owner bit check for RX channel 3.

0: Disable

1: Enable

(R/W)

H264_DMA_IN_MEM_BURST_LENGTH_CH3 Configures the burst length for RX channel 3.

0: 8 bytes

1: 16 bytes

2: 32 bytes

3: 64 bytes

4: 128 bytes

5 ~ 7: Invalid

(R/W)

H264_DMA_IN_PAGE_BOUND_EN_CH3 Configures whether to make sure AXI read data doesn't cross the address boundary which define by mem_burst_length.

0: AXI read data can cross the address boundary

1: AXI read data doesn't cross the address boundary

(R/W)

H264_DMA_IN_RST_CH3 Write 1 then write 0 to reset RX channel 3. (R/W)

Continued on the next page...

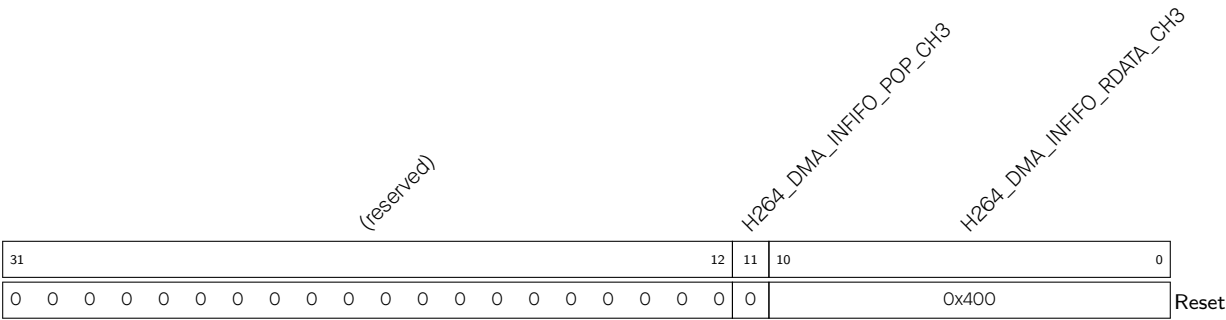
Register 39.102. H264_DMA_IN_CONFO_CH3_REG (0x0800)

Continued from the previous page...

H264_DMA_IN_CMD_DISABLE_CH3 Configures whether to disable command on RX channel 3.
0: Disable
1: Enable
(R/W)

H264_DMA_IN_ARB_WEIGHT_OPT_DIS_CH3 Configures whether to disable weight arbitration for RX channel 3.
0: Enable
1: Disable
(R/W)

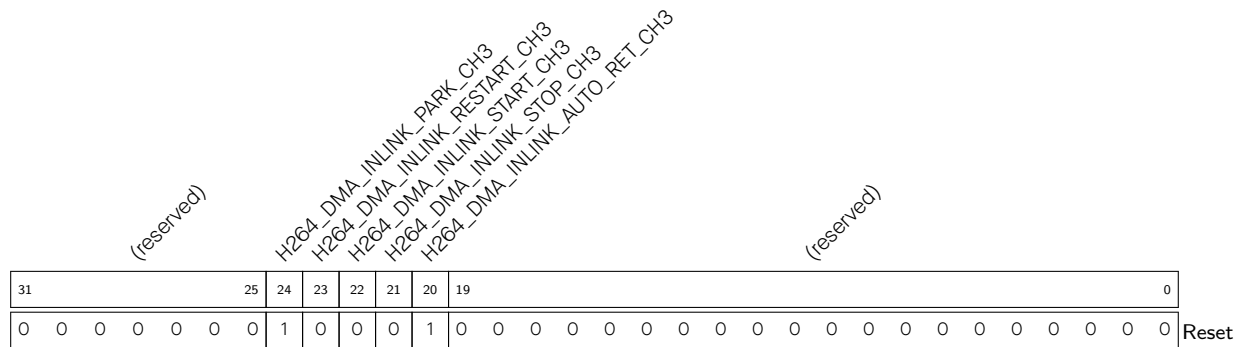
Register 39.103. H264_DMA_IN_POP_CH3_REG (0x0818)



H264_DMA_INFIFO_RDATA_CH3 Represents the data popped from DMA RX FIFO. (RO)

H264_DMA_INFIFO_POP_CH3 Configures whether to pop data from DMA RX FIFO.
0: No effect
1: Pop
(R/W/SC)

Register 39.104. H264_DMA_IN_LINK_CONF_CH3_REG (0x081C)



H264_DMA_INLINK_AUTO_RET_CH3 Configures whether to return to the address of the current receive descriptor when there are errors during data reception.

0: Not return

1: Return

(R/W)

H264_DMA_INLINK_STOP_CH3 Configures whether to stop processing the inlink descriptors.

0: No effect

1: Stop

(R/W/SC)

H264_DMA_INLINK_START_CH3 Configures whether to start processing the inlink descriptors.

0: No effect

1: Start

(R/W/SC)

H264_DMA_INLINK_RESTART_CH3 Configures whether to restart a new inlink from the last address.

0: No effect

1: Restart

(R/W/SC)

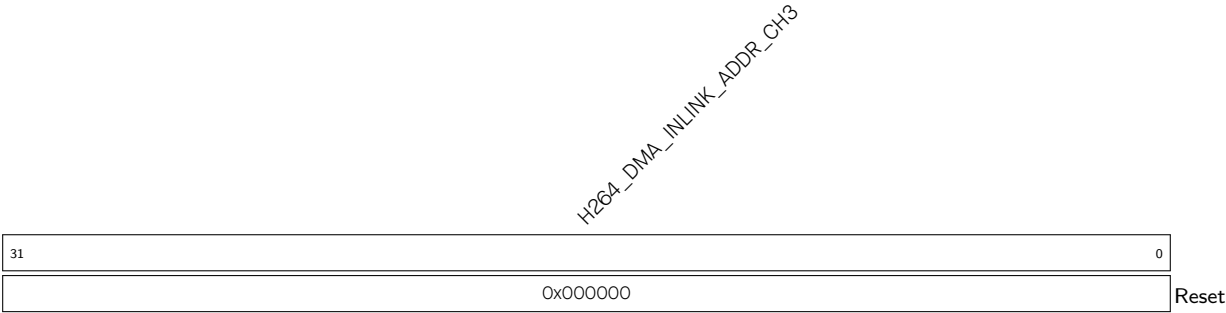
H264_DMA_INLINK_PARK_CH3 Represents the status of the inlink descriptor's FSM.

0: The inlink descriptor's FSM is working

1: The inlink descriptor's FSM is in idle state

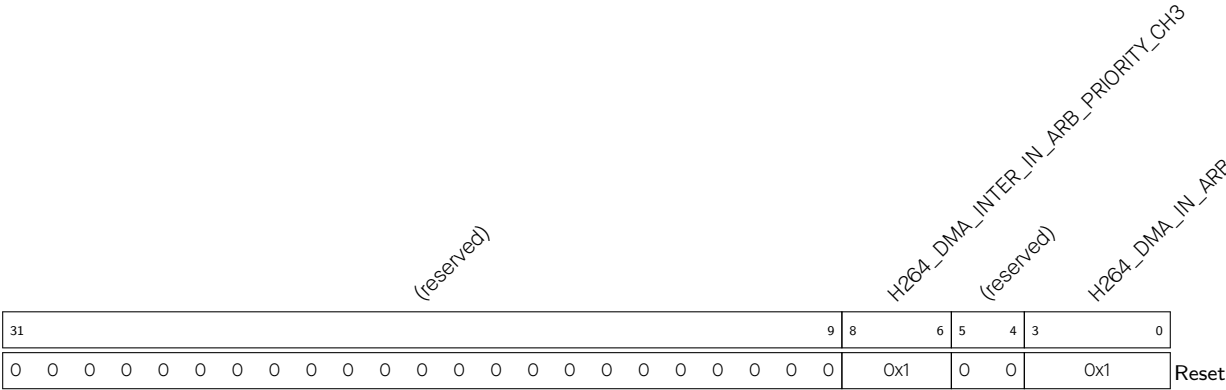
(RO)

Register 39.105. H264_DMA_IN_LINK_ADDR_CH3_REG (0x0820)



H264_DMA_INLINK_ADDR_CH3 Configures the first inlink descriptor’s address. (R/W)

Register 39.106. H264_DMA_IN_ARB_CH3_REG (0x0840)



H264_DMA_IN_ARB_TOKEN_NUM_CH3 Configures the maximum token count for the arbiter configuration. (R/W)

H264_DMA_INTER_IN_ARB_PRIORITY_CH3 Configures the priority of internal memory access for RX channel 3. (R/W)

Register 39.107. H264_DMA_IN_CONFO_CH4_REG (0x0900)

[illegible]

H264_DMA_INDSR_BURST_EN_CH4 Configures whether to enable INCR burst transfer for RX channel 4 to read descriptors.

0: Disable

1: Enable

(R/W)

H264_DMA_IN_ECC_AES_EN_CH4 Configures whether DMA can access external memory space for ECC and AES via RX channel 4.

0: Not access

1: Access. In this case, the start address of the memory region should be 16-byte aligned. The product of the width and the bytes per pixel must also be 16-byte aligned.

(R/W)

H264_DMA_IN_CHECK_OWNER_CH4 Configures whether to enable owner bit check for RX channel 4.

0: Disable

1: Enable

(R/W)

H264_DMA_IN_MEM_BURST_LENGTH_CH4 Configures the burst length for RX channel 4.

0: 8 bytes

1: 16 bytes

2: 32 bytes

3: 64 bytes

4: 128 bytes

5 ~ 7: Invalid

(R/W)

H264_DMA_IN_PAGE_BOUND_EN_CH4 Configures whether to make sure AXI read data doesn't cross the address boundary which define by mem_burst_length.

0: AXI read data can cross the address boundary

1: AXI read data doesn't cross the address boundary

(R/W)

H264_DMA_IN_RST_CH4 Write 1 then write 0 to reset RX channel 4. (R/W)

Continued on the next page...

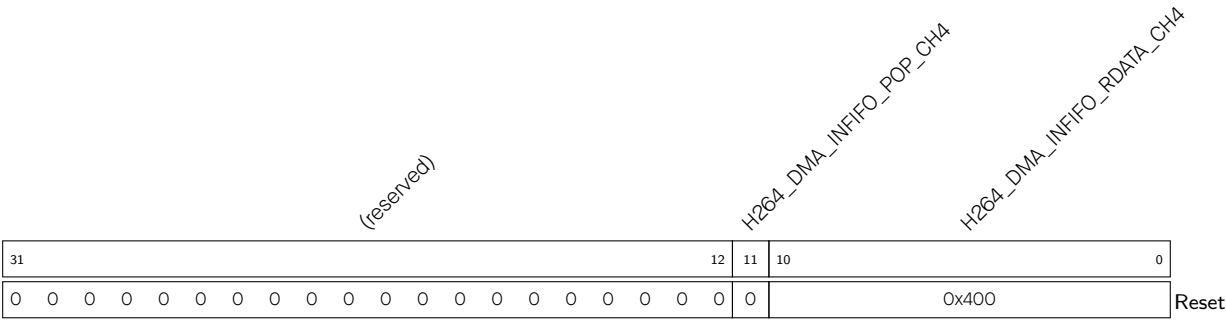
Register 39.107. H264_DMA_IN_CONFO_CH4_REG (0x0900)

Continued from the previous page...

H264_DMA_IN_CMD_DISABLE_CH4 Configures whether to disable command on RX channel 4.
0: Disable
1: Enable
(R/W)

H264_DMA_IN_ARB_WEIGHT_OPT_DIS_CH4 Configures whether to disable weight arbitration for RX channel 4.
0: Enable
1: Disable
(R/W)

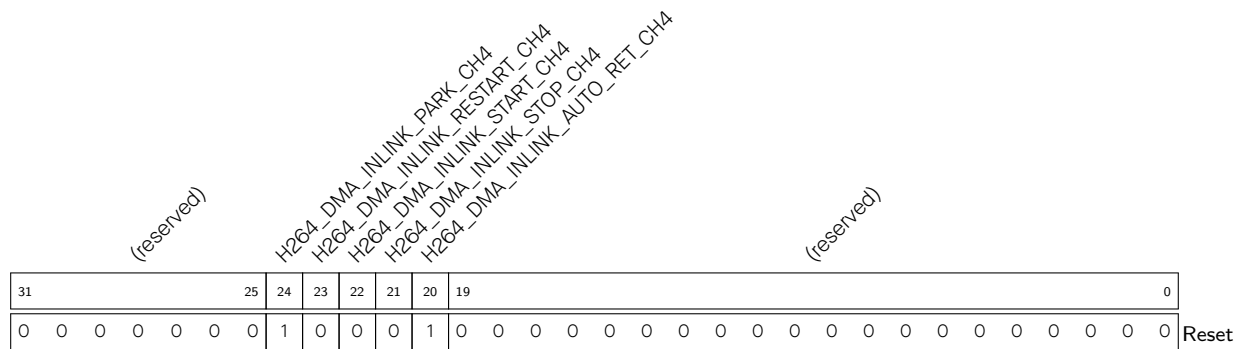
Register 39.108. H264_DMA_IN_POP_CH4_REG (0x0918)



H264_DMA_INFIFO_RDATA_CH4 Represents the data popped from DMA RX FIFO. (RO)

H264_DMA_INFIFO_POP_CH4 Configures whether to pop data from DMA RX FIFO.
0: No effect
1: Pop
(R/W/SC)

Register 39.109. H264_DMA_IN_LINK_CONF_CH4_REG (0x091C)



H264_DMA_INLINK_AUTO_RET_CH4 Configures whether to return to the address of the current receive descriptor when there are errors during data reception.

0: Not return

1: Return

(R/W)

H264_DMA_INLINK_STOP_CH4 Configures whether to stop processing the inlink descriptors.

0: No effect

1: Stop

(R/W/SC)

H264_DMA_INLINK_START_CH4 Configures whether to start processing the inlink descriptors.

0: No effect

1: Start

(R/W/SC)

H264_DMA_INLINK_RESTART_CH4 Configures whether to restart a new inlink from the last address.

0: No effect

1: Restart

(R/W/SC)

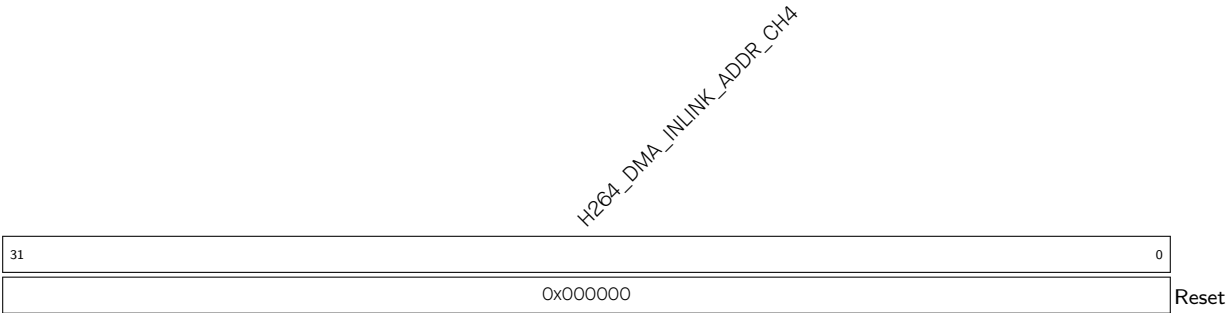
H264_DMA_INLINK_PARK_CH4 Represents the status of the inlink descriptor's FSM.

0: The inlink descriptor's FSM is working

1: The inlink descriptor's FSM is in idle state

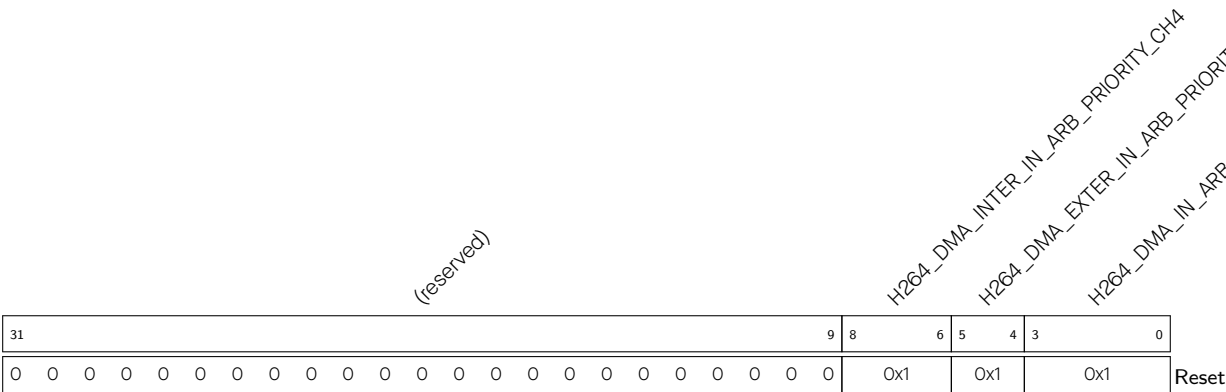
(RO)

Register 39.110. H264_DMA_IN_LINK_ADDR_CH4_REG (0x0920)



H264_DMA_INLINK_ADDR_CH4 Configures the first inlink descriptor’s address. (R/W)

Register 39.111. H264_DMA_IN_ARB_CH4_REG (0x0940)

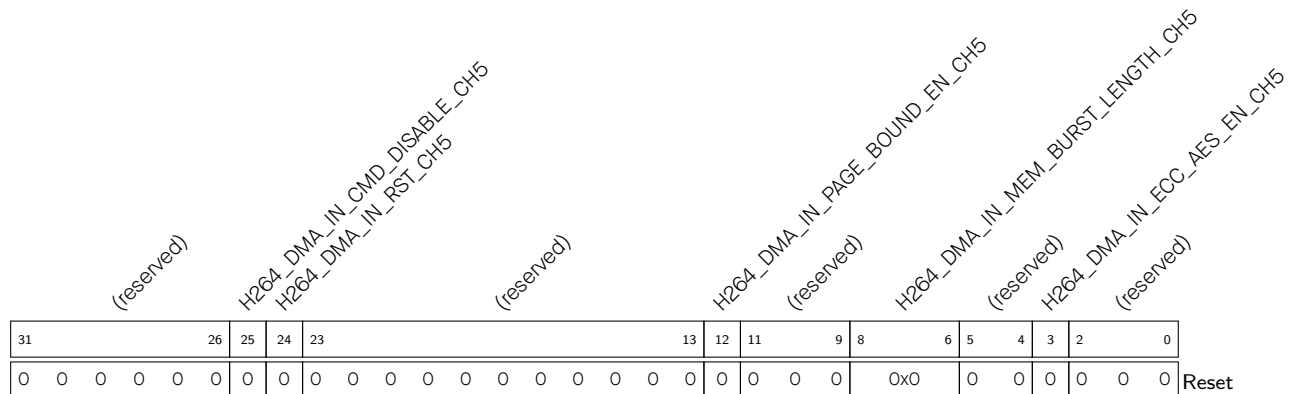


H264_DMA_IN_ARB_TOKEN_NUM_CH4 Configures the maximum token count for the arbiter configuration. (R/W)

H264_DMA_EXTER_IN_ARB_PRIORITY_CH4 Configures the priority of external memory access for RX channel 4. (R/W)

H264_DMA_INTER_IN_ARB_PRIORITY_CH4 Configures the priority of internal memory access for RX channel 4. (R/W)

Register 39.112. H264_DMA_IN_CONFO_CH5_REG (0x0A00)



H264_DMA_IN_ECC_AES_EN_CH5 Configures whether DMA can access external memory space for ECC and AES via RX channel 5.

0: Not access

1: Access. In this case, the start address of the memory region should be 16-byte aligned. The product of the width and the bytes per pixel must also be 16-byte aligned.

(R/W)

H264_DMA_IN_MEM_BURST_LENGTH_CH5 Configures the burst length for RX channel 5.

0: 8 bytes

1: 16 bytes

2: 32 bytes

3: 64 bytes

4: 128 bytes

5 ~ 7: Invalid

(R/W)

H264_DMA_IN_PAGE_BOUND_EN_CH5 Configures whether to make sure AXI read data doesn't cross the address boundary which define by mem_burst_length.

0: AXI read data can cross the address boundary

1: AXI read data doesn't cross the address boundary

(R/W)

H264_DMA_IN_RST_CH5 Write 1 then write 0 to reset RX channel 5. (R/W)

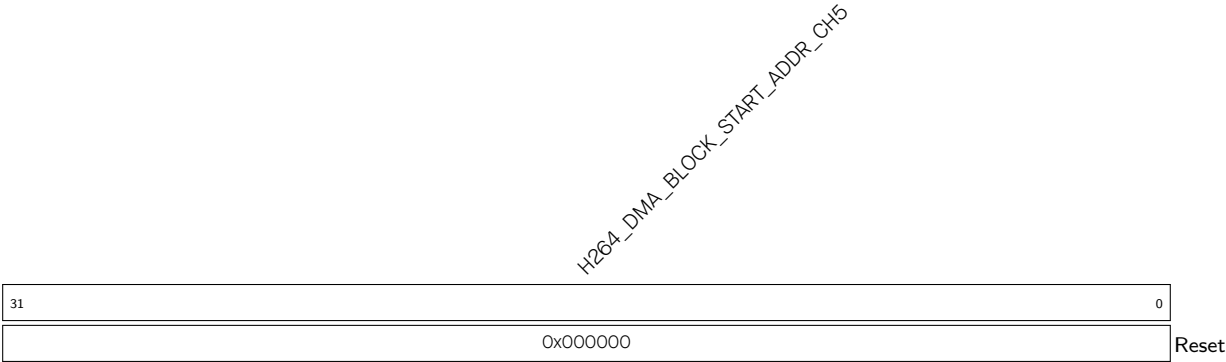
H264_DMA_IN_CMD_DISABLE_CH5 Configures whether to disable command on RX channel 5.

0: Disable

1: Enable

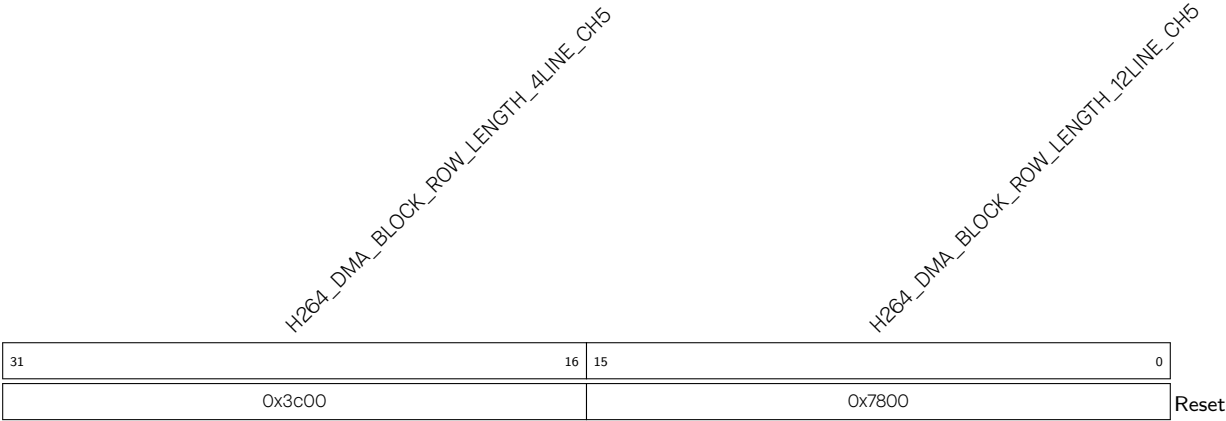
(R/W)

Register 39.113. H264_DMA_IN_CONF1_CH5_REG (0x0A04)



H264_DMA_BLOCK_START_ADDR_CH5 Configures the destination start address of RX channel 5.
(R/W)

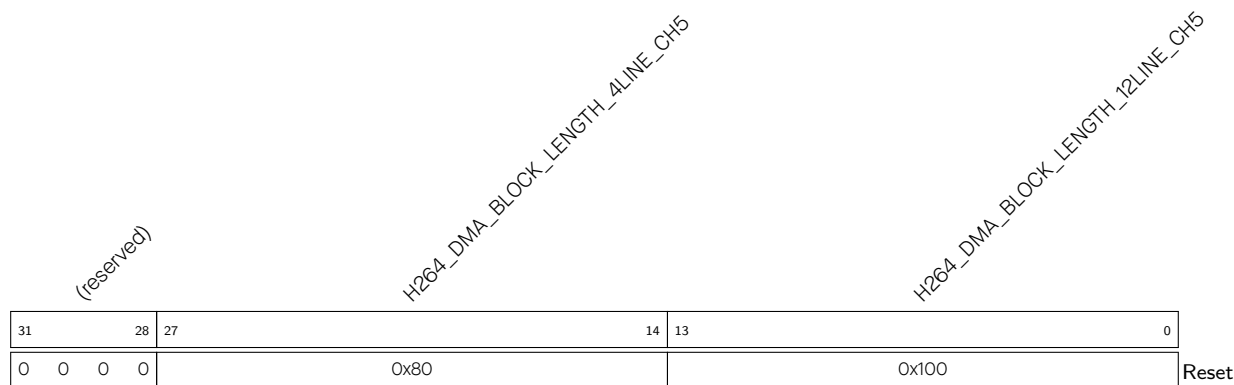
Register 39.114. H264_DMA_IN_CONF2_CH5_REG (0x0A08)



H264_DMA_BLOCK_ROW_LENGTH_12LINE_CH5 Configures the number of bytes contained in a row block 12line. (R/W)

H264_DMA_BLOCK_ROW_LENGTH_4LINE_CH5 Configures the number of bytes contained in a row block 4line. (R/W)

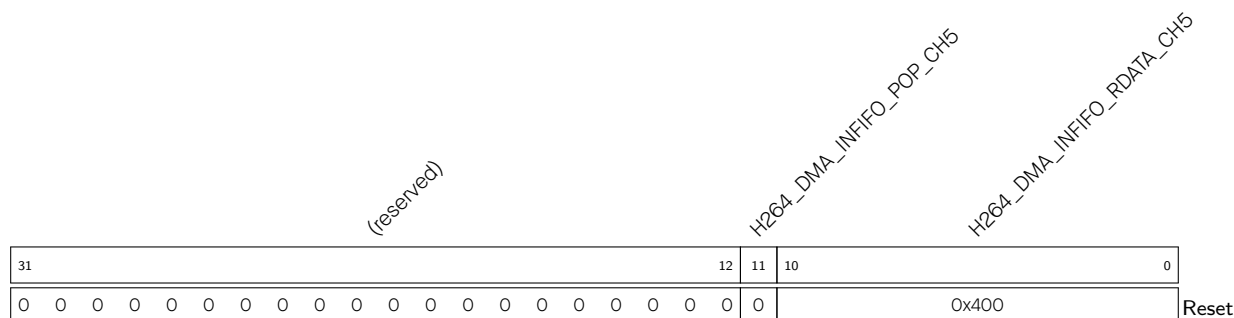
Register 39.115. H264_DMA_IN_CONF3_CH5_REG (0x0A0C)



H264_DMA_BLOCK_LENGTH_12LINE_CH5 Configures the number of bytes contained in a block 12line (R/W)

H264_DMA_BLOCK_LENGTH_4LINE_CH5 Configures the number of bytes contained in a block 4line (R/W)

Register 39.116. H264_DMA_IN_POP_CH5_REG (0x0A24)



H264_DMA_INFIFO_RDATA_CH5 Represents the data popped from DMA RX FIFO. (RO)

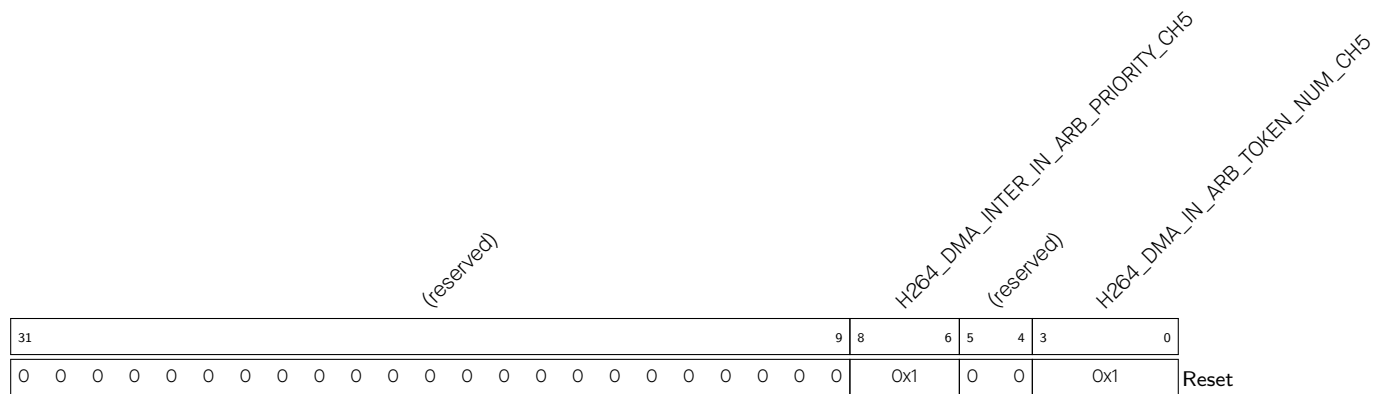
H264_DMA_INFIFO_POP_CH5 Configures whether to pop data from DMA RX FIFO.

0: No effect

1: Pop

(R/W/SC)

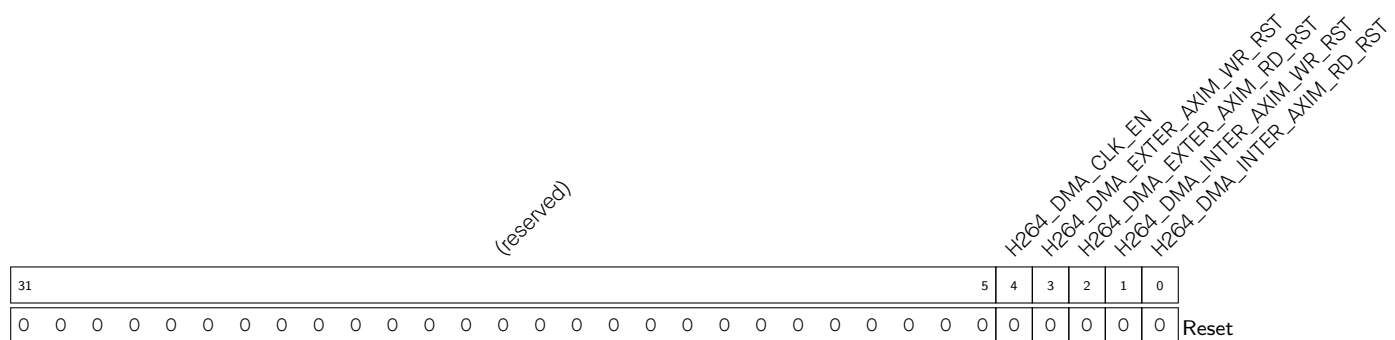
Register 39.117. H264_DMA_IN_ARB_CH5_REG (0x0A40)



H264_DMA_IN_ARB_TOKEN_NUM_CH5 Configures the maximum token count for the arbiter configuration. (R/W)

H264_DMA_INTER_IN_ARB_PRIORITY_CH5 Configures the priority of internal memory access for RX channel 5. (R/W)

Register 39.118. H264_DMA_RST_CONF_REG (0x0B08)



H264_DMA_INTER_AXIM_RD_RST Write 1 and then 0 to reset the inter AXI read FSM. (R/W)

H264_DMA_INTER_AXIM_WR_RST Write 1 and then 0 to reset the inter AXI write FSM. (R/W)

H264_DMA_EXTER_AXIM_RD_RST Write 1 and then 0 to reset the exter AXI read FSM. (R/W)

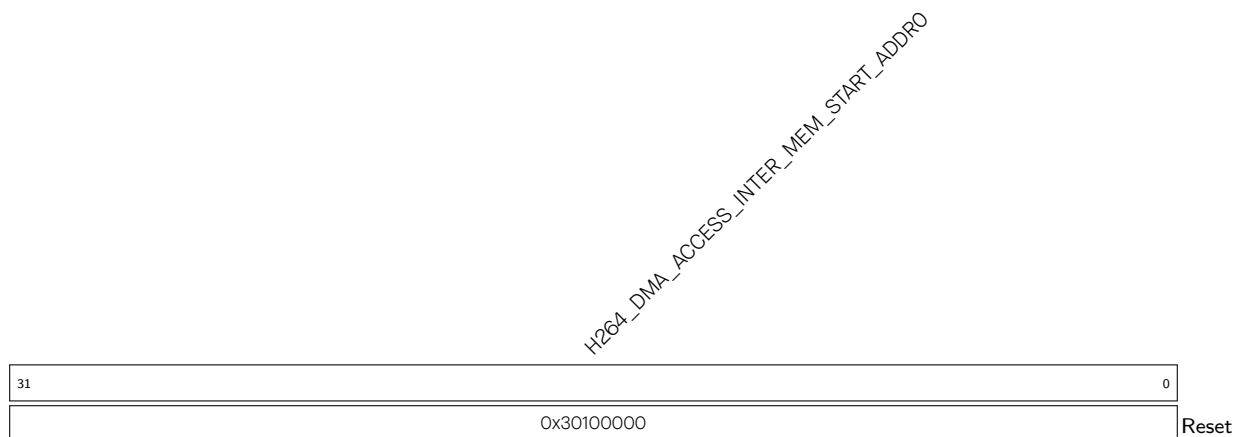
H264_DMA_EXTER_AXIM_WR_RST Write 1 and then 0 to reset the exter AXI write FSM. (R/W)

H264_DMA_CLK_EN Configures DMA clock gating.

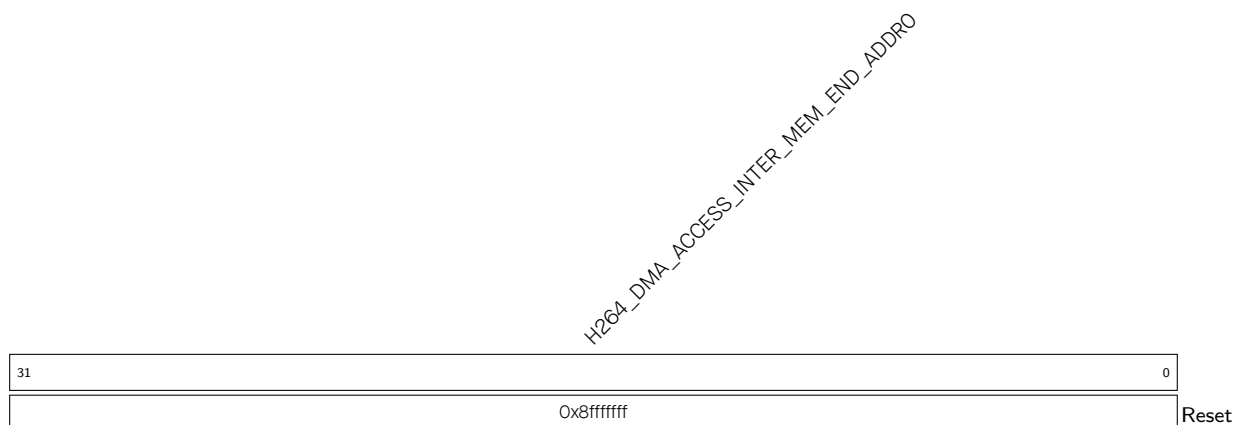
0: Enable the clock only when the application writes registers

1: Always enable the clock for registers

(R/W)

Register 39.119. H264_DMA_INTER_MEM_START_ADDRO_REG (0x0B0C)

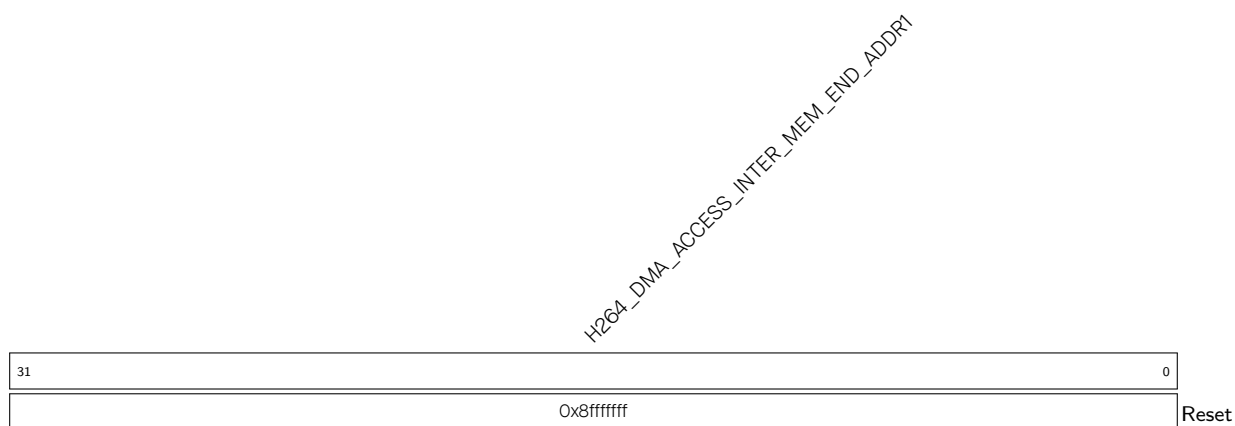
H264_DMA_ACCESS_INTER_MEM_START_ADDRO Configures the start address of the accessible internal memory address0 space. (R/W)

Register 39.120. H264_DMA_INTER_MEM_END_ADDRO_REG (0x0B10)

H264_DMA_ACCESS_INTER_MEM_END_ADDRO Configures the end address of the accessible internal memory address0 space. (R/W)

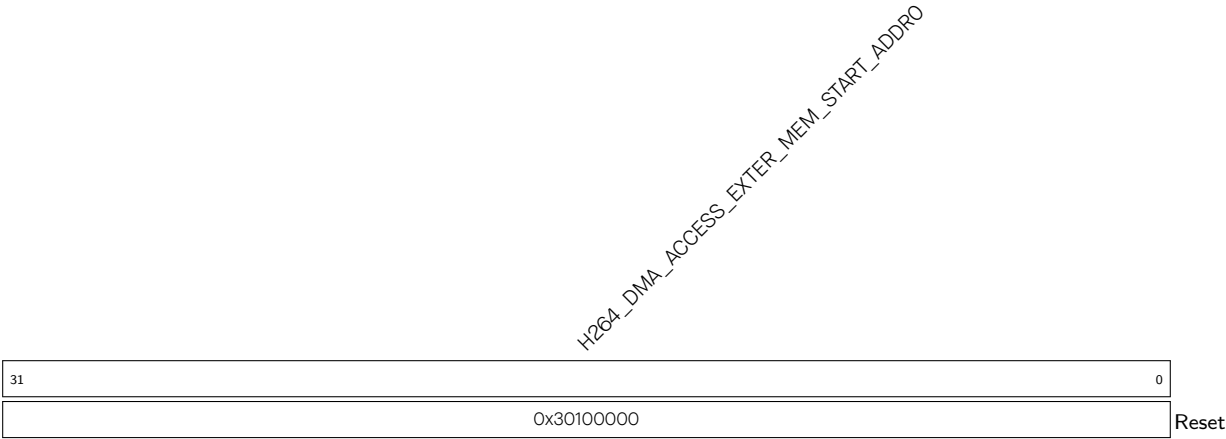
Register 39.121. H264_DMA_INTER_MEM_START_ADDR1_REG (0x0B14)

H264_DMA_ACCESS_INTER_MEM_START_ADDR1 Configures the start address of the accessible internal memory address1 space. (R/W)

Register 39.122. H264_DMA_INTER_MEM_END_ADDR1_REG (0x0B18)

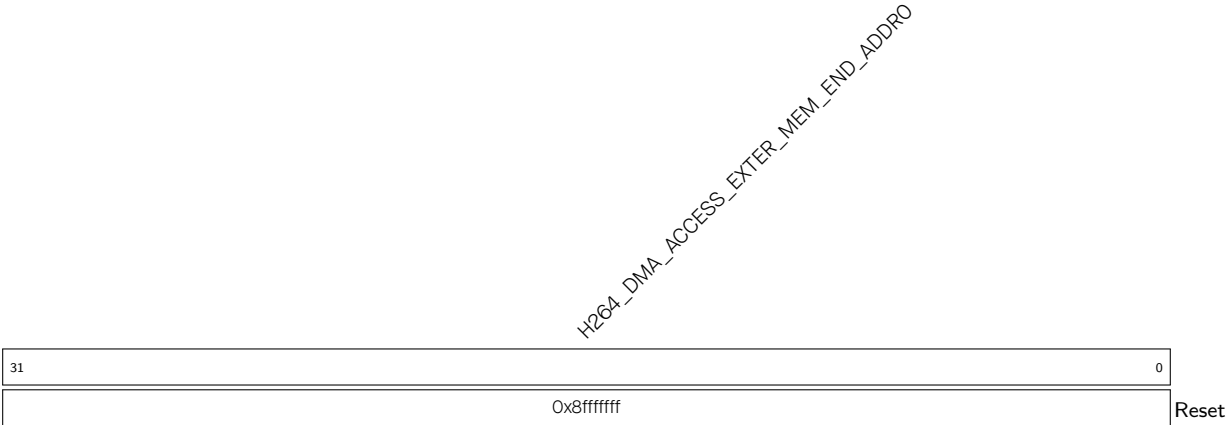
H264_DMA_ACCESS_INTER_MEM_END_ADDR1 Configures the end address of the accessible internal memory address1 space. (R/W)

Register 39.123. H264_DMA_EXTER_MEM_START_ADDRO_REG (0x0B20)

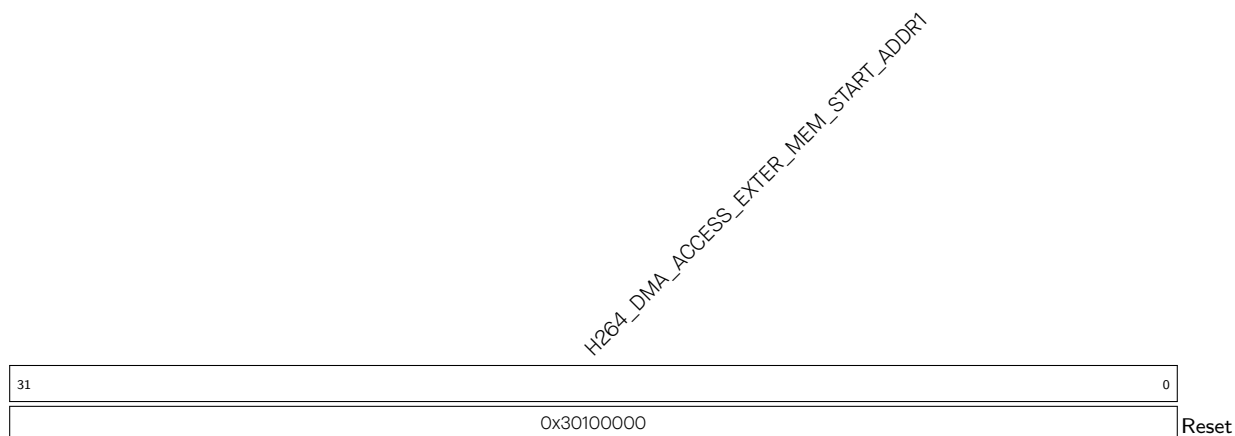


H264_DMA_ACCESS_EXTER_MEM_START_ADDRO Configures the start address of the accessible external memory address0 space. (R/W)

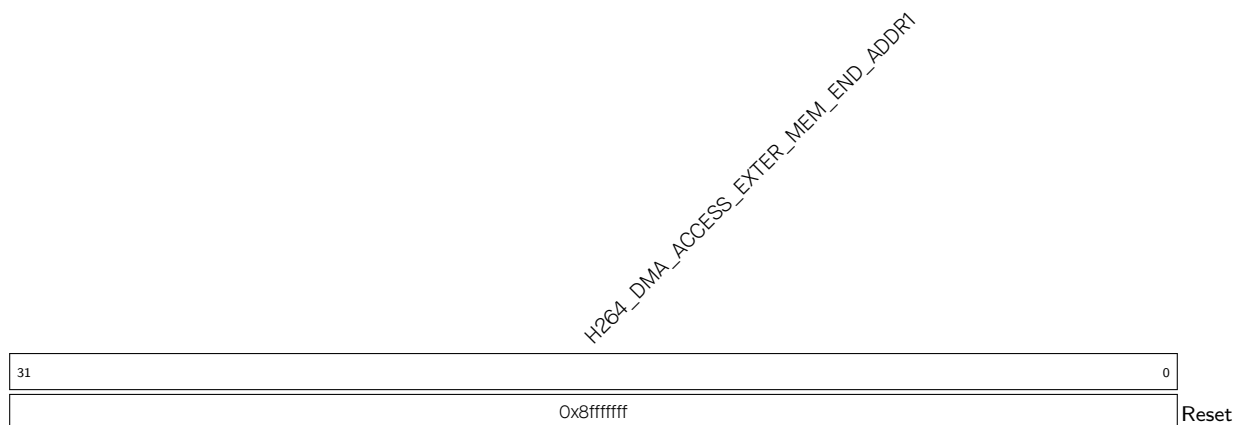
Register 39.124. H264_DMA_EXTER_MEM_END_ADDRO_REG (0x0B24)



H264_DMA_ACCESS_EXTER_MEM_END_ADDRO Configures the end address of the accessible external memory address0 space. (R/W)

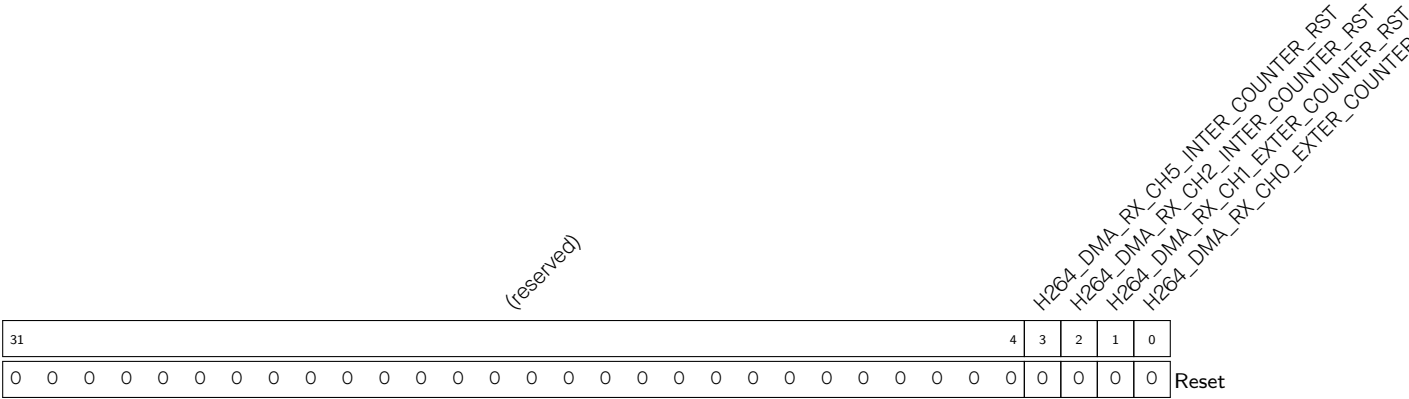
Register 39.125. H264_DMA_EXTER_MEM_START_ADDR1_REG (0x0B28)

H264_DMA_ACCESS_EXTER_MEM_START_ADDR1 Configures the start address of the accessible external memory address1 space. (R/W)

Register 39.126. H264_DMA_EXTER_MEM_END_ADDR1_REG (0x0B2C)

H264_DMA_ACCESS_EXTER_MEM_END_ADDR1 Configures the end address of the accessible external memory address1 space. (R/W)

Register 39.129. H264_DMA_COUNTER_RST_REG (0x0B50)



- H264_DMA_RX_CH0_EXTER_COUNTER_RST Write 1 then write 0 to reset RX ch0 counter. (R/W)
- H264_DMA_RX_CH1_EXTER_COUNTER_RST Write 1 then write 0 to reset RX ch1 counter. (R/W)
- H264_DMA_RX_CH2_INTER_COUNTER_RST Write 1 then write 0 to reset RX ch2 counter. (R/W)
- H264_DMA_RX_CH5_INTER_COUNTER_RST Write 1 then write 0 to reset RX ch5 counter. (R/W)

Register 39.130. H264_DMA_OUT_INT_RAW_CHO_REG (0x0004)

(reserved)																								H264_DMA_OUT_DSCR_TASK_OVF_CHO_INT_RAW H264_DMA_OUT_FIFO_UDF_L2_CHO_INT_RAW H264_DMA_OUT_FIFO_OVF_L2_CHO_INT_RAW H264_DMA_OUT_FIFO_UDF_L1_CHO_INT_RAW H264_DMA_OUT_FIFO_OVF_L1_CHO_INT_RAW H264_DMA_OUT_TOTAL_EOF_CHO_INT_RAW H264_DMA_OUT_DSCR_ERR_CHO_INT_RAW H264_DMA_OUT_DONE_CHO_INT_RAW										
31																								9	8	7	6	5	4	3	2	1	0	Reset
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0											

H264_DMA_OUT_DONE_CHO_INT_RAW The raw interrupt status of H264_DMA_OUT_DONE_CHO_INT. (R/WTC/SS)

H264_DMA_OUT_EOF_CHO_INT_RAW The raw interrupt status of H264_DMA_OUT_EOF_CHO_INT. (R/WTC/SS)

H264_DMA_OUT_DSCR_ERR_CHO_INT_RAW The raw interrupt status of H264_DMA_OUT_DSCR_ERR_CHO_INT. (R/WTC/SS)

H264_DMA_OUT_TOTAL_EOF_CHO_INT_RAW The raw interrupt status of H264_DMA_OUT_TOTAL_EOF_CHO_INT. (R/WTC/SS)

H264_DMA_OUT_FIFO_OVF_L1_CHO_INT_RAW The raw interrupt status of H264_DMA_OUT_FIFO_OVF_L1_CHO_INT. (R/WTC/SS)

H264_DMA_OUT_FIFO_UDF_L1_CHO_INT_RAW The raw interrupt status of H264_DMA_OUT_FIFO_UDF_L1_CHO_INT. (R/WTC/SS)

H264_DMA_OUT_FIFO_OVF_L2_CHO_INT_RAW The raw interrupt status of H264_DMA_OUT_FIFO_OVF_L2_CHO_INT. (R/WTC/SS)

H264_DMA_OUT_FIFO_UDF_L2_CHO_INT_RAW The raw interrupt status of H264_DMA_OUT_FIFO_UDF_L2_CHO_INT. (R/WTC/SS)

H264_DMA_OUT_DSCR_TASK_OVF_CHO_INT_RAW The raw interrupt status of H264_DMA_OUT_DSCR_TASK_OVF_CHO_INT. (R/WTC/SS)

Register 39.131. H264_DMA_OUT_INT_ENA_CHO_REG (0x0008)

(reserved)																H264_DMA_OUT_DSCR_TASK_OVF_CHO_INT_ENA				
																H264_DMA_OUT_DSCR_ERR_CHO_INT_ENA				
																H264_DMA_OUT_EOF_CHO_INT_ENA				
																H264_DMA_OUT_TOTAL_EOF_CHO_INT_ENA				
																H264_DMA_OUTFIFO_OVF_L1_CHO_INT_ENA				
																H264_DMA_OUTFIFO_OVF_L2_CHO_INT_ENA				
																H264_DMA_OUTFIFO_UDF_L1_CHO_INT_ENA				
																H264_DMA_OUTFIFO_UDF_L2_CHO_INT_ENA				
																H264_DMA_OUT_DONE_CHO_INT_ENA				
31									9	8	7	6	5	4	3	2	1	0	Reset	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

H264_DMA_OUT_DONE_CHO_INT_ENA Write 1 to enable H264_DMA_OUT_DONE_CHO_INT. (R/W)

H264_DMA_OUT_EOF_CHO_INT_ENA Write 1 to enable H264_DMA_OUT_EOF_CHO_INT. (R/W)

H264_DMA_OUT_DSCR_ERR_CHO_INT_ENA Write 1 to enable H264_DMA_OUT_DSCR_ERR_CHO_INT.
(R/W)

H264_DMA_OUT_TOTAL_EOF_CHO_INT_ENA Write 1 to enable H264_DMA_OUT_TOTAL_EOF_CHO_INT.
(R/W)

H264_DMA_OUTFIFO_OVF_L1_CHO_INT_ENA Write 1 to enable H264_DMA_OUTFIFO_OVF_L1_CHO_INT.
(R/W)

H264_DMA_OUTFIFO_UDF_L1_CHO_INT_ENA Write 1 to enable H264_DMA_OUTFIFO_UDF_L1_CHO_INT.
(R/W)

H264_DMA_OUTFIFO_OVF_L2_CHO_INT_ENA Write 1 to enable H264_DMA_OUTFIFO_OVF_L2_CHO_INT.
(R/W)

H264_DMA_OUTFIFO_UDF_L2_CHO_INT_ENA Write 1 to enable H264_DMA_OUTFIFO_UDF_L2_CHO_INT.
(R/W)

H264_DMA_OUT_DSCR_TASK_OVF_CHO_INT_ENA Write 1 to enable H264_DMA_OUT_DSCR_TASK_OVF_CHO_INT. (R/W)

Register 39.132. H264_DMA_OUT_INT_ST_CHO_REG (0x000C)

(reserved)																H264_DMA_OUT_DSCR_TASK_OVF_CHO_INT_ST H264_DMA_OUT_DSCR_ERR_CHO_INT_ST H264_DMA_OUT_TOTAL_EOF_CHO_INT_ST H264_DMA_OUT_FIFO_OVF_L1_CHO_INT_ST H264_DMA_OUT_FIFO_UDF_L1_CHO_INT_ST H264_DMA_OUT_FIFO_OVF_L2_CHO_INT_ST H264_DMA_OUT_FIFO_UDF_L2_CHO_INT_ST				
31										9	8	7	6	5	4	3	2	1	0	Reset
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

H264_DMA_OUT_DONE_CHO_INT_ST The masked interrupt status of H264_DMA_OUT_DONE_CHO_INT. (RO)

H264_DMA_OUT_EOF_CHO_INT_ST The masked interrupt status of H264_DMA_OUT_EOF_CHO_INT. (RO)

H264_DMA_OUT_DSCR_ERR_CHO_INT_ST The masked interrupt status of H264_DMA_OUT_DSCR_ERR_CHO_INT. (RO)

H264_DMA_OUT_TOTAL_EOF_CHO_INT_ST The masked interrupt status of H264_DMA_OUT_TOTAL_EOF_CHO_INT. (RO)

H264_DMA_OUT_FIFO_OVF_L1_CHO_INT_ST The masked interrupt status of H264_DMA_OUT_FIFO_OVF_L1_CHO_INT. (RO)

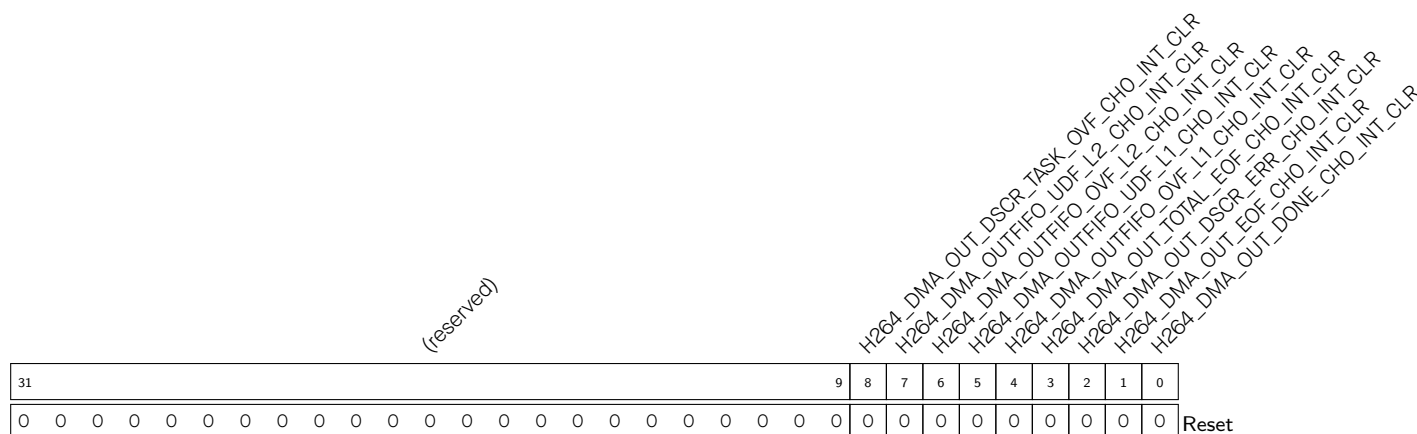
H264_DMA_OUT_FIFO_UDF_L1_CHO_INT_ST The masked interrupt status of H264_DMA_OUT_FIFO_UDF_L1_CHO_INT. (RO)

H264_DMA_OUT_FIFO_OVF_L2_CHO_INT_ST The masked interrupt status of H264_DMA_OUT_FIFO_OVF_L2_CHO_INT. (RO)

H264_DMA_OUT_FIFO_UDF_L2_CHO_INT_ST The masked interrupt status of H264_DMA_OUT_FIFO_UDF_L2_CHO_INT. (RO)

H264_DMA_OUT_DSCR_TASK_OVF_CHO_INT_ST The masked interrupt status of H264_DMA_OUT_DSCR_TASK_OVF_CHO_INT. (RO)

Register 39.133. H264_DMA_OUT_INT_CLR_CHO_REG (0x0010)



H264 DMA OUT DONE CHO INT CLR Write 1 to clear H264 DMA OUT DONE CHO INT. (WT)

H264_DMA_OUT_EOF_CHO_INT_CLR Write 1 to clear H264_DMA_OUT_EOF_CHO_INT. (WT)

H264_DMA_OUT_DSCR_ERR_CHO_INT_CLR Write 1 to clear H264_DMA_OUT_DSCR_ERR_CHO_INT.
(WT)

H264_DMA_OUT_TOTAL_EOF_CHO_INT_CLR Write 1 to clear H264_DMA_OUT_TOTAL_EOF_CHO_INT.
(WT)

H264_DMA_OUTFIFO_OVF_L1_CHO_INT_CLR Write 1 to clear H264_DMA_OUTFIFO_OVF_L1_CHO_INT.
(WT)

H264_DMA_OUTFIFO_UDF_L1_CHO_INT_CLR Write 1 to clear H264_DMA_OUTFIFO_UDF_L1_CHO_INT.
(WT)

H264_DMA_OUTFIFO_OVF_L2_CHO_INT_CLR Write 1 to clear H264_DMA_OUTFIFO_OVF_L2_CHO_INT.
(WT)

H264_DMA_OUTFIFO_UDF_L2_CHO_INT_CLR Write 1 to clear H264_DMA_OUTFIFO_UDF_L2_CHO_INT.
(WT)

H264_DMA_OUT_DSCR_TASK_OVF_CHO_INT_CLR	Write	1	to	clear
H264 DMA OUT DSCR TASK OVF CHO INT. (WT)				

Register 39.134. H264_DMA_OUT_INT_RAW_CH1_REG (0x0104)

(reserved)																H264_DMA_OUT_DONE_CH1_INT_RAW				
																H264_DMA_OUT_EOF_CH1_INT_RAW				
																H264_DMA_OUT_DSCR_ERR_CH1_INT_RAW				
																H264_DMA_OUT_TOTAL_EOF_CH1_INT_RAW				
																H264_DMA_OUT_FIFO_OVF_L1_CH1_INT_RAW				
																H264_DMA_OUT_FIFO_UDF_L1_CH1_INT_RAW				
																H264_DMA_OUT_FIFO_OVF_L2_CH1_INT_RAW				
																H264_DMA_OUT_FIFO_UDF_L2_CH1_INT_RAW				
																H264_DMA_OUT_DSCR_TASK_OVF_CH1_INT_RAW				
31									9	8	7	6	5	4	3	2	1	0		
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset

H264_DMA_OUT_DONE_CH1_INT_RAW The raw interrupt status of H264_DMA_OUT_DONE_CH1_INT. (R/WTC/SS)

H264_DMA_OUT_EOF_CH1_INT_RAW The raw interrupt status of H264_DMA_OUT_EOF_CH1_INT. (R/WTC/SS)

H264_DMA_OUT_DSCR_ERR_CH1_INT_RAW The raw interrupt status of H264_DMA_OUT_DSCR_ERR_CH1_INT. (R/WTC/SS)

H264_DMA_OUT_TOTAL_EOF_CH1_INT_RAW The raw interrupt status of H264_DMA_OUT_TOTAL_EOF_CH1_INT. (R/WTC/SS)

H264_DMA_OUTFIFO_OVF_L1_CH1_INT_RAW The raw interrupt status of H264_DMA_OUTFIFO_OVF_L1_CH1_INT. (R/WTC/SS)

H264_DMA_OUTFIFO_UDF_L1_CH1_INT_RAW The raw interrupt status of H264_DMA_OUTFIFO_UDF_L1_CH1_INT. (R/WTC/SS)

H264_DMA_OUTFIFO_OVF_L2_CH1_INT_RAW The raw interrupt status of H264_DMA_OUTFIFO_OVF_L2_CH1_INT. (R/WTC/SS)

H264_DMA_OUTFIFO_UDF_L2_CH1_INT_RAW The raw interrupt status of H264_DMA_OUTFIFO_UDF_L2_CH1_INT. (R/WTC/SS)

H264_DMA_OUT_DSCR_TASK_OVF_CH1_INT_RAW The raw interrupt status of H264_DMA_OUT_DSCR_TASK_OVF_CH1_INT. (R/WTC/SS)

Register 39.136. H264_DMA_OUT_INT_ST_CH1_REG (0x010C)

(reserved)																H264_DMA_OUT_DSCR_TASK_OVF_CH1_INT_ST			
																H264_DMA_OUT_FIFO_UDF_L2_CH1_INT_ST			
																H264_DMA_OUT_FIFO_OVF_L2_CH1_INT_ST			
																H264_DMA_OUT_FIFO_UDF_L1_CH1_INT_ST			
																H264_DMA_OUT_TOTAL_EOF_CH1_INT_ST			
																H264_DMA_OUT_DSCR_ERR_CH1_INT_ST			
																H264_DMA_OUT_DONE_CH1_INT_ST			
31										9	8	7	6	5	4	3	2	1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Reset

H264_DMA_OUT_DONE_CH1_INT_ST The masked interrupt status of H264_DMA_OUT_DONE_CH1_INT. (RO)

H264_DMA_OUT_EOF_CH1_INT_ST The masked interrupt status of H264_DMA_OUT_EOF_CH1_INT. (RO)

H264_DMA_OUT_DSCR_ERR_CH1_INT_ST The masked interrupt status of H264_DMA_OUT_DSCR_ERR_CH1_INT. (RO)

H264_DMA_OUT_TOTAL_EOF_CH1_INT_ST The masked interrupt status of H264_DMA_OUT_TOTAL_EOF_CH1_INT. (RO)

H264_DMA_OUTFIFO_OVF_L1_CH1_INT_ST The masked interrupt status of H264_DMA_OUTFIFO_OVF_L1_CH1_INT. (RO)

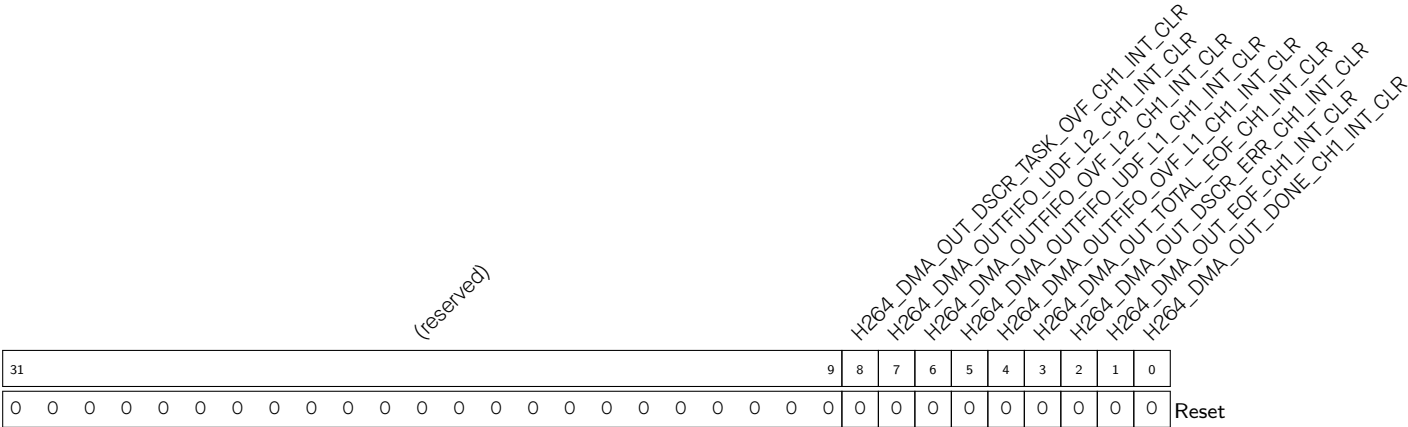
H264_DMA_OUTFIFO_UDF_L1_CH1_INT_ST The masked interrupt status of H264_DMA_OUTFIFO_UDF_L1_CH1_INT. (RO)

H264_DMA_OUTFIFO_OVF_L2_CH1_INT_ST The masked interrupt status of H264_DMA_OUTFIFO_OVF_L2_CH1_INT. (RO)

H264_DMA_OUTFIFO_UDF_L2_CH1_INT_ST The masked interrupt status of H264_DMA_OUTFIFO_UDF_L2_CH1_INT. (RO)

H264_DMA_OUT_DSCR_TASK_OVF_CH1_INT_ST The masked interrupt status of H264_DMA_OUT_DSCR_TASK_OVF_CH1_INT. (RO)

Register 39.137. H264_DMA_OUT_INT_CLR_CH1_REG (0x0110)



H264_DMA_OUT_DONE_CH1_INT_CLR Write 1 to clear H264_DMA_OUT_DONE_CH1_INT. (WT)

H264_DMA_OUT_EOF_CH1_INT_CLR Write 1 to clear H264_DMA_OUT_EOF_CH1_INT. (WT)

H264_DMA_OUT_DSCR_ERR_CH1_INT_CLR Write 1 to clear H264_DMA_OUT_DSCR_ERR_CH1_INT.
(WT)

H264_DMA_OUT_TOTAL_EOF_CH1_INT_CLR Write 1 to clear H264_DMA_OUT_TOTAL_EOF_CH1_INT.
(WT)

H264_DMA_OUTFIFO_OVF_L1_CH1_INT_CLR Write 1 to clear H264_DMA_OUTFIFO_OVF_L1_CH1_INT.
(WT)

H264_DMA_OUTFIFO_UDF_L1_CH1_INT_CLR Write 1 to clear H264_DMA_OUTFIFO_UDF_L1_CH1_INT.
(WT)

H264_DMA_OUTFIFO_OVF_L2_CH1_INT_CLR Write 1 to clear H264_DMA_OUTFIFO_OVF_L2_CH1_INT.
(WT)

H264_DMA_OUTFIFO_UDF_L2_CH1_INT_CLR Write 1 to clear H264_DMA_OUTFIFO_UDF_L2_CH1_INT.
(WT)

H264_DMA_OUT_DSCR_TASK_OVF_CH1_INT_CLR Write 1 to clear
H264_DMA_OUT_DSCR_TASK_OVF_CH1_INT. (WT)

Register 39.138. H264_DMA_OUT_INT_RAW_CH2_REG (0x0204)

(reserved)																H264_DMA_OUT_DSCR_TASK_OVF_CH2_INT_RAW				
																H264_DMA_OUT_FIFO_OVF_L2_CH2_INT_RAW				
																H264_DMA_OUT_FIFO_OVF_L2_CH2_INT_RAW				
																H264_DMA_OUT_FIFO_OVF_L1_CH2_INT_RAW				
																H264_DMA_OUT_TOTAL_EOF_CH2_INT_RAW				
																H264_DMA_OUT_DSCR_ERR_CH2_INT_RAW				
																H264_DMA_OUT_DONE_CH2_INT_RAW				
31									9	8	7	6	5	4	3	2	1	0		
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset	

H264_DMA_OUT_DONE_CH2_INT_RAW The raw interrupt status of H264_DMA_OUT_DONE_CH2_INT. Triggered when all data corresponding to a transmit descriptor has been sent via TX channel 2. (R/WTC/SS)

H264_DMA_OUT_EOF_CH2_INT_RAW The raw interrupt status of H264_DMA_OUT_EOF_CH2_INT. (R/WTC/SS)

H264_DMA_OUT_DSCR_ERR_CH2_INT_RAW The raw interrupt status of H264_DMA_OUT_DSCR_ERR_CH2_INT. (R/WTC/SS)

H264_DMA_OUT_TOTAL_EOF_CH2_INT_RAW The raw interrupt status of H264_DMA_OUT_TOTAL_EOF_CH2_INT. (R/WTC/SS)

H264_DMA_OUTFIFO_OVF_L1_CH2_INT_RAW The raw interrupt status of H264_DMA_OUTFIFO_OVF_L1_CH2_INT. (R/WTC/SS)

H264_DMA_OUTFIFO_UDF_L1_CH2_INT_RAW The raw interrupt status of H264_DMA_OUTFIFO_UDF_L1_CH2_INT. (R/WTC/SS)

H264_DMA_OUTFIFO_OVF_L2_CH2_INT_RAW The raw interrupt status of H264_DMA_OUTFIFO_OVF_L2_CH2_INT. (R/WTC/SS)

H264_DMA_OUTFIFO_UDF_L2_CH2_INT_RAW The raw interrupt status of H264_DMA_OUTFIFO_UDF_L2_CH2_INT. (R/WTC/SS)

H264_DMA_OUT_DSCR_TASK_OVF_CH2_INT_RAW The raw interrupt status of H264_DMA_OUT_DSCR_TASK_OVF_CH2_INT. (R/WTC/SS)

Register 39.139. H264_DMA_OUT_INT_ENA_CH2_REG (0x0208)

(reserved)																				H264_DMA_OUT_DSCR_TASK_OVF_CH2_INT_ENA H264_DMA_OUT_FIFO_UDF_L2_CH2_INT_ENA H264_DMA_OUT_FIFO_OVF_L2_CH2_INT_ENA H264_DMA_OUT_FIFO_UDF_L1_CH2_INT_ENA H264_DMA_OUT_FIFO_OVF_L1_CH2_INT_ENA H264_DMA_OUT_TOTAL_EOF_CH2_INT_ENA H264_DMA_OUT_DSCR_ERR_CH2_INT_ENA H264_DMA_OUT_EOF_CH2_INT_ENA H264_DMA_OUT_DONE_CH2_INT_ENA									
31																			9	8	7	6	5	4	3	2	1	0	Reset
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0									

H264_DMA_OUT_DONE_CH2_INT_ENA Write 1 to enable H264_DMA_OUT_DONE_CH2_INT. (R/W)

H264_DMA_OUT_EOF_CH2_INT_ENA Write 1 to enable H264_DMA_OUT_EOF_CH2_INT. (R/W)

H264_DMA_OUT_DSCR_ERR_CH2_INT_ENA Write 1 to enable H264_DMA_OUT_DSCR_ERR_CH2_INT.
(R/W)

H264_DMA_OUT_TOTAL_EOF_CH2_INT_ENA Write 1 to enable H264_DMA_OUT_TOTAL_EOF_CH2_INT.
(R/W)

H264_DMA_OUTFIFO_OVF_L1_CH2_INT_ENA Write 1 to enable H264_DMA_OUTFIFO_OVF_L1_CH2_INT.
(R/W)

H264_DMA_OUTFIFO_UDF_L1_CH2_INT_ENA Write 1 to enable H264_DMA_OUTFIFO_UDF_L1_CH2_INT.
(R/W)

H264_DMA_OUTFIFO_OVF_L2_CH2_INT_ENA Write 1 to enable H264_DMA_OUTFIFO_OVF_L2_CH2_INT.
(R/W)

H264_DMA_OUTFIFO_UDF_L2_CH2_INT_ENA Write 1 to enable H264_DMA_OUTFIFO_UDF_L2_CH2_INT.
(R/W)

H264_DMA_OUT_DSCR_TASK_OVF_CH2_INT_ENA Write 1 to enable
H264_DMA_OUT_DSCR_TASK_OVF_CH2_INT. (R/W)

Register 39.140. H264_DMA_OUT_INT_ST_CH2_REG (0x020C)

(reserved)																H264_DMA_OUT_DSCR_TASK_OVF_CH2_INT_ST				
																H264_DMA_OUT_DSCR_ERR_CH2_INT_ST				
																H264_DMA_OUT_TOTAL_EOF_CH2_INT_ST				
																H264_DMA_OUT_FIFO_OVF_L1_CH2_INT_ST				
																H264_DMA_OUT_FIFO_UDF_L1_CH2_INT_ST				
																H264_DMA_OUT_FIFO_OVF_L2_CH2_INT_ST				
																H264_DMA_OUT_FIFO_UDF_L2_CH2_INT_ST				
																H264_DMA_OUT_DONE_CH2_INT_ST				
31									9	8	7	6	5	4	3	2	1	0	Reset	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

H264_DMA_OUT_DONE_CH2_INT_ST The masked interrupt status of H264_DMA_OUT_DONE_CH2_INT. (RO)

H264_DMA_OUT_EOF_CH2_INT_ST The masked interrupt status of H264_DMA_OUT_EOF_CH2_INT. (RO)

H264_DMA_OUT_DSCR_ERR_CH2_INT_ST The masked interrupt status of H264_DMA_OUT_DSCR_ERR_CH2_INT. (RO)

H264_DMA_OUT_TOTAL_EOF_CH2_INT_ST The masked interrupt status of H264_DMA_OUT_TOTAL_EOF_CH2_INT. (RO)

H264_DMA_OUTFIFO_OVF_L1_CH2_INT_ST The masked interrupt status of H264_DMA_OUTFIFO_OVF_L1_CH2_INT. (RO)

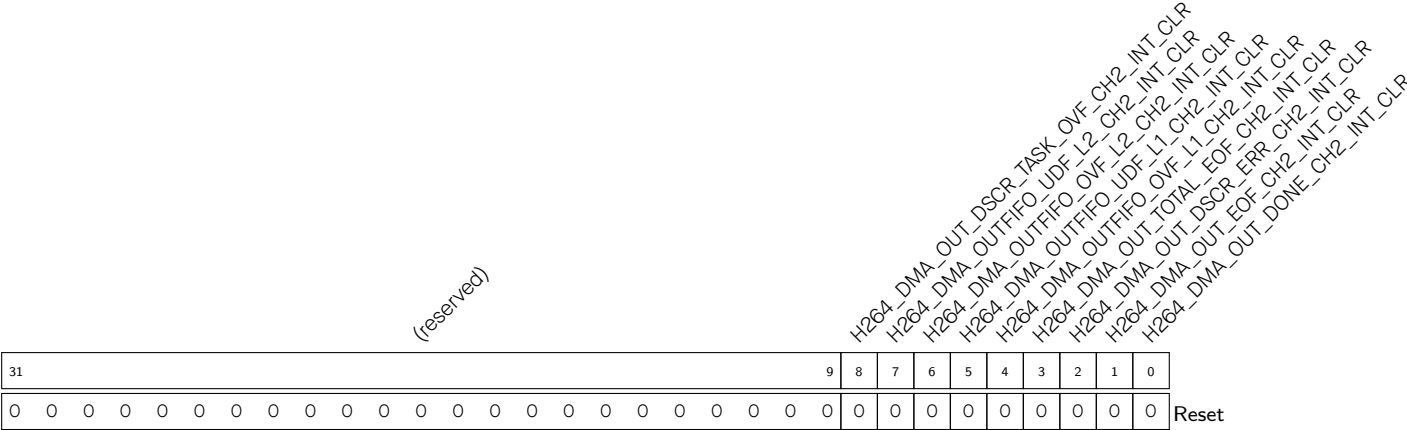
H264_DMA_OUTFIFO_UDF_L1_CH2_INT_ST The masked interrupt status of H264_DMA_OUTFIFO_UDF_L1_CH2_INT. (RO)

H264_DMA_OUTFIFO_OVF_L2_CH2_INT_ST The masked interrupt status of H264_DMA_OUTFIFO_OVF_L2_CH2_INT. (RO)

H264_DMA_OUTFIFO_UDF_L2_CH2_INT_ST The masked interrupt status of H264_DMA_OUTFIFO_UDF_L2_CH2_INT. (RO)

H264_DMA_OUT_DSCR_TASK_OVF_CH2_INT_ST The masked interrupt status of H264_DMA_OUT_DSCR_TASK_OVF_CH2_INT. (RO)

Register 39.141. H264_DMA_OUT_INT_CLR_CH2_REG (0x0210)



H264_DMA_OUT_DONE_CH2_INT_CLR Write 1 to clear H264_DMA_OUT_DONE_CH2_INT. (WT)

H264_DMA_OUT_EOF_CH2_INT_CLR Write 1 to clear H264_DMA_OUT_EOF_CH2_INT. (WT)

H264_DMA_OUT_DSCR_ERR_CH2_INT_CLR Write 1 to clear H264_DMA_OUT_DSCR_ERR_CH2_INT.
(WT)

H264_DMA_OUT_TOTAL_EOF_CH2_INT_CLR Write 1 to clear H264_DMA_OUT_TOTAL_EOF_CH2_INT.
(WT)

H264_DMA_OUTFIFO_OVF_L1_CH2_INT_CLR Write 1 to clear H264_DMA_OUTFIFO_OVF_L1_CH2_INT.
(WT)

H264_DMA_OUTFIFO_UDF_L1_CH2_INT_CLR Write 1 to clear H264_DMA_OUTFIFO_UDF_L1_CH2_INT.
(WT)

H264_DMA_OUTFIFO_OVF_L2_CH2_INT_CLR Write 1 to clear H264_DMA_OUTFIFO_OVF_L2_CH2_INT.
(WT)

H264_DMA_OUTFIFO_UDF_L2_CH2_INT_CLR Write 1 to clear H264_DMA_OUTFIFO_UDF_L2_CH2_INT.
(WT)

H264_DMA_OUT_DSCR_TASK_OVF_CH2_INT_CLR Write 1 to clear
H264_DMA_OUT_DSCR_TASK_OVF_CH2_INT. (WT)

Register 39.142. H264_DMA_OUT_INT_RAW_CH3_REG (0x0304)

(reserved)																								H264_DMA_OUT_DSCR_TASK_OVF_CH3_INT_RAW H264_DMA_OUT_FIFO_UDF_L2_CH3_INT_RAW H264_DMA_OUT_FIFO_OVF_L2_CH3_INT_RAW H264_DMA_OUT_FIFO_UDF_L1_CH3_INT_RAW H264_DMA_OUT_FIFO_OVF_L1_CH3_INT_RAW H264_DMA_OUT_TOTAL_EOF_CH3_INT_RAW H264_DMA_OUT_DSCR_ERR_CH3_INT_RAW H264_DMA_OUT_DONE_CH3_INT_RAW										
31																								9	8	7	6	5	4	3	2	1	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset									

H264_DMA_OUT_DONE_CH3_INT_RAW The raw interrupt status of H264_DMA_OUT_DONE_CH3_INT. (R/WTC/SS)

H264_DMA_OUT_EOF_CH3_INT_RAW The raw interrupt status of H264_DMA_OUT_EOF_CH3_INT. (R/WTC/SS)

H264_DMA_OUT_DSCR_ERR_CH3_INT_RAW The raw interrupt status of H264_DMA_OUT_DSCR_ERR_CH3_INT. (R/WTC/SS)

H264_DMA_OUT_TOTAL_EOF_CH3_INT_RAW The raw interrupt status of H264_DMA_OUT_TOTAL_EOF_CH3_INT. (R/WTC/SS)

H264_DMA_OUTFIFO_OVF_L1_CH3_INT_RAW The raw interrupt status of H264_DMA_OUTFIFO_OVF_L1_CH3_INT. (R/WTC/SS)

H264_DMA_OUTFIFO_UDF_L1_CH3_INT_RAW The raw interrupt status of H264_DMA_OUTFIFO_UDF_L1_CH3_INT. (R/WTC/SS)

H264_DMA_OUTFIFO_OVF_L2_CH3_INT_RAW The raw interrupt status of H264_DMA_OUTFIFO_OVF_L2_CH3_INT. (R/WTC/SS)

H264_DMA_OUTFIFO_UDF_L2_CH3_INT_RAW The raw interrupt status of H264_DMA_OUTFIFO_UDF_L2_CH3_INT. (R/WTC/SS)

H264_DMA_OUT_DSCR_TASK_OVF_CH3_INT_RAW The raw interrupt status of H264_DMA_OUT_DSCR_TASK_OVF_CH3_INT. (R/WTC/SS)

Register 39.143. H264_DMA_OUT_INT_ENA_CH3_REG (0x0308)

(reserved)																				H264_DMA_OUT_DSCR_TASK_OVF_CH3_INT_ENA H264_DMA_OUT_FIFO_UDF_L2_CH3_INT_ENA H264_DMA_OUT_FIFO_OVF_L2_CH3_INT_ENA H264_DMA_OUT_FIFO_UDF_L1_CH3_INT_ENA H264_DMA_OUT_FIFO_OVF_L1_CH3_INT_ENA H264_DMA_OUT_TOTAL_EOF_CH3_INT_ENA H264_DMA_OUT_DSCR_ERR_CH3_INT_ENA H264_DMA_OUT_DONE_CH3_INT_ENA										
31																				9	8	7	6	5	4	3	2	1	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset									

H264_DMA_OUT_DONE_CH3_INT_ENA Write 1 to enable H264_DMA_OUT_DONE_CH3_INT. (R/W)

H264_DMA_OUT_EOF_CH3_INT_ENA Write 1 to enable H264_DMA_OUT_EOF_CH3_INT. (R/W)

H264_DMA_OUT_DSCR_ERR_CH3_INT_ENA Write 1 to enable H264_DMA_OUT_DSCR_ERR_CH3_INT.
(R/W)

H264_DMA_OUT_TOTAL_EOF_CH3_INT_ENA Write 1 to enable H264_DMA_OUT_TOTAL_EOF_CH3_INT.
(R/W)

H264_DMA_OUTFIFO_OVF_L1_CH3_INT_ENA Write 1 to enable H264_DMA_OUTFIFO_OVF_L1_CH3_INT.
(R/W)

H264_DMA_OUTFIFO_UDF_L1_CH3_INT_ENA Write 1 to enable H264_DMA_OUTFIFO_UDF_L1_CH3_INT.
(R/W)

H264_DMA_OUTFIFO_OVF_L2_CH3_INT_ENA Write 1 to enable H264_DMA_OUTFIFO_OVF_L3_CH3_INT.
(R/W)

H264_DMA_OUTFIFO_UDF_L2_CH3_INT_ENA Write 1 to enable H264_DMA_OUTFIFO_UDF_L3_CH3_INT.
(R/W)

H264_DMA_OUT_DSCR_TASK_OVF_CH3_INT_ENA Write 1 to enable H264_DMA_OUT_DSCR_TASK_OVF_CH3_INT. (R/W)

Register 39.144. H264_DMA_OUT_INT_ST_CH3_REG (0x030C)

(reserved)																H264_DMA_OUT_DSCR_TASK_OVF_CH3_INT_ST				
																H264_DMA_OUT_DSCR_ERR_CH3_INT_ST				
																H264_DMA_OUT_TOTAL_EOF_CH3_INT_ST				
																H264_DMA_OUT_FIFO_OVF_L1_CH3_INT_ST				
																H264_DMA_OUT_FIFO_UDF_L1_CH3_INT_ST				
																H264_DMA_OUT_FIFO_OVF_L2_CH3_INT_ST				
																H264_DMA_OUT_FIFO_UDF_L2_CH3_INT_ST				
																H264_DMA_OUT_DONE_CH3_INT_ST				
31									9	8	7	6	5	4	3	2	1	0	Reset	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		

H264_DMA_OUT_DONE_CH3_INT_ST The masked interrupt status of H264_DMA_OUT_DONE_CH3_INT. (RO)

H264_DMA_OUT_EOF_CH3_INT_ST The masked interrupt status of H264_DMA_OUT_EOF_CH3_INT. (RO)

H264_DMA_OUT_DSCR_ERR_CH3_INT_ST The masked interrupt status of H264_DMA_OUT_DSCR_ERR_CH3_INT. (RO)

H264_DMA_OUT_TOTAL_EOF_CH3_INT_ST The masked interrupt status of H264_DMA_OUT_TOTAL_EOF_CH3_INT. (RO)

H264_DMA_OUT_FIFO_OVF_L1_CH3_INT_ST The masked interrupt status of H264_DMA_OUT_FIFO_OVF_L1_CH3_INT. (RO)

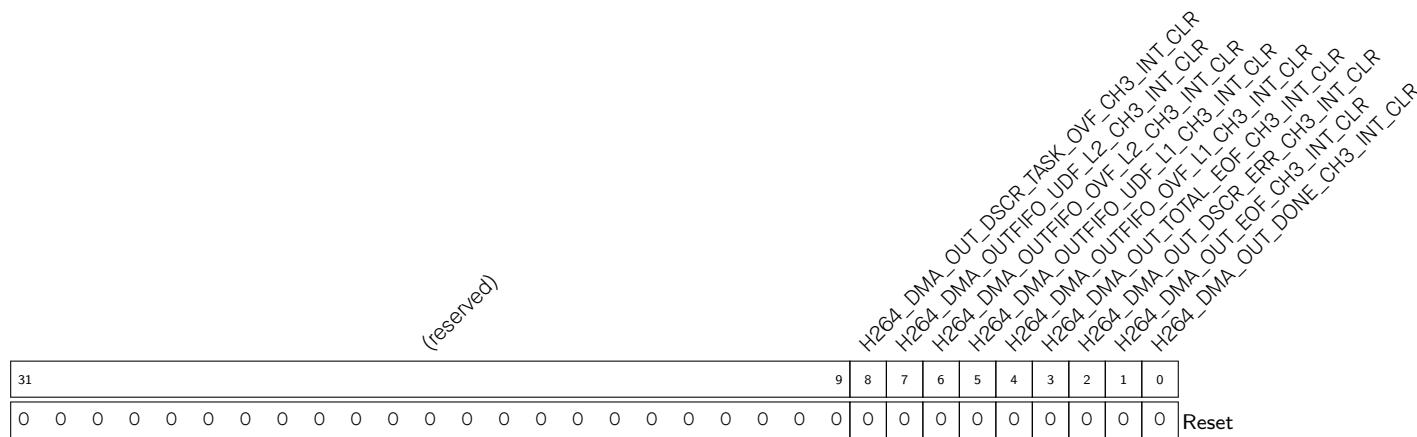
H264_DMA_OUT_FIFO_UDF_L1_CH3_INT_ST The masked interrupt status of H264_DMA_OUT_FIFO_UDF_L1_CH3_INT. (RO)

H264_DMA_OUT_FIFO_OVF_L2_CH3_INT_ST The masked interrupt status of H264_DMA_OUT_FIFO_OVF_L2_CH3_INT. (RO)

H264_DMA_OUT_FIFO_UDF_L2_CH3_INT_ST The masked interrupt status of H264_DMA_OUT_FIFO_UDF_L2_CH3_INT. (RO)

H264_DMA_OUT_DSCR_TASK_OVF_CH3_INT_ST The masked interrupt status of H264_DMA_OUT_DSCR_TASK_OVF_CH3_INT. (RO)

Register 39.145. H264_DMA_OUT_INT_CLR_CH3_REG (0x0310)



H264_DMA_OUT_DONE_CH3_INT_CLR Write 1 to clear H264_DMA_OUT_DONE_CH3_INT. (WT)

H264_DMA_OUT_EOF_CH3_INT_CLR Write 1 to clear H264_DMA_OUT_EOF_CH3_INT. (WT)

H264_DMA_OUT_DSCR_ERR_CH3_INT_CLR Write 1 to clear H264_DMA_OUT_DSCR_ERR_CH3_INT.
(WT)

H264_DMA_OUT_TOTAL_EOF_CH3_INT_CLR Write 1 to clear H264_DMA_OUT_TOTAL_EOF_CH3_INT.
(WT)

H264_DMA_OUTFIFO_OVF_L1_CH3_INT_CLR Write 1 to clear H264_DMA_OUTFIFO_OVF_L1_CH3_INT.
(WT)

H264_DMA_OUTFIFO_UDF_L1_CH3_INT_CLR Write 1 to clear H264_DMA_OUTFIFO_UDF_L1_CH3_INT.
(WT)

H264_DMA_OUTFIFO_OVF_L2_CH3_INT_CLR Write 1 to clear H264_DMA_OUTFIFO_OVF_L2_CH3_INT.
(WT)

H264_DMA_OUTFIFO_UDF_L2_CH3_INT_CLR Write 1 to clear H264_DMA_OUTFIFO_UDF_L2_CH3_INT.
(WT)

H264_DMA_OUT_DSCR_TASK_OVF_CH3_INT_CLR Write 1 to clear
H264_DMA_OUT_DSCR_TASK_OVF_CH3_INT. (WT)

Register 39.146. H264_DMA_OUT_INT_RAW_CH4_REG (0x0404)

(reserved)																								H264_DMA_OUT_DSCR_TASK_OVF_CH4_INT_RAW H264_DMA_OUT_FIFO_UDF_L2_CH4_INT_RAW H264_DMA_OUT_FIFO_OVF_L2_CH4_INT_RAW H264_DMA_OUT_FIFO_UDF_L1_CH4_INT_RAW H264_DMA_OUT_FIFO_OVF_L1_CH4_INT_RAW H264_DMA_OUT_TOTAL_EOF_CH4_INT_RAW H264_DMA_OUT_DSCR_ERR_CH4_INT_RAW H264_DMA_OUT_DONE_CH4_INT_RAW										
31																								9	8	7	6	5	4	3	2	1	0	Reset
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0												

H264_DMA_OUT_DONE_CH4_INT_RAW The raw interrupt status of H264_DMA_OUT_DONE_CH4_INT. (R/WTC/SS)

H264_DMA_OUT_EOF_CH4_INT_RAW The raw interrupt status of H264_DMA_OUT_EOF_CH4_INT. (R/WTC/SS)

H264_DMA_OUT_DSCR_ERR_CH4_INT_RAW The raw interrupt status of H264_DMA_OUT_DSCR_ERR_CH4_INT. 4. (R/WTC/SS)

H264_DMA_OUT_TOTAL_EOF_CH4_INT_RAW The raw interrupt status of H264_DMA_OUT_TOTAL_EOF_CH4_INT. (R/WTC/SS)

H264_DMA_OUTFIFO_OVF_L1_CH4_INT_RAW The raw interrupt status of H264_DMA_OUTFIFO_OVF_L1_CH4_INT. (R/WTC/SS)

H264_DMA_OUTFIFO_UDF_L1_CH4_INT_RAW The raw interrupt status of H264_DMA_OUTFIFO_UDF_L1_CH4_INT. (R/WTC/SS)

H264_DMA_OUTFIFO_OVF_L2_CH4_INT_RAW The raw interrupt status of H264_DMA_OUTFIFO_OVF_L2_CH4_INT. (R/WTC/SS)

H264_DMA_OUTFIFO_UDF_L2_CH4_INT_RAW The raw interrupt status of H264_DMA_OUTFIFO_UDF_L2_CH4_INT. (R/WTC/SS)

H264_DMA_OUT_DSCR_TASK_OVF_CH4_INT_RAW The raw interrupt status of H264_DMA_OUT_DSCR_TASK_OVF_CH4_INT. (R/WTC/SS)

Register 39.147. H264_DMA_OUT_INT_ENA_CH4_REG (0x0408)

(reserved)																												H264_DMA_OUT_DSCR_TASK_OVF_CH4_INT_ENA H264_DMA_OUT_FIFO_UDF_L2_CH4_INT_ENA H264_DMA_OUT_FIFO_OVF_L2_CH4_INT_ENA H264_DMA_OUT_FIFO_UDF_L1_CH4_INT_ENA H264_DMA_OUT_FIFO_OVF_L1_CH4_INT_ENA H264_DMA_OUT_TOTAL_EOF_CH4_INT_ENA H264_DMA_OUT_DSCR_ERR_CH4_INT_ENA H264_DMA_OUT_DONE_CH4_INT_ENA											
31																												9	8	7	6	5	4	3	2	1	0		
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset									

H264_DMA_OUT_DONE_CH4_INT_ENA Write 1 to enable H264_DMA_OUT_DONE_CH4_INT. (R/W)

H264_DMA_OUT_EOF_CH4_INT_ENA Write 1 to enable H264_DMA_OUT_EOF_CH4_INT. (R/W)

H264_DMA_OUT_DSCR_ERR_CH4_INT_ENA Write 1 to enable H264_DMA_OUT_DSCR_ERR_CH4_INT.
(R/W)

H264_DMA_OUT_TOTAL_EOF_CH4_INT_ENA Write 1 to enable H264_DMA_OUT_TOTAL_EOF_CH4_INT.
(R/W)

H264_DMA_OUT_FIFO_OVF_L1_CH4_INT_ENA Write 1 to enable H264_DMA_OUT_FIFO_OVF_L1_CH4_INT.
(R/W)

H264_DMA_OUT_FIFO_UDF_L1_CH4_INT_ENA Write 1 to enable H264_DMA_OUT_FIFO_UDF_L1_CH4_INT.
(R/W)

H264_DMA_OUT_FIFO_OVF_L2_CH4_INT_ENA Write 1 to enable H264_DMA_OUT_FIFO_OVF_L3_CH4_INT.
(R/W)

H264_DMA_OUT_FIFO_UDF_L2_CH4_INT_ENA Write 1 to enable H264_DMA_OUT_FIFO_UDF_L3_CH4_INT.
(R/W)

H264_DMA_OUT_DSCR_TASK_OVF_CH4_INT_ENA Write 1 to enable
H264_DMA_OUT_DSCR_TASK_OVF_CH4_INT. (R/W)

Register 39.148. H264_DMA_OUT_INT_ST_CH4_REG (0x040C)

(reserved)																H264_DMA_OUT_DSCR_TASK_OVF_CH4_INT_ST H264_DMA_OUT_FIFO_UDF_L2_CH4_INT_ST H264_DMA_OUT_FIFO_OVF_L2_CH4_INT_ST H264_DMA_OUT_FIFO_UDF_L1_CH4_INT_ST H264_DMA_OUT_FIFO_OVF_L1_CH4_INT_ST H264_DMA_OUT_TOTAL_EOF_CH4_INT_ST H264_DMA_OUT_DSCR_ERR_CH4_INT_ST H264_DMA_OUT_DONE_CH4_INT_ST				
31										9	8	7	6	5	4	3	2	1	0	Reset
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

H264_DMA_OUT_DONE_CH4_INT_ST The masked interrupt status of H264_DMA_OUT_DONE_CH4_INT. (RO)

H264_DMA_OUT_EOF_CH4_INT_ST The masked interrupt status of H264_DMA_OUT_EOF_CH4_INT. (RO)

H264_DMA_OUT_DSCR_ERR_CH4_INT_ST The masked interrupt status of H264_DMA_OUT_DSCR_ERR_CH4_INT. (RO)

H264_DMA_OUT_TOTAL_EOF_CH4_INT_ST The masked interrupt status of H264_DMA_OUT_TOTAL_EOF_CH4_INT. (RO)

H264_DMA_OUT_FIFO_OVF_L1_CH4_INT_ST The masked interrupt status of H264_DMA_OUT_FIFO_OVF_L1_CH4_INT. (RO)

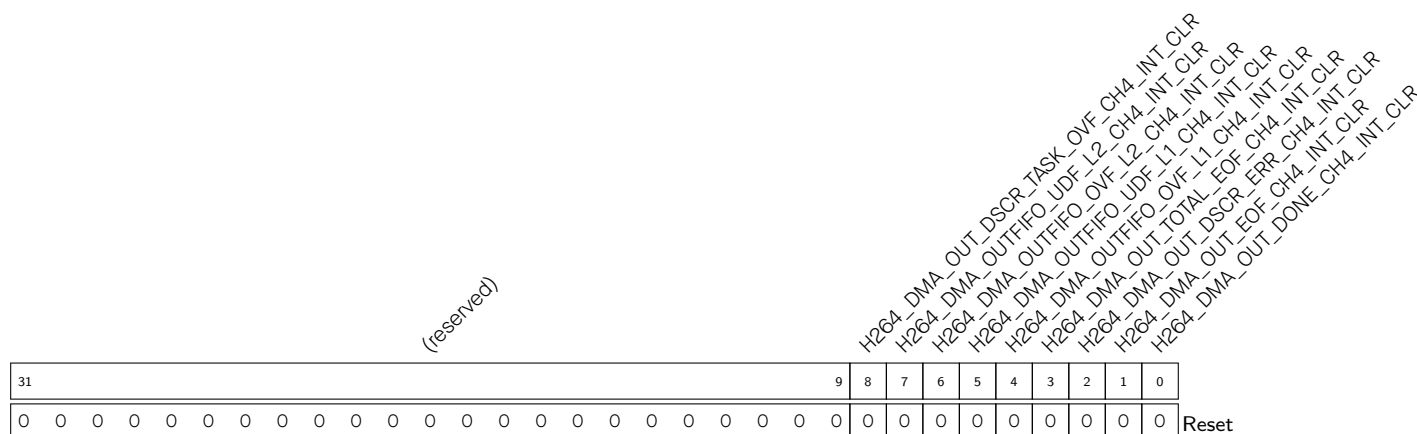
H264_DMA_OUT_FIFO_UDF_L1_CH4_INT_ST The masked interrupt status of H264_DMA_OUT_FIFO_UDF_L1_CH4_INT. (RO)

H264_DMA_OUT_FIFO_OVF_L2_CH4_INT_ST The masked interrupt status of H264_DMA_OUT_FIFO_OVF_L2_CH4_INT. (RO)

H264_DMA_OUT_FIFO_UDF_L2_CH4_INT_ST The masked interrupt status of H264_DMA_OUT_FIFO_UDF_L2_CH4_INT. (RO)

H264_DMA_OUT_DSCR_TASK_OVF_CH4_INT_ST The masked interrupt status of H264_DMA_OUT_DSCR_TASK_OVF_CH4_INT. (RO)

Register 39.149. H264_DMA_OUT_INT_CLR_CH4_REG (0x0410)



H264_DMA_OUT_DONE_CH4_INT_CLR Write 1 to clear H264_DMA_OUT_DONE_CH4_INT. (WT)

H264_DMA_OUT_EOF_CH4_INT_CLR Write 1 to clear H264_DMA_OUT_EOF_CH4_INT. (WT)

H264_DMA_OUT_DSCR_ERR_CH4_INT_CLR Write 1 to clear H264_DMA_OUT_DSCR_ERR_CH4_INT.
(WT)

H264_DMA_OUT_TOTAL_EOF_CH4_INT_CLR Write 1 to clear H264_DMA_OUT_TOTAL_EOF_CH4_INT.
(WT)

H264_DMA_OUTFIFO_OVF_L1_CH4_INT_CLR Write 1 to clear H264_DMA_OUTFIFO_OVF_L1_CH4_INT.
(WT)

H264_DMA_OUTFIFO_UDF_L1_CH4_INT_CLR Write 1 to clear H264_DMA_OUTFIFO_UDF_L1_CH4_INT.
(WT)

H264_DMA_OUTFIFO_OVF_L2_CH4_INT_CLR Write 1 to clear H264_DMA_OUTFIFO_OVF_L2_CH4_INT.
(WT)

H264_DMA_OUTFIFO_UDF_L2_CH4_INT_CLR Write 1 to clear H264_DMA_OUTFIFO_UDF_L2_CH4_INT.
(WT)

H264_DMA_OUT_DSCR_TASK_OVF_CH4_INT_CLR Write 1 to clear
H264_DMA_OUT_DSCR_TASK_OVF_CH4_INT. (WT)

Register 39.150. H264_DMA_IN_INT_RAW_CHO_REG (0x0504)

(reserved)																						H264_DMA_IN_DSCR_TASK_OVF_CHO_INT_RAW H264_DMA_IN_DSCR_EMPTY_CHO_INT_RAW H264_DMA_INFIFO_OVF_L2_CHO_INT_RAW H264_DMA_INFIFO_UDF_L2_CHO_INT_RAW H264_DMA_INFIFO_OVF_L1_CHO_INT_RAW H264_DMA_INFIFO_UDF_L1_CHO_INT_RAW H264_DMA_IN_DSCR_ERR_CHO_INT_RAW H264_DMA_IN_ERR_EOF_CHO_INT_RAW H264_DMA_IN_SUC_EOF_CHO_INT_RAW H264_DMA_IN_DONE_CHO_INT_RAW										
31										10										9	8	7	6	5	4	3	2	1	0			
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset									

H264_DMA_IN_DONE_CHO_INT_RAW The raw interrupt status of H264_DMA_IN_DONE_CHO_INT. (R/WTC/SS)

H264_DMA_IN_SUC_EOF_CHO_INT_RAW The raw interrupt status of H264_DMA_IN_SUC_EOF_CHO_INT. (R/WTC/SS)

H264_DMA_IN_ERR_EOF_CHO_INT_RAW The raw interrupt status of H264_DMA_IN_ERR_EOF_CHO_INT. (R/WTC/SS)

H264_DMA_IN_DSCR_ERR_CHO_INT_RAW The raw interrupt status of H264_DMA_IN_DSCR_ERR_CHO_INT. (R/WTC/SS)

H264_DMA_INFIFO_OVF_L1_CHO_INT_RAW The raw interrupt status of H264_DMA_INFIFO_OVF_L1_CHO_INT. (R/WTC/SS)

H264_DMA_INFIFO_UDF_L1_CHO_INT_RAW The raw interrupt status of H264_DMA_INFIFO_UDF_L1_CHO_INT. (R/WTC/SS)

H264_DMA_INFIFO_OVF_L2_CHO_INT_RAW The raw interrupt status of H264_DMA_INFIFO_OVF_L2_CHO_INT. (R/WTC/SS)

H264_DMA_INFIFO_UDF_L2_CHO_INT_RAW The raw interrupt status of H264_DMA_INFIFO_UDF_L2_CHO_INT. (R/WTC/SS)

H264_DMA_IN_DSCR_EMPTY_CHO_INT_RAW The raw interrupt status of H264_DMA_IN_DSCR_EMPTY_CHO_INT. (R/WTC/SS)

H264_DMA_IN_DSCR_TASK_OVF_CHO_INT_RAW The raw interrupt status of H264_DMA_IN_DSCR_TASK_OVF_CHO_INT. (R/WTC/SS)

Register 39.151. H264_DMA_IN_INT_ENA_CHO_REG (0x0508)

(reserved)																						H264_DMA_IN_DSCR_TASK_OVF_CHO_INT_ENA H264_DMA_IN_DSCR_EMPTY_CHO_INT_ENA H264_DMA_INFIFO_OVF_L2_CHO_INT_ENA H264_DMA_INFIFO_UDF_L2_CHO_INT_ENA H264_DMA_INFIFO_OVF_L1_CHO_INT_ENA H264_DMA_INFIFO_UDF_L1_CHO_INT_ENA H264_DMA_IN_DSCR_ERR_CHO_INT_ENA H264_DMA_IN_SUC_EOF_CHO_INT_ENA H264_DMA_IN_DONE_CHO_INT_ENA									
31																				10	9	8	7	6	5	4	3	2	1	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset				

- H264_DMA_IN_DONE_CHO_INT_ENA Write 1 to enable H264_DMA_IN_DONE_CHO_INT. (R/W)
- H264_DMA_IN_SUC_EOF_CHO_INT_ENA Write 1 to enable H264_DMA_IN_SUC_EOF_CHO_INT. (R/W)
- H264_DMA_IN_ERR_EOF_CHO_INT_ENA Write 1 to enable H264_DMA_IN_ERR_EOF_CHO_INT. (R/W)
- H264_DMA_IN_DSCR_ERR_CHO_INT_ENA Write 1 to enable H264_DMA_IN_DSCR_ERR_CHO_INT. (R/W)
- H264_DMA_INFIFO_OVF_L1_CHO_INT_ENA Write 1 to enable H264_DMA_INFIFO_OVF_L1_CHO_INT. (R/W)
- H264_DMA_INFIFO_UDF_L1_CHO_INT_ENA Write 1 to enable H264_DMA_INFIFO_UDF_L1_CHO_INT. (R/W)
- H264_DMA_INFIFO_OVF_L2_CHO_INT_ENA Write 1 to enable H264_DMA_INFIFO_OVF_L2_CHO_INT. (R/W)
- H264_DMA_INFIFO_UDF_L2_CHO_INT_ENA Write 1 to enable H264_DMA_INFIFO_UDF_L2_CHO_INT. (R/W)
- H264_DMA_IN_DSCR_EMPTY_CHO_INT_ENA Write 1 to enable H264_DMA_IN_DSCR_EMPTY_CHO_INT. (R/W)
- H264_DMA_IN_DSCR_TASK_OVF_CHO_INT_ENA Write 1 to enable H264_DMA_IN_DSCR_TASK_OVF_CHO_INT. (R/W)

Register 39.152. H264_DMA_IN_INT_ST_CHO_REG (0x050C)

(reserved)																						H264_DMA_IN_DSCR_TASK_OVF_CHO_INT_ST H264_DMA_IN_DSCR_EMPTY_CHO_INT_ST H264_DMA_INFIFO_UDF_L2_CHO_INT_ST H264_DMA_INFIFO_OVF_L2_CHO_INT_ST H264_DMA_INFIFO_UDF_L1_CHO_INT_ST H264_DMA_INFIFO_OVF_L1_CHO_INT_ST H264_DMA_IN_DSCR_ERR_CHO_INT_ST H264_DMA_IN_SUC_EOF_CHO_INT_ST H264_DMA_IN_DONE_CHO_INT_ST																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
31																						10		9	8	7	6	5	4	3	2	1	0																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																					
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

H264_DMA_IN_DONE_CHO_INT_ST The masked interrupt status of H264_DMA_IN_DONE_CHO_INT. (RO)

H264_DMA_IN_SUC_EOF_CHO_INT_ST The masked interrupt status of H264_DMA_IN_SUC_EOF_CHO_INT. (RO)

H264_DMA_IN_ERR_EOF_CHO_INT_ST The masked interrupt status of H264_DMA_IN_ERR_EOF_CHO_INT. (RO)

H264_DMA_IN_DSCR_ERR_CHO_INT_ST The masked interrupt status of H264_DMA_IN_DSCR_ERR_CHO_INT. (RO)

H264_DMA_INFIFO_OVF_L1_CHO_INT_ST The masked interrupt status of H264_DMA_INFIFO_OVF_L1_CHO_INT. (RO)

H264_DMA_INFIFO_UDF_L1_CHO_INT_ST The masked interrupt status of H264_DMA_INFIFO_UDF_L1_CHO_INT. (RO)

H264_DMA_INFIFO_OVF_L2_CHO_INT_ST The masked interrupt status of H264_DMA_INFIFO_OVF_L2_CHO_INT. (RO)

H264_DMA_INFIFO_UDF_L2_CHO_INT_ST The masked interrupt status of H264_DMA_INFIFO_UDF_L2_CHO_INT. (RO)

H264_DMA_IN_DSCR_EMPTY_CHO_INT_ST The masked interrupt status of H264_DMA_IN_DSCR_EMPTY_CHO_INT. (RO)

H264_DMA_IN_DSCR_TASK_OVF_CHO_INT_ST The masked interrupt status of H264_DMA_IN_DSCR_TASK_OVF_CHO_INT. (RO)

Register 39.153. H264_DMA_IN_INT_CLR_CHO_REG (0x0510)

(reserved)																						H264_DMA_IN_DSCR_TASK_OVF_CHO_INT_CLR H264_DMA_IN_DSCR_EMPTY_CHO_INT_CLR H264_DMA_INFIFO_UDF_L2_CHO_INT_CLR H264_DMA_INFIFO_OVF_L2_CHO_INT_CLR H264_DMA_INFIFO_UDF_L1_CHO_INT_CLR H264_DMA_INFIFO_OVF_L1_CHO_INT_CLR H264_DMA_IN_DSCR_ERR_CHO_INT_CLR H264_DMA_IN_SUC_EOF_CHO_INT_CLR H264_DMA_IN_DONE_CHO_INT_CLR												
31																						10		9	8	7	6	5	4	3	2	1	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset												

H264_DMA_IN_DONE_CHO_INT_CLR Write 1 to clear H264_DMA_IN_DONE_CHO_INT. (WT)

H264_DMA_IN_SUC_EOF_CHO_INT_CLR Write 1 to clear H264_DMA_IN_SUC_EOF_CHO_INT. (WT)

H264_DMA_IN_ERR_EOF_CHO_INT_CLR Write 1 to clear H264_DMA_IN_ERR_EOF_CHO_INT. (WT)

H264_DMA_IN_DSCR_ERR_CHO_INT_CLR Write 1 to clear H264_DMA_IN_DSCR_ERR_CHO_INT. (WT)

H264_DMA_INFIFO_OVF_L1_CHO_INT_CLR Write 1 to clear H264_DMA_INFIFO_OVF_L1_CHO_INT. (WT)

H264_DMA_INFIFO_UDF_L1_CHO_INT_CLR Write 1 to clear H264_DMA_INFIFO_UDF_L1_CHO_INT. (WT)

H264_DMA_INFIFO_OVF_L2_CHO_INT_CLR Write 1 to clear H264_DMA_INFIFO_OVF_L2_CHO_INT. (WT)

H264_DMA_INFIFO_UDF_L2_CHO_INT_CLR Write 1 to clear H264_DMA_INFIFO_UDF_L2_CHO_INT. (WT)

H264_DMA_IN_DSCR_EMPTY_CHO_INT_CLR Write 1 to clear H264_DMA_IN_DSCR_EMPTY_CHO_INT. (WT)

H264_DMA_IN_DSCR_TASK_OVF_CHO_INT_CLR Write 1 to clear H264_DMA_IN_DSCR_TASK_OVF_CHO_INT. (WT)

Register 39.154. H264_DMA_IN_INT_RAW_CH1_REG (0x0604)

(reserved)																						H264_DMA_IN_DSCR_TASK_OVF_CH1_INT_RAW H264_DMA_IN_DSCR_EMPTY_CH1_INT_RAW H264_DMA_INFIFO_OVF_L2_CH1_INT_RAW H264_DMA_INFIFO_OVF_L1_CH1_INT_RAW H264_DMA_IN_DSCR_ERR_CH1_INT_RAW H264_DMA_IN_SUC_EOF_CH1_INT_RAW H264_DMA_IN_DONE_CH1_INT_RAW												
31																						10	9	8	7	6	5	4	3	2	1	0		
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset												

H264_DMA_IN_DONE_CH1_INT_RAW The raw interrupt status of H264_DMA_IN_DONE_CH1_INT. (R/WTC/SS)

H264_DMA_IN_SUC_EOF_CH1_INT_RAW The raw interrupt status of H264_DMA_IN_SUC_EOF_CH1_INT. (R/WTC/SS)

H264_DMA_IN_ERR_EOF_CH1_INT_RAW The raw interrupt status of H264_DMA_IN_ERR_EOF_CH1_INT. (R/WTC/SS)

H264_DMA_IN_DSCR_ERR_CH1_INT_RAW The raw interrupt status of H264_DMA_IN_DSCR_ERR_CH1_INT. (R/WTC/SS)

H264_DMA_IN_FIFO_OVF_L1_CH1_INT_RAW The raw interrupt status of H264_DMA_IN_FIFO_OVF_L1_CH1_INT. (R/WTC/SS)

H264_DMA_IN_FIFO_UDF_L1_CH1_INT_RAW The raw interrupt status of H264_DMA_IN_FIFO_UDF_L1_CH1_INT. (R/WTC/SS)

H264_DMA_IN_FIFO_OVF_L2_CH1_INT_RAW The raw interrupt status of H264_DMA_IN_FIFO_OVF_L2_CH1_INT. (R/WTC/SS)

H264_DMA_IN_FIFO_UDF_L2_CH1_INT_RAW The raw interrupt status of H264_DMA_IN_FIFO_UDF_L2_CH1_INT. (R/WTC/SS)

H264_DMA_IN_DSCR_EMPTY_CH1_INT_RAW The raw interrupt status of H264_DMA_IN_DSCR_EMPTY_CH1_INT. (R/WTC/SS)

H264_DMA_IN_DSCR_TASK_OVF_CH1_INT_RAW The raw interrupt status of H264_DMA_IN_DSCR_TASK_OVF_CH1_INT. (R/WTC/SS)

Register 39.155. H264_DMA_IN_INT_ENA_CH1_REG (0x0608)

(reserved)																						H264_DMA_IN_DSCR_TASK_OVF_CH1_INT_ENA H264_DMA_IN_DSCR_EMPTY_CH1_INT_ENA H264_DMA_IN_FIFO_UDF_L2_CH1_INT_ENA H264_DMA_IN_FIFO_OVF_L2_CH1_INT_ENA H264_DMA_IN_FIFO_UDF_L1_CH1_INT_ENA H264_DMA_IN_DSCR_ERR_CH1_INT_ENA H264_DMA_IN_SUC_EOF_CH1_INT_ENA H264_DMA_IN_DONE_CH1_INT_ENA																		
31																						10	9	8	7	6	5	4	3	2	1	0								
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset									

H264_DMA_IN_DONE_CH1_INT_ENA Write 1 to enable H264_DMA_IN_DONE_CH1_INT. (R/W)

H264_DMA_IN_SUC_EOF_CH1_INT_ENA Write 1 to enable H264_DMA_IN_SUC_EOF_CH1_INT.
(R/W)

H264_DMA_IN_ERR_EOF_CH1_INT_ENA Write 1 to enable H264_DMA_IN_ERR_EOF_CH1_INT.
(R/W)

H264_DMA_IN_DSCR_ERR_CH1_INT_ENA Write 1 to enable H264_DMA_IN_DSCR_ERR_CH1_INT.
(R/W)

H264_DMA_IN_FIFO_OVF_L1_CH1_INT_ENA Write 1 to enable H264_DMA_IN_FIFO_OVF_L1_CH1_INT.
(R/W)

H264_DMA_IN_FIFO_UDF_L1_CH1_INT_ENA Write 1 to enable H264_DMA_IN_FIFO_UDF_L1_CH1_INT.
(R/W)

H264_DMA_IN_FIFO_OVF_L2_CH1_INT_ENA Write 1 to enable H264_DMA_IN_FIFO_OVF_L2_CH1_INT.
(R/W)

H264_DMA_IN_FIFO_UDF_L2_CH1_INT_ENA Write 1 to enable H264_DMA_IN_FIFO_UDF_L2_CH1_INT.
(R/W)

H264_DMA_IN_DSCR_EMPTY_CH1_INT_ENA Write 1 to enable H264_DMA_IN_DSCR_EMPTY_CH1_INT.
(R/W)

H264_DMA_IN_DSCR_TASK_OVF_CH1_INT_ENA Write 1 to enable H264_DMA_IN_DSCR_TASK_OVF_CH1_INT. (R/W)

Register 39.156. H264_DMA_IN_INT_ST_CH1_REG (0x060C)

(reserved)										H264_DMA_IN_DSCR_TASK_OVF_CH1_INT_ST H264_DMA_IN_DSCR_EMPTY_CH1_INT_ST H264_DMA_INFIFO_OVF_L2_CH1_INT_ST H264_DMA_INFIFO_UDF_L2_CH1_INT_ST H264_DMA_INFIFO_OVF_L1_CH1_INT_ST H264_DMA_INFIFO_UDF_L1_CH1_INT_ST H264_DMA_IN_DSCR_ERR_CH1_INT_ST H264_DMA_IN_ERR_EOF_CH1_INT_ST H264_DMA_IN_SUC_EOF_CH1_INT_ST H264_DMA_IN_DONE_CH1_INT_ST											
31										10	9	8	7	6	5	4	3	2	1	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset

H264_DMA_IN_DONE_CH1_INT_ST The masked interrupt status of H264_DMA_IN_DONE_CH1_INT. (RO)

H264_DMA_IN_SUC_EOF_CH1_INT_ST The masked interrupt status of H264_DMA_IN_SUC_EOF_CH1_INT. (RO)

H264_DMA_IN_ERR_EOF_CH1_INT_ST The masked interrupt status of H264_DMA_IN_ERR_EOF_CH1_INT. (RO)

H264_DMA_IN_DSCR_ERR_CH1_INT_ST The masked interrupt status of H264_DMA_IN_DSCR_ERR_CH1_INT. (RO)

H264_DMA_INFIFO_OVF_L1_CH1_INT_ST The masked interrupt status of H264_DMA_INFIFO_OVF_L1_CH1_INT. (RO)

H264_DMA_INFIFO_UDF_L1_CH1_INT_ST The masked interrupt status of H264_DMA_INFIFO_UDF_L1_CH1_INT. (RO)

H264_DMA_INFIFO_OVF_L2_CH1_INT_ST The masked interrupt status of H264_DMA_INFIFO_OVF_L2_CH1_INT. (RO)

H264_DMA_INFIFO_UDF_L2_CH1_INT_ST The masked interrupt status of H264_DMA_INFIFO_UDF_L2_CH1_INT. (RO)

H264_DMA_IN_DSCR_EMPTY_CH1_INT_ST The masked interrupt status of H264_DMA_IN_DSCR_EMPTY_CH1_INT. (RO)

H264_DMA_IN_DSCR_TASK_OVF_CH1_INT_ST The masked interrupt status of H264_DMA_IN_DSCR_TASK_OVF_CH1_INT. (RO)

Register 39.157. H264_DMA_IN_INT_CLR_CH1_REG (0x0610)

(reserved)																						H264_DMA_IN_DSCR_TASK_OVF_CH1_INT_CLR H264_DMA_IN_DSCR_EMPTY_CH1_INT_CLR H264_DMA_INFIFO_UDF_L2_CH1_INT_CLR H264_DMA_INFIFO_OVF_L2_CH1_INT_CLR H264_DMA_INFIFO_UDF_L1_CH1_INT_CLR H264_DMA_INFIFO_OVF_L1_CH1_INT_CLR H264_DMA_IN_DSCR_ERR_CH1_INT_CLR H264_DMA_IN_SUC_EOF_CH1_INT_CLR H264_DMA_IN_DONE_CH1_INT_CLR													
31																						10		9	8	7	6	5	4	3	2	1	0		
0 0																						0	0	0	0	0	0	0	0	0	0	0	0	0	Reset

H264_DMA_IN_DONE_CH1_INT_CLR Write 1 to clear H264_DMA_IN_DONE_CH1_INT. (WT)

H264_DMA_IN_SUC_EOF_CH1_INT_CLR Write 1 to clear H264_DMA_IN_SUC_EOF_CH1_INT. (WT)

H264_DMA_IN_ERR_EOF_CH1_INT_CLR Write 1 to clear H264_DMA_IN_ERR_EOF_CH1_INT. (WT)

H264_DMA_IN_DSCR_ERR_CH1_INT_CLR Write 1 to clear H264_DMA_IN_DSCR_ERR_CH1_INT.
(WT)

H264_DMA_INFIFO_OVF_L1_CH1_INT_CLR Write 1 to clear H264_DMA_INFIFO_OVF_L1_CH1_INT.
(WT)

H264_DMA_INFIFO_UDF_L1_CH1_INT_CLR Write 1 to clear H264_DMA_INFIFO_UDF_L1_CH1_INT.
(WT)

H264_DMA_INFIFO_OVF_L2_CH1_INT_CLR Write 1 to clear H264_DMA_INFIFO_OVF_L2_CH1_INT.
(WT)

H264_DMA_INFIFO_UDF_L2_CH1_INT_CLR Write 1 to clear H264_DMA_INFIFO_UDF_L2_CH1_INT.
(WT)

H264_DMA_IN_DSCR_EMPTY_CH1_INT_CLR Write 1 to clear H264_DMA_IN_DSCR_EMPTY_CH1_INT.
(WT)

H264_DMA_IN_DSCR_TASK_OVF_CH1_INT_CLR Write 1 to clear H264_DMA_IN_DSCR_TASK_OVF_CH1_INT.
(WT)

Register 39.158. H264_DMA_IN_INT_RAW_CH2_REG (0x0704)

(reserved)																						H264_DMA_IN_DSCR_TASK_OVF_CH2_INT_RAW H264_DMA_IN_DSCR_EMPTY_CH2_INT_RAW H264_DMA_INFIFO_UDF_L2_CH2_INT_RAW H264_DMA_INFIFO_OVF_L2_CH2_INT_RAW H264_DMA_INFIFO_UDF_L1_CH2_INT_RAW H264_DMA_INFIFO_OVF_L1_CH2_INT_RAW H264_DMA_IN_ERR_EOF_CH2_INT_RAW H264_DMA_IN_DSCR_ERR_CH2_INT_RAW H264_DMA_IN_SUC_EOF_CH2_INT_RAW H264_DMA_IN_DONE_CH2_INT_RAW										
31										10										9	8	7	6	5	4	3	2	1	0			
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset									

H264_DMA_IN_DONE_CH2_INT_RAW The raw interrupt status of H264_DMA_IN_DONE_CH2_INT. (R/WTC/SS)

H264_DMA_IN_SUC_EOF_CH2_INT_RAW The raw interrupt status of H264_DMA_IN_SUC_EOF_CH2_INT. (R/WTC/SS)

H264_DMA_IN_ERR_EOF_CH2_INT_RAW The raw interrupt status of H264_DMA_IN_ERR_EOF_CH2_INT. (R/WTC/SS)

H264_DMA_IN_DSCR_ERR_CH2_INT_RAW The raw interrupt status of H264_DMA_IN_DSCR_ERR_CH2_INT. (R/WTC/SS)

H264_DMA_INFIFO_OVF_L1_CH2_INT_RAW The raw interrupt status of H264_DMA_INFIFO_OVF_L1_CH2_INT. (R/WTC/SS)

H264_DMA_INFIFO_UDF_L1_CH2_INT_RAW The raw interrupt status of H264_DMA_INFIFO_UDF_L1_CH2_INT. (R/WTC/SS)

H264_DMA_INFIFO_OVF_L2_CH2_INT_RAW The raw interrupt status of H264_DMA_INFIFO_OVF_L2_CH2_INT. (R/WTC/SS)

H264_DMA_INFIFO_UDF_L2_CH2_INT_RAW The raw interrupt status of H264_DMA_INFIFO_UDF_L2_CH2_INT. (R/WTC/SS)

H264_DMA_IN_DSCR_EMPTY_CH2_INT_RAW The raw interrupt status of H264_DMA_IN_DSCR_EMPTY_CH2_INT. (R/WTC/SS)

H264_DMA_IN_DSCR_TASK_OVF_CH2_INT_RAW The raw interrupt status of H264_DMA_IN_DSCR_TASK_OVF_CH2_INT. (R/WTC/SS)

Register 39.159. H264_DMA_IN_INT_ENA_CH2_REG (0x0708)

(reserved)										H264_DMA_IN_DSCR_TASK_OVF_CH2_INT_ENA H264_DMA_IN_DSCR_EMPTY_CH2_INT_ENA H264_DMA_INFIFO_UDF_L2_CH2_INT_ENA H264_DMA_INFIFO_OVF_L2_CH2_INT_ENA H264_DMA_INFIFO_UDF_L1_CH2_INT_ENA H264_DMA_INFIFO_OVF_L1_CH2_INT_ENA H264_DMA_IN_DSCR_ERR_CH2_INT_ENA H264_DMA_IN_ERR_EOF_CH2_INT_ENA H264_DMA_IN_SUC_EOF_CH2_INT_ENA H264_DMA_IN_DONE_CH2_INT_ENA											
31										10	9	8	7	6	5	4	3	2	1	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset

H264_DMA_IN_DONE_CH2_INT_ENA Write 1 to enable H264_DMA_IN_DONE_CH2_INT. (R/W)

H264_DMA_IN_SUC_EOF_CH2_INT_ENA Write 1 to enable H264_DMA_IN_SUC_EOF_CH2_INT.
(R/W)

H264_DMA_IN_ERR_EOF_CH2_INT_ENA Write 1 to enable H264_DMA_IN_ERR_EOF_CH2_INT.
(R/W)

H264_DMA_IN_DSCR_ERR_CH2_INT_ENA Write 1 to enable H264_DMA_IN_DSCR_ERR_CH2_INT.
(R/W)

H264_DMA_INFIFO_OVF_L1_CH2_INT_ENA Write 1 to enable H264_DMA_INFIFO_OVF_L1_CH2_INT.
(R/W)

H264_DMA_INFIFO_UDF_L1_CH2_INT_ENA Write 1 to enable H264_DMA_INFIFO_UDF_L1_CH2_INT.
(R/W)

H264_DMA_INFIFO_OVF_L2_CH2_INT_ENA Write 1 to enable H264_DMA_INFIFO_OVF_L2_CH2_INT.
(R/W)

H264_DMA_INFIFO_UDF_L2_CH2_INT_ENA Write 1 to enable H264_DMA_INFIFO_UDF_L2_CH2_INT.
(R/W)

H264_DMA_IN_DSCR_EMPTY_CH2_INT_ENA Write 1 to enable H264_DMA_IN_DSCR_EMPTY_CH2_INT.
(R/W)

H264_DMA_IN_DSCR_TASK_OVF_CH2_INT_ENA Write 1 to enable H264_DMA_IN_DSCR_TASK_OVF_CH2_INT. (R/W)

Register 39.160. H264_DMA_IN_INT_ST_CH2_REG (0x070C)

(reserved)										H264_DMA_IN_DSCR_TASK_OVF_CH2_INT_ST H264_DMA_IN_DSCR_EMPTY_CH2_INT_ST H264_DMA_IN_FIFO_UDF_L2_CH2_INT_ST H264_DMA_IN_FIFO_OVF_L2_CH2_INT_ST H264_DMA_IN_FIFO_UDF_L1_CH2_INT_ST H264_DMA_IN_FIFO_OVF_L1_CH2_INT_ST H264_DMA_IN_DSCR_ERR_CH2_INT_ST H264_DMA_IN_SUC_EOF_CH2_INT_ST H264_DMA_IN_DONE_CH2_INT_ST											
31										10	9	8	7	6	5	4	3	2	1	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset

H264_DMA_IN_DONE_CH2_INT_ST The masked interrupt status of H264_DMA_IN_DONE_CH2_INT. (RO)

H264_DMA_IN_SUC_EOF_CH2_INT_ST The masked interrupt status of H264_DMA_IN_SUC_EOF_CH2_INT. (RO)

H264_DMA_IN_ERR_EOF_CH2_INT_ST The masked interrupt status of H264_DMA_IN_ERR_EOF_CH2_INT. (RO)

H264_DMA_IN_DSCR_ERR_CH2_INT_ST The masked interrupt status of H264_DMA_IN_DSCR_ERR_CH2_INT. (RO)

H264_DMA_IN_FIFO_OVF_L1_CH2_INT_ST The masked interrupt status of H264_DMA_IN_FIFO_OVF_L1_CH2_INT. (RO)

H264_DMA_IN_FIFO_UDF_L1_CH2_INT_ST The masked interrupt status of H264_DMA_IN_FIFO_UDF_L1_CH2_INT. (RO)

H264_DMA_IN_FIFO_OVF_L2_CH2_INT_ST The masked interrupt status of H264_DMA_IN_FIFO_OVF_L2_CH2_INT. (RO)

H264_DMA_IN_FIFO_UDF_L2_CH2_INT_ST The masked interrupt status of H264_DMA_IN_FIFO_UDF_L2_CH2_INT. (RO)

H264_DMA_IN_DSCR_EMPTY_CH2_INT_ST The masked interrupt status of H264_DMA_IN_DSCR_EMPTY_CH2_INT. (RO)

H264_DMA_IN_DSCR_TASK_OVF_CH2_INT_ST The masked interrupt status of H264_DMA_IN_DSCR_TASK_OVF_CH2_INT. (RO)

Register 39.161. H264_DMA_IN_INT_CLR_CH2_REG (0x0710)

[illegible]

H264_DMA_IN_DONE_CH2_INT_CLR Write 1 to clear H264_DMA_IN_DONE_CH2_INT. (WT)

H264_DMA_IN_SUC_EOF_CH2_INT_CLR Write 1 to clear H264_DMA_IN_SUC_EOF_CH2_INT.
(WT)

H264_DMA_IN_ERR_EOF_CH2_INT_CLR Write 1 to clear H264_DMA_IN_ERR_EOF_CH2_INT. (WT)

H264_DMA_IN_DSCR_ERR_CH2_INT_CLR Write 1 to clear H264_DMA_IN_DSCR_ERR_CH2_INT.
(WT)

H264_DMA_INFIFO_OVF_L1_CH2_INT_CLR Write 1 to clear H264_DMA_INFIFO_OVF_L1_CH2_INT.
(WT)

H264_DMA_INFIFO_UDF_L1_CH2_INT_CLR Write 1 to clear H264_DMA_INFIFO_UDF_L1_CH2_INT.
(WT)

H264_DMA_INFIFO_OVF_L2_CH2_INT_CLR Write 1 to clear H264_DMA_INFIFO_OVF_L2_CH2_INT.
(WT)

H264_DMA_INFIFO_UDF_L2_CH2_INT_CLR Write 1 to clear H264_DMA_INFIFO_UDF_L2_CH2_INT.
(WT)

H264_DMA_IN_DSCR_EMPTY_CH2_INT_CLR Write 1 to clear H264_DMA_IN_DSCR_EMPTY_CH2_INT.
(WT)

H264_DMA_IN_DSCR_TASK_OVF_CH2_INT_CLR Write 1 to clear H264 DMA IN DSCR TASK OVF CH2 INT. (WT)

Register 39.162. H264_DMA_IN_INT_RAW_CH3_REG (0x0804)

(reserved)										H264_DMA_IN_DSCR_TASK_OVF_CH3_INT_RAW H264_DMA_IN_DSCR_EMPTY_CH3_INT_RAW H264_DMA_INFIFO_OVF_L2_CH3_INT_RAW H264_DMA_INFIFO_UDF_L2_CH3_INT_RAW H264_DMA_INFIFO_OVF_L1_CH3_INT_RAW H264_DMA_INFIFO_UDF_L1_CH3_INT_RAW H264_DMA_IN_DSCR_ERR_CH3_INT_RAW H264_DMA_IN_ERR_EOF_CH3_INT_RAW H264_DMA_IN_SUC_EOF_CH3_INT_RAW H264_DMA_IN_DONE_CH3_INT_RAW												
31											10	9	8	7	6	5	4	3	2	1	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset

H264_DMA_IN_DONE_CH3_INT_RAW The raw interrupt status of H264_DMA_IN_DONE_CH3_INT. (R/WTC/SS)

H264_DMA_IN_SUC_EOF_CH3_INT_RAW The raw interrupt status of H264_DMA_IN_SUC_EOF_CH3_INT. (R/WTC/SS)

H264_DMA_IN_ERR_EOF_CH3_INT_RAW The raw interrupt status of H264_DMA_IN_ERR_EOF_CH3_INT. (R/WTC/SS)

H264_DMA_IN_DSCR_ERR_CH3_INT_RAW The raw interrupt status of H264_DMA_IN_DSCR_ERR_CH3_INT. channel 3. (R/WTC/SS)

H264_DMA_INFIFO_OVF_L1_CH3_INT_RAW The raw interrupt status of H264_DMA_INFIFO_OVF_L1_CH3_INT. (R/WTC/SS)

H264_DMA_INFIFO_UDF_L1_CH3_INT_RAW The raw interrupt status of H264_DMA_INFIFO_UDF_L1_CH3_INT. (R/WTC/SS)

H264_DMA_INFIFO_OVF_L2_CH3_INT_RAW The raw interrupt status of H264_DMA_INFIFO_OVF_L2_CH3_INT. (R/WTC/SS)

H264_DMA_INFIFO_UDF_L2_CH3_INT_RAW The raw interrupt status of H264_DMA_INFIFO_UDF_L2_CH3_INT. (R/WTC/SS)

H264_DMA_IN_DSCR_EMPTY_CH3_INT_RAW The raw interrupt status of H264_DMA_IN_DSCR_EMPTY_CH3_INT. (R/WTC/SS)

H264_DMA_IN_DSCR_TASK_OVF_CH3_INT_RAW The raw interrupt status of H264_DMA_IN_DSCR_TASK_OVF_CH3_INT. (R/WTC/SS)

Register 39.163. H264_DMA_IN_INT_ENA_CH3_REG (0x0808)

(reserved)																						H264_DMA_IN_DSCR_TASK_OVF_CH3_INT_ENA H264_DMA_IN_DSCR_EMPTY_CH3_INT_ENA H264_DMA_INFIFO_OVF_L2_CH3_INT_ENA H264_DMA_INFIFO_UDF_L2_CH3_INT_ENA H264_DMA_INFIFO_OVF_L1_CH3_INT_ENA H264_DMA_INFIFO_UDF_L1_CH3_INT_ENA H264_DMA_IN_DSCR_ERR_CH3_INT_ENA H264_DMA_IN_ERR_EOF_CH3_INT_ENA H264_DMA_IN_SUC_EOF_CH3_INT_ENA H264_DMA_IN_DONE_CH3_INT_ENA												
31																						10		9	8	7	6	5	4	3	2	1	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset												

H264_DMA_IN_DONE_CH3_INT_ENA Write 1 to enable H264_DMA_IN_DONE_CH3_INT. (R/W)

H264_DMA_IN_SUC_EOF_CH3_INT_ENA Write 1 to enable H264_DMA_IN_SUC_EOF_CH3_INT.
(R/W)

H264_DMA_IN_ERR_EOF_CH3_INT_ENA Write 1 to enable H264_DMA_IN_ERR_EOF_CH3_INT.
(R/W)

H264_DMA_IN_DSCR_ERR_CH3_INT_ENA Write 1 to enable H264_DMA_IN_DSCR_ERR_CH3_INT.
(R/W)

H264_DMA_INFIFO_OVF_L1_CH3_INT_ENA Write 1 to enable H264_DMA_INFIFO_OVF_L1_CH3_INT.
(R/W)

H264_DMA_INFIFO_UDF_L1_CH3_INT_ENA Write 1 to enable H264_DMA_INFIFO_UDF_L1_CH3_INT.
(R/W)

H264_DMA_INFIFO_OVF_L2_CH3_INT_ENA Write 1 to enable H264_DMA_INFIFO_OVF_L2_CH3_INT.
(R/W)

H264_DMA_INFIFO_UDF_L2_CH3_INT_ENA Write 1 to enable H264_DMA_INFIFO_UDF_L2_CH3_INT.
(R/W)

H264_DMA_IN_DSCR_EMPTY_CH3_INT_ENA Write 1 to enable H264_DMA_IN_DSCR_EMPTY_CH3_INT.
(R/W)

H264_DMA_IN_DSCR_TASK_OVF_CH3_INT_ENA Write 1 to enable H264_DMA_IN_DSCR_TASK_OVF_CH3_INT. (R/W)

Register 39.164. H264_DMA_IN_INT_ST_CH3_REG (0x080C)

(reserved)																						H264_DMA_IN_DSCR_TASK_OVF_CH3_INT_ST H264_DMA_IN_DSCR_EMPTY_CH3_INT_ST H264_DMA_INFIFO_UDF_L2_CH3_INT_ST H264_DMA_INFIFO_OVF_L2_CH3_INT_ST H264_DMA_INFIFO_UDF_L1_CH3_INT_ST H264_DMA_INFIFO_OVF_L1_CH3_INT_ST H264_DMA_IN_DSCR_ERR_CH3_INT_ST H264_DMA_IN_ERR_EOF_CH3_INT_ST H264_DMA_IN_SUC_EOF_CH3_INT_ST H264_DMA_IN_DONE_CH3_INT_ST													
31																						10		9	8	7	6	5	4	3	2	1	0		
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset												

H264_DMA_IN_DONE_CH3_INT_ST The masked interrupt status of H264_DMA_IN_DONE_CH3_INT. (RO)

H264_DMA_IN_SUC_EOF_CH3_INT_ST The masked interrupt status of H264_DMA_IN_SUC_EOF_CH3_INT. (RO)

H264_DMA_IN_ERR_EOF_CH3_INT_ST The masked interrupt status of H264_DMA_IN_ERR_EOF_CH3_INT. (RO)

H264_DMA_IN_DSCR_ERR_CH3_INT_ST The masked interrupt status of H264_DMA_IN_DSCR_ERR_CH3_INT. (RO)

H264_DMA_INFIFO_OVF_L1_CH3_INT_ST The masked interrupt status of H264_DMA_INFIFO_OVF_L1_CH3_INT. (RO)

H264_DMA_INFIFO_UDF_L1_CH3_INT_ST The masked interrupt status of H264_DMA_INFIFO_UDF_L1_CH3_INT. (RO)

H264_DMA_INFIFO_OVF_L2_CH3_INT_ST The masked interrupt status of H264_DMA_INFIFO_OVF_L2_CH3_INT. (RO)

H264_DMA_INFIFO_UDF_L2_CH3_INT_ST The masked interrupt status of H264_DMA_INFIFO_UDF_L2_CH3_INT. (RO)

H264_DMA_IN_DSCR_EMPTY_CH3_INT_ST The masked interrupt status of H264_DMA_IN_DSCR_EMPTY_CH3_INT. (RO)

H264_DMA_IN_DSCR_TASK_OVF_CH3_INT_ST The masked interrupt status of H264_DMA_IN_DSCR_TASK_OVF_CH3_INT. (RO)

Register 39.165. H264_DMA_IN_INT_CLR_CH3_REG (0x0810)

(reserved)																						H264_DMA_IN_DSCR_TASK_OVF_CH3_INT_CLR H264_DMA_IN_DSCR_EMPTY_CH3_INT_CLR H264_DMA_INFIFO_UDF_L2_CH3_INT_CLR H264_DMA_INFIFO_OVF_L2_CH3_INT_CLR H264_DMA_INFIFO_UDF_L1_CH3_INT_CLR H264_DMA_INFIFO_OVF_L1_CH3_INT_CLR H264_DMA_IN_DSCR_ERR_CH3_INT_CLR H264_DMA_IN_ERR_EOF_CH3_INT_CLR H264_DMA_IN_SUC_EOF_CH3_INT_CLR H264_DMA_IN_DONE_CH3_INT_CLR											
31																						10		9	8	7	6	5	4	3	2	1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset											

H264_DMA_IN_DONE_CH3_INT_CLR Write 1 to clear H264_DMA_IN_DONE_CH3_INT. (WT)

H264_DMA_IN_SUC_EOF_CH3_INT_CLR Write 1 to clear H264_DMA_IN_SUC_EOF_CH3_INT. (WT)

H264_DMA_IN_ERR_EOF_CH3_INT_CLR Write 1 to clear H264_DMA_IN_ERR_EOF_CH3_INT. (WT)

H264_DMA_IN_DSCR_ERR_CH3_INT_CLR Write 1 to clear H264_DMA_IN_DSCR_ERR_CH3_INT.
(WT)

H264_DMA_INFIFO_OVF_L1_CH3_INT_CLR Write 1 to clear H264_DMA_INFIFO_OVF_L1_CH3_INT.
(WT)

H264_DMA_INFIFO_UDF_L1_CH3_INT_CLR Write 1 to clear H264_DMA_INFIFO_UDF_L1_CH3_INT.
(WT)

H264_DMA_INFIFO_OVF_L2_CH3_INT_CLR Write 1 to clear H264_DMA_INFIFO_OVF_L2_CH3_INT.
(WT)

H264_DMA_INFIFO_UDF_L2_CH3_INT_CLR Write 1 to clear H264_DMA_INFIFO_UDF_L2_CH3_INT.
(WT)

H264_DMA_IN_DSCR_EMPTY_CH3_INT_CLR Write 1 to clear H264_DMA_IN_DSCR_EMPTY_CH3_INT.
(WT)

H264_DMA_IN_DSCR_TASK_OVF_CH3_INT_CLR Write 1 to clear H264_DMA_IN_DSCR_TASK_OVF_CH3_INT.
(WT)

Register 39.166. H264_DMA_IN_INT_RAW_CH4_REG (0x0904)

(reserved)										H264_DMA_IN_DSCR_TASK_OVF_CH4_INT_RAW H264_DMA_IN_DSCR_EMPTY_CH4_INT_RAW H264_DMA_INFIFO_OVF_L2_CH4_INT_RAW H264_DMA_INFIFO_UDF_L2_CH4_INT_RAW H264_DMA_INFIFO_OVF_L1_CH4_INT_RAW H264_DMA_INFIFO_UDF_L1_CH4_INT_RAW H264_DMA_IN_DSCR_ERR_CH4_INT_RAW H264_DMA_IN_ERR_EOF_CH4_INT_RAW H264_DMA_IN_SUC_EOF_CH4_INT_RAW H264_DMA_IN_DONE_CH4_INT_RAW												
31											10	9	8	7	6	5	4	3	2	1	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset

H264_DMA_IN_DONE_CH4_INT_RAW The raw interrupt status of H264_DMA_IN_DONE_CH4_INT. (R/WTC/SS)

H264_DMA_IN_SUC_EOF_CH4_INT_RAW The raw interrupt status of H264_DMA_IN_SUC_EOF_CH4_INT. (R/WTC/SS)

H264_DMA_IN_ERR_EOF_CH4_INT_RAW The raw interrupt status of H264_DMA_IN_ERR_EOF_CH4_INT. (R/WTC/SS)

H264_DMA_IN_DSCR_ERR_CH4_INT_RAW The raw interrupt status of H264_DMA_IN_DSCR_ERR_CH4_INT. (R/WTC/SS)

H264_DMA_INFIFO_OVF_L1_CH4_INT_RAW The raw interrupt status of H264_DMA_INFIFO_OVF_L1_CH4_INT. (R/WTC/SS)

H264_DMA_INFIFO_UDF_L1_CH4_INT_RAW The raw interrupt status of H264_DMA_INFIFO_UDF_L1_CH4_INT. (R/WTC/SS)

H264_DMA_INFIFO_OVF_L2_CH4_INT_RAW The raw interrupt status of H264_DMA_INFIFO_OVF_L2_CH4_INT. (R/WTC/SS)

H264_DMA_INFIFO_UDF_L2_CH4_INT_RAW The raw interrupt status of H264_DMA_INFIFO_UDF_L2_CH4_INT. (R/WTC/SS)

H264_DMA_IN_DSCR_EMPTY_CH4_INT_RAW The raw interrupt status of H264_DMA_IN_DSCR_EMPTY_CH4_INT. (R/WTC/SS)

H264_DMA_IN_DSCR_TASK_OVF_CH4_INT_RAW The raw interrupt status of H264_DMA_IN_DSCR_TASK_OVF_CH4_INT. (R/WTC/SS)

Register 39.167. H264_DMA_IN_INT_ENA_CH4_REG (0x0908)

(reserved)										H264_DMA_IN_DSCR_TASK_OVF_CH4_INT_ENA H264_DMA_IN_DSCR_EMPTY_CH4_INT_ENA H264_DMA_IN_FIFO_OVF_L2_CH4_INT_ENA H264_DMA_IN_FIFO_UDF_L2_CH4_INT_ENA H264_DMA_IN_DSCR_ERR_CH4_INT_ENA H264_DMA_IN_SUC_EOF_CH4_INT_ENA H264_DMA_IN_DONE_CH4_INT_ENA											
31										10	9	8	7	6	5	4	3	2	1	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset

H264_DMA_IN_DONE_CH4_INT_ENA Write 1 to enable H264_DMA_IN_DONE_CH4_INT. (R/W)

H264_DMA_IN_SUC_EOF_CH4_INT_ENA Write 1 to enable H264_DMA_IN_SUC_EOF_CH4_INT.
(R/W)

H264_DMA_IN_ERR_EOF_CH4_INT_ENA Write 1 to enable H264_DMA_IN_ERR_EOF_CH4_INT.
(R/W)

H264_DMA_IN_DSCR_ERR_CH4_INT_ENA Write 1 to enable H264_DMA_IN_DSCR_ERR_CH4_INT.
(R/W)

H264_DMA_IN_FIFO_OVF_L1_CH4_INT_ENA Write 1 to enable H264_DMA_IN_FIFO_OVF_L1_CH4_INT.
(R/W)

H264_DMA_IN_FIFO_UDF_L1_CH4_INT_ENA Write 1 to enable H264_DMA_IN_FIFO_UDF_L1_CH4_INT.
(R/W)

H264_DMA_IN_FIFO_OVF_L2_CH4_INT_ENA Write 1 to enable H264_DMA_IN_FIFO_OVF_L2_CH4_INT.
(R/W)

H264_DMA_IN_FIFO_UDF_L2_CH4_INT_ENA Write 1 to enable H264_DMA_IN_FIFO_UDF_L2_CH4_INT.
(R/W)

H264_DMA_IN_DSCR_EMPTY_CH4_INT_ENA Write 1 to enable H264_DMA_IN_DSCR_EMPTY_CH4_INT.
(R/W)

H264_DMA_IN_DSCR_TASK_OVF_CH4_INT_ENA Write 1 to enable H264_DMA_IN_DSCR_TASK_OVF_CH4_INT. (R/W)

Register 39.168. H264_DMA_IN_INT_ST_CH4_REG (0x090C)

(reserved)										H264_DMA_IN_DSCR_TASK_OVF_CH4_INT_ST H264_DMA_IN_DSCR_EMPTY_CH4_INT_ST H264_DMA_INFIFO_UDF_L2_CH4_INT_ST H264_DMA_INFIFO_OVF_L2_CH4_INT_ST H264_DMA_INFIFO_UDF_L1_CH4_INT_ST H264_DMA_INFIFO_OVF_L1_CH4_INT_ST H264_DMA_IN_DSCR_ERR_CH4_INT_ST H264_DMA_IN_SUC_EOF_CH4_INT_ST H264_DMA_IN_DONE_CH4_INT_ST											
31										10	9	8	7	6	5	4	3	2	1	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset

H264_DMA_IN_DONE_CH4_INT_ST The masked interrupt status of H264_DMA_IN_DONE_CH4_INT. (RO)

H264_DMA_IN_SUC_EOF_CH4_INT_ST The masked interrupt status of H264_DMA_IN_SUC_EOF_CH4_INT. (RO)

H264_DMA_IN_ERR_EOF_CH4_INT_ST The masked interrupt status of H264_DMA_IN_ERR_EOF_CH4_INT. (RO)

H264_DMA_IN_DSCR_ERR_CH4_INT_ST The masked interrupt status of H264_DMA_IN_DSCR_ERR_CH4_INT. (RO)

H264_DMA_INFIFO_OVF_L1_CH4_INT_ST The masked interrupt status of H264_DMA_INFIFO_OVF_L1_CH4_INT. (RO)

H264_DMA_INFIFO_UDF_L1_CH4_INT_ST The masked interrupt status of H264_DMA_INFIFO_UDF_L1_CH4_INT. (RO)

H264_DMA_INFIFO_OVF_L2_CH4_INT_ST The masked interrupt status of H264_DMA_INFIFO_OVF_L2_CH4_INT. (RO)

H264_DMA_INFIFO_UDF_L2_CH4_INT_ST The masked interrupt status of H264_DMA_INFIFO_UDF_L2_CH4_INT. (RO)

H264_DMA_IN_DSCR_EMPTY_CH4_INT_ST The masked interrupt status of H264_DMA_IN_DSCR_EMPTY_CH4_INT. (RO)

H264_DMA_IN_DSCR_TASK_OVF_CH4_INT_ST The masked interrupt status of H264_DMA_IN_DSCR_TASK_OVF_CH4_INT. (RO)

Register 39.169. H264_DMA_IN_INT_CLR_CH4_REG (0x0910)

(reserved)																						H264_DMA_IN_DSCR_TASK_OVF_CH4_INT_CLR H264_DMA_IN_DSCR_EMPTY_CH4_INT_CLR H264_DMA_INFIFO_UDF_L2_CH4_INT_CLR H264_DMA_INFIFO_OVF_L2_CH4_INT_CLR H264_DMA_INFIFO_UDF_L1_CH4_INT_CLR H264_DMA_INFIFO_OVF_L1_CH4_INT_CLR H264_DMA_IN_DSCR_ERR_CH4_INT_CLR H264_DMA_IN_ERR_EOF_CH4_INT_CLR H264_DMA_IN_SUC_EOF_CH4_INT_CLR H264_DMA_IN_DONE_CH4_INT_CLR												
31																						10		9	8	7	6	5	4	3	2	1	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset												

H264_DMA_IN_DONE_CH4_INT_CLR Write 1 to clear H264_DMA_IN_DONE_CH4_INT. (WT)

H264_DMA_IN_SUC_EOF_CH4_INT_CLR Write 1 to clear H264_DMA_IN_SUC_EOF_CH4_INT. (WT)

H264_DMA_IN_ERR_EOF_CH4_INT_CLR Write 1 to clear H264_DMA_IN_ERR_EOF_CH4_INT. (WT)

H264_DMA_IN_DSCR_ERR_CH4_INT_CLR Write 1 to clear H264_DMA_IN_DSCR_ERR_CH4_INT.
(WT)

H264_DMA_INFIFO_OVF_L1_CH4_INT_CLR Write 1 to clear H264_DMA_INFIFO_OVF_L1_CH4_INT.
(WT)

H264_DMA_INFIFO_UDF_L1_CH4_INT_CLR Write 1 to clear H264_DMA_INFIFO_UDF_L1_CH4_INT.
(WT)

H264_DMA_INFIFO_OVF_L2_CH4_INT_CLR Write 1 to clear H264_DMA_INFIFO_OVF_L2_CH4_INT.
(WT)

H264_DMA_INFIFO_UDF_L2_CH4_INT_CLR Write 1 to clear H264_DMA_INFIFO_UDF_L2_CH4_INT.
(WT)

H264_DMA_IN_DSCR_EMPTY_CH4_INT_CLR Write 1 to clear H264_DMA_IN_DSCR_EMPTY_CH4_INT.
(WT)

H264_DMA_IN_DSCR_TASK_OVF_CH4_INT_CLR Write 1 to clear H264_DMA_IN_DSCR_TASK_OVF_CH4_INT.
(WT)

Register 39.170. H264_DMA_IN_INT_RAW_CH5_REG (0x0A10)

31	5	4	3	2	1	0
Reset						

(reserved)

H264_DMA_FETCH_MB_COL_CNT_OVF_CH5_INT_RAW
H264_DMA_INFIFO_UDF_L1_CH5_INT_RAW
H264_DMA_INFIFO_OVF_L1_CH5_INT_RAW
H264_DMA_IN_SUC_EOF_CH5_INT_RAW
H264_DMA_IN_DONE_CH5_INT_RAW

H264_DMA_IN_DONE_CH5_INT_RAW The raw interrupt status of H264_DMA_IN_DONE_CH5_INT.
(R/WTC/SS)

H264_DMA_IN_SUC_EOF_CH5_INT_RAW The raw interrupt status of H264_DMA_IN_SUC_EOF_CH5_INT. (R/WTC/SS)

H264_DMA_INFIFO_OVF_L1_CH5_INT_RAW The raw interrupt status of H264_DMA_INFIFO_OVF_L1_CH5_INT. (R/WTC/SS)

H264_DMA_INFIFO_UDF_L1_CH5_INT_RAW The raw interrupt status of H264_DMA_INFIFO_UDF_L1_CH5_INT. (R/WTC/SS)

H264_DMA_FETCH_MB_COL_CNT_OVF_CH5_INT_RAW The raw interrupt status of H264_DMA_FETCH_MB_COL_CNT_OVF_CH5_INT. (R/WTC/SS)

Register 39.171. H264_DMA_IN_INT_ENA_CH5_REG (0x0A14)

Diagram illustrating the structure of the **H264_DMA_IN_DONE_CH5_INT_ENA** register (32 bits):

- Bits 31 to 5: (reserved)
- Bit 5: Reset
- Bits 4 to 0: H264_DMA_IN_DONE_CH5_INT_ENA

H264_DMA_IN_DONE_CH5_INT_ENA Write 1 to enable H264_DMA_IN_DONE_CH5_INT. (R/W)

H264_DMA_IN_SUC_EOF_CH5_INT_ENA Write 1 to enable H264_DMA_IN_SUC_EOF_CH5_INT.
(R/W)

H264_DMA_INFIFO_OVF_L1_CH5_INT_ENA Write 1 to enable H264_DMA_INFIFO_OVF_L1_CH5_INT.
(R/W)

H264_DMA_INFIFO_UDF_L1_CH5_INT_ENA Write 1 to enable H264_DMA_INFIFO_UDF_L1_CH5_INT.
(R/W)

H264_DMA_FETCH_MB_COL_CNT_OVF_CH5_INT_ENA	Write	1	to	enable
H264_DMA_FETCH_MB_COL_CNT_OVF_CH5_INT. (R/W)				

Register 39.172. H264_DMA_IN_INT_ST_CH5_REG (0x0A18)

(reserved)																								H264_DMA_FETCH_MB_COL_CNT_OVF_CH5_INT_ST						
																								H264_DMA_INFIFO_UDF_L1_CH5_INT_ST						
																								H264_DMA_INFIFO_OVF_L1_CH5_INT_ST						
																								H264_DMA_IN_SUC_EOF_CH5_INT_ST						
																								H264_DMA_IN_DONE_CH5_INT_ST						
31																								5	4	3	2	1	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset
H264_DMA_IN_DONE_CH5_INT_ST The masked interrupt status of H264_DMA_IN_DONE_CH5_INT. (RO)																														
H264_DMA_IN_SUC_EOF_CH5_INT_ST The masked interrupt status of H264_DMA_IN_SUC_EOF_CH5_INT. (RO)																														
H264_DMA_INFIFO_OVF_L1_CH5_INT_ST The masked interrupt status of H264_DMA_INFIFO_OVF_L1_CH5_INT. (RO)																														
H264_DMA_INFIFO_UDF_L1_CH5_INT_ST The masked interrupt status of H264_DMA_INFIFO_UDF_L1_CH5_INT. (RO)																														
H264_DMA_FETCH_MB_COL_CNT_OVF_CH5_INT_ST The masked interrupt status of H264_DMA_FETCH_MB_COL_CNT_OVF_CH5_INT. (RO)																														

Register 39.173. H264_DMA_IN_INT_CLR_CH5_REG (0x0A1C)

Diagram of the H264_DMA_IN_DONE_CH5_INT_CLR register structure:

Bit Range	Field Name
31	(reserved)
5	(reserved)
4	H264_DMA_FETCH_MB_COL_CNT_OVF_CH5_INT_CLR
3	H264_DMA_INFIFO_OVF_L1_CH5_INT_CLR
2	H264_DMA_INFIFO_OVF_L1_CH5_INT_CLR
1	H264_DMA_INFIFO_OVF_L1_CH5_INT_CLR
0	H264_DMA_IN_SUC_EOF_CH5_INT_CLR

Reset

H264_DMA_IN_DONE_CH5_INT_CLR Write 1 to clear H264_DMA_IN_DONE_CH5_INT. (WT)

H264_DMA_IN_SUC_EOF_CH5_INT_CLR Write 1 to clear H264_DMA_IN_SUC_EOF_CH5_INT. (WT)

H264_DMA_INFIFO_OVF_L1_CH5_INT_CLR Write 1 to clear H264_DMA_INFIFO_OVF_L1_CH5_INT.
(WT)

H264_DMA_INFIFO_UDF_L1_CH5_INT_CLR Write 1 to clear H264_DMA_INFIFO_UDF_L1_CH5_INT.
(WT)

H264_DMA_FETCH_MB_COL_CNT_OVF_CH5_INT_CLR Write 1 to clear
H264 DMA FETCH MB COL CNT OVF CH5. (WT)

Register 39.174. H264_DMA_OUTFIFO_STATUS_CHO_REG (0x0014)

(reserved)												H264_DMA_OUTFIFO_CNT_L3_CHO				H264_DMA_OUTFIFO_EMPTY_L3_CHO				H264_DMA_OUTFIFO_FULL_L3_CHO				(reserved)				H264_DMA_OUTFIFO_CNT_L1_CHO				H264_DMA_OUTFIFO_EMPTY_L1_CHO				H264_DMA_OUTFIFO_FULL_L1_CHO				H264_DMA_OUTFIFO_CNT_L2_CHO				H264_DMA_OUTFIFO_EMPTY_L2_CHO				H264_DMA_OUTFIFO_FULL_L2_CHO																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
31												20												19	18	17	16	15	13				12												8				7	6	5	2				1	0																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																							
0												0												0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

H264_DMA_OUTFIFO_FULL_L2_CHO Represents whether l2 FIFO of TX channel 0 is full.

0: Not Full

1: Full

(RO)

H264_DMA_OUTFIFO_EMPTY_L2_CHO Represents whether l2 FIFO of TX channel 0 is empty.

0: Not empty

1: Empty

(RO)

H264_DMA_OUTFIFO_CNT_L2_CHO Represents the data quantity in l2 FIFO for TX channel 0. Measurement unit: 16 bytes. (RO)

H264_DMA_OUTFIFO_FULL_L1_CHO Represents whether l1 FIFO of TX channel 0 is full.

0: Not Full

1: Full

(RO)

H264_DMA_OUTFIFO_EMPTY_L1_CHO Represents whether l1 FIFO of TX channel 0 is empty.

0: Not empty

1: Empty

(RO)

H264_DMA_OUTFIFO_CNT_L1_CHO Represents the data quantity in l1 FIFO for TX channel 0. Measurement unit: byte. (RO)

H264_DMA_OUTFIFO_FULL_L3_CHO Represents whether l3 FIFO of TX channel 0 is full.

0: Not Full

1: Full

(RO)

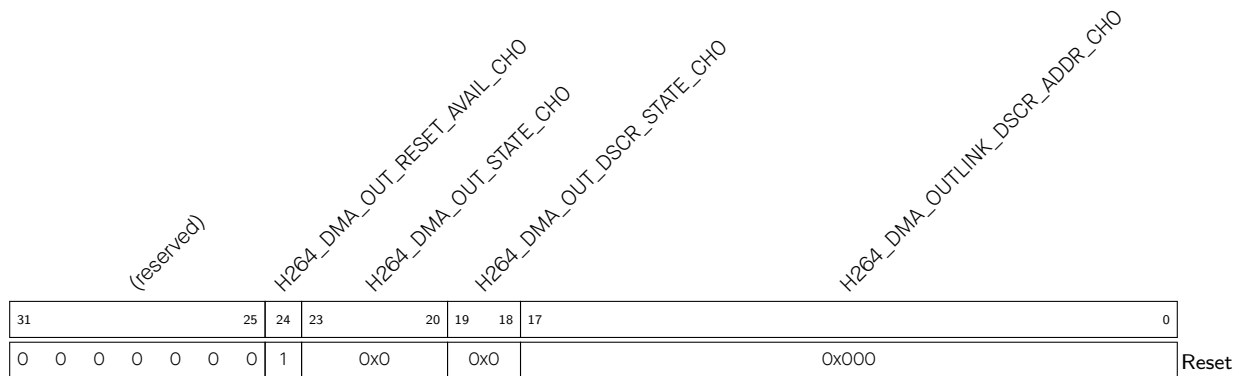
H264_DMA_OUTFIFO_EMPTY_L3_CHO Represents whether l3 FIFO of TX channel 0 is empty.

0: Not empty

1: Empty

(RO)

H264_DMA_OUTFIFO_CNT_L3_CHO Represents the data quantity in l3 FIFO for TX channel 0. Measurement unit: 8 bytes. (RO)

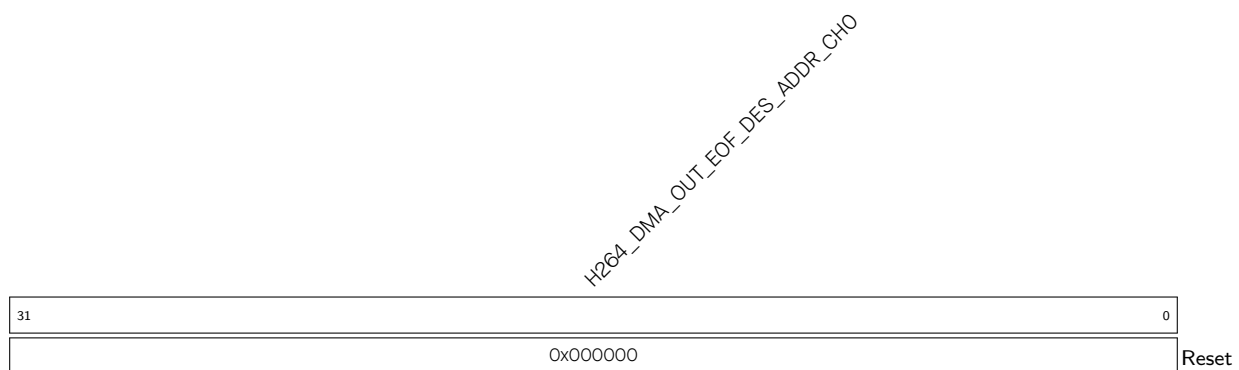
Register 39.175. H264_DMA_OUT_STATE_CHO_REG (0x0024)

H264_DMA_OUTLINK_DSCR_ADDR_CHO Represents the current outlink descriptor's address for TX channel 0. (RO)

H264_DMA_OUT_DSCR_STATE_CHO Represents the current state of the descriptor state machine for TX channel 0. (RO)

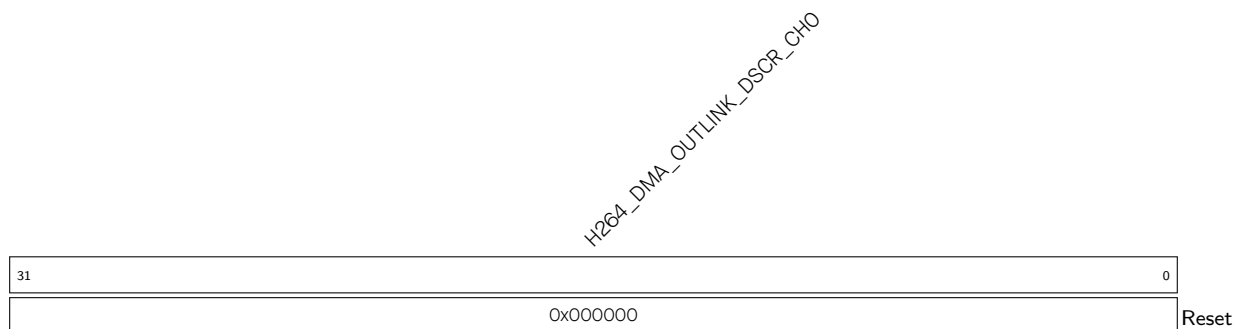
H264_DMA_OUT_STATE_CHO Represents the current control module state machine state for TX channel 0. (RO)

H264_DMA_OUT_RESET_AVAIL_CHO Represents whether it is safe to reset the channel.
 0: Unsafe
 1: Safe
 (RO)

Register 39.176. H264_DMA_OUT_EOF_DES_ADDR_CHO_REG (0x0028)

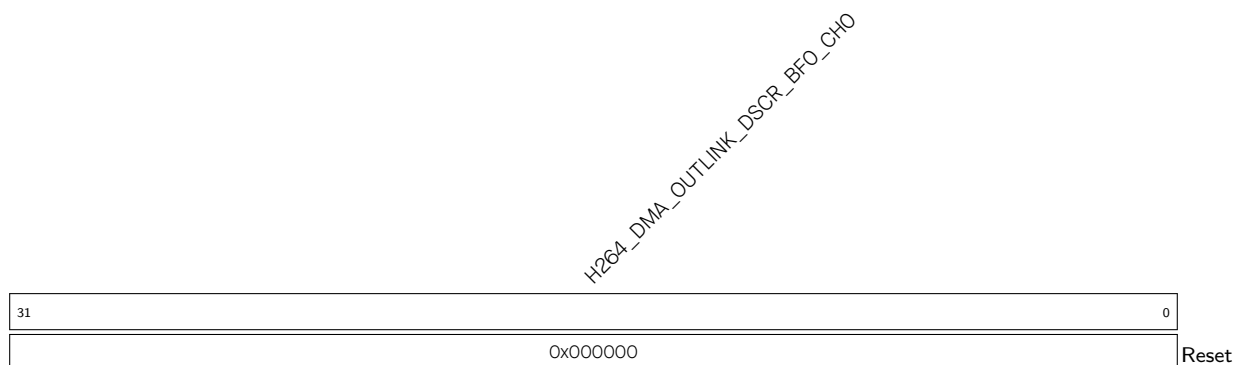
H264_DMA_OUT_EOF_DES_ADDR_CHO Represents the address of the outlink descriptor when the EOF bit in this descriptor is 1. (RO)

Register 39.177. H264_DMA_OUT_DSCR_CHO_REG (0x002C)



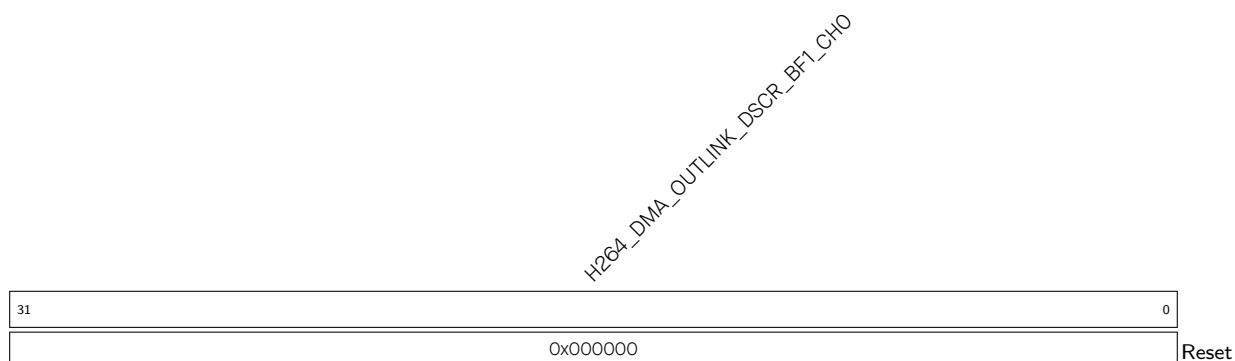
H264_DMA_OUTLINK_DSCR_CHO Represents the address of the next descriptor y+1 pointed by the current descriptor that is pre-read. (RO)

Register 39.178. H264_DMA_OUT_DSCR_BFO_CHO_REG (0x0030)



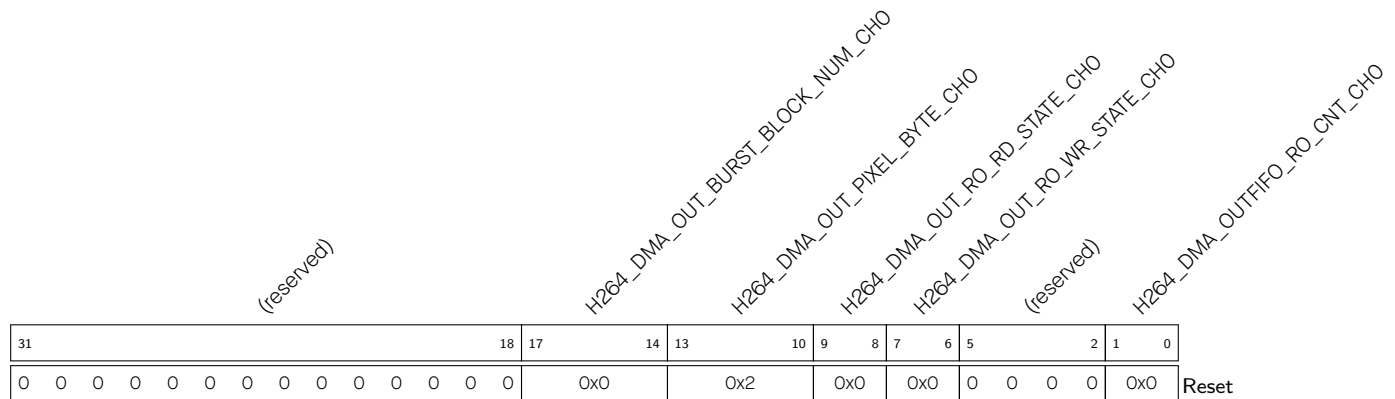
H264_DMA_OUTLINK_DSCR_BFO_CHO Represents the address of the current descriptor y that is pre-read. (RO)

Register 39.179. H264_DMA_OUT_DSCR_BF1_CHO_REG (0x0034)



H264_DMA_OUTLINK_DSCR_BF1_CHO Represents the address of the previous descriptor y-1 that is pre-read. (RO)

Register 39.180. H264_DMA_OUT_RO_STATUS_CHO_REG (0x0040)



H264_DMA_OUTFIFO_RO_CNT_CHO Represents the 8-byte number of the data in the reordered TX FIFO for channel 0. (RO)

H264_DMA_OUT_RO_WR_STATE_CHO Represents the state of writing RAM for reorder. (RO)

H264_DMA_OUT_RO_RD_STATE_CHO Represents the state of reading RAM for reorder. (RO)

H264_DMA_OUT_PIXEL_BYTE_CHO Represents the number of bytes contained in a pixel at TX channel 0.

0: 1 byte

1: 1.5 bytes

2: 2 bytes

3: 2.5 bytes

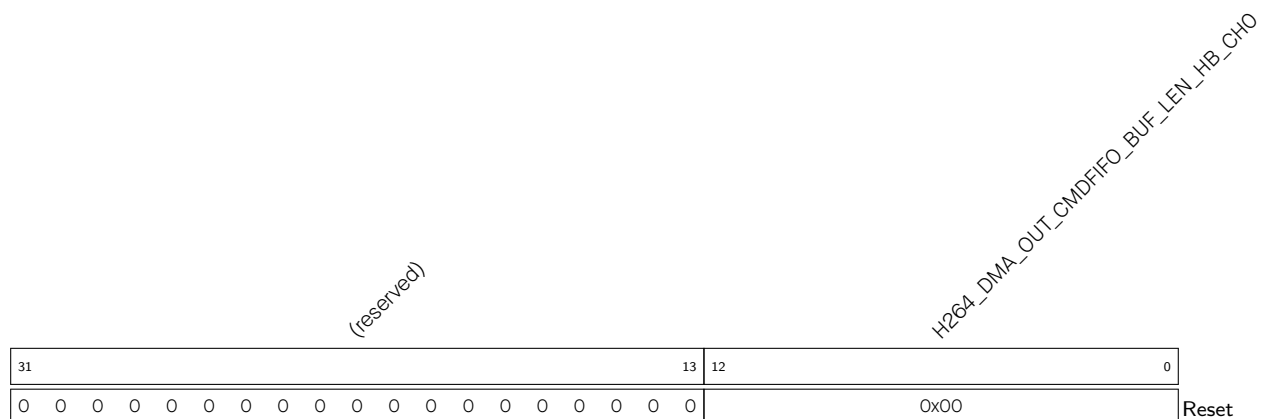
4: 3 bytes

5: 4 bytes

(RO)

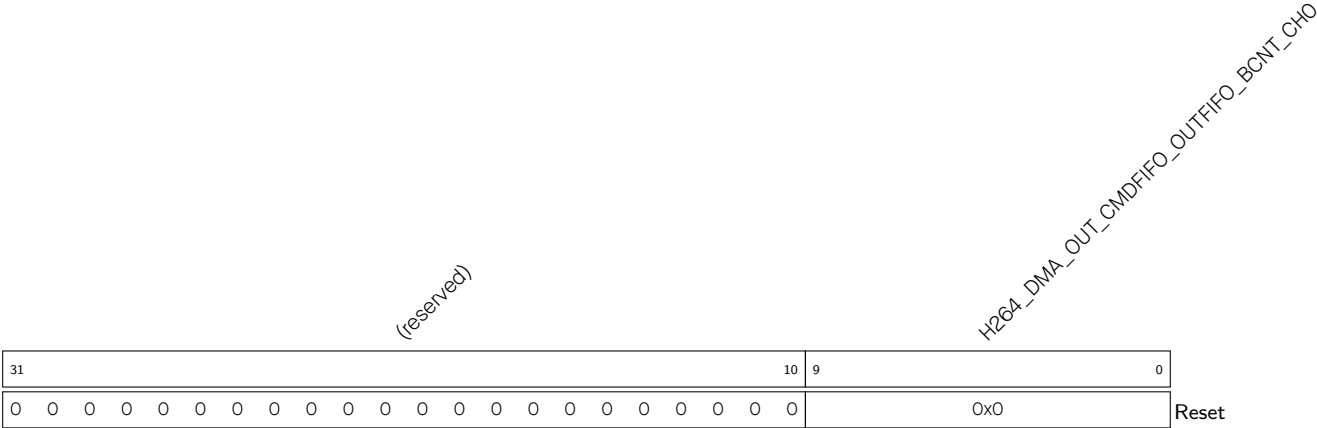
H264_DMA_OUT_BURST_BLOCK_NUM_CHO Reserved. (RO)

Register 39.181. H264_DMA_OUT_BUF_LEN_CHO_REG (0x0070)



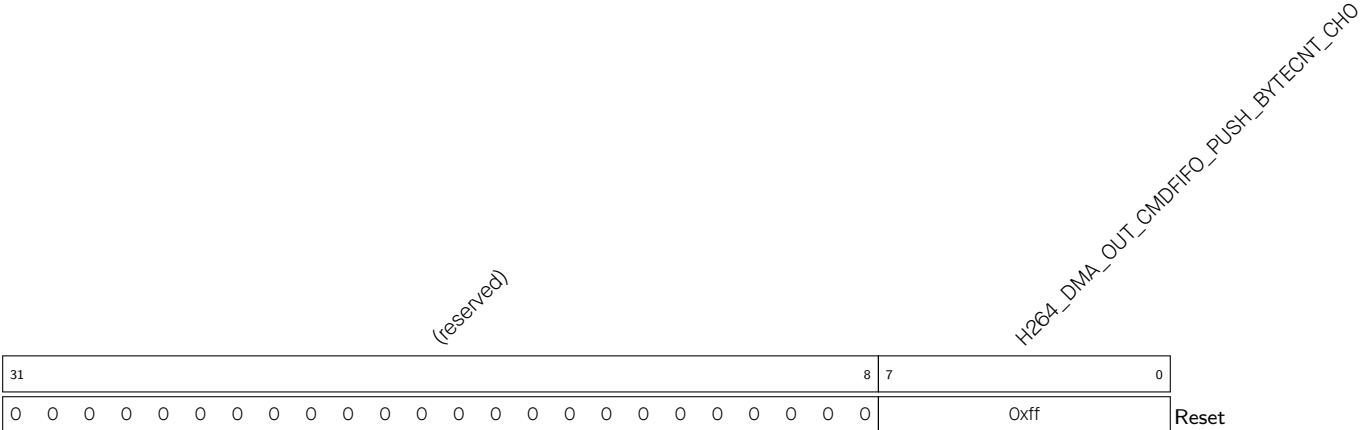
H264_DMA_OUT_CMDFIFO_BUF_LEN_HB_CHO Represents the remaining request data. (RO)

Register 39.182. H264_DMA_OUT_FIFO_BCNT_CHO_REG (0x0074)



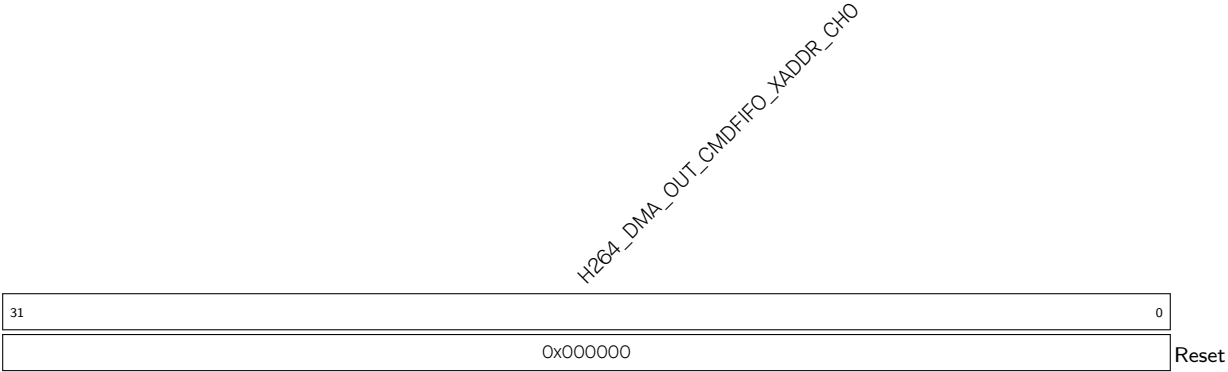
H264_DMA_OUT_CMDFIFO_OUTFIFO_BCNT_CHO Represents that the data space has been used in FIFO. (RO)

Register 39.183. H264_DMA_OUT_PUSH_BYTECNT_CHO_REG (0x0078)



H264_DMA_OUT_CMDFIFO_PUSH_BYTECNT_CHO Represents the data quantity in the current AXI request. (RO)

Register 39.184. H264_DMA_OUT_XADDR_CHO_REG (0x007C)



H264_DMA_OUT_CMDFIFO_XADDR_CHO Represents the address of the current AXI request. (RO)

Register 39.185. H264_DMA_OUTFIFO_STATUS_CH1_REG (0x0114)

(reserved)												H264_DMA_OUTFIFO_CNT_L3_CH1				H264_DMA_OUTFIFO_EMPTY_L3_CH1				H264_DMA_OUTFIFO_FULL_L3_CH1				(reserved)				H264_DMA_OUTFIFO_CNT_L1_CH1				H264_DMA_OUTFIFO_EMPTY_L1_CH1				H264_DMA_OUTFIFO_FULL_L1_CH1				H264_DMA_OUTFIFO_CNT_L2_CH1				H264_DMA_OUTFIFO_EMPTY_L2_CH1				H264_DMA_OUTFIFO_FULL_L2_CH1			
31												20		19	18	17	16	15	13		12	8		7	6	5	2		1	0																					
0 0 0 0 0 0 0 0 0 0 0 0												0x0		1	0	0	0	0	0x0		1	0	0x0		1	0	Reset																								

H264_DMA_OUTFIFO_FULL_L2_CH1 Represents whether I2 FIFO of TX channel 1 is full.

0: Not Full

1: Full

(RO)

H264_DMA_OUTFIFO_EMPTY_L2_CH1 Represents whether I2 FIFO of TX channel 1 is empty.

0: Not empty

1: Empty

(RO)

H264_DMA_OUTFIFO_CNT_L2_CH1	Represents the data quantity in l2 FIFO for TX channel 1. Measurement unit: 16 bytes. (RO)
------------------------------------	--

H264_DMA_OUTFIFO_FULL_L1_CH1 Represents whether l1 FIFO of TX channel 1 is full.

0: Not Full

1: Full

(RO)

H264_DMA_OUTFIFO_EMPTY_L1_CH1 Represents whether l1 FIFO of TX channel 1 is empty.

0: Not empty

1: Empty

(RO)

H264_DMA_OUTFIFO_CNT_L1_CH1 Represents the data quantity in l1 FIFO for TX channel 1. Measurement unit: byte. (RO)

H264_DMA_OUTFIFO_FULL_L3_CH1 Represents whether L3 FIFO of TX channel 1 is full.

0: Not Full

1: Full

(RO)

H264_DMA_OUTFIFO_EMPTY_L3_CH1 Represents whether I3 FIFO of TX channel 1 is empty.

0: Not empty

1: Empty

(RO)

H264_DMA_OUTFIFO_CNT_L3_CH1	Represents the data quantity in I3 FIFO for TX channel 1. Measurement unit: 8 bytes. (RO)
------------------------------------	---

Register 39.186. H264_DMA_OUT_STATE_CH1_REG (0x0124)

(reserved)								H264_DMA_OUT_RESET_AVAIL_CH1								H264_DMA_OUT_STATE_CH1								H264_DMA_OUT_DSCR_STATE_CH1								H264_DMA_OUTLINK_DSCR_ADDR_CH1							
31					25	24	23					20	19	18	17																				0				
0	0	0	0	0	0	0	0	1	0x0				0x0				0x000																			Reset			

H264_DMA_OUTLINK_DSCR_ADDR_CH1 Represents the current outlink descriptor's address for TX channel 1. (RO)

H264_DMA_OUT_DSCR_STATE_CH1 Represents the current state of the descriptor state machine for TX channel 1. (RO)

H264_DMA_OUT_STATE_CH1 Represents the current control module state machine state for TX channel 1. (RO)

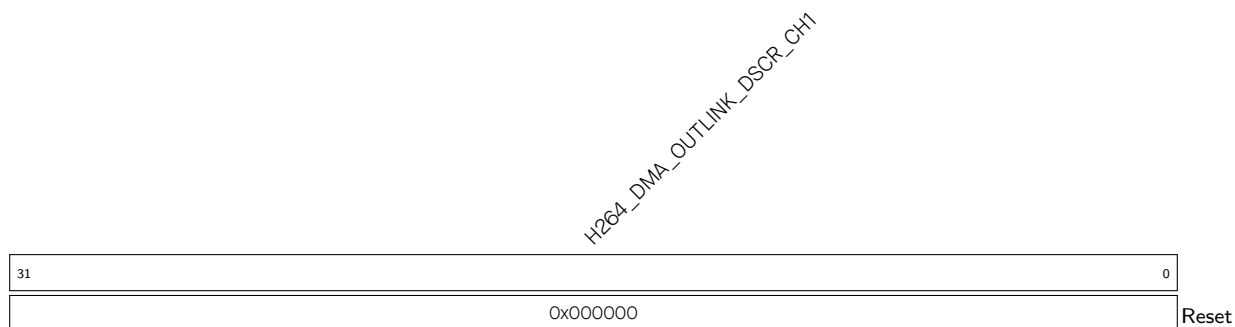
H264_DMA_OUT_RESET_AVAIL_CH1 Represents whether it is safe to reset the channel.
 0: Unsafe
 1: Safe
 (RO)

Register 39.187. H264_DMA_OUT_EOF_DES_ADDR_CH1_REG (0x0128)

H264_DMA_OUT_EOF_DES_ADDR_CH1																															
31																															0
0x000000																															
Reset																															

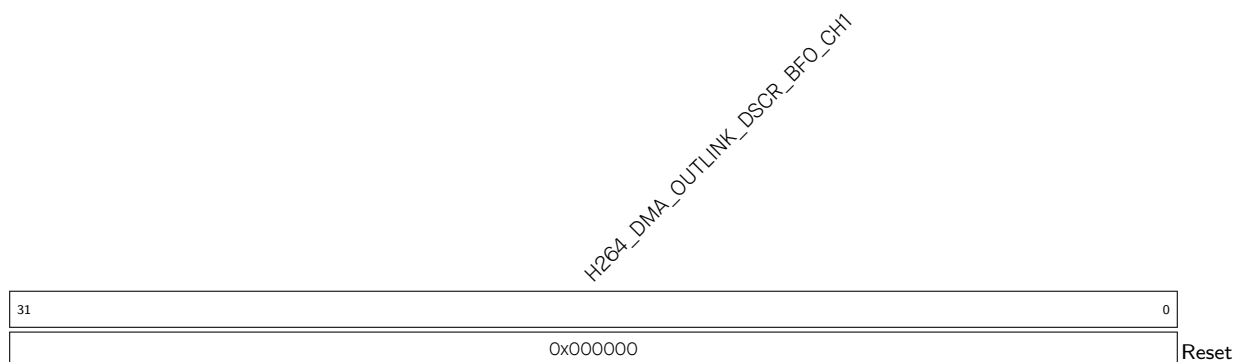
H264_DMA_OUT_EOF_DES_ADDR_CH1 Represents the address of the outlink descriptor when the EOF bit in this descriptor is 1. (RO)

Register 39.188. H264_DMA_OUT_DSCR_CH1_REG (0x012C)



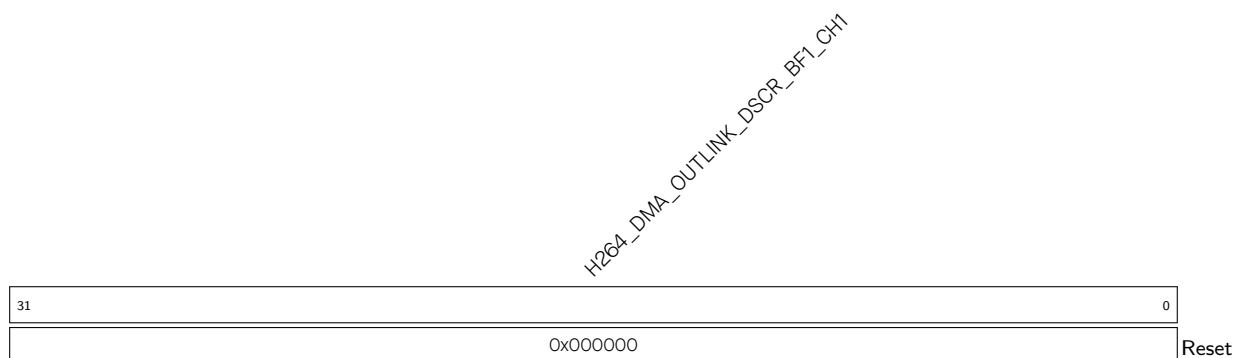
H264_DMA_OUTLINK_DSCR_CH1 Represents the address of the next descriptor y+1 pointed by the current descriptor that is pre-read. (RO)

Register 39.189. H264_DMA_OUT_DSCR_BFO_CH1_REG (0x0130)



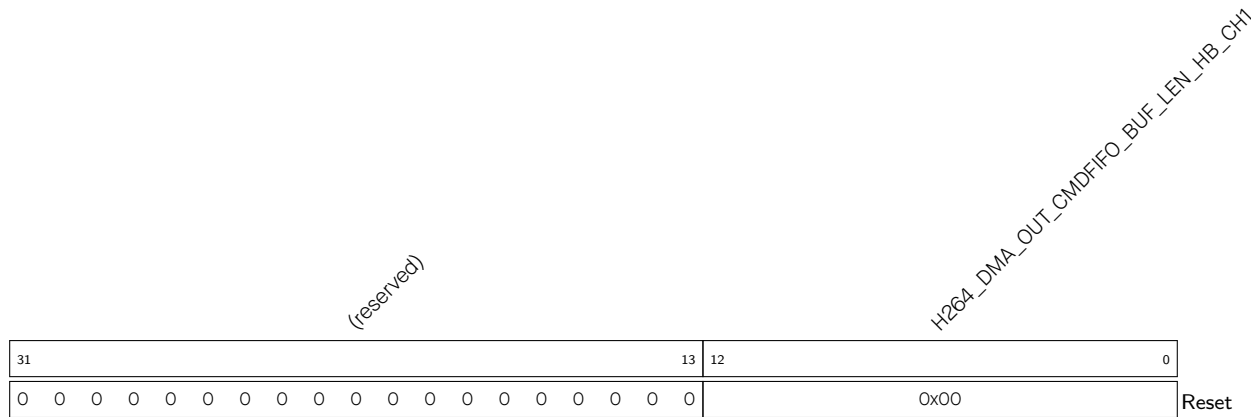
H264_DMA_OUTLINK_DSCR_BFO_CH1 Represents the address of the current descriptor y that is pre-read. (RO)

Register 39.190. H264_DMA_OUT_DSCR_BF1_CH1_REG (0x0134)



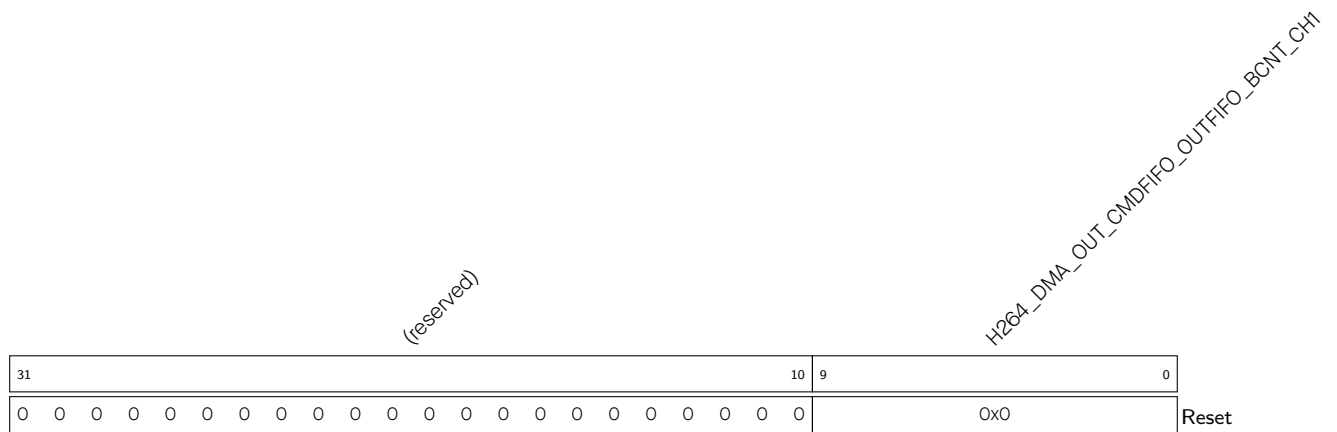
H264_DMA_OUTLINK_DSCR_BF1_CH1 Represents the address of the previous descriptor y-1 that is pre-read. (RO)

Register 39.191. H264_DMA_OUT_BUF_LEN_CH1_REG (0x0170)



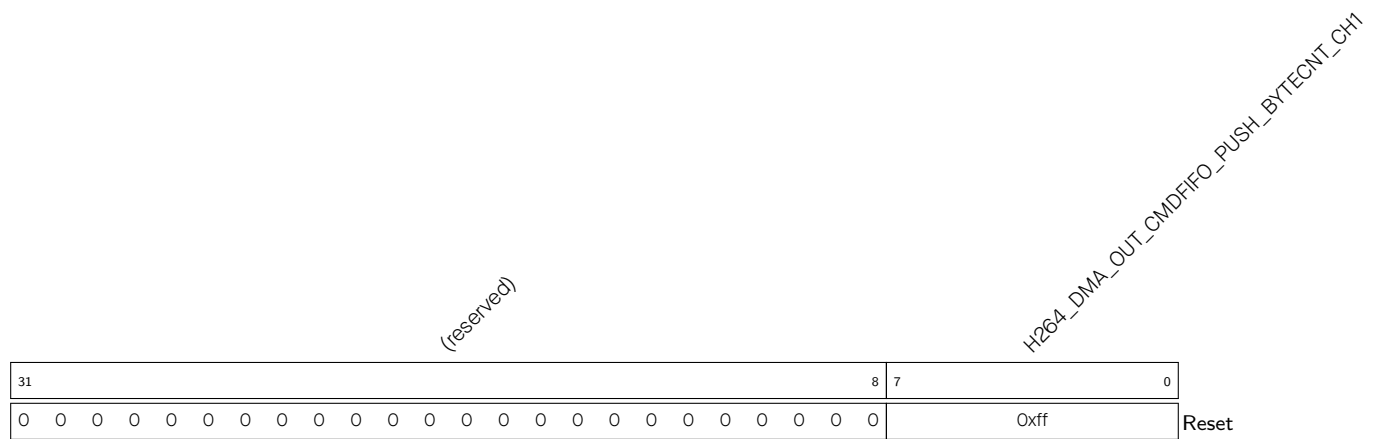
H264_DMA_OUT_CMDFIFO_BUF_LEN_HB_CH1 Represents the remaining request data. (RO)

Register 39.192. H264_DMA_OUT_FIFO_BCNT_CH1_REG (0x0174)



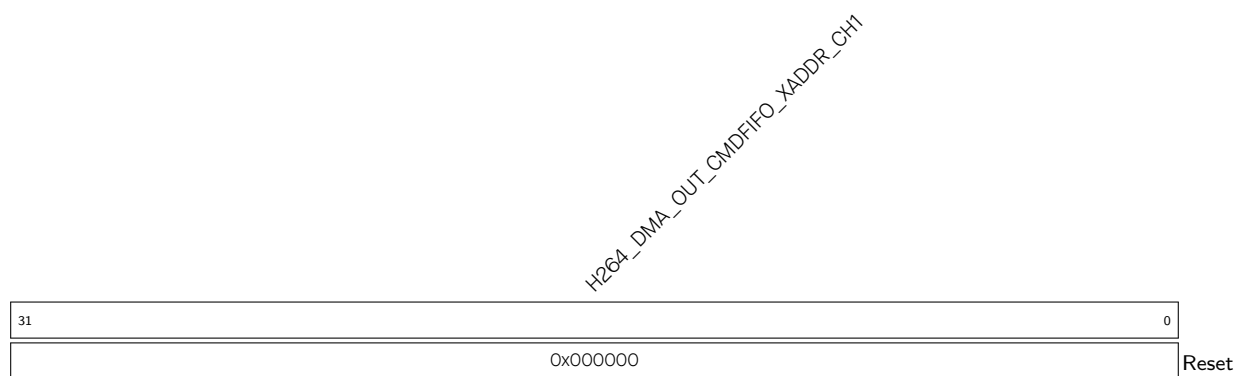
H264_DMA_OUT_CMDFIFO_OUTFIFO_BCNT_CH1 Represents that the data space has been used in FIFO. (RO)

Register 39.193. H264_DMA_OUT_PUSH_BYTECNT_CH1_REG (0x0178)



H264_DMA_OUT_CMDFIFO_PUSH_BYTECNT_CH1 Represents the data quantity in the current AXI request. (RO)

Register 39.194. H264_DMA_OUT_XADDR_CH1_REG (0x017C)



H264_DMA_OUT_CMDFIFO_XADDR_CH1 Represents the address of the current AXI request. (RO)

Register 39.195. H264_DMA_OUTFIFO_STATUS_CH2_REG (0x0214)

(reserved)												H264_DMA_OUTFIFO_CNT_L3_CH2			H264_DMA_OUTFIFO_EMPTY_L3_CH2			H264_DMA_OUTFIFO_FULL_L3_CH2			(reserved)			H264_DMA_OUTFIFO_CNT_L1_CH2			H264_DMA_OUTFIFO_EMPTY_L1_CH2			H264_DMA_OUTFIFO_FULL_L1_CH2			H264_DMA_OUTFIFO_CNT_L2_CH2			H264_DMA_OUTFIFO_EMPTY_L2_CH2			H264_DMA_OUTFIFO_FULL_L2_CH2		
31												20	19	18	17	16	15	13	12	8	7	6	5	2	1	0	Reset														
0 0 0 0 0 0 0 0 0 0 0 0												0x0	1	0	0	0	0	0x0	1	0	0x0	1	0																		

H264_DMA_OUTFIFO_FULL_L2_CH2 Represents whether l2 FIFO of TX channel 2 is full.

0: Not Full

1: Full

(RO)

H264_DMA_OUTFIFO_EMPTY_L2_CH2 Represents whether l2 FIFO of TX channel 2 is empty.

0: Not empty

1: Empty

(RO)

H264_DMA_OUTFIFO_CNT_L2_CH2 Represents the data quantity in I2 FIFO for TX channel 2. Measurement unit: 16 bytes. (RO)

64_DMA_OUTFIFO_FULL_L1_CH2 Represents whether l1 FIFO of TX channel 2 is full.

0: Not Full

1: Full

(RO)

H264_DMA_OUTFIFO_EMPTY_L1_CH2 Represents whether l1 FIFO of TX channel 2 is empty.

0: Not empty

1: Empty

(RO)

H264_DMA_OUTFIFO_CNT_L1_CH2 Represents the data quantity in l1 FIFO for TX channel 2. Measurement unit: byte. (RO)

64 DMA_OUTFIFO_FULL_L3_CH2 Represents whether l3 FIFO of TX channel 2 is full.

0: Not Full

1: Full

(RO)

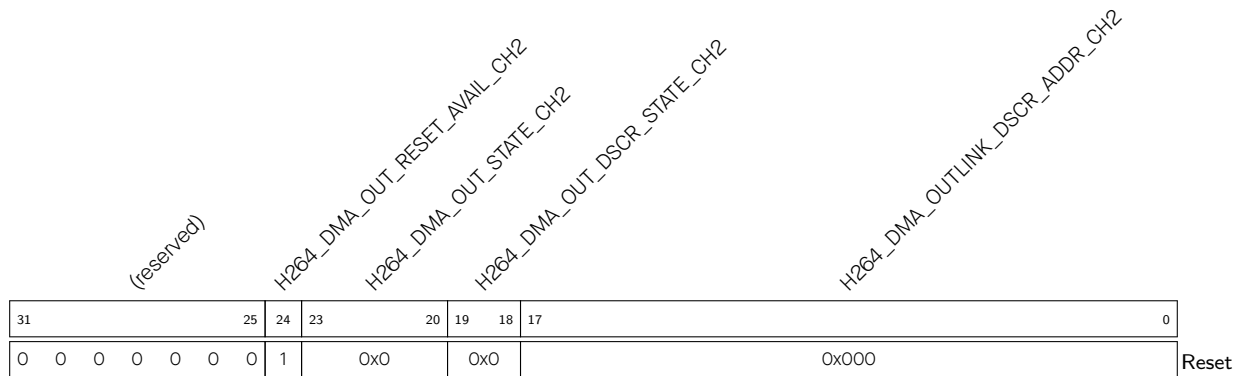
H264_DMA_OUTFIFO_EMPTY_L3_CH2 Represents whether l3 FIFO of TX channel 2 is empty.

0: Not empty

1: Empty

(RO)

H264_DMA_OUTFIFO_CNT_L3_CH2 Represents the data quantity in I3 FIFO for TX channel 2. Measurement unit: 8 bytes. (RO)

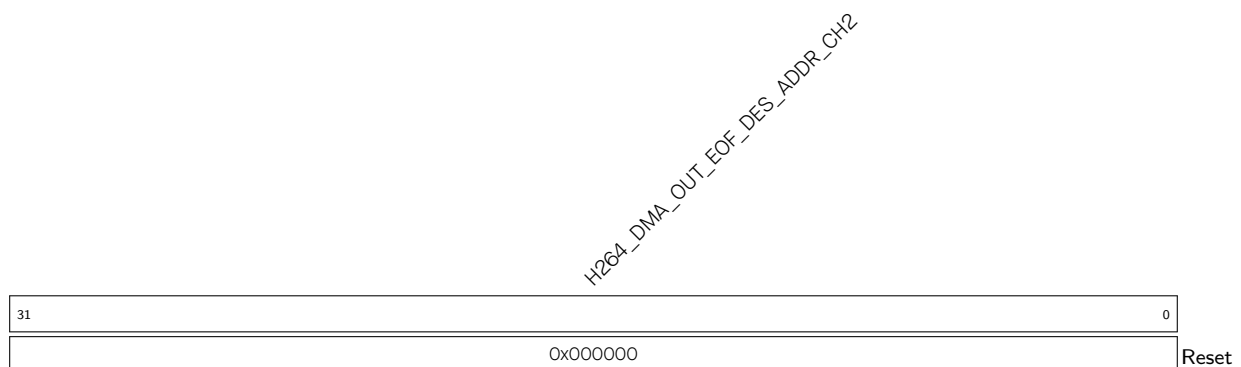
Register 39.196. H264_DMA_OUT_STATE_CH2_REG (0x0224)

H264_DMA_OUTLINK_DSCR_ADDR_CH2 Represents the current outlink descriptor's address for TX channel 2. (RO)

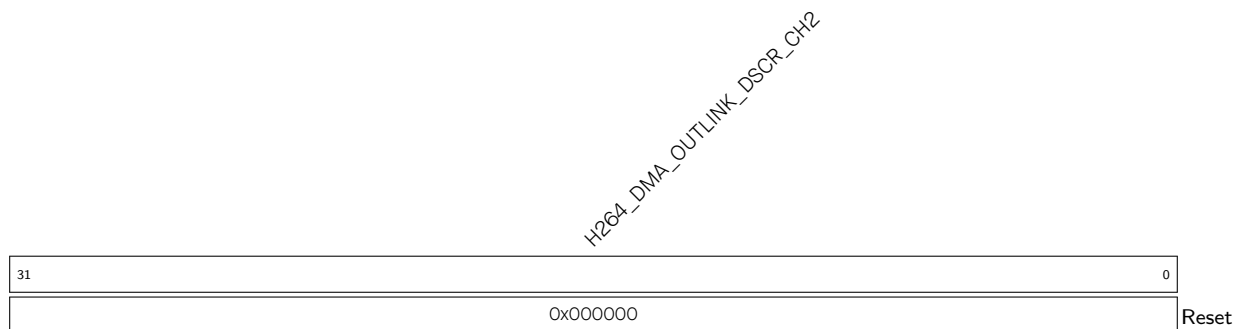
H264_DMA_OUT_DSCR_STATE_CH2 Represents the current state of the descriptor state machine for TX channel 2. (RO)

H264_DMA_OUT_STATE_CH2 Represents the current control module state machine state for TX channel 2. (RO)

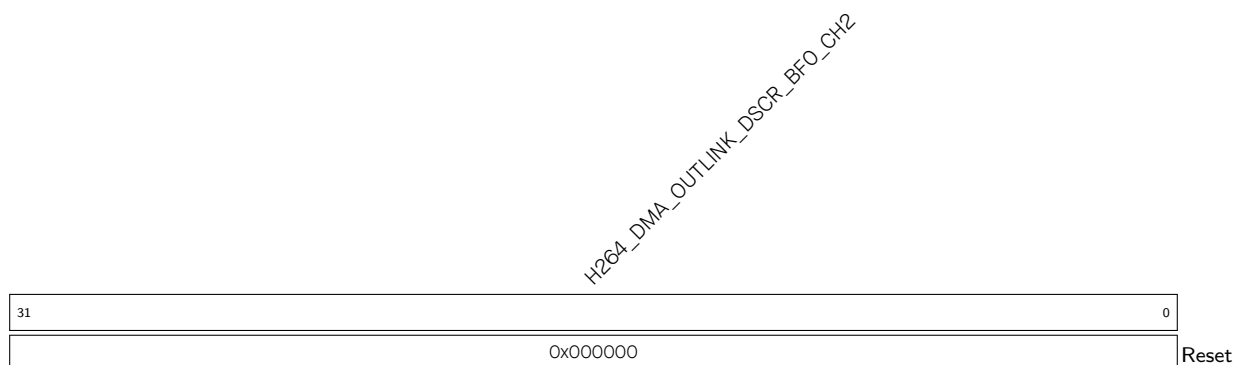
H264_DMA_OUT_RESET_AVAIL_CH2 Represents whether it is safe to reset the channel.
 0: Unsafe
 1: Safe
 (RO)

Register 39.197. H264_DMA_OUT_EOF_DES_ADDR_CH2_REG (0x0228)

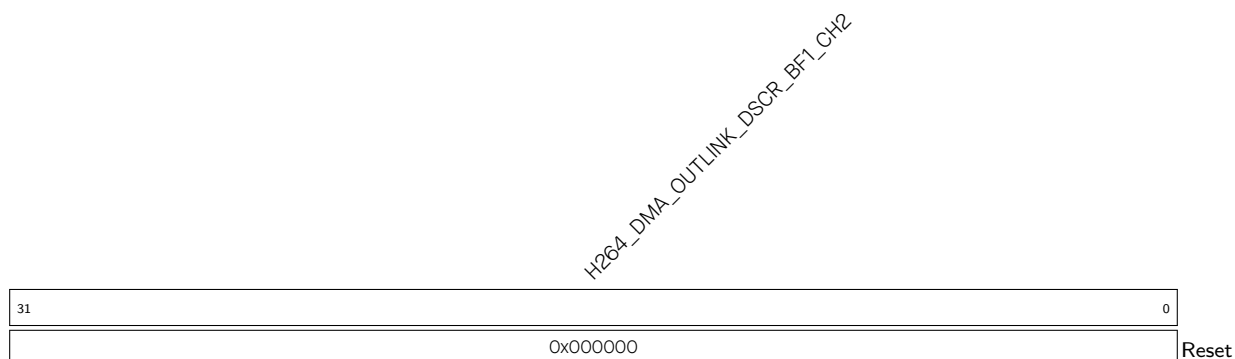
H264_DMA_OUT_EOF_DES_ADDR_CH2 Represents the address of the inlink descriptor when the EOF bit in this descriptor is 1. (RO)

Register 39.198. H264_DMA_OUT_DSCR_CH2_REG (0x022C)

H264_DMA_OUTLINK_DSCR_CH2 Represents the address of the next descriptor y+1 pointed by the current descriptor that is pre-read.. (RO)

Register 39.199. H264_DMA_OUT_DSCR_BFO_CH2_REG (0x0230)

H264_DMA_OUTLINK_DSCR_BFO_CH2 Represents the address of the current descriptor y that is pre-read. (RO)

Register 39.200. H264_DMA_OUT_DSCR_BF1_CH2_REG (0x0234)

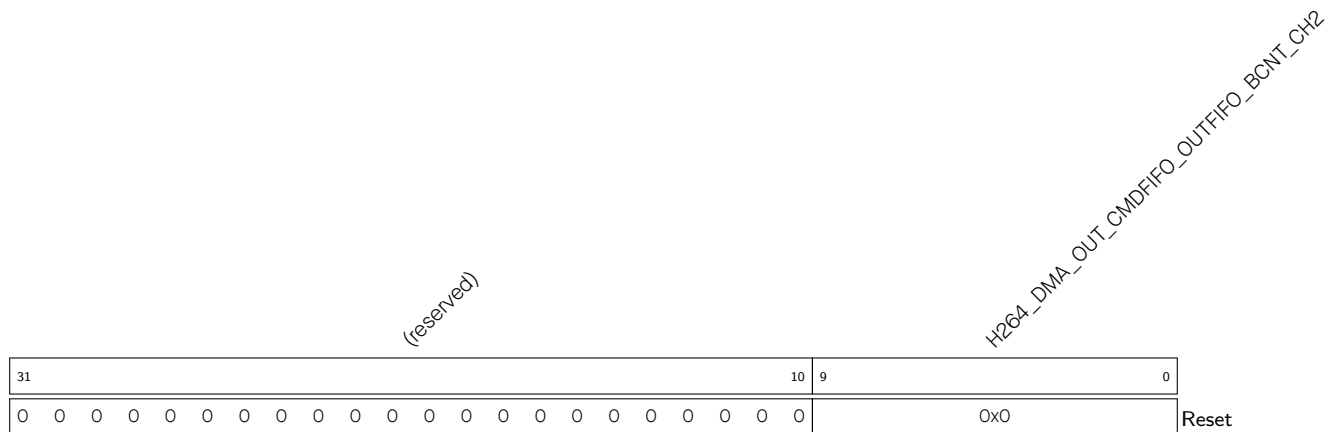
H264_DMA_OUTLINK_DSCR_BF1_CH2 Represents the address of the previous descriptor y-1 that is pre-read. (RO)

Register 39.201. H264_DMA_OUT_BUF_LEN_CH2_REG (0x0270)



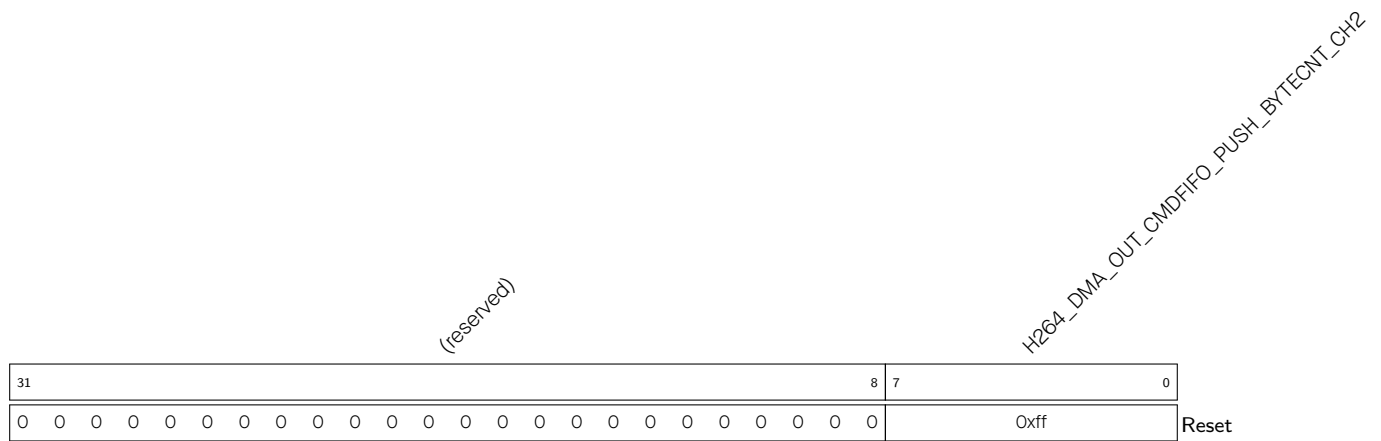
H264_DMA_OUT_CMDFIFO_BUF_LEN_HB_CH2 Represents the remaining request data. (RO)

Register 39.202. H264_DMA_OUT_FIFO_BCNT_CH2_REG (0x0274)



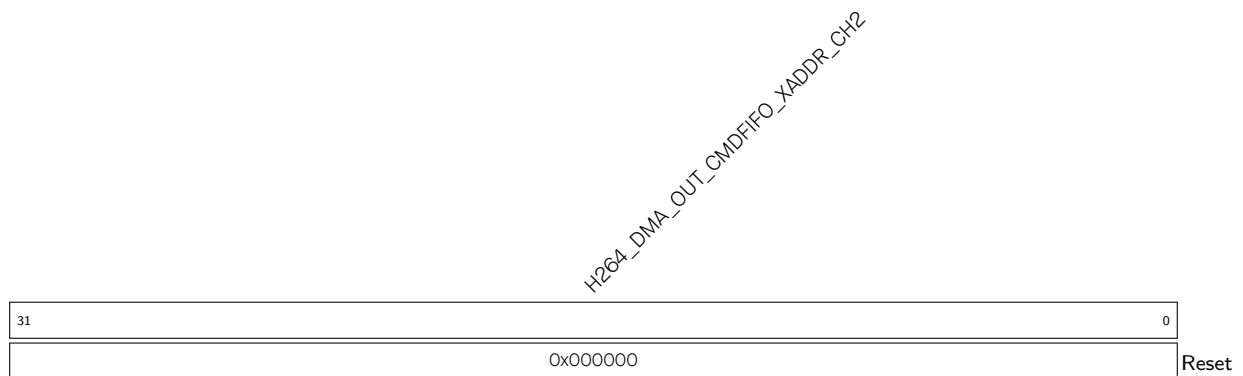
H264_DMA_OUT_CMDFIFO_OUTFIFO_BCNT_CH2 Represents that the data space has been used in FIFO. (RO)

Register 39.203. H264_DMA_OUT_PUSH_BYTECNT_CH2_REG (0x0278)



H264_DMA_OUT_CMDFIFO_PUSH_BYTECNT_CH2	Represents the data quantity in the current AXI request. (RO)
--	---

Register 39.204. H264_DMA_OUT_XADDR_CH2_REG (0x027C)



H264_DMA_OUT_CMDFIFO_XADDR_CH2 Represents the address of the current AXI request. (RO)

Register 39.205. H264_DMA_OUTFIFO_STATUS_CH3_REG (0x0314)

(reserved)												H264_DMA_OUTFIFO_CNT_L3_CH3			H264_DMA_OUTFIFO_EMPTY_L3_CH3			H264_DMA_OUTFIFO_FULL_L3_CH3			(reserved)			H264_DMA_OUTFIFO_CNT_L1_CH3			H264_DMA_OUTFIFO_EMPTY_L1_CH3			H264_DMA_OUTFIFO_FULL_L1_CH3			H264_DMA_OUTFIFO_CNT_L2_CH3			H264_DMA_OUTFIFO_EMPTY_L2_CH3			H264_DMA_OUTFIFO_FULL_L2_CH3		
31												20	19	18	17	16	15	13	12	8	7	6	5	2	1	0	Reset														
0 0 0 0 0 0 0 0 0 0 0 0												0x0	1	0	0	0	0	0x0	1	0	0x0	1	0																		

H264_DMA_OUTFIFO_FULL_L2_CH3 Represents whether I2 FIFO of TX channel 3 is full.

0: Not Full

1: Full

(RO)

H264_DMA_OUTFIFO_EMPTY_L2_CH3 Represents whether l2 FIFO of TX channel 3 is empty.

0: Not empty

1: Empty

(RO)

H264_DMA_OUTFIFO_CNT_L2_CH3 Represents the data quantity in I2 FIFO for TX channel 3. Measurement unit: 16 bytes. (RO)

H264_DMA_OUTFIFO_FULL_L1_CH3 Represents whether l1 FIFO of TX channel 3 is full.

0: Not Full

1: Full

(RO)

H264_DMA_OUTFIFO_EMPTY_L1_CH3 Represents whether l1 FIFO of TX channel 3 is empty.

0: Not empty

1: Empty

(RO)

H264_DMA_OUTFIFO_CNT_L1_CH3 Represents the data quantity in l1 FIFO for TX channel 3. Measurement unit: byte. (RO)

H264_DMA_OUTFIFO_FULL_L3_CH3 Represents whether L3 FIFO of TX channel 3 is full.

0: Not Full

1: Full

(RO)

H264_DMA_OUTFIFO_EMPTY_L3_CH3 Represents whether L3 FIFO of TX channel 3 is empty.

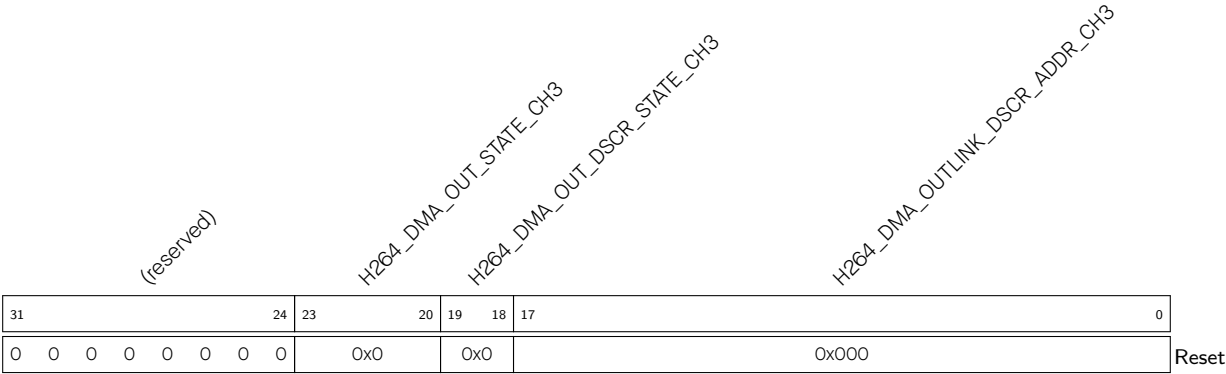
0: Not empty

1: Empty

(RO)

H264_DMA_OUTFIFO_CNT_L3_CH3 Represents the data quantity in I3 FIFO for TX channel 3. Measurement unit: 8 bytes. (RO)

Register 39.206. H264_DMA_OUT_STATE_CH3_REG (0x0324)

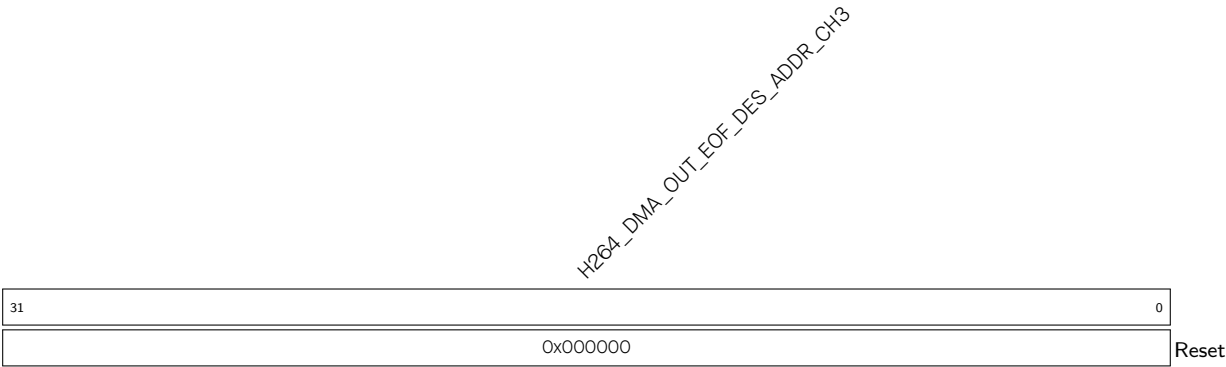


H264_DMA_OUTLINK_DSCR_ADDR_CH3 Represents the current outlink descriptor’s address for TX channel 3. (RO)

H264_DMA_OUT_DSCR_STATE_CH3 Represents the current state of the descriptor state machine for TX channel 3. (RO)

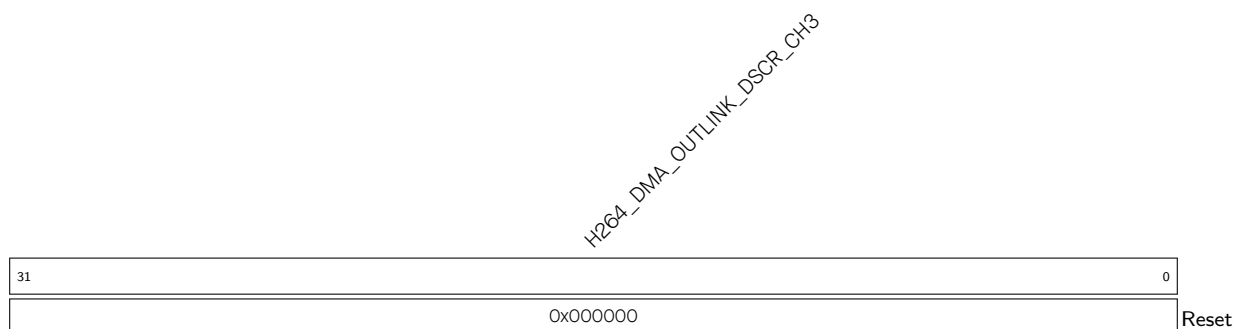
H264_DMA_OUT_STATE_CH3 Represents the current control module state machine state for TX channel 3. (RO)

Register 39.207. H264_DMA_OUT_EOF_DES_ADDR_CH3_REG (0x0328)



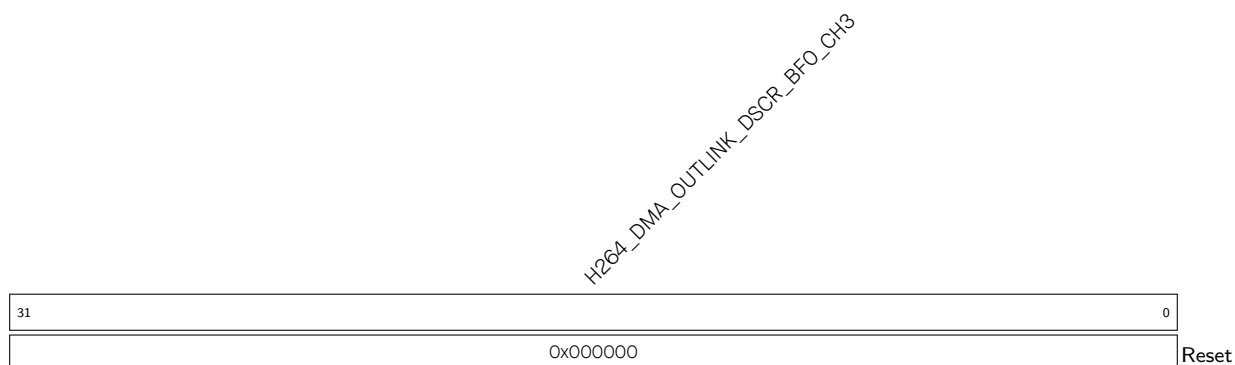
H264_DMA_OUT_EOF_DES_ADDR_CH3 Represents the address of the inlink descriptor when the EOF bit in this descriptor is 1. (RO)

Register 39.208. H264_DMA_OUT_DSCR_CH3_REG (0x032C)



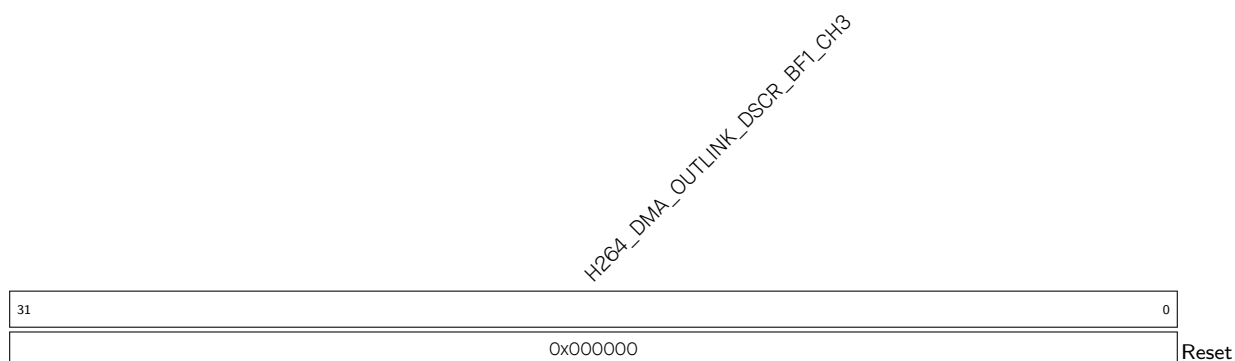
H264_DMA_OUTLINK_DSCR_CH3 Represents the address of the next descriptor y+1 pointed by the current descriptor that is pre-read.. (RO)

Register 39.209. H264_DMA_OUT_DSCR_BFO_CH3_REG (0x0330)



H264_DMA_OUTLINK_DSCR_BFO_CH3 Represents the address of the current descriptor y that is pre-read. (RO)

Register 39.210. H264_DMA_OUT_DSCR_BF1_CH3_REG (0x0334)



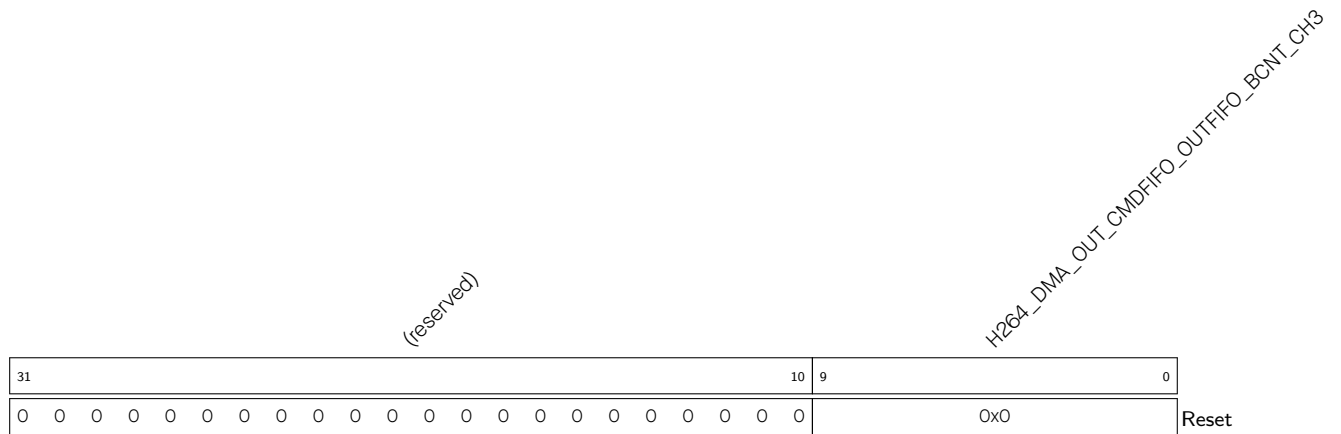
H264_DMA_OUTLINK_DSCR_BF1_CH3 Represents the address of the previous descriptor y-1 that is pre-read. (RO)

Register 39.211. H264_DMA_OUT_BUF_LEN_CH3_REG (0x0370)



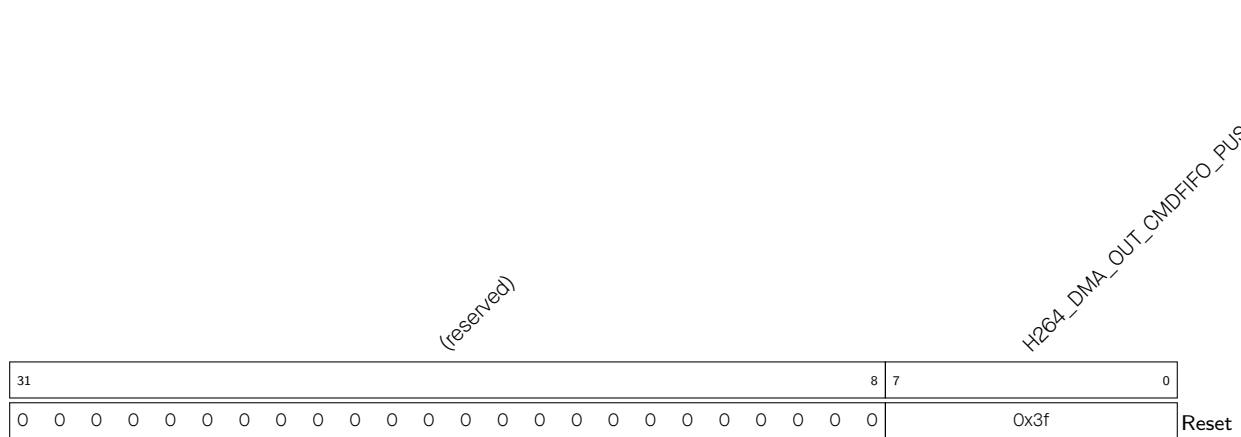
H264_DMA_OUT_CMDFIFO_BUF_LEN_HB_CH3 Represents the remaining request data. (RO)

Register 39.212. H264_DMA_OUT_FIFO_BCNT_CH3_REG (0x0374)



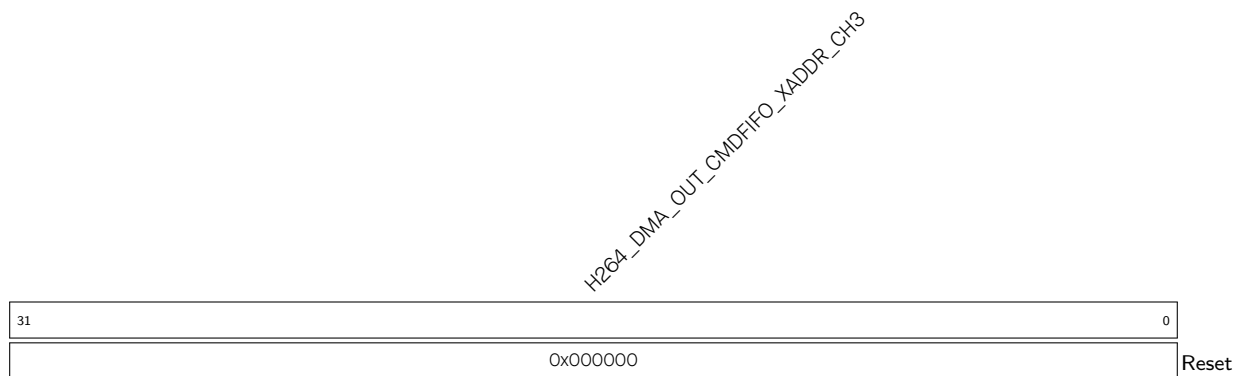
H264_DMA_OUT_CMDFIFO_OUTFIFO_BCNT_CH3 Represents that the data space has been used in FIFO. (RO)

Register 39.213. H264_DMA_OUT_PUSH_BYTECNT_CH3_REG (0x0378)



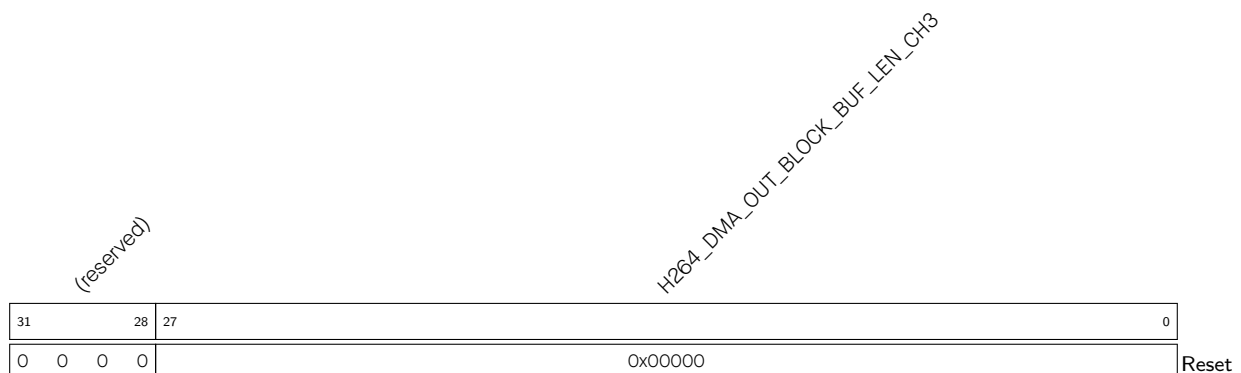
H264_DMA_OUT_CMDFIFO_PUSH_BYTECNT_CH3 Represents the data quantity in the current AXI request. (RO)

Register 39.214. H264_DMA_OUT_XADDR_CH3_REG (0x037C)



H264_DMA_OUT_CMDFIFO_XADDR_CH3 Represents the address of the current AXI request. (RO)

Register 39.215. H264_DMA_OUT_BLOCK_BUF_LEN_CH3_REG (0x0380)



H264_DMA_OUT_BLOCK_BUF_LEN_CH3 Represents the remaining block request data. (RO)

Register 39.216. H264_DMA_OUTFIFO_STATUS_CH4_REG (0x0414)

(reserved)																H264_DMA_OUTFIFO_CNT_L3_CH4				H264_DMA_OUTFIFO_EMPTY_L3_CH4				H264_DMA_OUTFIFO_FULL_L3_CH4				(reserved)																H264_DMA_OUTFIFO_CNT_L1_CH4				H264_DMA_OUTFIFO_EMPTY_L1_CH4				H264_DMA_OUTFIFO_FULL_L1_CH4				H264_DMA_OUTFIFO_CNT_L2_CH4				H264_DMA_OUTFIFO_EMPTY_L2_CH4				H264_DMA_OUTFIFO_FULL_L2_CH4			
31																20				19	18	17	16	15	13				12				8				7	6	5	2				1	0																						
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0																0x0				1	0	0	0	0					0x0				1	0	0x0				1	0	Reset																										

H264_DMA_OUTFIFO_FULL_L2_CH4 Represents whether I2 FIFO of TX channel 4 is full.

0: Not Full

1: Full

(RO)

H264_DMA_OUTFIFO_EMPTY_L2_CH4 Represents whether I2 FIFO of TX channel 4 is empty.

0: Not empty

1: Empty

(RO)

H264_DMA_OUTFIFO_CNT_L2_CH4 Represents the data quantity in I2 FIFO for TX channel 4. Measurement unit: 16 bytes. (RO)

H264_DMA_OUTFIFO_FULL_L1_CH4 Represents whether l1 FIFO of TX channel 4 is full.

0: Not Full

1: Full

(RO)

H264_DMA_OUTFIFO_EMPTY_L1_CH4 Represents whether l1 FIFO of TX channel 4 is empty.

0: Not empty

1: Empty

(RO)

H264_DMA_OUTFIFO_CNT_L1_CH4 Represents the data quantity in l1 FIFO for TX channel 4. Measurement unit: byte. (RO)

H264_DMA_OUTFIFO_FULL_L3_CH4 Represents whether L3 FIFO of TX channel 4 is full.

0: Not Full

1: Full

(RO)

H264_DMA_OUTFIFO_EMPTY_L3_CH4 Represents whether I3 FIFO of TX channel 4 is empty.

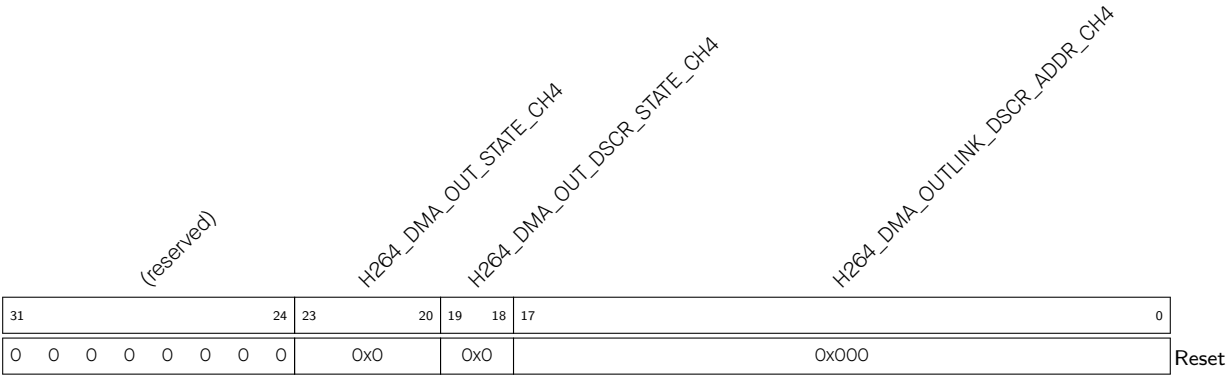
0: Not empty

1: Empty

(RO)

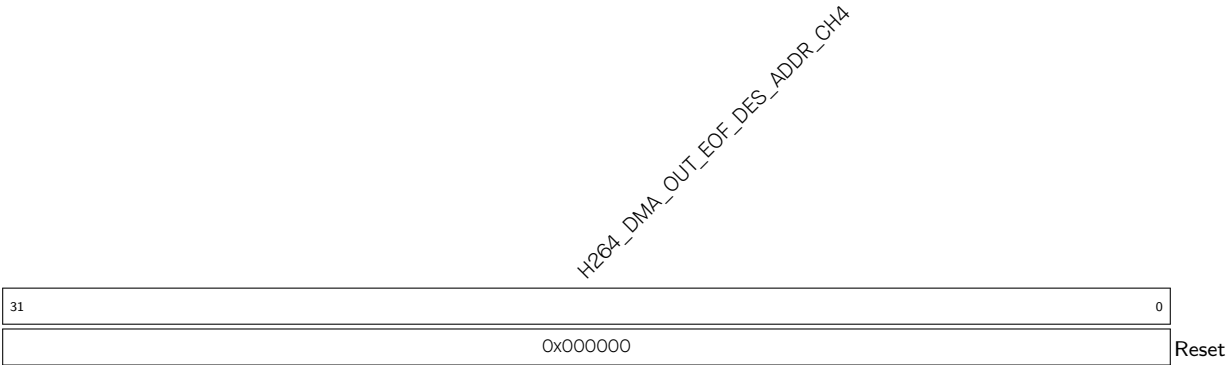
H264_DMA_OUTFIFO_CNT_L3_CH4 Represents the data quantity in I3 FIFO for TX channel 4. Measurement unit: 8 bytes. (RO)

Register 39.217. H264_DMA_OUT_STATE_CH4_REG (0x0424)



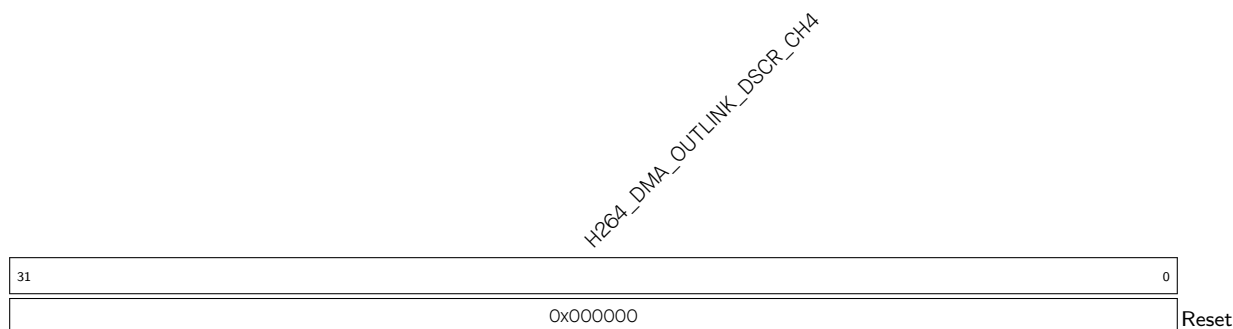
- H264_DMA_OUTLINK_DSCR_ADDR_CH4** Represents the current outlink descriptor’s address for TX channel 4. (RO)
- H264_DMA_OUT_DSCR_STATE_CH4** Represents the current state of the descriptor state machine for TX channel 4. (RO)
- H264_DMA_OUT_STATE_CH4** Represents the current control module state machine state for TX channel 4. (RO)

Register 39.218. H264_DMA_OUT_EOF_DES_ADDR_CH4_REG (0x0428)



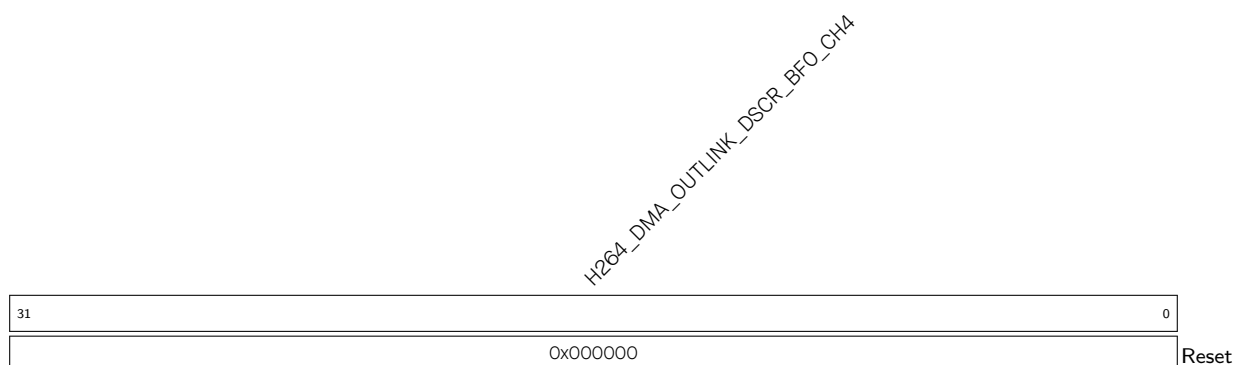
- H264_DMA_OUT_EOF_DES_ADDR_CH4** Represents the address of the inlink descriptor when the EOF bit in this descriptor is 1. (RO)

Register 39.219. H264_DMA_OUT_DSCR_CH4_REG (0x042C)



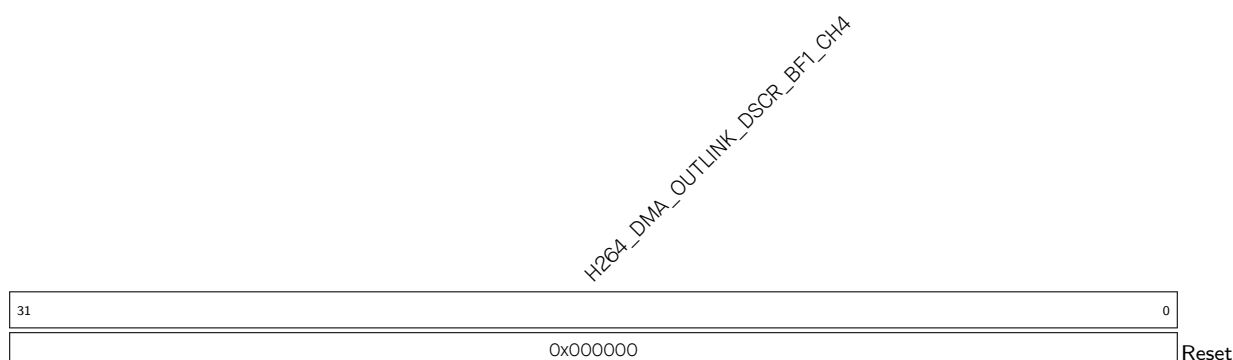
H264_DMA_OUTLINK_DSCR_CH4 Represents the address of the next descriptor y+1 pointed by the current descriptor that is pre-read. (RO)

Register 39.220. H264_DMA_OUT_DSCR_BFO_CH4_REG (0x0430)



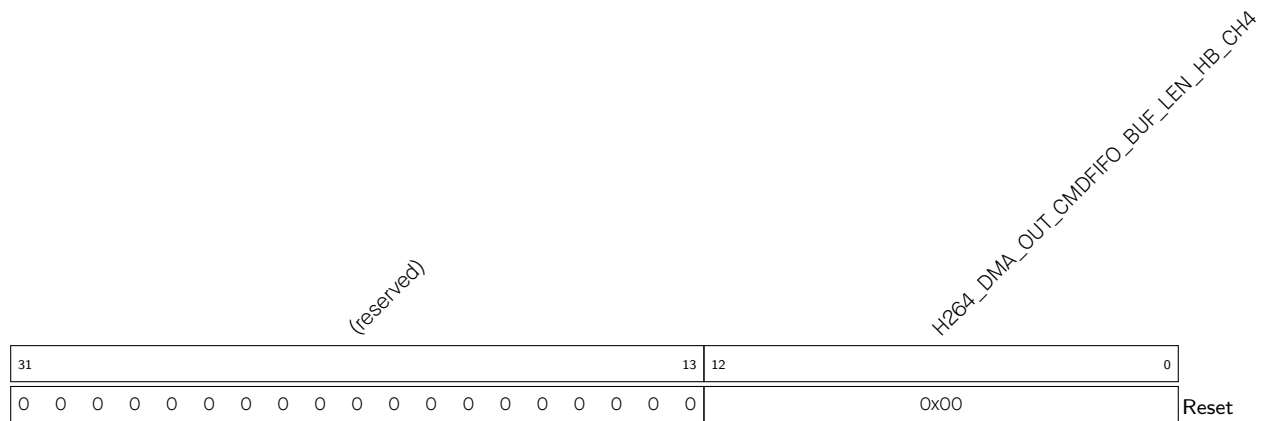
H264_DMA_OUTLINK_DSCR_BFO_CH4 Represents the address of the current descriptor y that is pre-read. (RO)

Register 39.221. H264_DMA_OUT_DSCR_BF1_CH4_REG (0x0434)



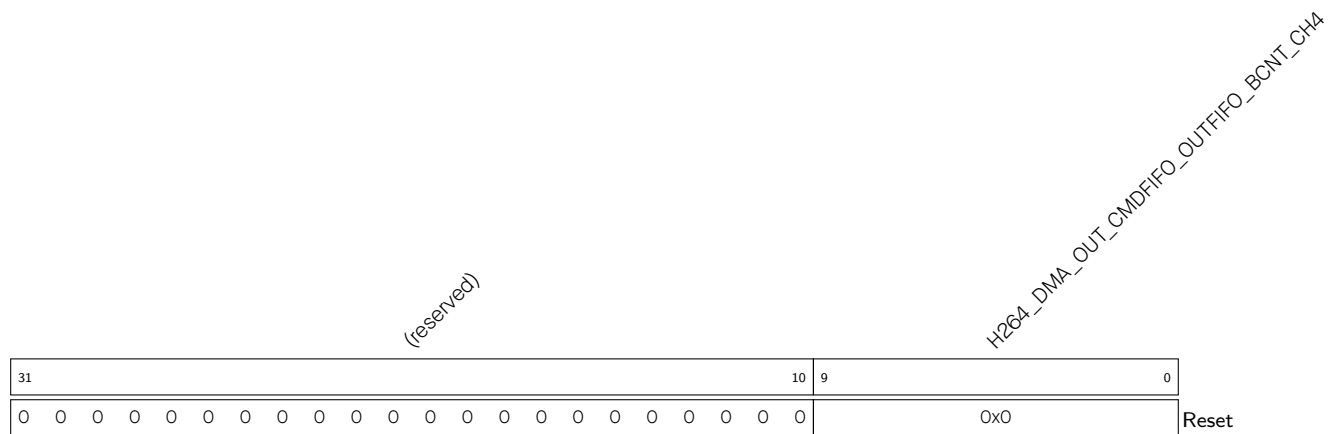
H264_DMA_OUTLINK_DSCR_BF1_CH4 Represents the address of the previous descriptor y-1 that is pre-read. (RO)

Register 39.222. H264_DMA_OUT_BUF_LEN_CH4_REG (0x0470)



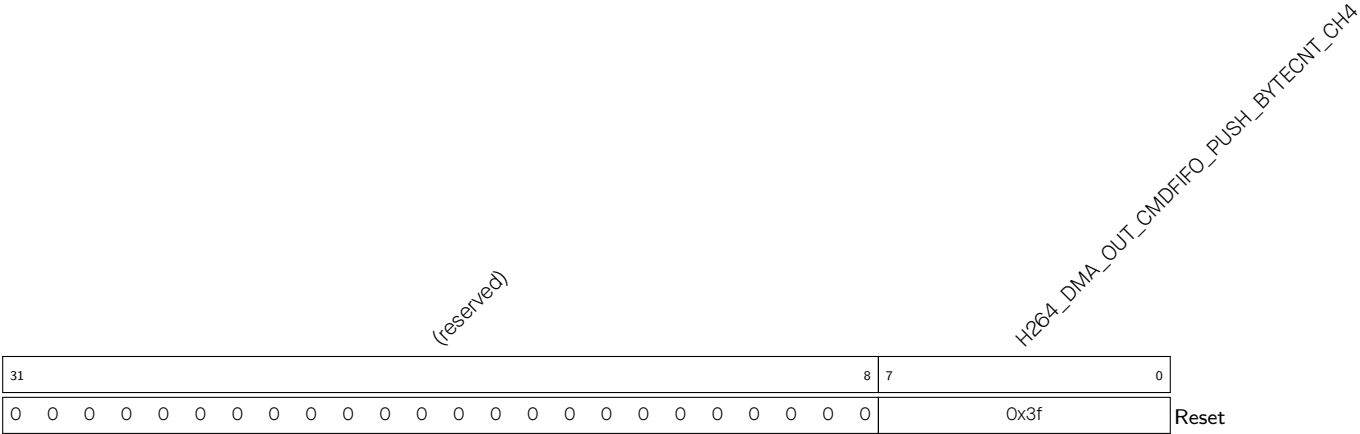
H264_DMA_OUT_CMDFIFO_BUF_LEN_HB_CH4 Represents the remaining request data. (RO)

Register 39.223. H264_DMA_OUT_FIFO_BCNT_CH4_REG (0x0474)



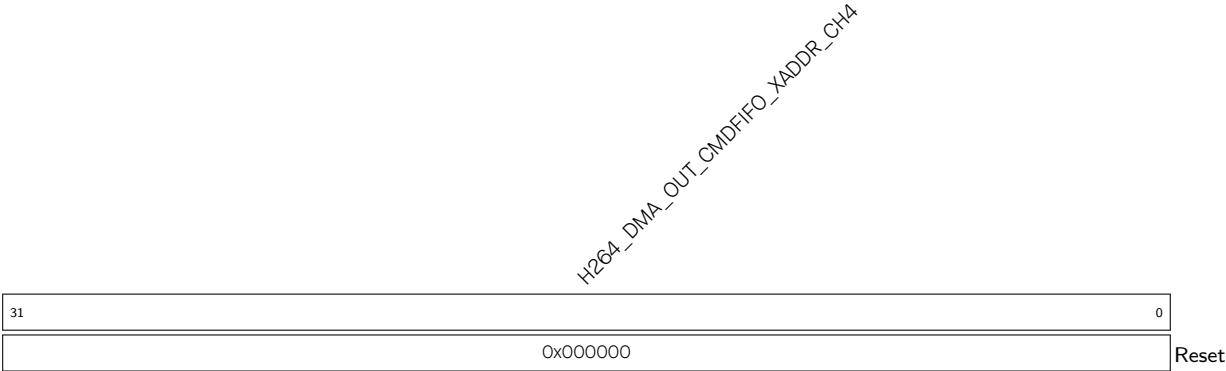
H264_DMA_OUT_CMDFIFO_OUTFIFO_BCNT_CH4 Represents that the data space has been used in FIFO. (RO)

Register 39.224. H264_DMA_OUT_PUSH_BYTECNT_CH4_REG (0x0478)



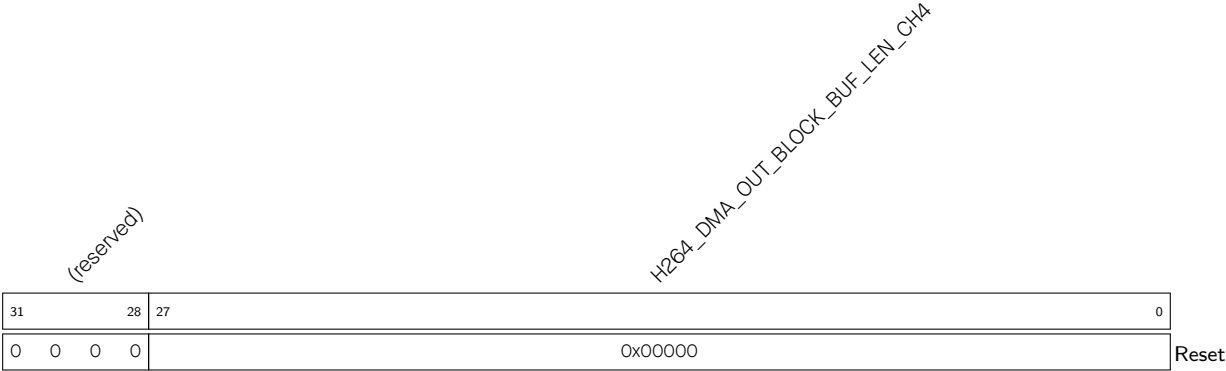
H264_DMA_OUT_CMDFIFO_PUSH_BYTECNT_CH4 Represents the data quantity in the current AXI request. (RO)

Register 39.225. H264_DMA_OUT_XADDR_CH4_REG (0x047C)



H264_DMA_OUT_CMDFIFO_XADDR_CH4 Represents the address of the current AXI request. (RO)

Register 39.226. H264_DMA_OUT_BLOCK_BUF_LEN_CH4_REG (0x0480)



H264_DMA_OUT_BLOCK_BUF_LEN_CH4 Represents the remaining block request data. (RO)

Register 39.227. H264_DMA_INFIFO_STATUS_CHO_REG (0x0514)

(reserved)												H264_DMA_INFIFO_CNT_L3_CHO				H264_DMA_INFIFO_EMPTY_L3_CHO				H264_DMA_INFIFO_FULL_L3_CHO				(reserved)				H264_DMA_INFIFO_CNT_L1_CHO				H264_DMA_INFIFO_EMPTY_L1_CHO				H264_DMA_INFIFO_FULL_L1_CHO				H264_DMA_INFIFO_CNT_L2_CHO				H264_DMA_INFIFO_EMPTY_L2_CHO				H264_DMA_INFIFO_FULL_L2_CHO			
31												20												19	18	17	16	15	13			12	8			7	6	5	2			1	0								
0 0 0 0 0 0 0 0 0 0 0 0												0x0												1	0	0	0	0	0x0			1	0	0x0			1	0	Reset												

Reset

H264_DMA_INFIFO_FULL_L2_CHO Represents whether l2 FIFO of RX channel 0 is full.

0: Not Full

1: Full

(RO)

H264_DMA_INFIFO_EMPTY_L2_CHO Represents whether l2 FIFO of RX channel 0 is empty.

0: Not empty

1: Empty

(RO)

H264_DMA_INFIFO_CNT_L2_CHO Represents the data quantity in l2 FIFO for RX channel 0. Measurement unit: 16 bytes. (RO)**H264_DMA_INFIFO_FULL_L1_CHO** Represents whether l1 FIFO of RX channel 0 is full.

0: Not Full

1: Full

(RO)

H264_DMA_INFIFO_EMPTY_L1_CHO Represents whether l1 FIFO of RX channel 0 is empty.

0: Not empty

1: Empty

(RO)

H264_DMA_INFIFO_CNT_L1_CHO Represents the data quantity in l1 FIFO for RX channel 0. Measurement unit: byte. (RO)**H264_DMA_INFIFO_FULL_L3_CHO** Represents whether l3 FIFO of RX channel 0 is full.

0: Not Full

1: Full

(RO)

H264_DMA_INFIFO_EMPTY_L3_CHO Represents whether l3 FIFO of RX channel 0 is empty.

0: Not empty

1: Empty

(RO)

H264_DMA_INFIFO_CNT_L3_CHO Represents the data quantity in l3 FIFO for RX channel 0. Measurement unit: 8 bytes. (RO)

Register 39.228. H264_DMA_IN_STATE_CHO_REG (0x0524)

(reserved)								H264_DMA_IN_RESET_AVAIL_CHO								H264_DMA_IN_STATE_CHO								H264_DMA_IN_DSCR_STATE_CHO								H264_DMA_INLINK_DSCR_ADDR_CHO										
31								24								23	22				20				19	18	17								0							
0 0 0 0 0 0 0 0								1								0x0				0x0				0x000								Reset										

H264_DMA_INLINK_DSCR_ADDR_CHO Represents the current inlink descriptor's address for RX channel 0. (RO)

H264_DMA_IN_DSCR_STATE_CHO Represents the current state of the descriptor state machine for RX channel 0. (RO)

H264_DMA_IN_STATE_CHO Represents the current control module state machine state for RX channel 0. (RO)

H264_DMA_IN_RESET_AVAIL_CHO Represents whether it is safe to reset the channel.

0: Unsafe

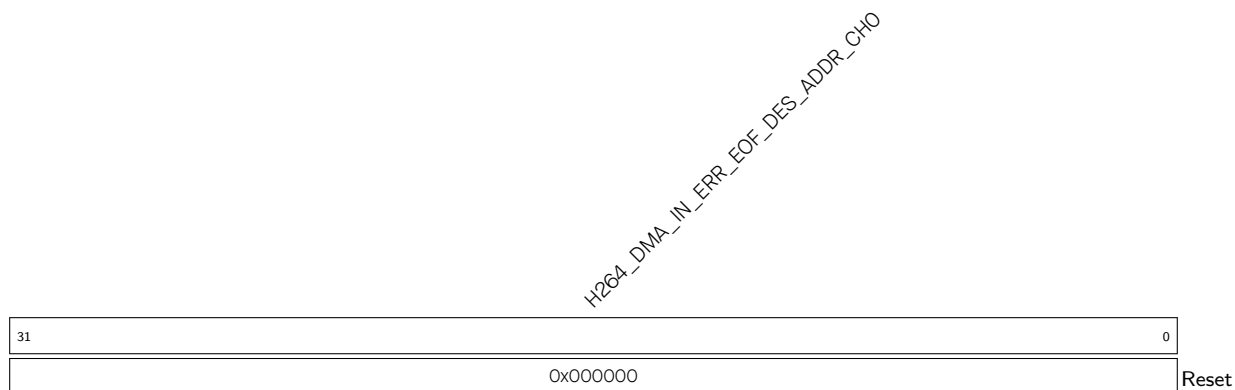
1: Safe

(RO)

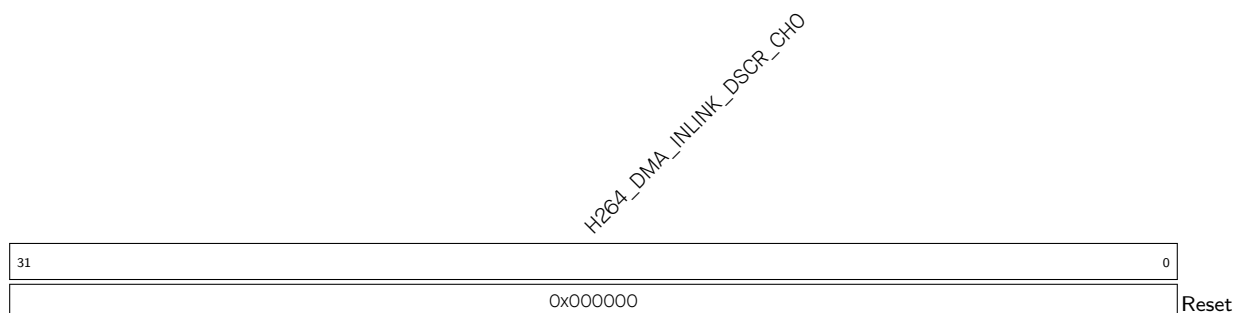
Register 39.229. H264_DMA_IN_SUC_EOF_DES_ADDR_CHO_REG (0x0528)

H264_DMA_IN_SUC_EOF_DES_ADDR_CHO																															
31																															0
0x000000																															
Reset																															

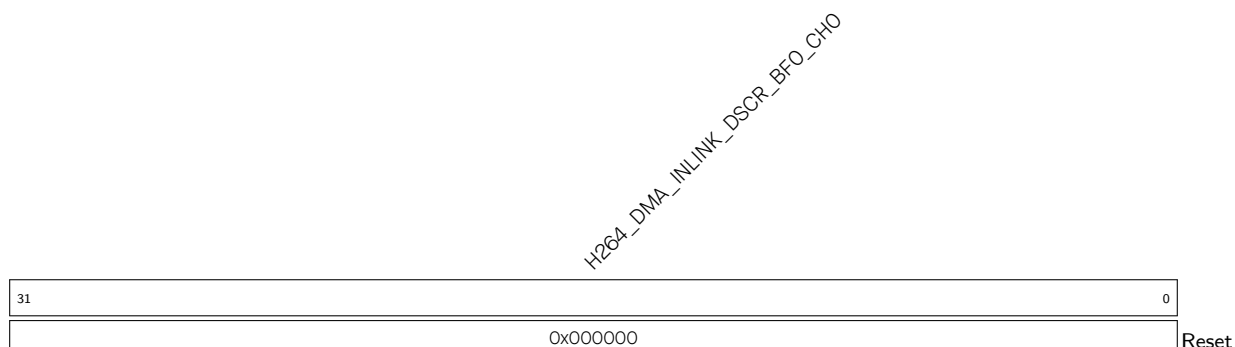
H264_DMA_IN_SUC_EOF_DES_ADDR_CHO Represents the address of the inlink descriptor when the EOF bit in this descriptor is 1. (RO)

Register 39.230. H264_DMA_IN_ERR_EOF_DES_ADDR_CHO_REG (0x052C)

H264_DMA_IN_ERR_EOF_DES_ADDR_CHO Represents the address of the inlink descriptor when there are errors during the current data reception. (RO)

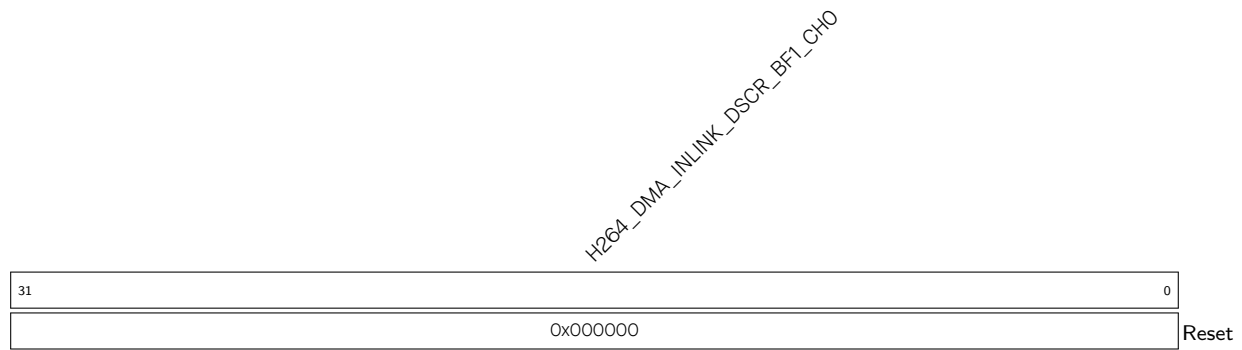
Register 39.231. H264_DMA_IN_DSCR_CHO_REG (0x0530)

H264_DMA_INLINK_DSCR_CHO Represents the address of the next descriptor x+1 pointed by the current descriptor that is pre-read. (RO)

Register 39.232. H264_DMA_IN_DSCR_BFO_CHO_REG (0x0534)

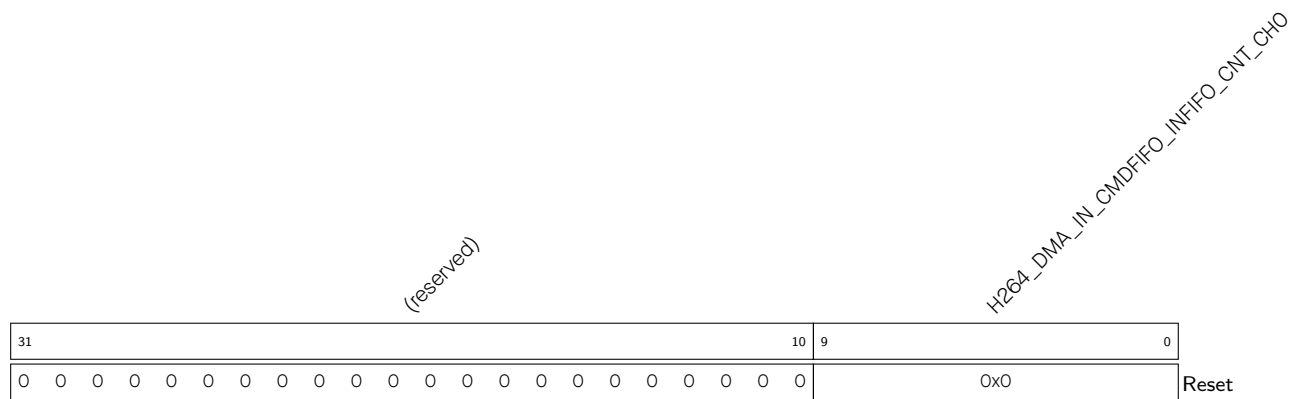
H264_DMA_INLINK_DSCR_BFO_CHO Represents the address of the current descriptor x that is pre-read. (RO)

Register 39.233. H264_DMA_IN_DSCR_BF1_CHO_REG (0x0538)



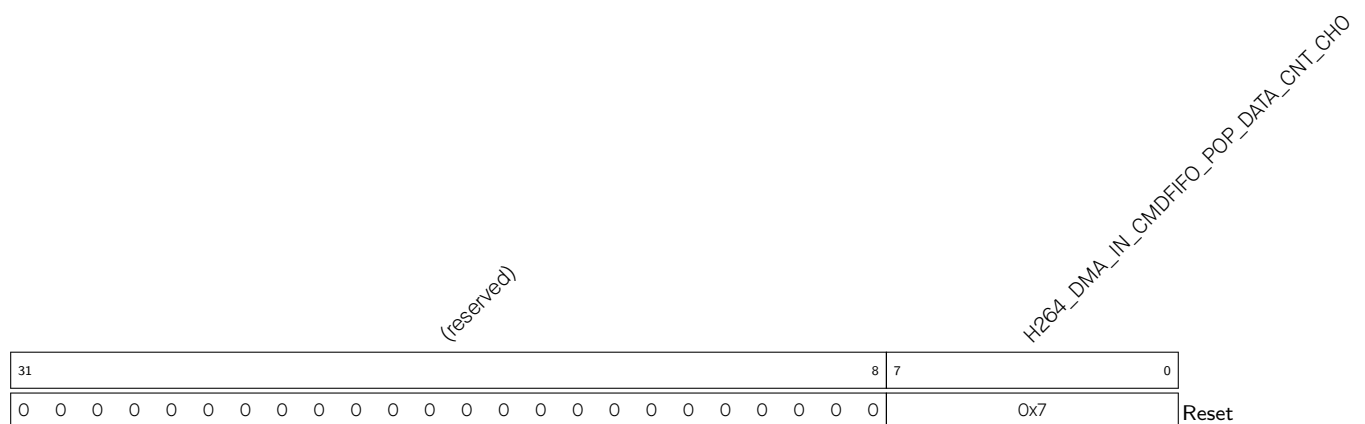
H264_DMA_INLINK_DSCR_BF1_CHO	Represents the address of the previous descriptor x-1 that is pre-read. (RO)
-------------------------------------	--

Register 39.234. H264_DMA_IN_FIFO_CNT_CHO_REG (0x0580)



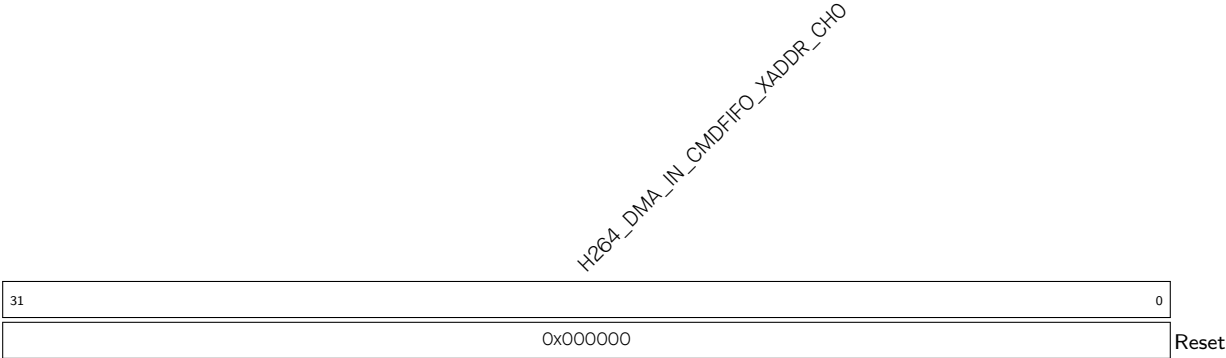
H264_DMA_IN_CMDFIFO_INFIFO_CNT_CHO Represents the remaining request data in FIFO. (RO)

Register 39.235. H264_DMA_IN_POP_DATA_CNT_CHO_REG (0x0584)



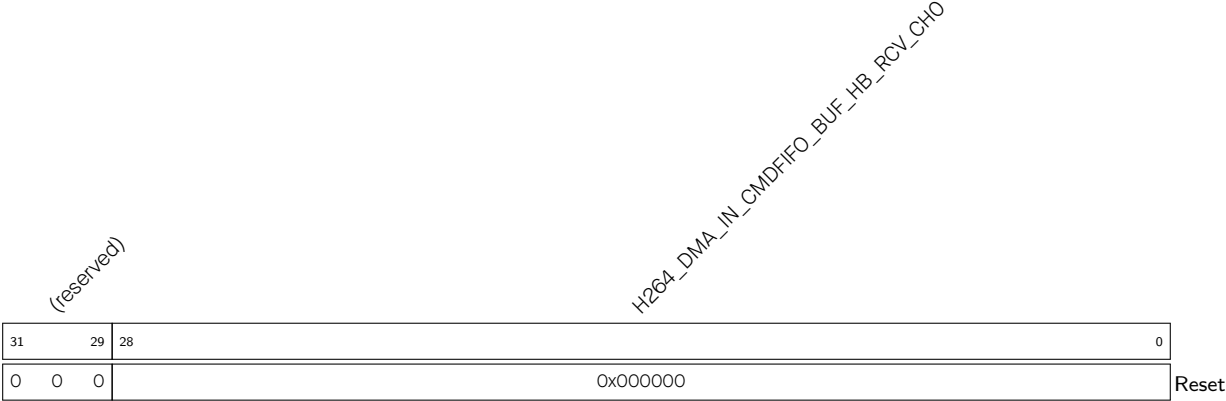
H264_DMA_IN_CMDFIFO_POP_DATA_CNT_CHO Represents the data quantity in the current AXI request. (RO)

Register 39.236. H264_DMA_IN_XADDR_CHO_REG (0x0588)



H264_DMA_IN_CMDFIFO_XADDR_CHO Represents the address of the current AXI request. (RO)

Register 39.237. H264_DMA_IN_BUF_HB_RCV_CHO_REG (0x058C)



H264_DMA_IN_CMDFIFO_BUF_HB_RCV_CHO Represents the data quantity that has been requested. (RO)

Register 39.238. H264_DMA_INFIFO_STATUS_CH1_REG (0x0614)

(reserved)												H264_DMA_INFIFO_CNT_L3_CH1				H264_DMA_INFIFO_EMPTY_L3_CH1				H264_DMA_INFIFO_FULL_L3_CH1				(reserved)				H264_DMA_INFIFO_CNT_L1_CH1				H264_DMA_INFIFO_EMPTY_L1_CH1				H264_DMA_INFIFO_FULL_L1_CH1				H264_DMA_INFIFO_CNT_L2_CH1				H264_DMA_INFIFO_EMPTY_L2_CH1				H264_DMA_INFIFO_FULL_L2_CH1			
31												20		19	18	17	16	15	13		12	8		7	6	5	2		1	0	Reset																				
0 0 0 0 0 0 0 0 0 0 0 0												0x0		1	0	0	0	0	0x0		1	0	0x0		1	0																									

Reset

H264_DMA_INFIFO_FULL_L2_CH1 Represents whether I2 FIFO of RX channel 1 is full.

0: Not Full

1: Full

(RO)

H264_DMA_INFIFO_EMPTY_L2_CH1 Represents whether I2 FIFO of RX channel 1 is empty.

0: Not empty

1: Empty

(RO)

H264_DMA_INFIFO_CNT_L2_CH1 Represents the data quantity in I2 FIFO for RX channel 1. Measurement unit: 16 bytes. (RO)

H264_DMA_INFIFO_FULL_L1_CH1 Represents whether I1 FIFO of RX channel 1 is full.

0: Not Full

1: Full

(RO)

H264_DMA_INFIFO_EMPTY_L1_CH1 Represents whether I1 FIFO of RX channel 1 is empty.

0: Not empty

1: Empty

(RO)

H264_DMA_INFIFO_CNT_L1_CH1 Represents the data quantity in I1 FIFO for RX channel 1. Measurement unit: byte. (RO)

H264_DMA_INFIFO_FULL_L3_CH1 Represents whether I3 FIFO of RX channel 1 is full.

0: Not Full

1: Full

(RO)

H264_DMA_INFIFO_EMPTY_L3_CH1 Represents whether I3 FIFO of RX channel 1 is empty.

0: Not empty

1: Empty

(RO)

H264_DMA_INFIFO_CNT_L3_CH1 Represents the data quantity in I3 FIFO for RX channel 1. Measurement unit: 8 bytes. (RO)

Register 39.239. H264_DMA_IN_STATE_CH1_REG (0x0624)

(reserved)								H264_DMA_IN_RESET_AVAIL_CH1				H264_DMA_IN_STATE_CH1				H264_DMA_IN_DSCR_STATE_CH1				H264_DMA_INLINK_DSCR_ADDR_CH1																					
31								24								23	22	20				19	18	17																	0
0								0	0	0	0	0	0	0	1	0x0				0x0				0x000																Reset	

H264_DMA_INLINK_DSCR_ADDR_CH1 Represents the current inlink descriptor's address for RX channel 1. (RO)

H264_DMA_IN_DSCR_STATE_CH1 Represents the current state of the descriptor state machine for RX channel 1. (RO)

H264_DMA_IN_STATE_CH1 Represents the current control module state machine state for RX channel 1. (RO)

H264_DMA_IN_RESET_AVAIL_CH1 Represents whether it is safe to reset the channel.

0: Unsafe

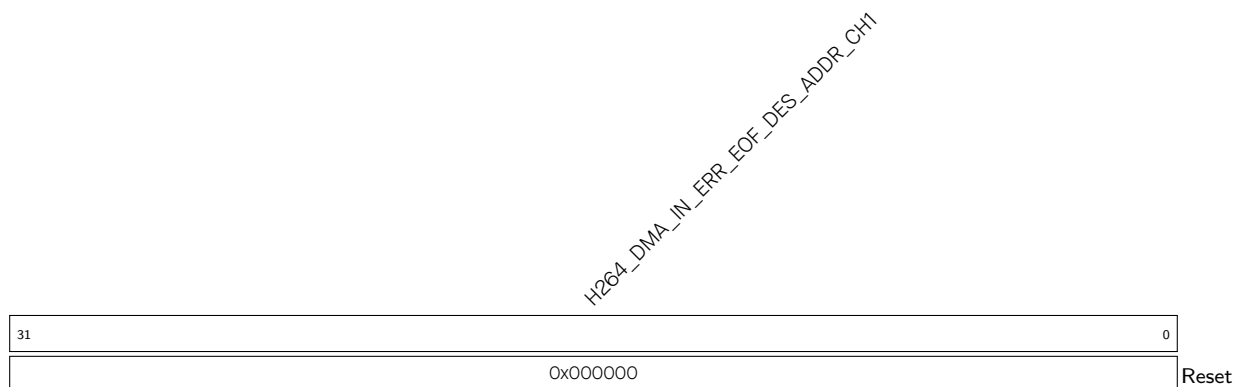
1: Safe

(RO)

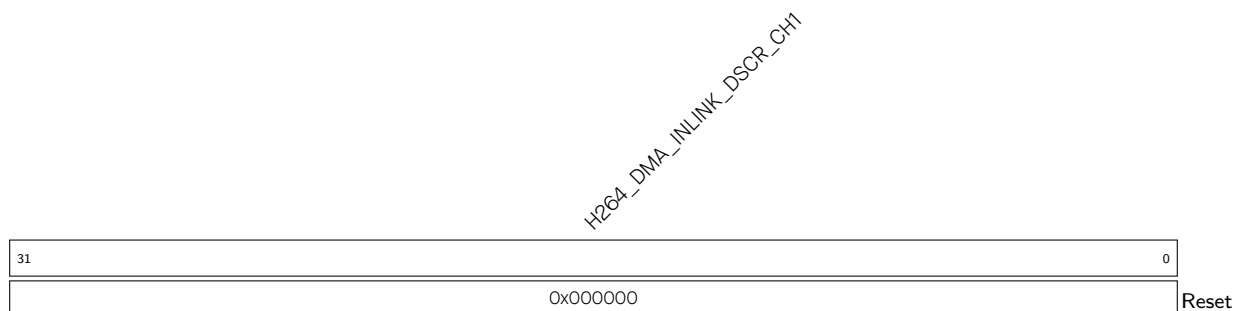
Register 39.240. H264_DMA_IN_SUC_EOF_DES_ADDR_CH1_REG (0x0628)

Diagram illustrating the H264 DMA_IN_SUC_EOF_DES_ADDR_CH1 register structure. The register is 32 bits wide, with bit positions 31 and 0 indicated. The reset value is 0x00000000.

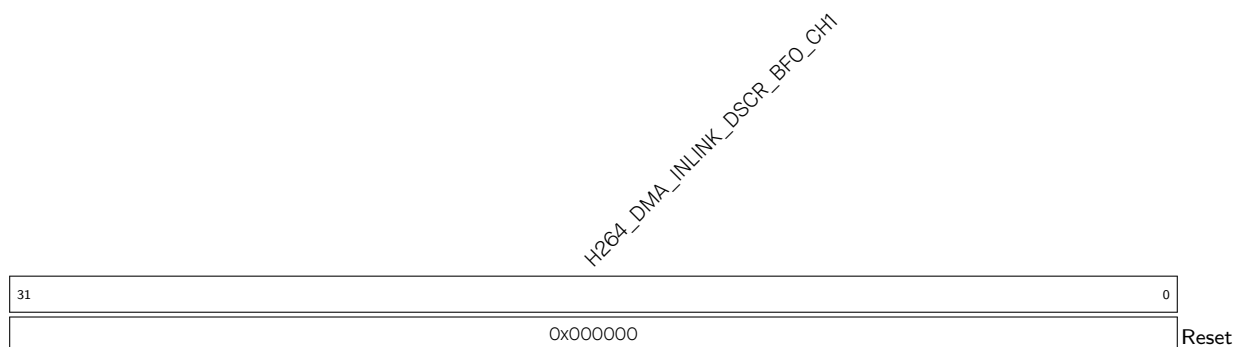
H264_DMA_IN_SUC_EOF_DES_ADDR_CH1 Represents the address of the inlink descriptor when the EOF bit in this descriptor is 1. (RO)

Register 39.241. H264_DMA_IN_ERR_EOF_DES_ADDR_CH1_REG (0x062C)

H264_DMA_IN_ERR_EOF_DES_ADDR_CH1 Represents the address of the inlink descriptor when there are errors during the current data reception. (RO)

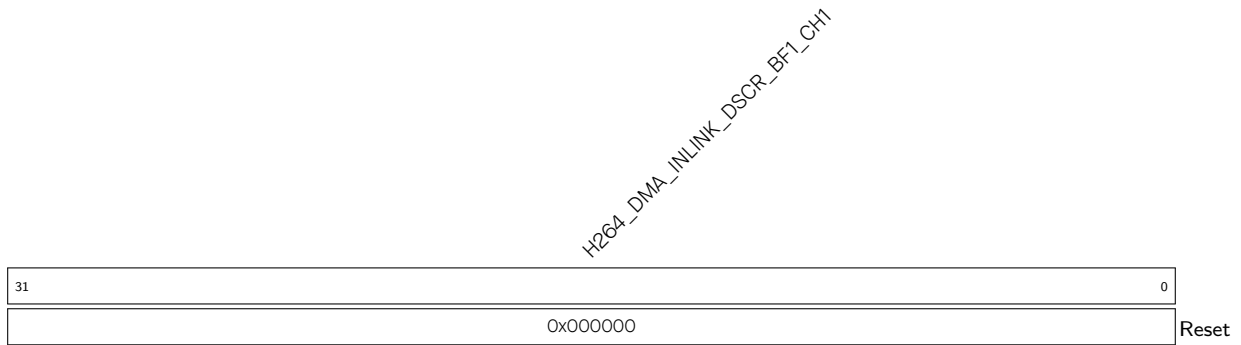
Register 39.242. H264_DMA_IN_DSCR_CH1_REG (0x0630)

H264_DMA_INLINK_DSCR_CH1 Represents the address of the next descriptor x+1 pointed by the current descriptor that is pre-read. (RO)

Register 39.243. H264_DMA_IN_DSCR_BFO_CH1_REG (0x0634)

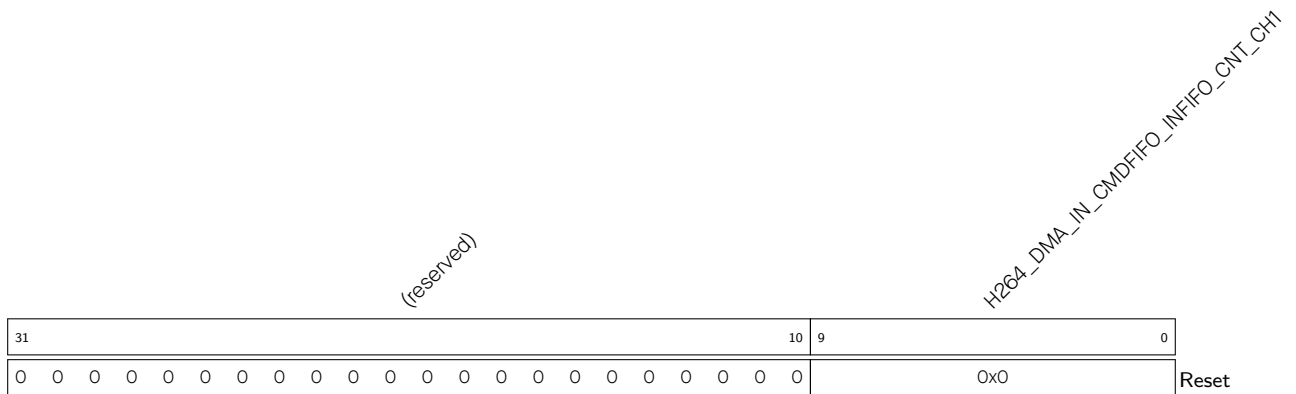
H264_DMA_INLINK_DSCR_BFO_CH1 Represents the address of the current descriptor x that is pre-read. (RO)

Register 39.244. H264_DMA_IN_DSCR_BF1_CH1_REG (0x0638)



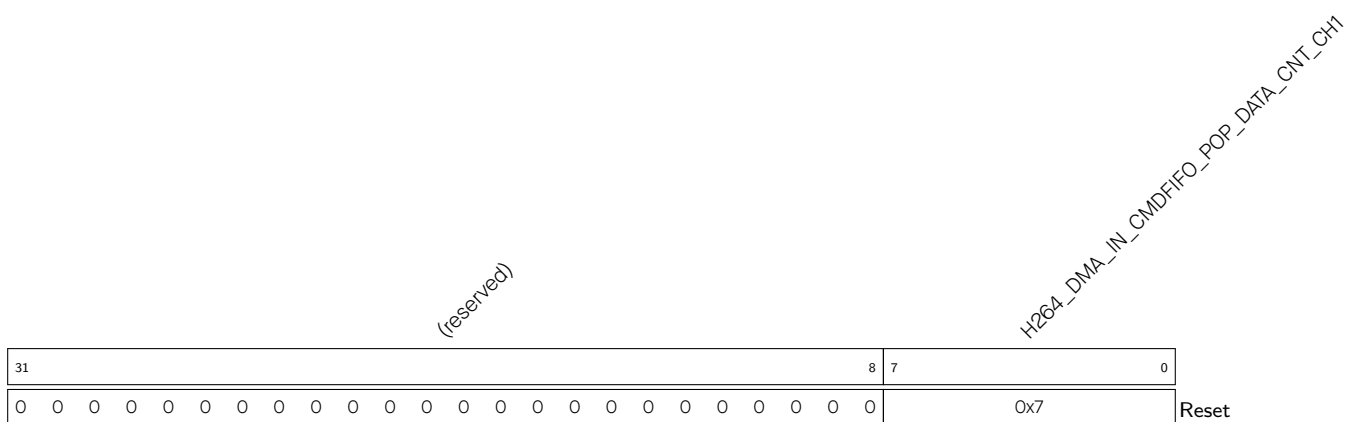
H264_DMA_INLINK_DSCR_BF1_CH1 Represents the address of the previous descriptor x-1 that is pre-read. (RO)

Register 39.245. H264_DMA_IN_FIFO_CNT_CH1_REG (0x0680)



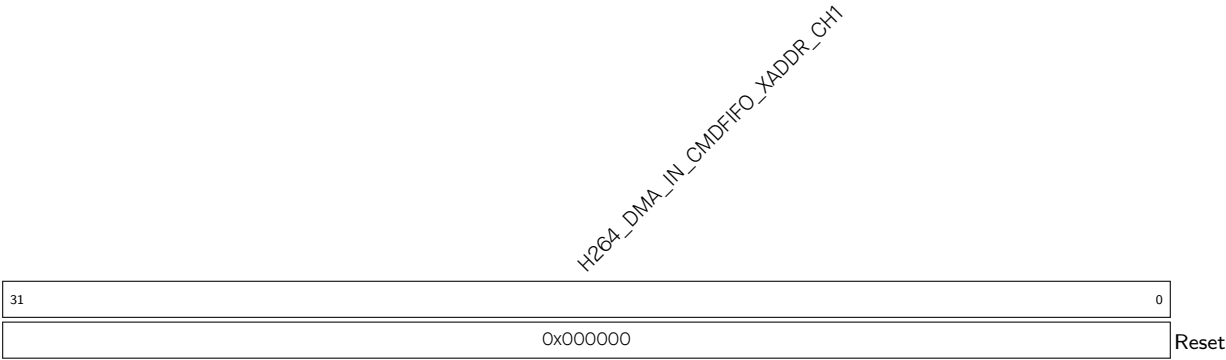
H264_DMA_IN_CMDFIFO_INFIFO_CNT_CH1 Represents the remaining request data in FIFO. (RO)

Register 39.246. H264_DMA_IN_POP_DATA_CNT_CH1_REG (0x0684)



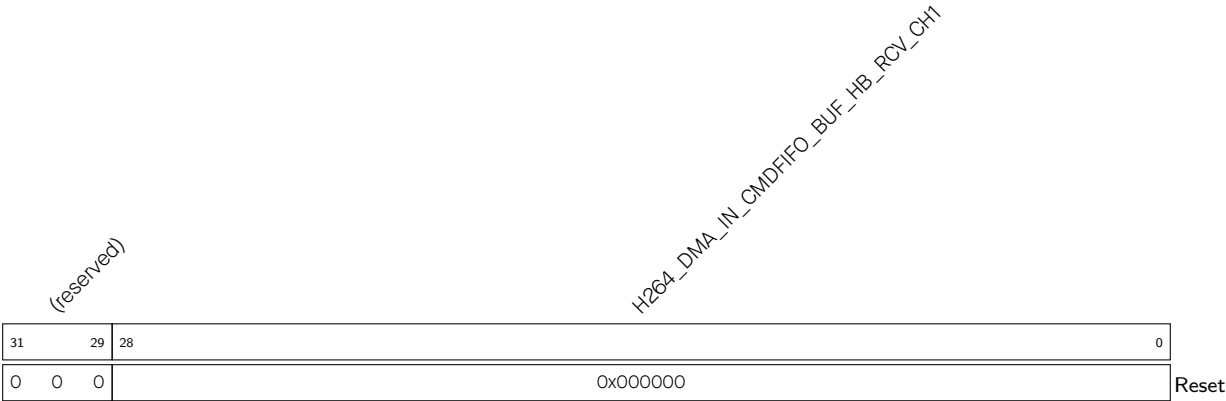
H264_DMA_IN_CMDFIFO_POP_DATA_CNT_CH1 Represents the data quantity in the current AXI request. (RO)

Register 39.247. H264_DMA_IN_XADDR_CH1_REG (0x0688)



H264_DMA_IN_CMDFIFO_XADDR_CH1 Represents the address of the current AXI request. (RO)

Register 39.248. H264_DMA_IN_BUF_HB_RCV_CH1_REG (0x068C)



H264_DMA_IN_CMDFIFO_BUF_HB_RCV_CH1 Represents the data quantity that has been requested. (RO)

Register 39.249. H264_DMA_INFIFO_STATUS_CH2_REG (0x0714)

(reserved)																H264_DMA_INFIFO_CNT_L3_CH2				H264_DMA_INFIFO_EMPTY_L3_CH2				H264_DMA_INFIFO_FULL_L3_CH2				(reserved)				H264_DMA_INFIFO_CNT_L1_CH2				H264_DMA_INFIFO_EMPTY_L1_CH2				H264_DMA_INFIFO_FULL_L1_CH2				H264_DMA_INFIFO_CNT_L2_CH2				H264_DMA_INFIFO_EMPTY_L2_CH2				H264_DMA_INFIFO_FULL_L2_CH2			
31												20				19	18	17	16	15	13				12	8				7	6	5	2				1	0																	
0												0				0	0	0	0	0	0	0x0				1	0	0	0	0	0x0				1	0	0x0				1	0	Reset												

H264_DMA_INFIFO_FULL_L2_CH2 Represents whether l2 FIFO of RX channel 2 is full.

0: Not Full

1: Full

(RO)

H264_DMA_INFIFO_EMPTY_L2_CH2 Represents whether l2 FIFO of RX channel 2 is empty.

0: Not empty

1: Empty

(RO)

H264_DMA_INFIFO_CNT_L2_CH2	Represents the data quantity in I2 FIFO for RX channel 2. Measurement unit: 16 bytes. (RO)
-----------------------------------	--

H264_DMA_INFIFO_FULL_L1_CH2 Represents whether l1 FIFO of RX channel 2 is full.

0: Not Full

1: Full

(RO)

H264_DMA_INFIFO_EMPTY_L1_CH2 Represents whether l1 FIFO of RX channel 2 is empty.

0: Not empty

1: Empty

(RO)

H264_DMA_INFIFO_CNT_L1_CH2 Represents the data quantity in I1 FIFO for RX channel 2. Measurement unit: byte. (RO)

H264 DMA INFIFO FULL L3 CH2 Represents whether L3 FIFO of RX channel 2 is full.

0: Not Full

1: Full

(RO)

H264_DMA_INFIFO_EMPTY_L3_CH2 Represents whether I3 FIFO of RX channel 2 is empty.

0: Not empty

1: Empty

(RO)

H264_DMA_INFIFO_CNT_L3_CH2 Represents the data quantity in I3 FIFO for RX channel 2. Measurement unit: 8 bytes. (RO)

Register 39.250. H264_DMA_IN_STATE_CH2_REG (0x0724)

(reserved)								H264_DMA_IN_RESET_AVAIL_CH2								H264_DMA_IN_STATE_CH2								H264_DMA_IN_DSCR_STATE_CH2								H264_DMA_INLINK_DSCR_ADDR_CH2							
31								24	23	22	20	19	18	17																	0								
0	0	0	0	0	0	0	0	1	0x0		0x0		0x000																Reset										

Reset

H264_DMA_INLINK_DSCR_ADDR_CH2 Represents the current inlink descriptor's address for RX channel 2. (RO)

H264_DMA_IN_DSCR_STATE_CH2 Represents the current state of the descriptor state machine for RX channel 2. (RO)

H264_DMA_IN_STATE_CH2 Represents the current control module state machine state for RX channel 2. (RO)

H264_DMA_IN_RESET_AVAIL_CH2 Represents whether it is safe to reset the channel.

0: Unsafe

1: Safe

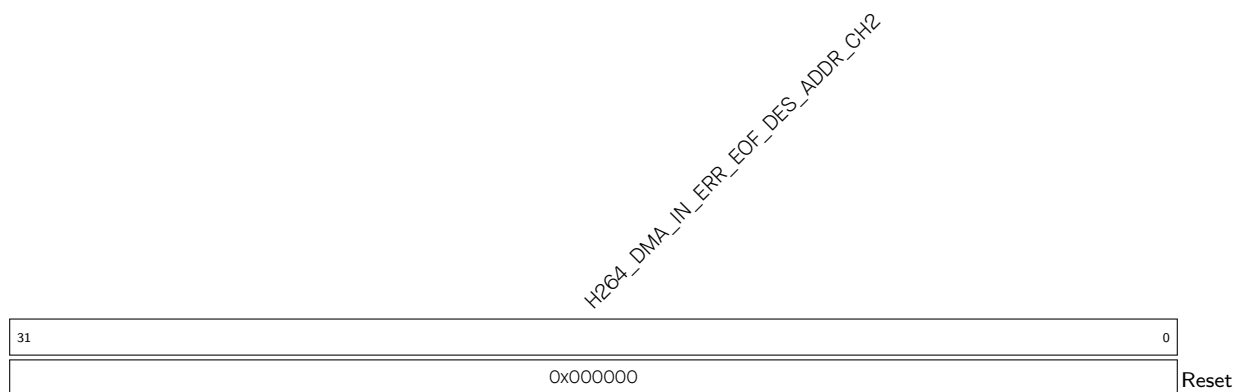
(RO)

Register 39.251. H264_DMA_IN_SUC_EOF_DES_ADDR_CH2_REG (0x0728)

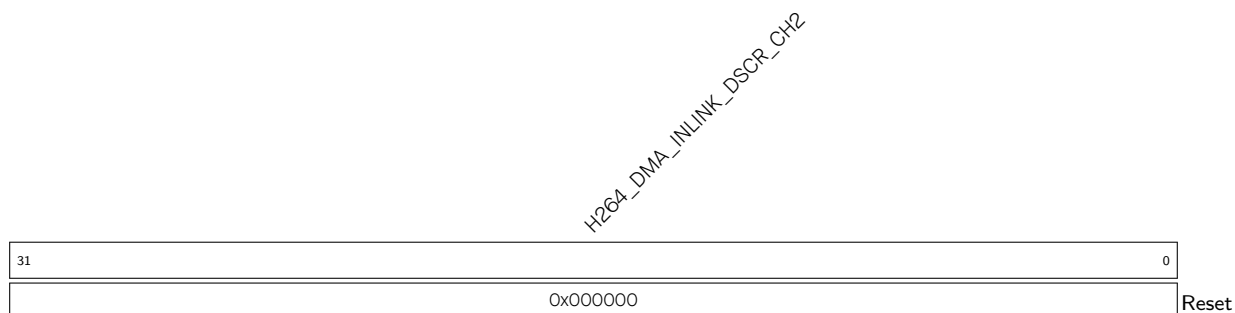
H264_DMA_IN_SUC_EOF_DES_ADDR_CH2																															
31																															0
0x0000000																															
Reset																															

Reset

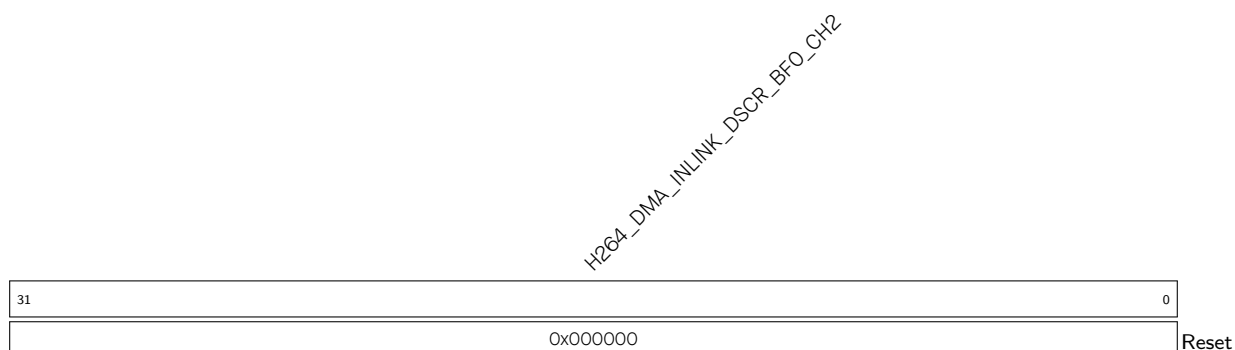
H264_DMA_IN_SUC_EOF_DES_ADDR_CH2 Represents the address of the inlink descriptor when the EOF bit in this descriptor is 1. (RO)

Register 39.252. H264_DMA_IN_ERR_EOF_DES_ADDR_CH2_REG (0x072C)

H264_DMA_IN_ERR_EOF_DES_ADDR_CH2 Represents the address of the inlink descriptor when there are errors during the current data reception. (RO)

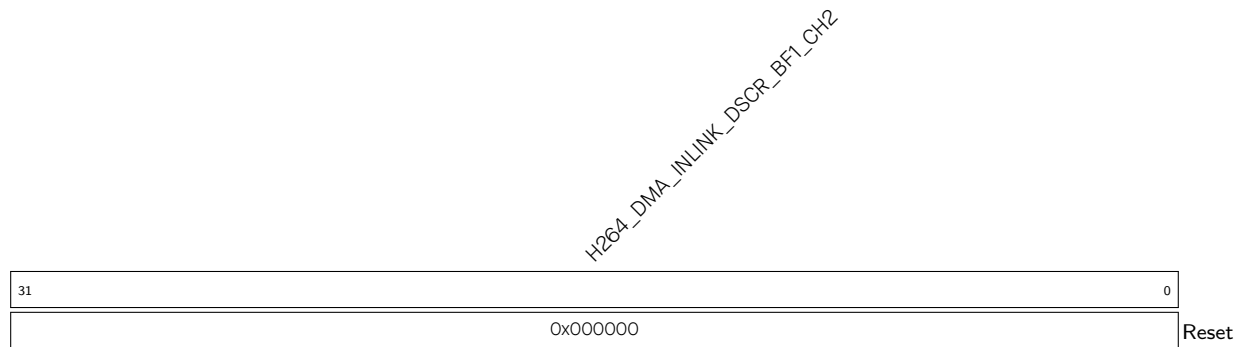
Register 39.253. H264_DMA_IN_DSCR_CH2_REG (0x0730)

H264_DMA_INLINK_DSCR_CH2 Represents the address of the next descriptor x+1 pointed by the current descriptor that is pre-read. (RO)

Register 39.254. H264_DMA_IN_DSCR_BFO_CH2_REG (0x0734)

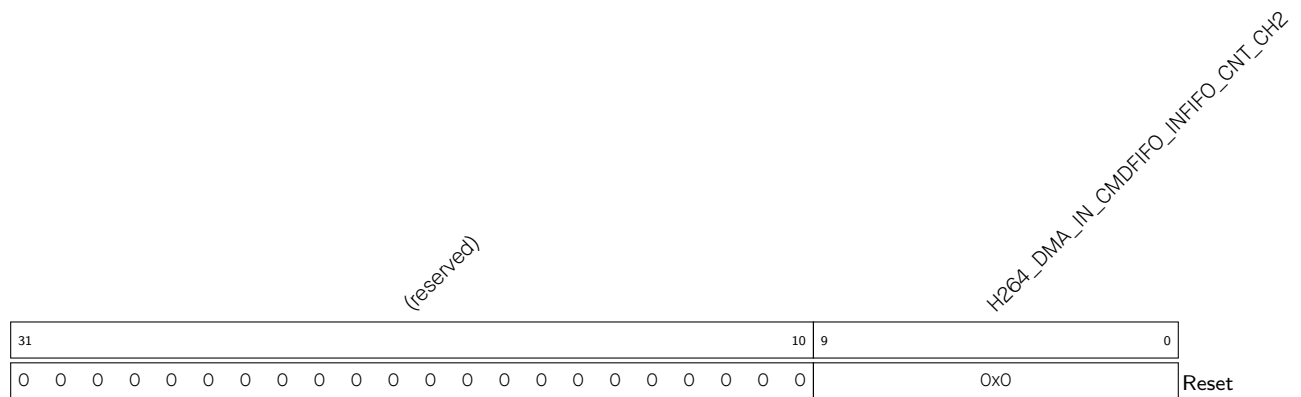
H264_DMA_INLINK_DSCR_BFO_CH2 Represents the address of the current descriptor x that is pre-read. (RO)

Register 39.255. H264_DMA_IN_DSCR_BF1_CH2_REG (0x0738)



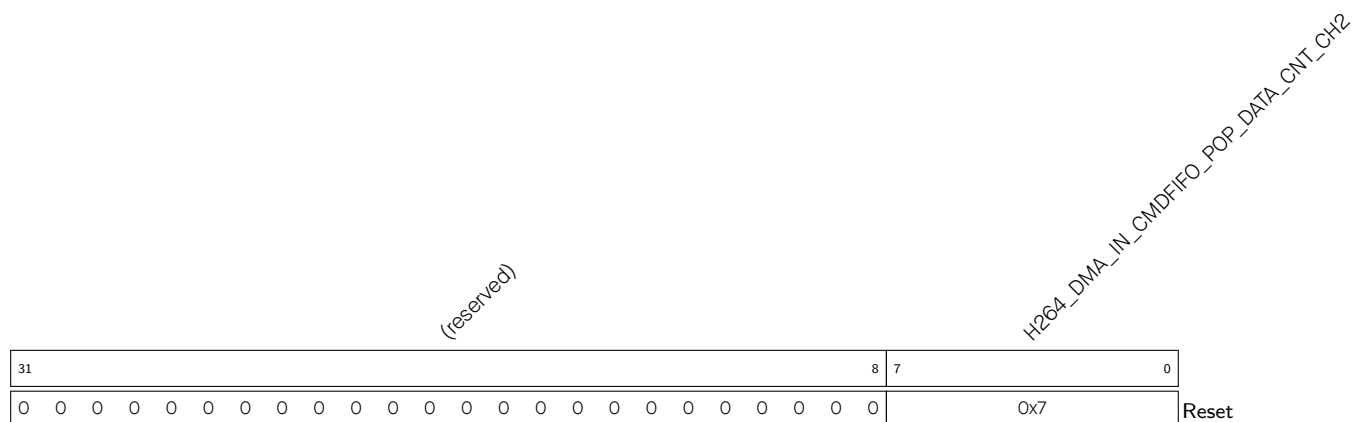
H264_DMA_INLINK_DSCR_BF1_CH2	Represents the address of the previous descriptor x-1 that is pre-read. (RO)
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Register 39.256. H264_DMA_IN_FIFO_CNT_CH2_REG (0x0780)



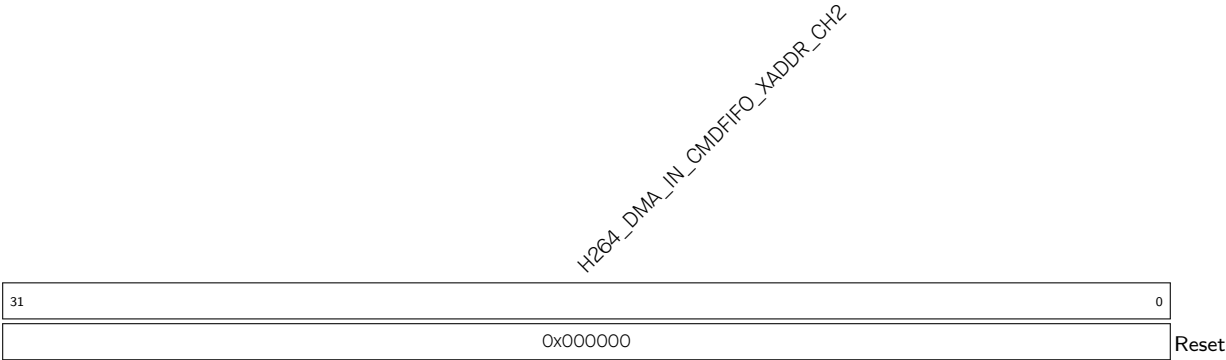
H264_DMA_IN_CMDFIFO_INFIFO_CNT_CH2 Represents the remaining request data in FIFO. (RO)

Register 39.257. H264_DMA_IN_POP_DATA_CNT_CH2_REG (0x0784)



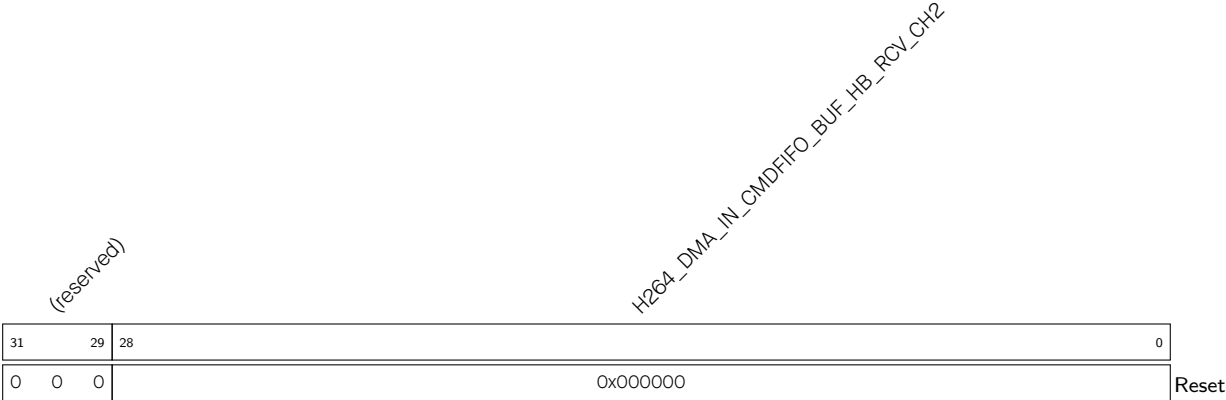
H264_DMA_IN_CMDFIFO_POP_DATA_CNT_CH2 Represents the data quantity in the current AXI request. (RO)

Register 39.258. H264_DMA_IN_XADDR_CH2_REG (0x0788)



H264_DMA_IN_CMDFIFO_XADDR_CH2 Represents the address of the current AXI request. (RO)

Register 39.259. H264_DMA_IN_BUF_HB_RCV_CH2_REG (0x078C)



H264_DMA_IN_CMDFIFO_BUF_HB_RCV_CH2 Represents the data quantity that has been requested. (RO)

Register 39.260. H264_DMA_INFIFO_STATUS_CH3_REG (0x0814)

(reserved)												H264_DMA_INFIFO_CNT_L3_CH3				H264_DMA_INFIFO_EMPTY_L3_CH3				H264_DMA_INFIFO_FULL_L3_CH3				(reserved)				H264_DMA_INFIFO_CNT_L1_CH3				H264_DMA_INFIFO_EMPTY_L1_CH3				H264_DMA_INFIFO_FULL_L1_CH3				H264_DMA_INFIFO_CNT_L2_CH3				H264_DMA_INFIFO_EMPTY_L2_CH3				H264_DMA_INFIFO_FULL_L2_CH3			
31													19	18	17	16	15					13	12					8	7	6	5					2	1	0													
0 0 0 0 0 0 0 0 0 0 0 0												0x0		1	0	0 0 0		0x0				1		0	0x0		1		0	Reset																					

Reset

H264_DMA_INFIFO_FULL_L2_CH3 Represents whether l2 FIFO of RX channel 3 is full.

0: Not Full

1: Full

(RO)

H264_DMA_INFIFO_EMPTY_L2_CH3 Represents whether l2 FIFO of RX channel 3 is empty.

0: Not empty

1: Empty

(RO)

H264_DMA_INFIFO_CNT_L2_CH3 Represents the data quantity in l2 FIFO for RX channel 3. Measurement unit: 16 bytes. (RO)**H264_DMA_INFIFO_FULL_L1_CH3** Represents whether l1 FIFO of RX channel 3 is full.

0: Not Full

1: Full

(RO)

H264_DMA_INFIFO_EMPTY_L1_CH3 Represents whether l1 FIFO of RX channel 3 is empty.

0: Not empty

1: Empty

(RO)

H264_DMA_INFIFO_CNT_L1_CH3 Represents the data quantity in l1 FIFO for RX channel 3. Measurement unit: byte. (RO)**H264_DMA_INFIFO_FULL_L3_CH3** Represents whether l3 FIFO of RX channel 3 is full.

0: Not Full

1: Full

(RO)

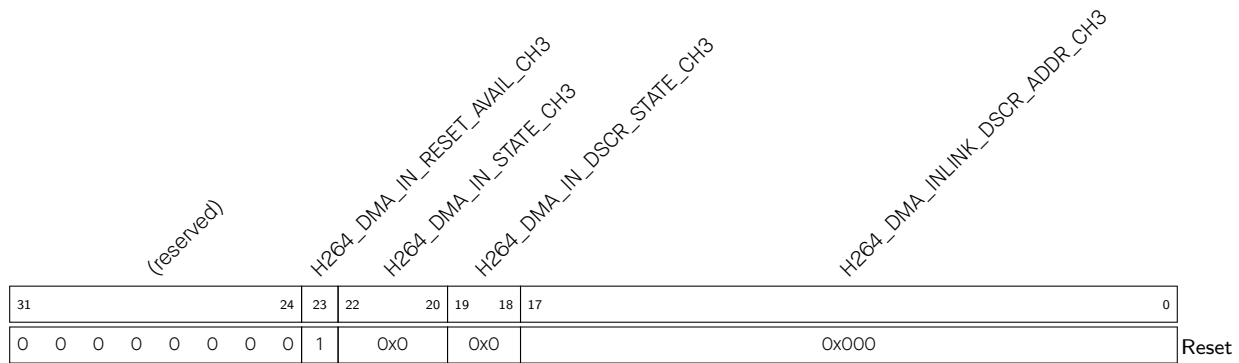
H264_DMA_INFIFO_EMPTY_L3_CH3 Represents whether l3 FIFO of RX channel 3 is empty.

0: Not empty

1: Empty

(RO)

H264_DMA_INFIFO_CNT_L3_CH3 Represents the data quantity in l3 FIFO for RX channel 3. Measurement unit: 8 bytes. (RO)

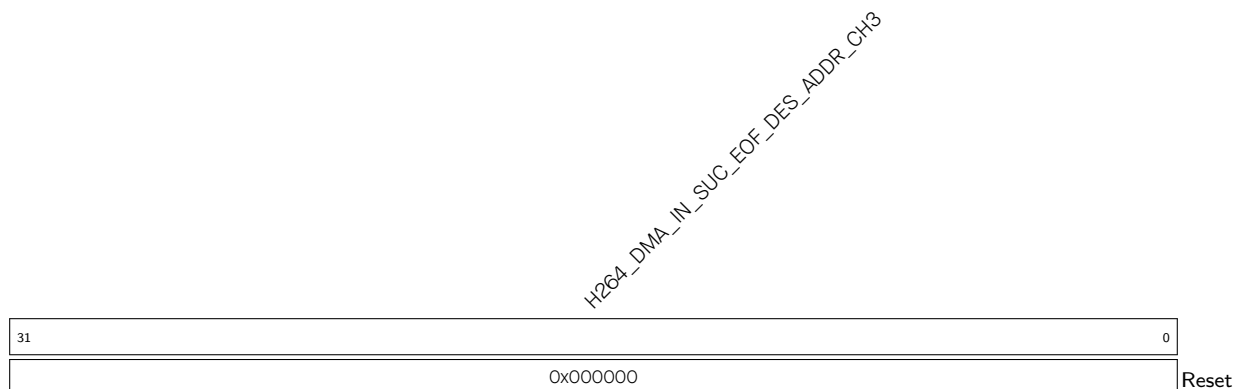
Register 39.261. H264_DMA_IN_STATE_CH3_REG (0x0824)

H264_DMA_INLINK_DSCR_ADDR_CH3 Represents the current inlink descriptor's address for RX channel 3. (RO)

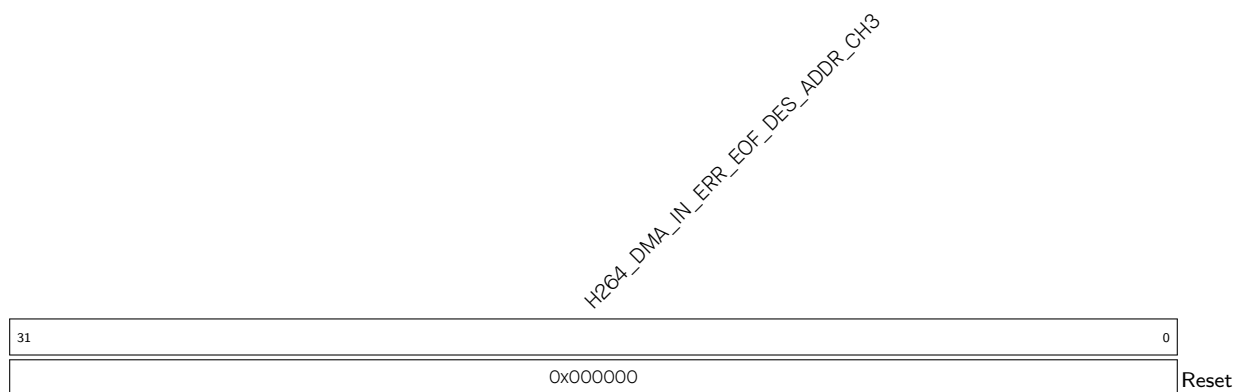
H264_DMA_IN_DSCR_STATE_CH3 Represents the current state of the descriptor state machine for RX channel 3. (RO)

H264_DMA_IN_STATE_CH3 Represents the current control module state machine state for RX channel 3. (RO)

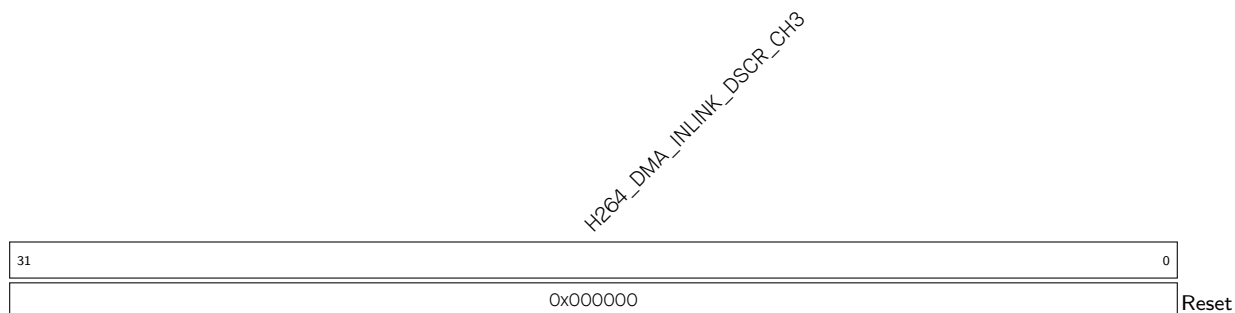
H264_DMA_IN_RESET_AVAIL_CH3 Represents whether it is safe to reset the channel.
 0: Unsafe
 1: Safe
 (RO)

Register 39.262. H264_DMA_IN_SUC_EOF_DES_ADDR_CH3_REG (0x0828)

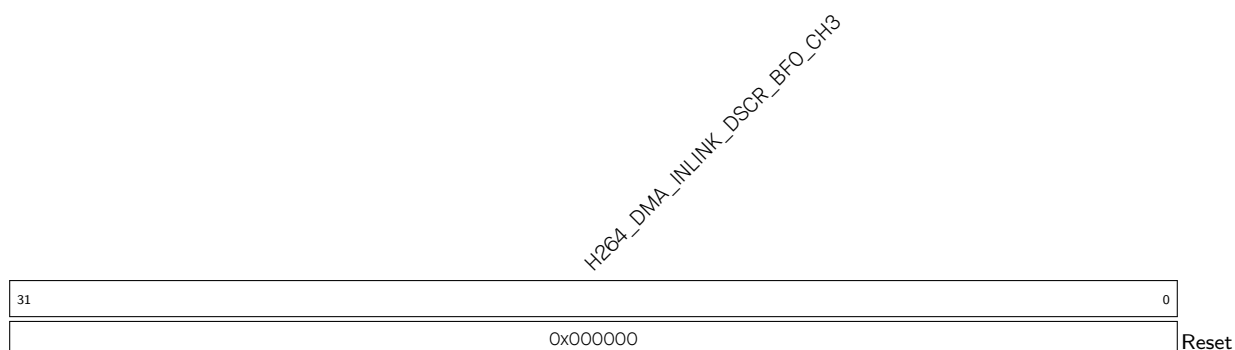
H264_DMA_IN_SUC_EOF_DES_ADDR_CH3 Represents the address of the inlink descriptor when the EOF bit in this descriptor is 1. (RO)

Register 39.263. H264_DMA_IN_ERR_EOF_DES_ADDR_CH3_REG (0x082C)

H264_DMA_IN_ERR_EOF_DES_ADDR_CH3 Represents the address of the inlink descriptor when there are errors during the current data reception. (RO)

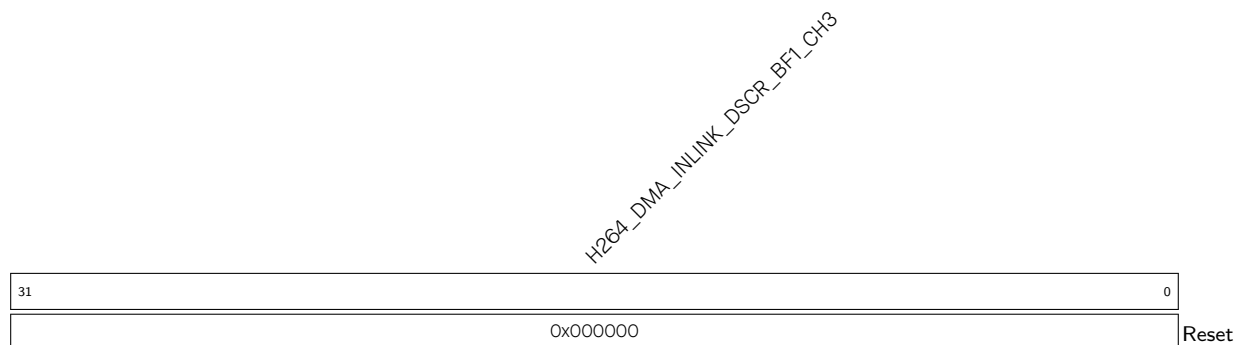
Register 39.264. H264_DMA_IN_DSCR_CH3_REG (0x0830)

H264_DMA_INLINK_DSCR_CH3 Represents the address of the next descriptor x+1 pointed by the current descriptor that is pre-read. (RO)

Register 39.265. H264_DMA_IN_DSCR_BFO_CH3_REG (0x0834)

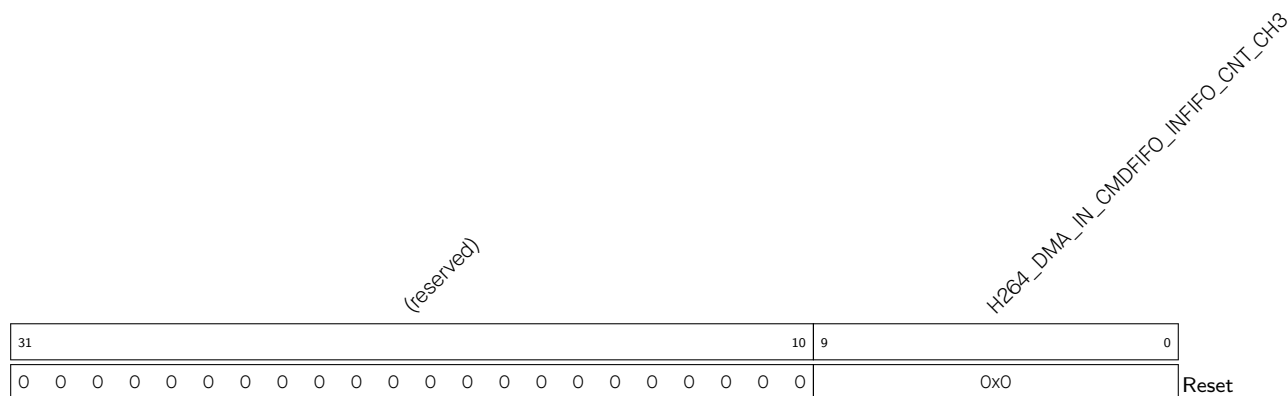
H264_DMA_INLINK_DSCR_BFO_CH3 Represents the address of the current descriptor x that is pre-read. (RO)

Register 39.266. H264_DMA_IN_DSCR_BF1_CH3_REG (0x0838)



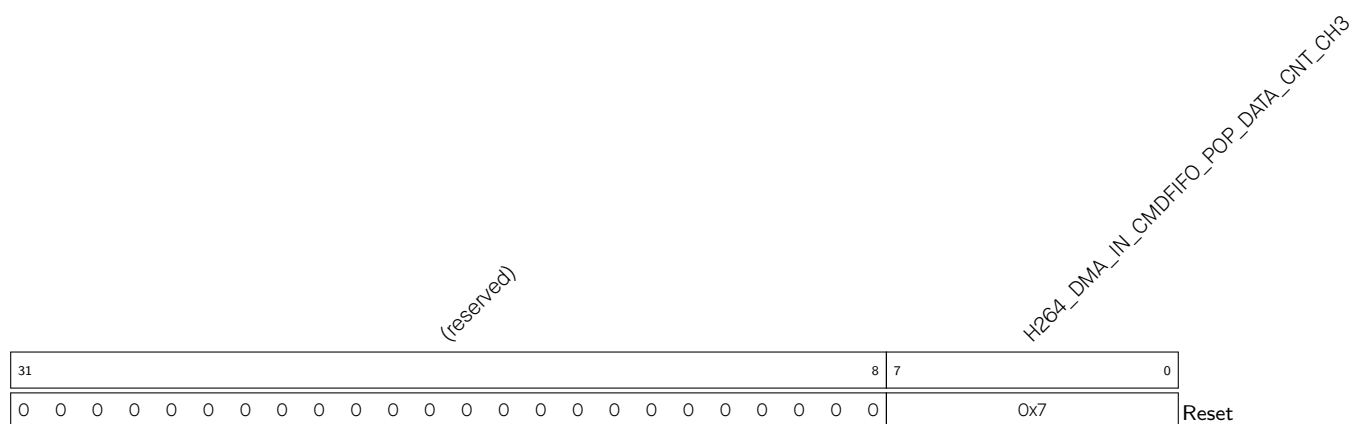
H264_DMA_INLINK_DSCR_BF1_CH3 Represents the address of the previous descriptor x-1 that is pre-read. (RO)

Register 39.267. H264_DMA_IN_FIFO_CNT_CH3_REG (0x0880)



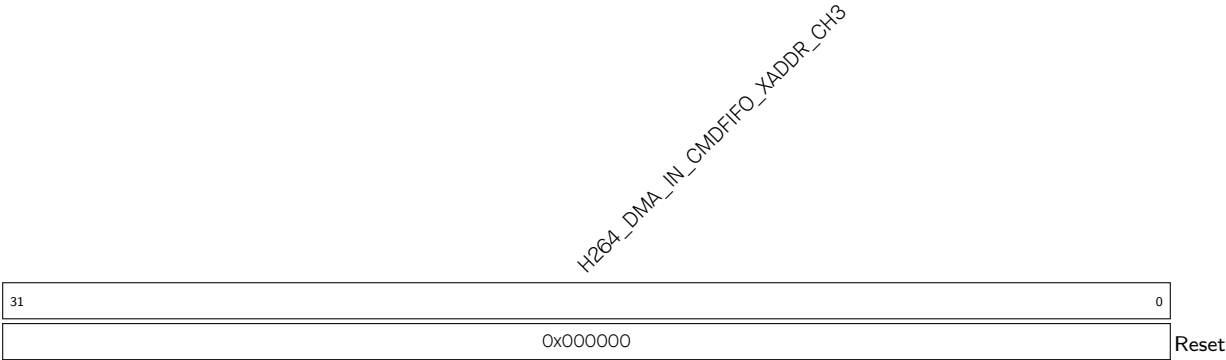
H264_DMA_IN_CMDFIFO_INFIFO_CNT_CH3 Represents the remaining request data in FIFO. (RO)

Register 39.268. H264_DMA_IN_POP_DATA_CNT_CH3_REG (0x0884)



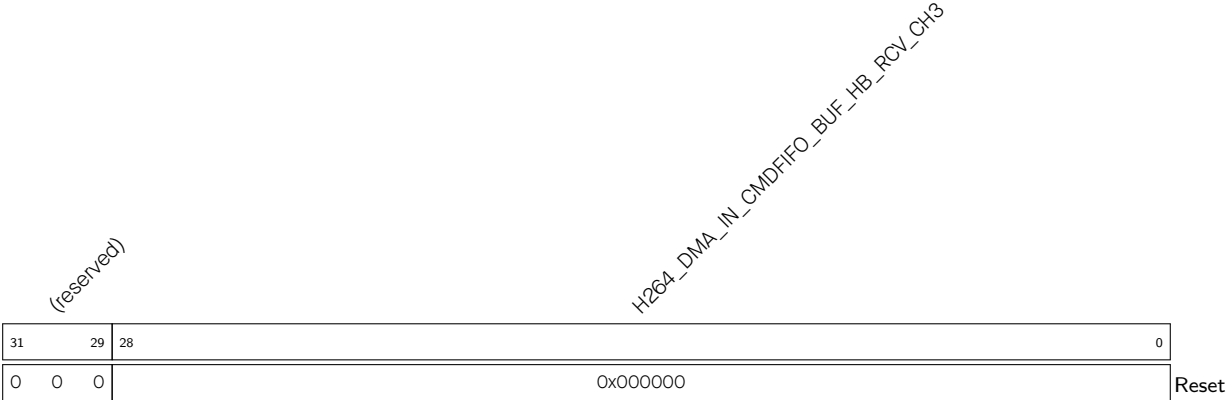
H264_DMA_IN_CMDFIFO_POP_DATA_CNT_CH3 Represents the data quantity in the current AXI request. (RO)

Register 39.269. H264_DMA_IN_XADDR_CH3_REG (0x0888)



H264_DMA_IN_CMDFIFO_XADDR_CH3 Represents the address of the current AXI request. (RO)

Register 39.270. H264_DMA_IN_BUF_HB_RCV_CH3_REG (0x088C)



H264_DMA_IN_CMDFIFO_BUF_HB_RCV_CH3 Represents the data quantity that has been requested. (RO)

Register 39.271. H264_DMA_INFIFO_STATUS_CH4_REG (0x0914)

(reserved)												H264_DMA_INFIFO_CNT_L3_CH4				H264_DMA_INFIFO_EMPTY_L3_CH4				H264_DMA_INFIFO_FULL_L3_CH4				(reserved)				H264_DMA_INFIFO_CNT_L1_CH4				H264_DMA_INFIFO_EMPTY_L1_CH4				H264_DMA_INFIFO_FULL_L1_CH4				H264_DMA_INFIFO_CNT_L2_CH4				H264_DMA_INFIFO_EMPTY_L2_CH4				H264_DMA_INFIFO_FULL_L2_CH4																								
31												20												19	18	17	16	15	13	12	8												7	6	5	2												1	0													
0												0												0	0	0	0	0	0	0	0	0	0x0												1	0	0x0												1	0	Reset											

H264_DMA_INFIFO_FULL_L2_CH4 Represents whether l2 FIFO of RX channel 4 is full.

0: Not Full

1: Full

(RO)

H264_DMA_INFIFO_EMPTY_L2_CH4 Represents whether l2 FIFO of RX channel 4 is empty.

0: Not empty

1: Empty

(RO)

H264_DMA_INFIFO_CNT_L2_CH4 Represents the data quantity in l2 FIFO for RX channel 4. Measurement unit: 16 bytes. (RO)

H264_DMA_INFIFO_FULL_L1_CH4 Represents whether l1 FIFO of RX channel 4 is full.

0: Not Full

1: Full

(RO)

H264_DMA_INFIFO_EMPTY_L1_CH4 Represents whether l1 FIFO of RX channel 4 is empty.

0: Not empty

1: Empty

(RO)

H264_DMA_INFIFO_CNT_L1_CH4 Represents the data quantity in l1 FIFO for RX channel 4. Measurement unit: byte. (RO)

H264_DMA_INFIFO_FULL_L3_CH4 Represents whether l3 FIFO of RX channel 4 is full.

0: Not Full

1: Full

(RO)

H264_DMA_INFIFO_EMPTY_L3_CH4 Represents whether l3 FIFO of RX channel 4 is empty.

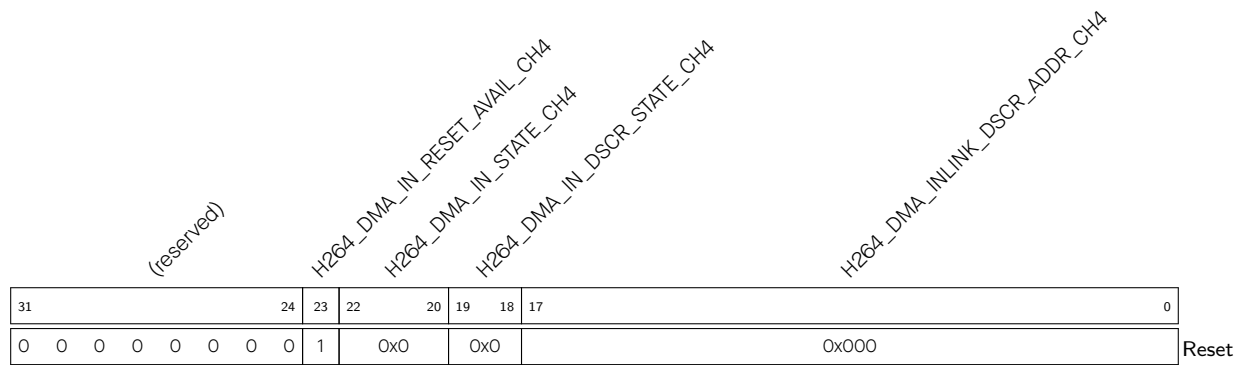
0: Not empty

1: Empty

(RO)

H264_DMA_INFIFO_CNT_L3_CH4 Represents the data quantity in l3 FIFO for RX channel 4. Measurement unit: 8 bytes. (RO)

Register 39.272. H264_DMA_IN_STATE_CH4_REG (0x0924)



H264_DMA_INLINK_DSCR_ADDR_CH4 Represents the current inlink descriptor's address for RX channel 4. (RO)

H264_DMA_IN_DSCR_STATE_CH4 Represents the current state of the descriptor state machine for RX channel 4. (RO)

H264_DMA_IN_STATE_CH4 Represents the current control module state machine state for RX channel 4. (RO)

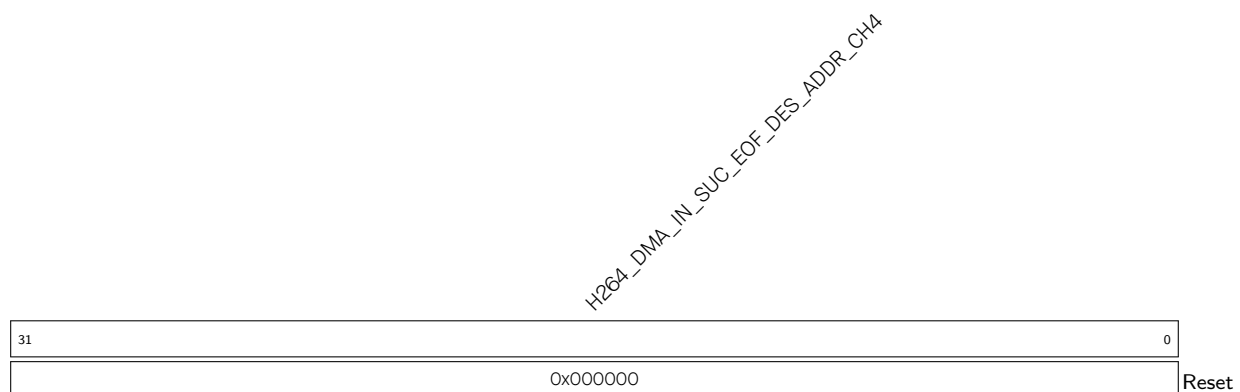
H264_DMA_IN_RESET_AVAIL_CH4 Represents whether it is safe to reset the channel.

0: Unsafe

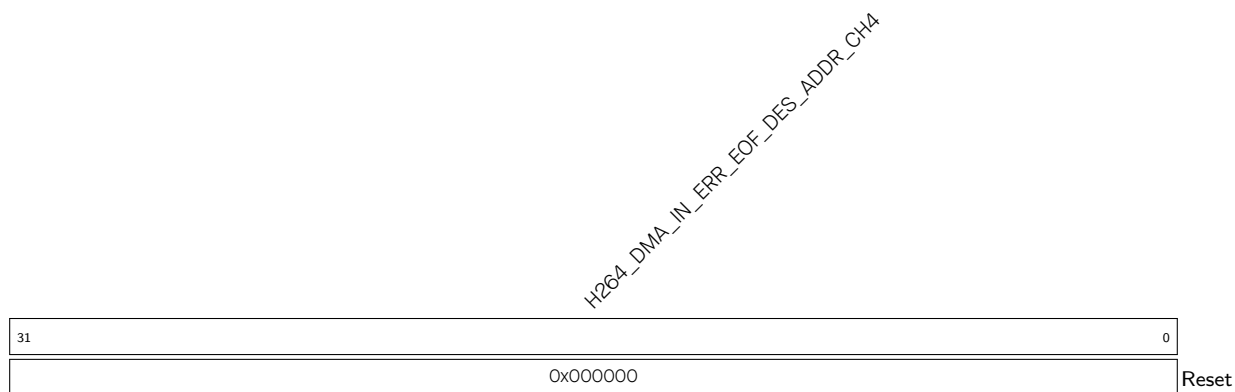
1: Safe

(RO)

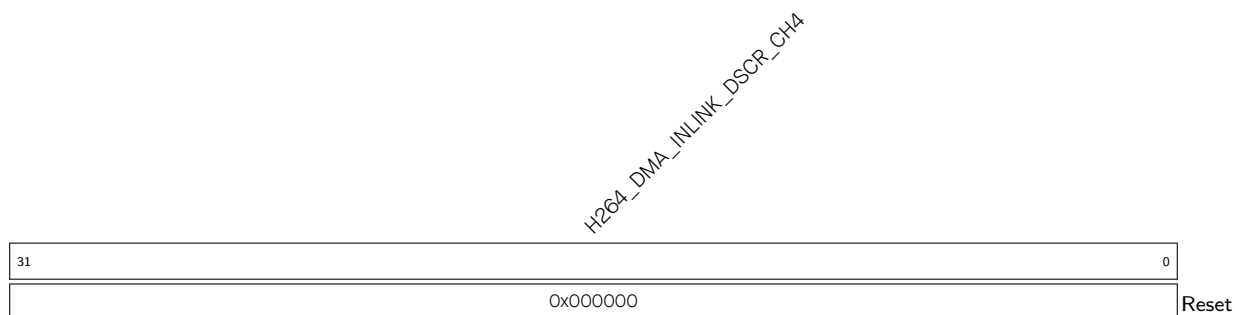
Register 39.273. H264_DMA_IN_SUC_EOF_DES_ADDR_CH4_REG (0x0928)



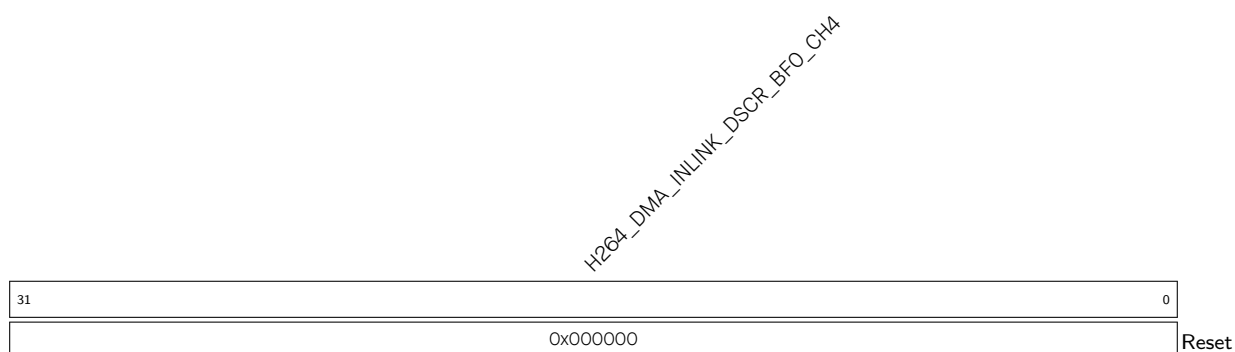
H264_DMA_IN_SUC_EOF_DES_ADDR_CH4 Represents the address of the inlink descriptor when the EOF bit in this descriptor is 1. (RO)

Register 39.274. H264_DMA_IN_ERR_EOF_DES_ADDR_CH4_REG (0x092C)

H264_DMA_IN_ERR_EOF_DES_ADDR_CH4 Represents the address of the inlink descriptor when there are errors during the current data reception. (RO)

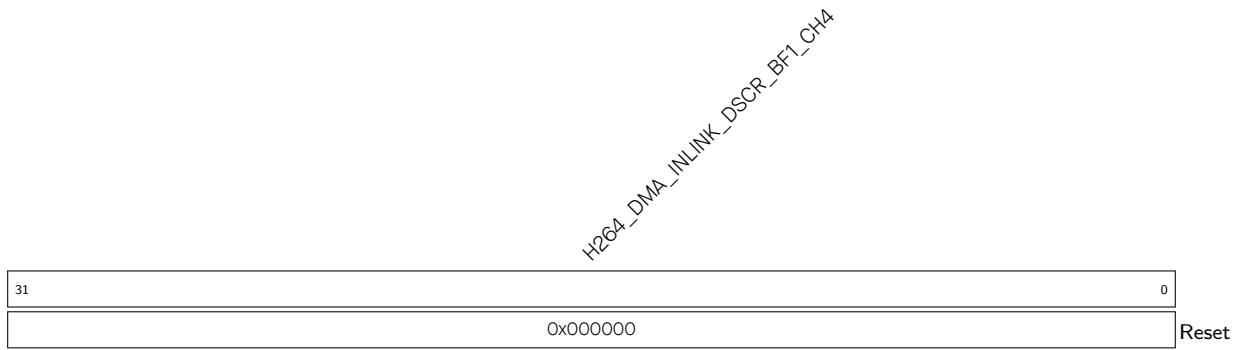
Register 39.275. H264_DMA_IN_DSCR_CH4_REG (0x0930)

H264_DMA_INLINK_DSCR_CH4 Represents the address of the next descriptor x+1 pointed by the current descriptor that is pre-read. (RO)

Register 39.276. H264_DMA_IN_DSCR_BFO_CH4_REG (0x0934)

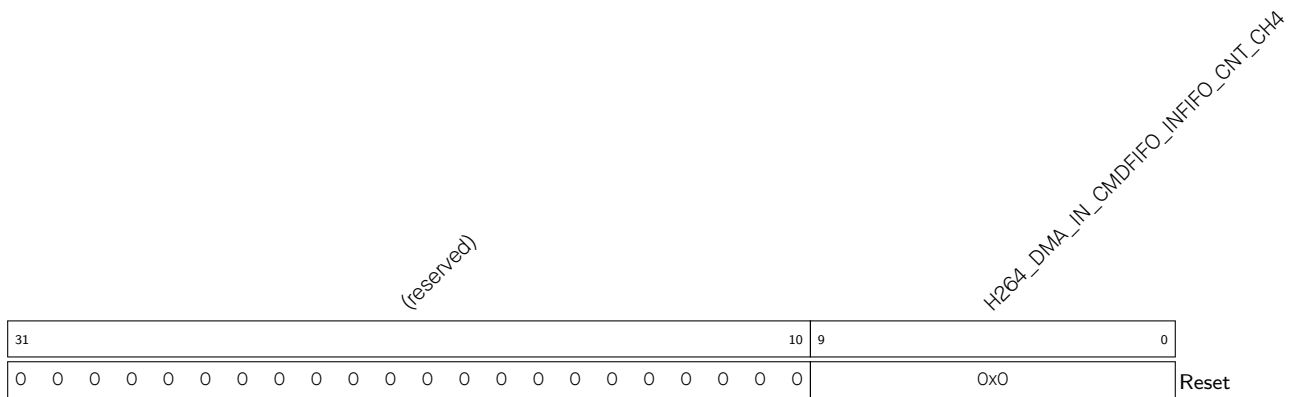
H264_DMA_INLINK_DSCR_BFO_CH4 Represents the address of the current descriptor x that is pre-read. (RO)

Register 39.277. H264_DMA_IN_DSCR_BF1_CH4_REG (0x0938)



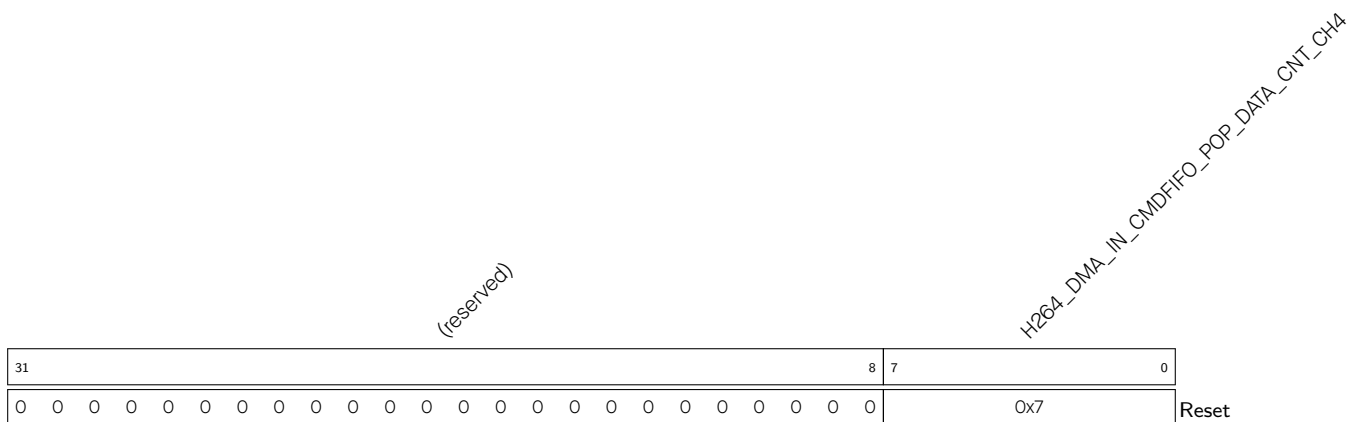
H264_DMA_INLINK_DSCR_BF1_CH4 Represents the address of the previous descriptor x-1 that is pre-read. (RO)

Register 39.278. H264_DMA_IN_FIFO_CNT_CH4_REG (0x0980)



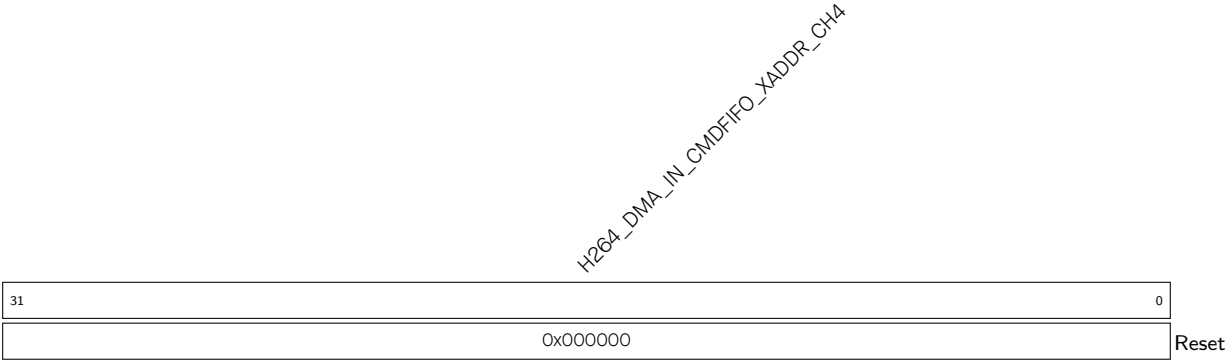
H264_DMA_IN_CMDFIFO_INFIFO_CNT_CH4 Represents the remaining request data in FIFO. (RO)

Register 39.279. H264_DMA_IN_POP_DATA_CNT_CH4_REG (0x0984)



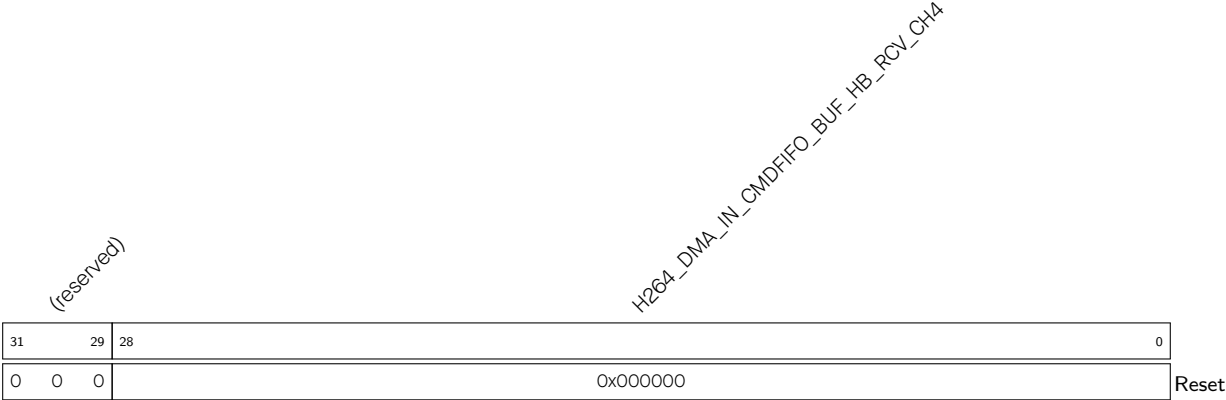
H264_DMA_IN_CMDFIFO_POP_DATA_CNT_CH4 Represents the data quantity in the current AXI request. (RO)

Register 39.280. H264_DMA_IN_XADDR_CH4_REG (0x0988)



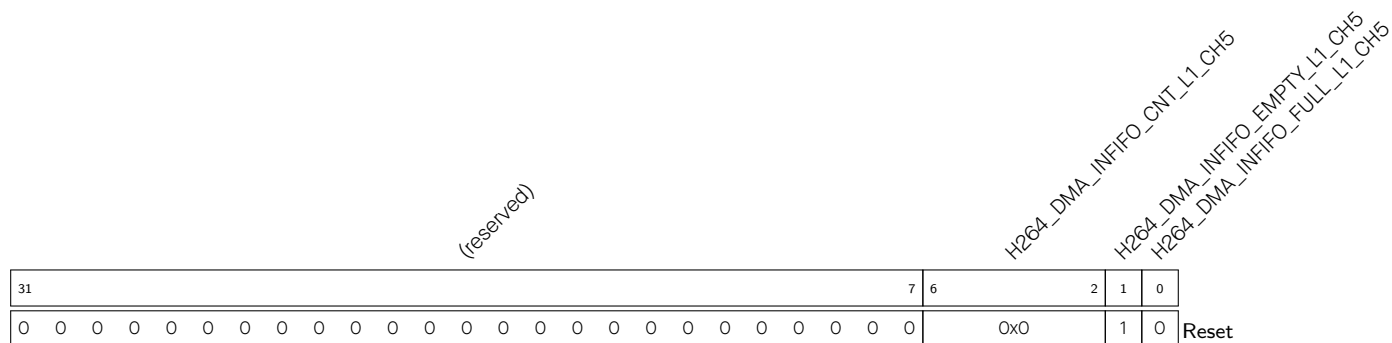
H264_DMA_IN_CMDFIFO_XADDR_CH4 Represents the address of the current AXI request. (RO)

Register 39.281. H264_DMA_IN_BUF_HB_RCV_CH4_REG (0x098C)



H264_DMA_IN_CMDFIFO_BUF_HB_RCV_CH4 Represents the data quantity that has been requested. (RO)

Register 39.282. H264_DMA_INFIFO_STATUS_CH5_REG (0x0A20)



H264_DMA_INFIFO_FULL_L1_CH5 Represents whether l1 FIFO of RX channel 5 is full.

0: Not Full

1: Full

(RO)

H264_DMA_INFIFO_EMPTY_L1_CH5 Represents whether l1 FIFO of RX channel 5 is empty.

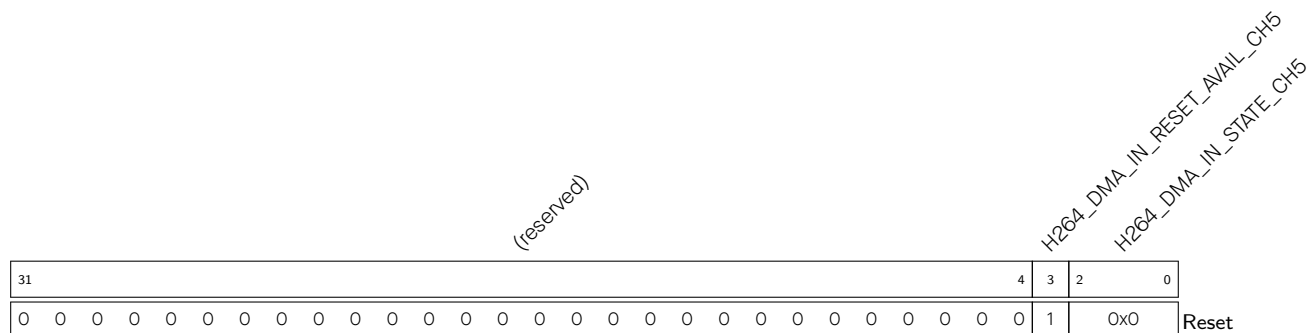
0: Not empty

1: Empty

(RO)

H264_DMA_INFIFO_CNT_L1_CH5 Represents the data quantity in l1 FIFO for RX channel 5. Measurement unit: byte. (RO)

Register 39.283. H264_DMA_IN_STATE_CH5_REG (0x0A28)



H264_DMA_IN_STATE_CH5 Represents the current control module state machine state for RX channel 5. (RO)

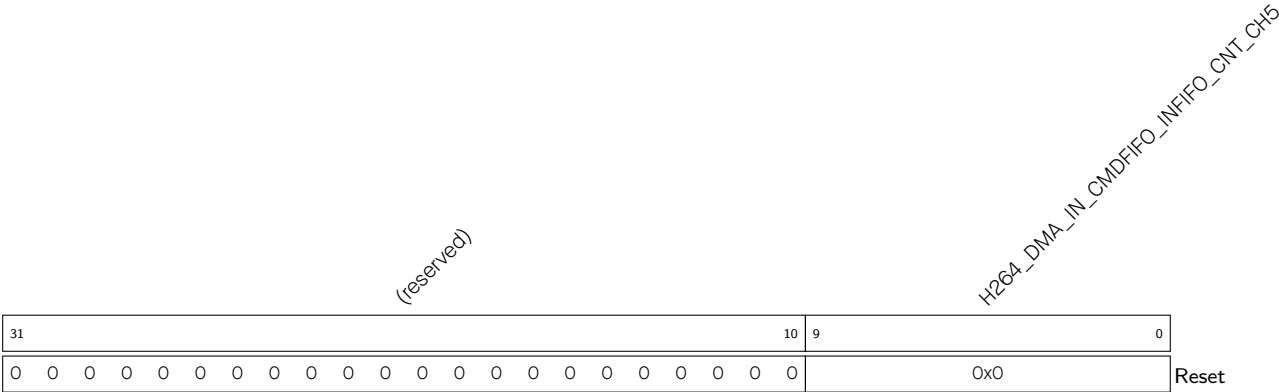
H264_DMA_IN_RESET_AVAIL_CH5 Represents whether it is safe to reset the channel.

0: Unsafe

1: Safe

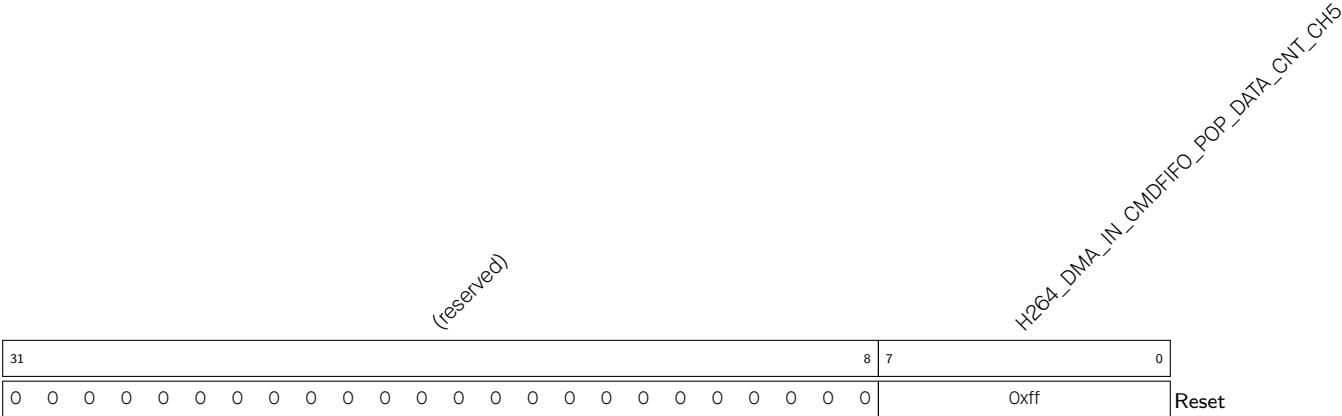
(RO)

Register 39.284. H264_DMA_IN_FIFO_CNT_CH5_REG (0x0A80)



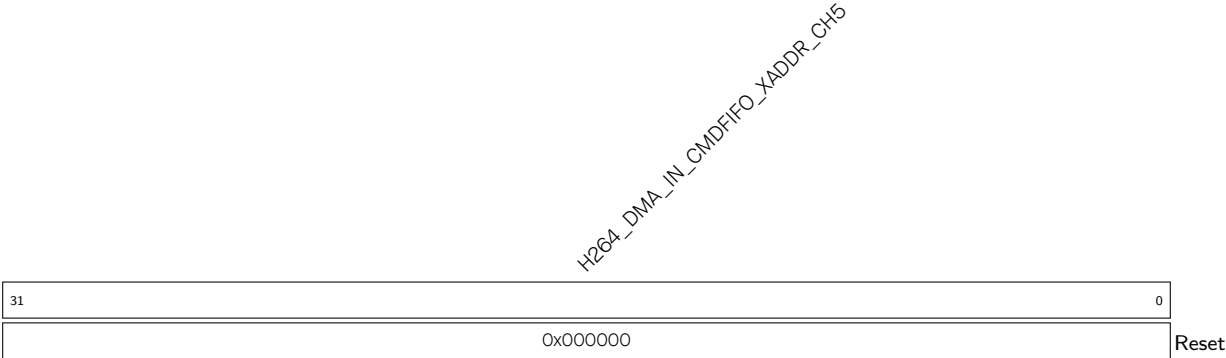
H264_DMA_IN_CMDFIFO_INFIFO_CNT_CH5 Represents the remaining request data in FIFO. (RO)

Register 39.285. H264_DMA_IN_POP_DATA_CNT_CH5_REG (0x0A84)



H264_DMA_IN_CMDFIFO_POP_DATA_CNT_CH5 Represents the data quantity in the current AXI request. (RO)

Register 39.286. H264_DMA_IN_XADDR_CH5_REG (0x0A88)



H264_DMA_IN_CMDFIFO_XADDR_CH5 Represents the address of the current AXI request. (RO)

Register 39.287. H264_DMA_IN_BUF_HB_RCV_CH5_REG (0x0A8C)

31	29	28											0
0	0	0	0x000000										Reset

H264_DMA_IN_CMDFIFO_BUF_HB_RCV_CH5 Represents the data quantity that has been requested. (RO)

Register 39.288. H264_DMA_INTER_AXI_ERR_REG (0x0B00)

[illegible]

H264_DMA_INTER_RID_ERR_CNT	Represents the count of errors in the internal memory ID for AXI read operations. (RO)
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H264_DMA_INTER_RRESP_ERR_CNT Represents the count of errors in the response for internal memory during AXI read operations. (RO)

H264_DMA_INTER_WRESP_ERR_CNT	Represents the count of AXI write internal memory resp err.
(RO)	

H264_DMA_INTER_RD_FIFO_CNT	Represents the count of cmd FIFO remain cmd in AXI read internal memory module. (RO)
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H264_DMA_INTER_RD_BAK_FIFO_CNT Represents the count of backup cmd FIFO remain cmd in AXI read internal memory module.(RO)

H264_DMA_INTER_WR_FIFO_CNT	Represents the count of cmd FIFO remain cmd in AXI write internal memory module. (RO)
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H264_DMA_INTER_WR_BAK_FIFO_CNT Represents the count of backup cmd FIFO remain cmd in AXI write internal memory module. (RO)

Register 39.289. H264_DMA_EXTER_AXI_ERR_REG (0x0B04)

[illegible]

H264_DMA_EXTER_RID_ERR_CNT	Represents the count of errors in the external memory ID for AXI read. (RO)
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H264_DMA_EXTER_RRESP_ERR_CNT Represents the count of errors in the response for external memory during AXI read operations. (RO)

H264_DMA_EXTER_WRESP_ERR_CNT Represents the count of AXI write external memory resp err.
(RO)

H264_DMA_EXTER_RD_FIFO_CNT	Represents the count of cmd FIFO remain cmd in AXI read external memory module. (RO)
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H264_DMA_EXTER_RD_BAK_FIFO_CNT Represents the count of backup cmd FIFO remain cmd in AXI read external memory module.(RO)

H264_DMA_EXTER_WR_FIFO_CNT	Represents the count of cmd FIFO remain cmd in AXI write external memory module. (RO)
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H264_DMA_EXTER_WR_BAK_FIFO_CNT Represents the count of backup cmd FIFO remain cmd in AXI write external memory module. (RO)

Register 39.290. H264_DMA_RX_CHO_COUNTER_REG (0x0B54)

(reserved)										H264_DMA_RX_CHO_CNT																							
31										23	22																						0
0 0 0 0 0 0 0 0 0 0										0x0000																						Reset	

H264_DMA_RX_CH0_CNT Represents the RX channel 0 counter register. (RO)

Register 39.291. H264_DMA_RX_CH1_COUNTER_REG (0x0B58)

(reserved)											H264_DMA_RX_CH1_CNT																					
31											21																					0
0	0	0	0	0	0	0	0	0	0	0	0x0000																				Reset	

H264_DMA_RX_CH1_CNT Represents the RX channel 1 counter register. (RO)

Register 39.292. H264_DMA_RX_CH2_COUNTER_REG (0x0B5C)

(reserved)	11	10	0
0 0			0x0

Reset

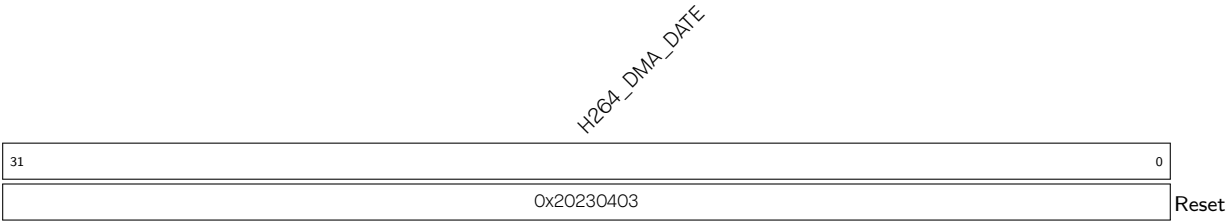
H264_DMA_RX_CH2_CNT Represents the RX channel 2 counter register. (RO)

Register 39.293. H264_DMA_RX_CH5_COUNTER_REG (0x0B60)

(reserved)																H264_DMA_RX_CH5_ONT																																																																																																																																																															
31																17																16																0																																																																																																																															
0																0																0																0																0																0																0																0																0																0x000																Reset															

H264_DMA_RX_CH5_CNT Represents the RX channel 5 counter register (RO)

Register 39.294. H264_DMA_DATE_REG (0x0B3C)



H264_DMA_DATE Version control register. (R/W)